Fundamental Theorem of Matrix Product States

A Formalization Blueprint

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Chapter 1

Introduction

This blueprint describes the fundamental theorem for periodic, translation-invariant matrix product vectors. The basic object is a family of matrices $\{A^i\}$, viewed as a tensor that generates vectors $|V^{(N)}(A)\rangle$ on chains of length N.

Chapters 2 and 4 describe the algebraic core. Chapter 2 introduces word evaluation, MPV coefficients, gauge equivalence, injectivity, blocking, overlap, and the block-injective canonical-form data. Chapter 4 proves the single-block fundamental theorem by a purely algebraic argument through trace pairings, linear extension, simplicity of matrix algebras, and Skolem–Noether.

The later chapters develop the spectral and canonical-form machinery needed for the full multi-block theorem. Chapter 5 studies positive maps and quantum channels. Chapters 8 and 9 supply Perron–Frobenius theory and overlap decay. Chapters 10 and 11 connect blocking, primitivity, irreducibility, and the normal/canonical-form reductions. Chapters 12 and 13 carry out block permutation and BNT theory; Chapter 14 proves the Fundamental Theorem.

Chapter 2

Matrix Product Vectors

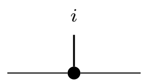
Fix a physical dimension d and a bond dimension D. This chapter studies the tensors that generate matrix product vector (MPV) families. All tensors here are translation-invariant with periodic boundary conditions (PBC).

2.1 Basic definitions

Definition 2.1 (MPS tensor)**.** A (translation-invariant, PBC) *MPS tensor* with physical dimension d and bond dimension D is a collection of matrices

$$\{A^i\}_{i=0}^{d-1}, \qquad A^i \in M_D(\mathbb{C}),$$

indexed by a physical index $i \in \{0, \dots, d-1\}$. Such a tensor defines an MPV family. Diagrammatically,

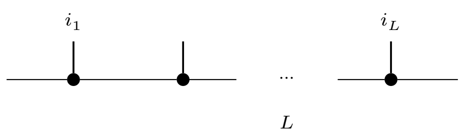


The black node denotes the tensor A, the horizontal legs are virtual, and the upper leg is the physical index i.

Definition 2.2 (Word evaluation)**.** Given a word $w = (i_1, \dots, i_L) \in \{0, \dots, d-1\}^L$, the *word evaluation* is the matrix product

$$A^w \ := \ A^{i_1} A^{i_2} \cdots A^{i_L} \ \in \ M_D(\mathbb{C}),$$

with the convention that the empty word evaluates to the identity: $A^{\emptyset} = \mathbb{1}_D$. In tensor-network notation,



in which each black node denotes the same local tensor A, the virtual legs remain open, and the physical legs record the word $(i_1, \dots, i_L)$.

Lemma 2.3 (Multiplicativity of word evaluation)**.** *For any words w_1, w_2, $A^{w_1 \cdot w_2} = A^{w_1} A^{w_2}$.*

Proof. Immediate from the definition of A^w. □

Definition 2.4 (Matrix product vector)**.** The *matrix product vector* (MPV) of a tensor A at system size N is the vector

$$|V^{(N)}(A)\rangle = \sum_{i_1,\dots,i_N=0}^{d-1} \operatorname{tr}(A^{i_1}A^{i_2}\cdots A^{i_N})|i_1,\dots,i_N\rangle \in (\mathbb{C}^d)^{\otimes N}.$$

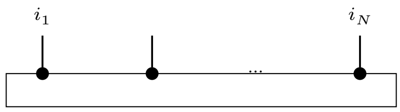
Equivalently, for a configuration $\sigma = (i_1,\dots,i_N) \in \{0,\dots,d-1\}^N$, we write

$$V^{(N)}(A)_\sigma := \operatorname{tr}(A^{i_1}A^{i_2}\cdots A^{i_N}).$$

The coefficient function $\sigma \mapsto V^{(N)}(A)_\sigma$ gives the components; the displayed ket is the corresponding vector in $(\mathbb{C}^d)^{\otimes N}$. For a general word w, we also write

$$c_w(A) := \operatorname{tr}(A^w),$$

The *MPV family* generated by A is the collection $\mathcal{V}(A) = \{|V^{(N)}(A)\rangle\}_{N\geq 1}$. The coefficient $V^{(N)}(A)_\sigma$ is the periodic contraction

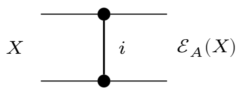


of N copies of the local tensor A, with the outer virtual legs closed by the trace.

Definition 2.5 (Transfer map)**.** The *transfer map* associated to a tensor A is the linear map $\mathcal{E}_A : M_D(\mathbb{C}) \to M_D(\mathbb{C})$ defined by

$$\mathcal{E}_A(X) \;=\; \sum_{i=0}^{d-1} A^i\, X\, (A^i)^\dagger.$$

Diagrammatically, the transfer map is the double-layer contraction



in which the upper node denotes A, the lower node denotes $A^\dagger$, and the physical index is summed over between them.

Remark 2.6. The transfer map is automatically positive: if $X \geq 0$ then $\mathcal{E}_A(X) \geq 0$, because each summand $A^iX(A^i)^\dagger$ is positive semidefinite. The abstract positive-map framework appears in Chapter 5.

2.2 Periodic-chain aliases

Definition 2.7 (Periodic-chain tensor alias)**.** For a fixed period m, a *periodic-chain tensor* is a site-dependent tensor family indexed by $\{0,\dots,m-1\}$, with local physical dimension d and bond dimension D.

Definition 2.8 (Periodic same-state and gauge-equivalence relations)**.** For periodic-chain tensors A, B at fixed period m:

- equality of the induced periodic-chain states;
- cyclic gauge equivalence.

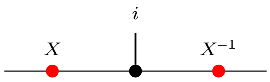
Lemma 2.9 (Equivalence structures for periodic relations)**.** *Both periodic relations are equivalence relations (reflexive, symmetric, transitive).*

2.3 Gauge equivalence and same MPV

Definition 2.10 (Gauge equivalence)**.** Two tensors A and B of the same bond dimension D are *gauge equivalent* if there exists an invertible matrix $X \in \mathrm{GL}_D(\mathbb{C})$ such that

$$B^i \;=\; X\, A^i\, X^{-1} \qquad \text{for all } i \in \{0, \dots, d{-}1\}.$$

In tensor-network notation,



the black node denotes the tensor named in the surrounding formula, the upper leg is the physical index i, and the two red side nodes denote the gauge matrices acting on the virtual legs.

Definition 2.11 (Same MPV)**.** Two tensors A and B (of the same bond dimension) *generate the same MPV family* if $|V^{(N)}(A)\rangle = |V^{(N)}(B)\rangle$ for every system size N.

Definition 2.12 (Same MPV — different bond dimensions)**.** For tensors A and B of possibly different bond dimensions D_1 and D_2, we write $\mathcal{V}(A) = \mathcal{V}(B)$ if $|V^{(N)}(A)\rangle = |V^{(N)}(B)\rangle$ for every N.

Remark 2.13. Definition 2.11 is the same-bond-dimension specialization of Definition 2.12. We keep both because later results use both the same-dimension and different-dimension forms.

Definition 2.14 (Proportional MPV)**.** Two tensors A and B (of possibly different bond dimensions) generate *proportional MPV families* if for every N there exists $c_N \in \mathbb{C}$ such that $|V^{(N)}(A)\rangle = c_N\, |V^{(N)}(B)\rangle$.

Lemma 2.15 (Scaling of word evaluation)**.** *Let ζA denote the tensor with matrices $\{\zeta\, A^i\}_{i=0}^{d-1}$. Then for any word w,*

$$(\zeta A)^w \;=\; \zeta^{|w|}\, A^w,$$

where $|w|$ is the length of w.

Proof. Induction on w: the base case gives $\zeta^0 = 1$, and each step picks up one factor of ζ. □

Definition 2.16 (Gauge-phase equivalence)**.** Two tensors A and B are *gauge-phase equivalent* if there exist $X \in \mathrm{GL}_D(\mathbb{C})$ and $\zeta \in \mathbb{C} \setminus \{0\}$ such that

$$B^i \;=\; \zeta\, X\, A^i\, X^{-1} \qquad \text{for all } i.$$

Remark 2.17. After spectral-radius normalization one may restrict ζ to a unit-modulus phase $e^{i\phi}$. We allow an arbitrary nonzero scalar because later canonical-form statements keep the block-scaling factors explicit.

Lemma 2.18 (Gauge covariance of word evaluation)**.** *If $B^i = X\,A^i\,X^{-1}$ for all i, then for every word w,*

$$B^w \;=\; X\,A^w\,X^{-1}.$$

Proof. Induction on the word length, using $XX^{-1} = \mathbb{1}$. ☐

Theorem 2.19 (Gauge invariance of MPVs)**.** *If A and B are gauge equivalent, then they generate the same MPV family.*

Proof. Let $B^i = XA^iX^{-1}$. By Lemma 2.18, $B^w = X\,A^w\,X^{-1}$ for every word w. Then $\mathrm{tr}(B^w) = \mathrm{tr}(X\,A^w\,X^{-1}) = \mathrm{tr}(A^w)$ by cyclicity of the trace. ☐

2.4 Injectivity and normality

Definition 2.20 (Injectivity)**.** A tensor A is *injective* if its matrices span the full matrix algebra: $\mathrm{span}_{\mathbb{C}}\{A^i : i = 0, \dots, d-1\} = M_D(\mathbb{C})$.

Definition 2.21 (L-block injectivity)**.** A tensor A is *L-block injective* if $\mathrm{span}_{\mathbb{C}}\{A^{i_1} \cdots A^{i_L} : (i_1, \dots, i_L) \in \{0, \dots, d-1\}^L\} = M_D(\mathbb{C})$. An injective tensor is 1-block injective.

Definition 2.22 (Normal tensor)**.** A tensor A is *normal* [CPGSV21] if there exists a finite L_0 such that A is L_0-block injective.

Remark 2.23*.* The algebraic characterization of normality in Definition 2.22 — eventual full matrix span, equivalently eventual block injectivity — is equivalent to the spectral characterization used in [CPGSV21, Definition IV.1]: after TP normalization, the transfer map $\mathcal{E}_A$ is a primitive channel, equivalently irreducible with trivial peripheral spectrum. This equivalence is proved in [SPGWC10, Proposition 3]. In what follows, the directions of the equivalence are supplied by Theorem 10.49, Theorem 10.59, and Theorem 11.28. In particular, every subsequent use of "normal" here refers to this single notion; Definition 11.30 employs the spectral formulation, but it describes the same class of tensors.

2.5 Canonical form

Definition 2.24 (Block-injective canonical form)**.** Here, a *block-injective canonical form* for a tensor consists of:

1. a number of blocks $r \geq 1$,
2. bond dimensions $D_1, \dots, D_r$,
3. injective tensors $\{A_k^i\}_{i=0}^{d-1}$ with $A_k^i \in M_{D_k}(\mathbb{C})$ for each block $k \in \{1, \dots, r\}$,
4. scaling factors $\mu_1, \dots, \mu_r \in \mathbb{C}$.

The associated tensor is the block-diagonal tensor

$$A^i \;=\; \bigoplus_{k=1}^{r} \mu_k\, A_k^i.$$

This is the block-injective version of the canonical-form data; see [PGVWC07] and [CPGSV21].

Remark 2.25. Here we use the block-injective (biCF-style) canonical form: each block is injective in the sense of Definition 2.20. This is stronger than the NT-block canonical form of [CPGSV21], where the blocks are only normal in the spectral sense. Chapter 11 treats the normal and periodic canonical forms needed for the general multi-block theory, whereas the present section isolates the stronger block-injective form used in the injective and block-injective arguments.

Definition 2.26 (Block-diagonal tensor from blocks)**.** Given r blocks with bond dimensions $(D_k)_{k=1}^r$, per-block tensors $\{A_k^i\}$, and scaling factors $(\mu_k)_{k=1}^r$, the associated *block-diagonal tensor* is the tensor of total bond dimension $D = \sum_{k=1}^r D_k$ defined by

$$A^i \;=\; \bigoplus_{k=1}^{r} \mu_k\, A_k^i.$$

Theorem 2.27 (MPV decomposition)**.** *The MPV of a block-diagonal tensor decomposes as*

$$|V^{(N)}(A)\rangle \;=\; \sum_{k=1}^{r} \mu_k^N\, |V^{(N)}(A_k)\rangle.$$

Equivalently, for each configuration σ one has

$$V^{(N)}(A)_\sigma \;=\; \sum_{k=1}^{r} \mu_k^N\, V^{(N)}(A_k)_\sigma.$$

Proof. For each physical index i, the block-diagonal tensor $A^i = \bigoplus_k \mu_k A_k^i$ is a block-diagonal matrix. By Lemma 2.15, the word evaluation of each block picks up a factor $\mu_k^{|w|}$, and the full word evaluation is block-diagonal with blocks $\mu_k^N A_k^w$. Taking the trace of a block-diagonal matrix gives the sum of the block traces: $\mathrm{tr}(A^w) = \sum_k \mu_k^N\, \mathrm{tr}(A_k^w)$. □

2.6 Blocking

Definition 2.28 (Blocked tensor)**.** Given a tensor A and a blocking length L, the L*-blocked tensor* is the tensor with physical index set $\{0, \dots, d-1\}^L$ (hence physical dimension d^L) and the same bond dimension D, whose matrices are indexed by words $(i_1, \dots, i_L) \in \{0, \dots, d-1\}^L$:

$$(A^{[L]})^{(i_1,\dots,i_L)} \;=\; A^{i_1} A^{i_2} \cdots A^{i_L}.$$

Lemma 2.29 (Blocked word evaluation)**.** *If $w = (J_1, \dots, J_m)$ is a word in the blocked alphabet, where each J_k is an L-tuple in $\{0, \dots, d-1\}^L$, then concatenating the block words gives a word $\widetilde{w}$ of length mL in the original alphabet and one has*

$$(A^{[L]})^w = A^{\widetilde{w}}.$$

Proof. Induct on the blocked word w and split off the first block word. Lemma 2.3 turns concatenation of blocks into multiplication of the corresponding evaluations. □

Lemma 2.30 (Blocking preserves MPV coefficients)**.** *For each basis state $\sigma = (\sigma_1, \dots, \sigma_N)$ of the blocked system, where each σ_k is an L-tuple in $\{0, \dots, d-1\}^L$, there exists a basis state $\tilde{\sigma} = (\tilde{\sigma}_1, \dots, \tilde{\sigma}_{NL})$ of the original system such that*

$$V^{(N)}(A^{[L]})_\sigma \;=\; V^{(NL)}(A)_{\tilde{\sigma}}.$$

Proof. Concatenate the block words and apply the blocked word evaluation lemma (Lemma 2.29). ☐

Theorem 2.31 (Blocking preserves same MPV)**.** *If A and B generate the same MPV family, then for any blocking length L, the blocked tensors $A^{[L]}$ and $B^{[L]}$ also generate the same MPV family.*

Proof. By Lemma 2.30, each MPV coefficient of the blocked tensor $A^{[L]}$ equals a coefficient of A at the corresponding concatenated configuration. The same holds for $B^{[L]}$ (with the same concatenated configuration). Since A and B agree on all configurations, so do $A^{[L]}$ and $B^{[L]}$. ☐

2.7 MPV overlap

Definition 2.32 (MPV as Hilbert-space vector)**.** We regard $|V^{(N)}(A)\rangle$ as an element of $\mathbb{C}^{d^N}$ indexed by configurations $\sigma \in \{0, \ldots, d-1\}^N$, with σ-component $V^{(N)}(A)_\sigma$ as in Definition 2.4. This allows us to use the standard Euclidean inner product on $\mathbb{C}^{d^N}$.

Definition 2.33 (MPV inner product)**.** The *MPV inner product* between two tensors A and B at system size N is

$$\langle V^{(N)}(A)|V^{(N)}(B)\rangle \;:=\; \sum_{\sigma\in\{0,\ldots,d-1\}^N} \overline{V^{(N)}(A)_\sigma}\, V^{(N)}(B)_\sigma,$$

i.e. the Euclidean inner product on $\mathbb{C}^{d^N}$ (conjugate-linear in the first argument).

Definition 2.34 (MPV overlap)**.** The *MPV overlap* between two tensors A (of bond dimension D_1) and B (of bond dimension D_2) at system size N is

$$O_{AB}(N) \;:=\; \sum_{\sigma\in\{0,\ldots,d-1\}^N} V^{(N)}(A)_\sigma\, \overline{V^{(N)}(B)_\sigma}.$$

This is the bilinear overlap (without complex conjugation) used later. By Lemma 2.35, it is the complex conjugate of the Hilbert-space inner product from Definition 2.33.

Lemma 2.35 (Overlap equals conjugate inner product)**.** $O_{AB}(N) = \overline{\langle V^{(N)}(A)|V^{(N)}(B)\rangle}$.

Proof. The overlap sums $V_\sigma \overline{W_\sigma}$ while the inner product sums $\overline{V_\sigma} W_\sigma$. Complex conjugation swaps these. ☐

Lemma 2.36 (Overlap determined by positive-length MPV)**.** *If $V^{(N)}(A)_\sigma = V^{(N)}(A')_\sigma$ and $V^{(N)}(B)_\sigma = V^{(N)}(B')_\sigma$ for all $N > 0$ and all σ, then $O_{AB}(N) = O_{A'B'}(N)$ for all positive N.*

Proof. Direct pointwise rewriting: if the MPV coefficients agree for all positive chain lengths, the overlap sums are identical. ☐

Chapter 3

Matrix Product Operators and Density Operators

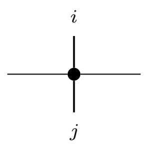
Fix a physical dimension d and a bond dimension D. An MPO tensor carries two physical indices, one ket index and one bra index, together with the virtual left and right indices.

3.1 MPO tensors

Definition 3.1 (MPO tensor)**.** An *MPO tensor* is a family of matrices

$$\{M^{ij}\}_{i,j=0}^{d-1}, \qquad M^{ij} \in M_D(\mathbb{C}),$$

indexed by a ket index i and a bra index j. Diagrammatically,



The black node denotes the tensor M, the horizontal legs are virtual, the upper leg is the ket index i, and the lower leg is the bra index j.

Definition 3.2 (Doubled-index MPS view)**.** By combining the pair (i, j) into a single physical index in $\{0, \dots, d^2 - 1\}$, one obtains an MPS tensor with physical dimension d^2 and bond dimension D.

Definition 3.3 (Word evaluation)**.** For words $u = (i_1, \dots, i_N)$ and $v = (j_1, \dots, j_N)$ in $\{0, \dots, d-1\}^N$, the *word evaluation* is

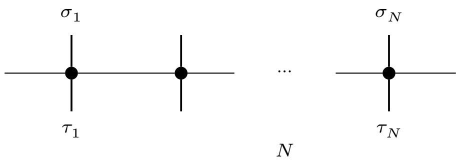
$$M^{u,v} := M^{i_1 j_1} M^{i_2 j_2} \cdots M^{i_N j_N} \in M_D(\mathbb{C}).$$

The empty pair of words evaluates to $\mathbb{1}_D$, and word pairs of different lengths evaluate to 0.

Definition 3.4 (MPO operator family)**.** The operator generated by M on N sites is the matrix $\rho^{(N)}(M)$ indexed by configurations $\sigma, \tau \in \{0, \dots, d-1\}^N$ and given by

$$\rho^{(N)}(M)_{\sigma,\tau} = \mathrm{tr}(M^{\sigma,\tau}).$$

Diagrammatically,



each black node denotes the same local tensor M, and the matrix entry $\rho^{(N)}(M)_{\sigma,\tau}$ is obtained by closing the outer virtual legs cyclically and taking the resulting trace.

Definition 3.5 (Hermitian MPO tensor)**.** An MPO tensor is *Hermitian* if

$$M^{ij} = (M^{ji})^\dagger \qquad \text{for all } i, j \in \{0, \dots, d-1\}.$$

3.2 Transfer maps

Definition 3.6 (MPO transfer map)**.** The *transfer map* associated to M is the linear map $\mathcal{E}_M : M_D(\mathbb{C}) \to M_D(\mathbb{C})$ defined by

$$\mathcal{E}_M(X) = \sum_{i,j=0}^{d-1} M^{ij} X (M^{ij})^\dagger.$$

Lemma 3.7 (Identification with the doubled-index MPS transfer map)**.** *The transfer map of an MPO tensor agrees with the transfer map of its doubled-index MPS view.*

Proof. After identifying the single physical index of the doubled tensor with a pair (i, j), the single sum defining the MPS transfer map is exactly the double sum defining $\mathcal{E}_M$. □

Theorem 3.8 (Positivity of the MPO transfer map)**.** *If $X \geq 0$, then $\mathcal{E}_M(X) \geq 0$.*

Proof. By Lemma 3.7, $\mathcal{E}_M$ is the transfer map of the doubled-index MPS tensor. The corresponding MPS transfer map is positive, so $\mathcal{E}_M(X) \geq 0$ whenever $X \geq 0$. □

3.3 MPDOs and LPDOs

Definition 3.9 (MPDO)**.** An MPO tensor is an *MPDO* if

$$\rho^{(N)}(M) \geq 0 \qquad \text{for every } N \in \mathbb{N}.$$

This is the global positivity condition of [CPGSV17a, §4.1].

Definition 3.10 (LPDO)**.** An MPO tensor is an *LPDO* if there exist an ancilla dimension d_K, an inner bond dimension D', an identification of bond indices $e : \{0, \dots, D-1\} \simeq \{0, \dots, D'-1\} \times \{0, \dots, D'-1\}$ (equivalently, D and $(D')^2$ have the same cardinality), and matrices $A^{(i,k)} \in M_{D'}(\mathbb{C})$ such that

$$M^{ij} = \Big(\sum_{k=0}^{d_K-1} A^{(i,k)} \otimes \overline{A^{(j,k)}} \Big)_{e,e} \qquad \text{for all } i, j,$$

where $\otimes$ denotes the Kronecker product, $\overline{(\cdot)}$ denotes entrywise complex conjugation, and $(\cdot)_{e,e}$ denotes reindexing along e from $\{0, \dots, D'-1\} \times \{0, \dots, D'-1\}$ back to $\{0, \dots, D-1\}$. This is the local purification form of [CPGSV17a, §4.3].

Theorem 3.11 (LPDOs are MPDOs)**.** *Every LPDO tensor is an MPDO.*

Proof. By the mixed-product property $(A \otimes B)(C \otimes D) = (AC) \otimes (BD)$, the N-site product decomposes as a sum of Kronecker products after transporting along the bond-space identification e:

$$M^{\sigma_1 \tau_1} \cdots M^{\sigma_N \tau_N} = \big(\sum_{\kappa} P_\kappa \otimes \overline{Q_\kappa}\big)\big|_e,$$

where $P_\kappa = A^{(\sigma_1,\kappa_1)} \cdots A^{(\sigma_N,\kappa_N)}$ and $Q_\kappa = A^{(\tau_1,\kappa_1)} \cdots A^{(\tau_N,\kappa_N)}$. Since the trace is invariant under the reindexing by e, we apply $\mathrm{tr}(X \otimes Y) = \mathrm{tr}(X)\,\mathrm{tr}(Y)$ to obtain

$$\rho^{(N)}(M)_{\sigma,\tau} = \sum_{\kappa} \mathrm{tr}(P_\kappa)\,\overline{\mathrm{tr}(Q_\kappa)} = \sum_{\kappa} \psi_\kappa(\sigma)\,\overline{\psi_\kappa(\tau)},$$

where $\psi_\kappa(\sigma) = \mathrm{tr}(P_\kappa)$. Hence $\rho^{(N)}(M) = \sum_\kappa |\psi_\kappa\rangle\langle\psi_\kappa| \geq 0$. $\square$

Chapter 4

The Single-Block Fundamental Theorem

This chapter proves that if an injective MPS tensor A generates the same MPV family as another tensor B of the same bond dimension, then A and B are gauge equivalent. The proof is purely algebraic and does not use the later overlap or normal-form reductions. The trace-pairing maps introduced below are auxiliary devices attached to the chosen generating families $\{A^i\}$ and $\{B^i\}$; they serve only to construct the linear map that is later shown to be a gauge transform.

4.1 Trace pairing

Theorem 4.1 (Nondegeneracy of the trace pairing)**.** *For $M \in M_D(\mathbb{C})$, $\operatorname{tr}(MN) = 0$ for all $N \in M_D(\mathbb{C})$ if and only if $M = 0$.*

Proof. Testing against the matrix units E_{ji} (the matrix with 1 in position (j, i) and 0 elsewhere) gives $\operatorname{tr}(E_{ji}M) = M_{ij}$. Hence if $\operatorname{tr}(MN) = 0$ for all N, then every entry of M vanishes. □

Lemma 4.2 (Same MPV implies trace agreement)**.** *If A and B generate the same MPV family, then $\operatorname{tr}(A^w) = \operatorname{tr}(B^w)$ for every word w.*

Proof. Viewing the word w as a configuration of length $|w|$, Definition 2.4 gives the corresponding MPV coefficient as the trace coefficient $c_w(A) = \operatorname{tr}(A^w)$. The same-MPV hypothesis gives equality at every system size. □

Definition 4.3 (Trace pairing map)**.** For a tensor A, attach to the chosen family $\{A^i\}$ the auxiliary linear map $\Phi_A : M_D(\mathbb{C}) \to \mathbb{C}^d$ defined by

$$\Phi_A(M)_i = \operatorname{tr}(M \cdot A^i).$$

Theorem 4.4 (Injectivity of the trace pairing map)**.** *If A is injective, then* $\ker \Phi_A = \{0\}$.

Proof. If $\Phi_A(M) = 0$, then $\operatorname{tr}(M \cdot A^i) = 0$ for all i. Since the $\{A^i\}$ span $M_D(\mathbb{C})$, this gives $\operatorname{tr}(MN) = 0$ for all N, so $M = 0$ by Theorem 4.1. □

Theorem 4.5 (Nonvanishing trace on injective tensors)**.** *If $D \geq 1$, A is injective, A and B generate the same MPV family, and $B^i = 0$ for all i, then we have a contradiction.*

Proof. If $B^i = 0$ for all i, then Lemma 4.2 gives $\operatorname{tr}(A^w) = 0$ for every word w, in particular for all length-1 words. Since the $\{A^i\}$ span $M_D(\mathbb{C})$, the trace functional vanishes identically, contradicting $\operatorname{tr}(\mathbb{1}_D) = D \neq 0$. $\square$

4.2 Linear extension

Theorem 4.6 (Existence and uniqueness of the linear extension)**.** *If A is injective and A, B generate the same MPV family, then there exists a unique linear map $T : M_D(\mathbb{C}) \to M_D(\mathbb{C})$ satisfying $T(A^i) = B^i$ for all i.*

Proof. Write $\ell_A : \mathbb{C}^d \to M_D(\mathbb{C})$ for the map that sends a coefficient vector c to $\sum_i c_i A^i$ (linear combination with the A^i). Applying Lemma 4.2 to all words of length 2 shows $\Phi_A \circ \ell_A = \Phi_B \circ \ell_B$. Since A is injective, ℓ_A is surjective, giving $\operatorname{range}(\Phi_A) \subseteq \operatorname{range}(\Phi_B)$. By Theorem 4.4, $\ker \Phi_A = \{0\}$, so rank–nullity gives $\dim(\operatorname{range}(\Phi_A)) = \dim(M_D(\mathbb{C}))$. Hence $\dim(\operatorname{range}(\Phi_B)) \geq \dim(M_D(\mathbb{C}))$, and rank–nullity again gives $\ker \Phi_B = \{0\}$. A left inverse g of Φ_B gives $T := g \circ \Phi_A$, and uniqueness follows because the $\{A^i\}$ span $M_D(\mathbb{C})$. $\square$

Theorem 4.7 (Multiplicativity of the linear extension)**.** *If A is injective and A, B generate the same MPV family, and $T : M_D(\mathbb{C}) \to M_D(\mathbb{C})$ is the unique linear map satisfying $T(A^i) = B^i$ for all i, then $T(MN) = T(M)T(N)$ for all $M, N \in M_D(\mathbb{C})$.*

Proof. **Step 1** ($\Phi_B \circ T = \Phi_A$). Applying Lemma 4.2 to words of length 2 gives $\Phi_B(T(A^i)) = \Phi_B(B^i) = \Phi_A(A^i)$ for all i. Since the $\{A^i\}$ span $M_D(\mathbb{C})$, $\Phi_B \circ T = \Phi_A$.

Step 2 ($T(A^i A^j) = B^i B^j$). Lemma 4.2 applied to the word (i, j, k) gives $\operatorname{tr}(A^i A^j A^k) = \operatorname{tr}(B^i B^j B^k)$ for all i, j, k. So $\Phi_B(T(A^i A^j))_k = \Phi_A(A^i A^j)_k = \operatorname{tr}(A^i A^j A^k) = \operatorname{tr}(B^i B^j B^k) = \Phi_B(B^i B^j)_k$. The same rank–nullity argument used in the proof of Theorem 4.6 gives $\ker \Phi_B = \{0\}$. Hence $T(A^i A^j) = B^i B^j$.

Step 3 (bilinear extension). For fixed j, the maps $M \mapsto T(M \cdot A^j)$ and $M \mapsto T(M) \cdot B^j$ agree on the spanning set $\{A^i\}$, hence on all of $M_D(\mathbb{C})$. Applying this once more with j replaced by a general matrix gives full multiplicativity. $\square$

4.3 Simplicity and Skolem–Noether

Theorem 4.8 (Simplicity of $M_D(\mathbb{C})$)**.** *A nonzero multiplicative $\mathbb{C}$-linear endomorphism of $M_D(\mathbb{C})$ is bijective.*

Proof. The kernel of a multiplicative linear map is a two-sided ideal. Since $M_D(\mathbb{C})$ is a simple ring (it has no nontrivial two-sided ideals), the kernel is either $\{0\}$ or $M_D(\mathbb{C})$. The latter would force the map to be zero, contradicting the hypothesis. Hence the map is injective, and surjectivity follows from finite-dimensionality. $\square$

Theorem 4.9 (Skolem–Noether)**.** *Every $\mathbb{C}$-algebra automorphism of $M_D(\mathbb{C})$ is inner: for any $f \in \operatorname{Aut}_{\mathbb{C}\text{-}alg}(M_D(\mathbb{C}))$, there exists $X \in \mathrm{GL}_D(\mathbb{C})$ such that $f(M) = XMX^{-1}$ for all M.*

Proof. Via the canonical identification $M_D(\mathbb{C}) \cong \operatorname{End}_{\mathbb{C}}(\mathbb{C}^D)$, the automorphism f induces an automorphism of $\operatorname{End}_{\mathbb{C}}(\mathbb{C}^D)$. Every algebra automorphism of $\operatorname{End}(V)$ is conjugation by an invertible linear map (this is a standard result in the theory of central simple algebras). Translating back to matrices gives the invertible X. $\square$

Lemma 4.10 (Promotion to algebra homomorphism)**.** *If $T : M_D(\mathbb{C}) \to M_D(\mathbb{C})$ is a multiplicative surjective $\mathbb{C}$-linear map, then $T(\mathbb{1}) = \mathbb{1}$. Equivalently, the same underlying function defines a $\mathbb{C}$-algebra homomorphism.*

Proof. By surjectivity, there exists X with $T(X) = \mathbb{1}$. Then $T(\mathbb{1}) = T(X) \cdot T(\mathbb{1}) = T(X \cdot \mathbb{1}) = T(X) = \mathbb{1}$, using multiplicativity and $T(X) = \mathbb{1}$. With $T(\mathbb{1}) = \mathbb{1}$ established, T preserves the unit and hence is a $\mathbb{C}$-algebra homomorphism. □

4.4 Proof of the single-block theorem

Theorem 4.11 (Single-block Fundamental Theorem)**.** *Let A be an injective MPV tensor and let B be an MPV tensor of the same bond dimension. If A and B generate the same MPV family, then they are gauge equivalent: there exists $X \in \mathrm{GL}_D(\mathbb{C})$ such that $B^i = XA^iX^{-1}$ for all i.*

Proof. By Theorem 4.6, there is a unique linear T with $T(A^i) = B^i$. By Theorem 4.7, T is multiplicative. T is nonzero: if $T = 0$ then $B^i = 0$ for all i, contradicting Theorem 4.5. By Theorem 4.8, T is bijective. By Lemma 4.10, T is a $\mathbb{C}$-algebra automorphism. Theorem 4.9 gives an invertible X with $T(M) = XMX^{-1}$, and evaluating at $M = A^i$ yields $B^i = XA^iX^{-1}$. □

4.5 Alternative fixed-length non-TI setup

The proof above treats a single translation-invariant tensor and equality of the full MPV family for all system sizes. The alternative algebraic proof of [MGRPG+18] instead keeps a fixed periodic chain length and allows the local tensors to vary from site to site. The first step is therefore a separate data layer for finite non-translation-invariant chains.

Definition 4.12 (Periodic non-TI chain data)**.** Fix a chain length n. A *non-translation-invariant periodic MPS chain* consists of sitewise tensors

$$A_k = \{A_k^i\}_{i=0}^{d-1}, \qquad k = 0, \dots, n-1,$$

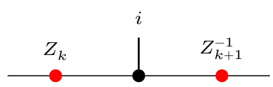
where each $A_k^i \in M_D(\mathbb{C})$. For a basis configuration $\sigma = (\sigma_0, \dots, \sigma_{n-1})$, its coefficient is

$$c_{A,\sigma} := \mathrm{tr}\big(A_0^{\sigma_0} A_1^{\sigma_1} \cdots A_{n-1}^{\sigma_{n-1}}\big).$$

Two such chains generate the *same chain state* if these coefficients agree for every σ. The chain is *injective* if each local tensor A_k is injective in the usual single-site sense. Finally, two chains are *cyclically gauge equivalent* if there are invertible bond matrices $Z_0, \dots, Z_{n-1}$ such that

$$B_k^i = Z_k A_k^i Z_{k+1}^{-1}$$

for every site k and physical index i, with indices understood modulo n so that $Z_n = Z_0$. Locally, this gauge relation is represented by



where the black node denotes the site tensor named in the surrounding formula and the two red side nodes denote the bond gauges on the neighboring virtual legs.

Lemma 4.13 (Cyclic gauge equivalence is an equivalence relation)**.** *On fixed-length periodic chains, cyclic gauge equivalence is reflexive, symmetric, and transitive.*

Proof. Reflexivity uses the identity gauge on every bond. Symmetry inverts each bond gauge, and transitivity multiplies the gauges pointwise along the chain. □

Chapter 5

Quantum Channels and Positive Maps

This chapter develops the theory of positive maps and quantum channels on matrix algebras, following [Wol12].

5.1 Positive maps and channels

Definition 5.1 (Positive map)**.** A linear map $E : M_D(\mathbb{C}) \to M_D(\mathbb{C})$ is *positive* if $E(X) \geq 0$ whenever $X \geq 0$, where $X \geq 0$ means that X is positive semidefinite.

Remark 5.2. We use the Loewner order on matrices: $X \leq Y$ means $Y - X \geq 0$.

Definition 5.3 (Trace-preserving map)**.** A linear map $E : M_D(\mathbb{C}) \to M_D(\mathbb{C})$ is *trace-preserving* if $\operatorname{tr}(E(X)) = \operatorname{tr}(X)$ for all X.

Definition 5.4 (Completely positive map)**.** A linear map $E : M_D(\mathbb{C}) \to M_D(\mathbb{C})$ is *completely positive* (CP) if it admits a *Kraus representation*: there exist operators $\{K_i\}_{i=0}^{r-1}$ with $K_i \in M_D(\mathbb{C})$ such that

$$E(X) = \sum_{i=0}^{r-1} K_i X K_i^\dagger \quad \text{for all } X \in M_D(\mathbb{C}).$$

We use the Kraus characterisation because it matches the later Kadison–Schwarz arguments. By Choi's theorem, this is equivalent to $E \otimes \mathrm{id}_n$ being positive for all n.

Theorem 5.5 (CP maps are positive)**.** *Every completely positive map is positive.*

Proof. If $X \geq 0$, then each summand $K_i X K_i^\dagger \geq 0$ (conjugation preserves positive semidefiniteness), so the sum is PSD. □

Definition 5.6 (Quantum channel)**.** A linear map $E : M_D(\mathbb{C}) \to M_D(\mathbb{C})$ is a *quantum channel* if it is both completely positive and trace-preserving (CPTP), following [Wol12].

Definition 5.7 (Density matrices)**.** The set of *density matrices* in $M_D(\mathbb{C})$ is

$$\mathcal{D}_D = \{\rho \in M_D(\mathbb{C}) : \rho \geq 0, \operatorname{tr}(\rho) = 1\}.$$

Theorem 5.8 (Density matrices are compact)**.** *$\mathcal{D}_D$ is compact (in the entry-wise topology on $M_D(\mathbb{C})$).*

Proof. The PSD cone is closed (preimage of the closed nonnegative cone under continuous quadratic forms), and if $\rho \geq 0$ with $\operatorname{tr}(\rho) = 1$ then each entry satisfies $|\rho_{ij}|^2 \leq \rho_{ii}\rho_{jj} \leq 1$: the diagonal entries are nonnegative and sum to 1, and the off-diagonal bound is the PSD Cauchy–Schwarz inequality. Hence the matrix entries are uniformly bounded, and Heine–Borel gives compactness. $\square$

Theorem 5.9 (Density matrices are convex)**.** *$\mathcal{D}_D$ is convex.*

Proof. Convex combination of PSD matrices is PSD, and trace is linear. $\square$

Theorem 5.10 (Density matrices are nonempty)**.** *For $D \geq 1$, $\mathcal{D}_D \neq \emptyset$.*

Proof. $\frac{1}{D}\mathbb{1}_D$ is a density matrix. $\square$

Theorem 5.11 (Channels preserve density matrices)**.** *If E is a quantum channel, then $E(\mathcal{D}_D) \subseteq \mathcal{D}_D$.*

Proof. Complete positivity implies positivity (Theorem 5.5), so $E(\rho) \geq 0$; trace preservation gives $\operatorname{tr}(E(\rho)) = \operatorname{tr}(\rho) = 1$. $\square$

5.2 Irreducibility

Definition 5.12 (Irreducible map)**.** A linear map $E : M_D(\mathbb{C}) \to M_D(\mathbb{C})$ is *irreducible* if whenever P is an orthogonal projection satisfying $E(PM_D(\mathbb{C})P) \subseteq PM_D(\mathbb{C})P$, then $P = 0$ or $P = \mathbb{1}$. This is [Wol12, Thm. 6.2, condition (1)]. The definition applies to any linear map; complete positivity is not required.

Theorem 5.13 (Injectivity implies irreducibility)**.** *If A is an injective MPS tensor, then its transfer map $\mathcal{E}_A$ is irreducible.*

Proof. If P is an invariant projection for $\mathcal{E}_A$, then $(\mathbb{1} - P)A^iP = 0$ for all i. Since the $\{A^i\}$ span $M_D(\mathbb{C})$, $(\mathbb{1} - P)MP = 0$ for all M, so $P = 0$ or $P = \mathbb{1}$. $\square$

Remark 5.14 (Transfer map is CP). The transfer map $\mathcal{E}_A(X) = \sum_i A^iX(A^i)^\dagger$ is completely positive: it is already written in Kraus form, with Kraus operators $\{A^i\}$.

Theorem 5.15 (Transfer map is a channel)**.** *If $\sum_i (A^i)^\dagger A^i = \mathbb{1}$, then the transfer map $\mathcal{E}_A(X) = \sum_i A^iX(A^i)^\dagger$ is a quantum channel (CPTP).*

Proof. Complete positivity is the observation of Remark 5.14. Trace preservation: $\operatorname{tr}(\mathcal{E}_A(X)) = \sum_i \operatorname{tr}(A^iX(A^i)^\dagger) = \operatorname{tr}((\sum_i (A^i)^\dagger A^i)X) = \operatorname{tr}(X)$ by cyclicity and the normalization hypothesis. $\square$

5.3 Cesàro fixed-point theorem

Definition 5.16 (Cesàro mean)**.** The *Cesàro mean* of a linear map E at order N, applied to a starting matrix ρ_0, is

$$\sigma_N = \frac{1}{N}\sum_{n=0}^{N-1} E^n(\rho_0).$$

Theorem 5.17 (Cesàro telescope)**.** $E(\sigma_N) - \sigma_N = \frac{1}{N}(E^N(\rho_0) - \rho_0)$.

Proof. Telescoping: $E(\sigma_N) = \frac{1}{N}\sum_{n=1}^{N} E^n(\rho_0)$ and subtracting σ_N cancels all but the first and last terms. □

Theorem 5.18 (Hermitian fixed-point decomposition)**.** *Let E be a quantum channel and $X = E(X)$ a Hermitian fixed point with decomposition $X = P_+ - P_-$ into orthogonal positive and negative parts ($P_\pm \geq 0$, $\mathrm{tr}(P_+P_-) = 0$). Then $E(P_+) = P_+$ and $E(P_-) = P_-$. This is [Wol12, Prop. 6.8].*

Proof. Let Q be the support projection of P_+. Trace preservation and positivity give $\mathrm{tr}(QE(P_-)) = 0$ and $E(P_-) \geq 0$, so $QE(P_-) = 0$. The fixed-point equation $E(P_+) - E(P_-) = P_+ - P_-$ then identifies $E(P_\pm) = P_\pm$. □

Theorem 5.19 (Cesàro fixed-point theorem)**.** *Every quantum channel $E : M_D(\mathbb{C}) \to M_D(\mathbb{C})$ (with $D \geq 1$) has a nonzero positive semidefinite fixed point: there exists $\rho \geq 0$, $\rho \neq 0$, with $E(\rho) = \rho$.*

Proof. Start with any $\rho_0 \in \mathcal{D}_D$ (Theorem 5.10). Each Cesàro mean σ_N lies in $\mathcal{D}_D$ (by convexity and channel preservation). By compactness (Theorem 5.8), extract a convergent subsequence $\sigma_{N_k} \to \rho$. The telescope identity (Theorem 5.17) gives $\|E(\sigma_N) - \sigma_N\| = \frac{1}{N}\|E^N(\rho_0) - \rho_0\| \to 0$, so $E(\rho) = \rho$ by continuity. In [Wol12, Thm. 6.11], existence is proved via Brouwer's fixed-point theorem; we use the Cesàro argument instead. □

5.4 Fixed-point projection

Definition 5.20 (Fixed-point projection)**.** Given a linear map $E : M_D(\mathbb{C}) \to M_D(\mathbb{C})$ and a fixed point ρ with $E(\rho) = \rho$ and $\mathrm{tr}(\rho) \neq 0$, the *fixed-point projection* is the rank-one map

$$P(X) = \frac{\mathrm{tr}(X)}{\mathrm{tr}(\rho)}\rho.$$

This projects onto the span of ρ along the kernel of the trace functional. When the fixed point is unique up to scaling, this span is the full fixed-point space.

Theorem 5.21 (Power decomposition)**.** *If E is trace-preserving and $E(\rho) = \rho$, then for $n \geq 1$,*

$$E^n = P + (E - P)^n,$$

where P is the fixed-point projection.

Proof. P is idempotent ($P^2 = P$), the fixed-point relation $E(\rho) = \rho$ gives $E \circ P = P$, and trace preservation gives $P \circ E = P$. These imply $(E - P) \circ P = 0$ and $P \circ (E - P) = 0$, so in the binomial expansion of $(P + (E - P))^n$ all cross terms vanish, leaving $E^n = P^n + (E - P)^n = P + (E - P)^n$. □

5.5 Peripheral spectrum and primitivity

Definition 5.22 (Peripheral spectrum)**.** The *peripheral spectrum* of a bounded linear operator T is the set of spectral values μ with $|\mu|$ equal to the spectral radius of T.

Definition 5.23 (Peripheral eigenvalues)**.** The *peripheral eigenvalues* of a linear map E on a finite-dimensional space are the eigenvalues λ on the unit circle: $|\lambda| = 1$. For a channel, the spectral radius is 1 [Wol12, Prop. 6.1], so these coincide with the eigenvalues of maximal modulus.

Note: The condition $|\lambda| = 1$ is appropriate for channels, since the spectral radius of a channel is 1. For a general linear map with spectral radius $r \neq 1$ the condition would generalise to $|\lambda| = r$.

Theorem 5.24 (1 is a peripheral eigenvalue)**.** *If E has a nonzero fixed point $\rho \neq 0$ with $E(\rho) = \rho$, then* 1 *is a peripheral eigenvalue of E.*

Proof. ρ is an eigenvector with eigenvalue 1, and $|1| = 1$. □

Remark 5.25 (Peripheral eigenvalues lie on the unit circle)*.* Every peripheral eigenvalue λ satisfies $|\lambda| = 1$. This is simply the defining condition in Definition 5.23.

Theorem 5.26 (Finiteness of peripheral eigenvalues)**.** *On a finite-dimensional space, the set of peripheral eigenvalues is finite.*

Proof. Peripheral eigenvalues are a subset of all eigenvalues, which is a finite set in finite dimensions. □

Theorem 5.27 (Power-stable peripheral eigenvalues are roots of unity)**.** *Let E be a linear endomorphism on a finite-dimensional space. If μ is a peripheral eigenvalue of E and μ^n is an eigenvalue of E for every $n \geq 1$, then μ is a root of unity.*

Proof. Since μ^n is an eigenvalue for every n, the pigeonhole principle on the finite eigenvalue set gives $\mu^a = \mu^b$ for some $a \neq b$, hence $\mu^{|a-b|} = 1$. □

Theorem 5.28 (Peripheral powers imply root of unity)**.** *If the set of peripheral eigenvalues of a linear endomorphism E on a finite-dimensional space is* closed under powers *($\mu \in \mathrm{peripheral}(E)$ and $n \geq 1$ imply $\mu^n \in \mathrm{peripheral}(E)$), then every peripheral eigenvalue is a root of unity.*

Proof. The closure hypothesis provides $\mu^n \in \mathrm{peripheral}(E)$ for all n, which in particular means μ^n is an eigenvalue. Theorem 5.27 then gives $\mu^p = 1$ for some $p > 0$. □

5.5.1 Periodicity removal by powering

Remark 5.29*.* The results in this subsection are purely spectral: they use only finiteness of a set of complex numbers and the spectral mapping theorem for powers. In particular, they do not rely on complete positivity. For transfer maps, replacing a tensor by its p-blocked tensor replaces $\mathcal{E}_A$ by $\mathcal{E}_A^p$, so this powering step is the operator-theoretic form of blocking.

Lemma 5.30 (Common exponent for a finite set of roots of unity)**.** *Let $s \subseteq \mathbb{C}$ be a finite set. Assume that for every $\mu \in s$ there exists an exponent $p_\mu > 0$ with $\mu^{p_\mu} = 1$. Then there exists $p > 0$ such that $\mu^p = 1$ for all $\mu \in s$.*

Proof. Take p to be the least common multiple of the exponents p_μ. □

Lemma 5.31 (Powering collapses peripheral eigenvalues)**.** *Let $E : M_D(\mathbb{C}) \to M_D(\mathbb{C})$ be a linear map, and assume that E has a nonzero fixed point $\rho \neq 0$. Let $p > 0$. If every peripheral eigenvalue μ of E satisfies $\mu^p = 1$, then* $\mathrm{peripheral}(E^p) = \{1\}$.

Proof. The fixed point ρ gives $1 \in \mathrm{peripheral}(E^p)$. For the reverse inclusion, let $\nu \in \mathrm{peripheral}(E^p)$. Spectral mapping gives $\nu = \mu^p$ for some $\mu \in \sigma(E)$. From $|\nu| = 1$ and $p > 0$ we obtain $|\mu| = 1$, hence $\mu \in \mathrm{peripheral}(E)$. The hypothesis $\mu^p = 1$ then forces $\nu = 1$. □

For a normalized MPS tensor, the transfer map is naturally trace-preserving but need not be unital. In general one should not expect a single gauge choice to make it both unital and trace-preserving.

Remark 5.32 (Bi-canonical special case). When a Kraus map happens to be both unital and trace-preserving (the "bi-canonical" case), the Kadison–Schwarz equality at peripheral eigenvectors (Theorem 7.24) shows directly that the peripheral eigenvalues are closed under powers, and hence are roots of unity by Theorem 5.28. This is the simplest argument, but it requires both normalizations simultaneously. The more general adjoint-fixed-point formulation below covers the case where only unitality and a positive definite adjoint fixed point are available.

5.5.2 Peripheral closure via adjoint fixed point

The following results replace the trace-preservation hypothesis by the existence of a positive definite fixed point for the *adjoint* Kraus map, matching the weighted-trace argument in [CPGSV17a, Appendix A].

Lemma 5.33 (Peripheral closure via adjoint fixed point). *Let $E(X) = \sum_i K_i X K_i^\dagger$ be a Kraus map that is unital ($\sum_i K_i K_i^\dagger = \mathbb{1}$), and assume:*

1. *the adjoint Kraus map $E^*(X) = \sum_i K_i^\dagger X K_i$ has a positive definite fixed point $\rho > 0$, and*
2. *E is irreducible.*

Then $\mathrm{peripheral}(E)$ *is closed under powers: if $\mu \in \mathrm{peripheral}(E)$ then $\mu^n \in \mathrm{peripheral}(E)$ for every $n \in \mathbb{N}$.*

Proof. Let $E(X) = \mu X$ with $|\mu| = 1$, and set $G := E(X^\dagger X) - E(X)^\dagger E(X)$. By Kadison–Schwarz, $G \geq 0$. Since $E^*(\rho) = \rho$,

$$\mathrm{tr}(\rho G) = \mathrm{tr}(\rho E(X^\dagger X)) - \mathrm{tr}(\rho E(X)^\dagger E(X)) = \mathrm{tr}(E^*(\rho) X^\dagger X) - \mathrm{tr}(\rho E(X)^\dagger E(X)) = \mathrm{tr}(\rho X^\dagger X) - |\mu|^2\,\mathrm{tr}(\rho X^\dagger X) = 0.$$

Thus the trace of the Kadison–Schwarz gap against the adjoint fixed point vanishes. Because $\rho > 0$ and $G \geq 0$, this forces $G = 0$, so $E(X^\dagger X) = E(X)^\dagger E(X)$. Hence $X^\dagger X$ is a PSD fixed point, which is positive definite by irreducibility (Theorem 8.3). So X is invertible. By Theorem 7.23 (KS equality implies Kraus commutation), X commutes with each Kraus operator; iterating via the left multiplicative identity (Theorem 7.25) gives $E(X^n) = \mu^n X^n$, so μ^n is peripheral. ☐

Theorem 5.34 (Roots of unity via adjoint fixed point). *Let $E(X) = \sum_i K_i X K_i^\dagger$ be a Kraus map that is unital, and assume that the adjoint Kraus map $E^*(X) = \sum_i K_i^\dagger X K_i$ has a positive definite fixed point and that E is irreducible. Then every peripheral eigenvalue of E is a root of unity.*

Proof. Combine the closure-under-powers result (Lemma 5.33) with Theorem 5.28. ☐

Remark 5.35. Lemma 5.33 and Theorem 5.34 cover the bi-canonical (unital + TP) case as well: trace preservation gives $E^*(\mathbb{1}) = \mathbb{1}$, which is a positive definite fixed point of the adjoint. The adjoint-fixed-point formulation matches the argument in [CPGSV17a, Appendix A], which uses a faithful invariant state rather than bi-canonicality.

Definition 5.36 (Channel period). The *period* of a channel E is the number of peripheral eigenvalues (counted without multiplicity): $p(E) := |\mathrm{peripheral}(E)|$.

Definition 5.37 (Primitive map)**.** A linear map is *primitive* if its only peripheral eigenvalue is $\lambda = 1$, i.e. $\mathrm{peripheral}(E) = \{1\}$. For channels this corresponds to [Wol12, Thm. 6.7]. This definition applies to any linear endomorphism, not only to channels.

Theorem 5.38 (Primitivity equals period one)**.** *A linear map with a nonzero fixed point is primitive if and only if its period is* 1*.*

Proof. If $\mathrm{peripheral}(E) = \{1\}$, then $|\mathrm{peripheral}(E)| = 1$. Conversely, if $|\mathrm{peripheral}(E)| = 1$, then since $1 \in \mathrm{peripheral}(E)$ (Theorem 5.24), the only element is 1. □

Theorem 5.39 (Primitive maps have a spectral gap on the complement)**.** *Let $E : M_D(\mathbb{C}) \to M_D(\mathbb{C})$ be trace-preserving, and let $\rho \neq 0$ satisfy $E(\rho) = \rho$ and $\mathrm{tr}(\rho) \neq 0$. Let*

$$P(X) = \frac{\mathrm{tr}(X)}{\mathrm{tr}(\rho)}\,\rho$$

be the fixed-point projection associated to ρ. Assume that:

1. *E is primitive;*
2. *every eigenvalue μ of E satisfies $|\mu| \leq 1$;*
3. *whenever $E(X) = X$ and $\mathrm{tr}(X) = 0$, one has $X = 0$.*

Then every eigenvalue ν of $E - P$ satisfies $|\nu| < 1$.

Proof. Let $(E - P)(X) = \nu X$ with $X \neq 0$. Then

$$E(X) = \nu X + P(X).$$

If $\mathrm{tr}(X) = 0$, then $P(X) = 0$, so $E(X) = \nu X$ and hence ν is an eigenvalue of E with $|\nu| \leq 1$. If $|\nu| = 1$, primitivity forces $\nu = 1$, so X is a trace-zero fixed point of E; by assumption this implies $X = 0$, a contradiction. Thus $|\nu| < 1$.

If instead $\mathrm{tr}(X) \neq 0$, trace preservation and $\mathrm{tr}(P(X)) = \mathrm{tr}(X)$ give

$$\mathrm{tr}(X) = \mathrm{tr}(E(X)) = \nu\ \mathrm{tr}(X) + \mathrm{tr}(P(X)) = \nu\ \mathrm{tr}(X) + \mathrm{tr}(X),$$

so $\nu\ \mathrm{tr}(X) = 0$ and therefore $\nu = 0$. In either case, $|\nu| < 1$. □

5.6 Representations

The following definitions support the Choi–Jamiolkowski isomorphism (Theorem 5.44) and the representation theorems of [Wol12, Ch. 2].

Definition 5.40 (Partial traces)**.** Let $X \in M_{d \cdot d'}(\mathbb{C})$ be a bipartite matrix indexed by $(\mathrm{Fin}\, d \times \mathrm{Fin}\, d')^2$. The *left partial trace* $\mathrm{tr}_A(X)$ and *right partial trace* $\mathrm{tr}_B(X)$ are the $d' \times d'$ and $d \times d$ matrices defined by

$$(\mathrm{tr}_A(X))_{ij} = \sum_k X_{(k,i),(k,j)}, \qquad (\mathrm{tr}_B(X))_{ij} = \sum_k X_{(i,k),(j,k)}.$$

Definition 5.41 (Maximally entangled state)**.** The *maximally entangled state* on $\mathbb{C}^d \otimes \mathbb{C}^d$ is the rank-one projector

$$|\Omega\rangle\langle\Omega|, \quad |\Omega\rangle = \frac{1}{\sqrt{d}} \sum_{j=0}^{d-1} |j, j\rangle.$$

Concretely, $(|\Omega\rangle\langle\Omega|)_{(i_1,i_2),(j_1,j_2)} = \frac{1}{d}\,\delta_{i_1 j_1} \delta_{i_2 j_2}$ and $\mathrm{tr}(|\Omega\rangle\langle\Omega|) = 1$.

Definition 5.42 (Tensor product of a map with identity)**.** Given a linear map $T : M_D(\mathbb{C}) \to M_D(\mathbb{C})$, the *tensor extension* $T \otimes \mathrm{id}$ acts on bipartite matrices $X \in M_{D\times D}(\mathbb{C})$ by applying T to each "slice":

$$(T \otimes \mathrm{id})(X)_{(i_1,i_2),(j_1,j_2)} = (T(X^{(i_2,j_2)}))_{i_1 j_1},$$

where $X^{(i_2,j_2)}_{ab} = X_{(a,i_2),(b,j_2)}$ is the bipartite slice.

5.7 Choi–Jamiołkowski isomorphism (Wolf Prop. 2.1)

Definition 5.43 (Choi matrix)**.** The *Choi matrix* of a linear map $T : M_D(\mathbb{C}) \to M_D(\mathbb{C})$ is

$$\tau = (T \otimes \mathrm{id})(|\Omega\rangle\langle\Omega|) \ \in \ M_{D\times D}(\mathbb{C}).$$

Concretely, $\tau_{(i_1,i_2),(j_1,j_2)} = \frac{1}{D}\,(T(E_{i_2 j_2}))_{i_1 j_1}$ where $E_{i_2 j_2}$ is the matrix unit. This is the Choi–Jamiolkowski convention of [Wol12, Prop. 2.1].

Theorem 5.44 (Prop. 2.1: CP iff Choi matrix is PSD)**.** *A linear map $T : M_D(\mathbb{C}) \to M_D(\mathbb{C})$ is completely positive (in the Kraus sense) if and only if its Choi matrix $\tau \geq 0$.*

Proof. ($\Rightarrow$) If $T(X) = \sum_i K_i X K_i^\dagger$, then $\tau = \sum_i |v_i\rangle\langle v_i|$ where $(v_i)_{(a,b)} = c\,K_i(a,b)$ and $c = 1/\sqrt{D}$. This is a sum of rank-one PSD matrices.

($\Leftarrow$) If $\tau \geq 0$, write $\tau = \sum_m |v_m\rangle\langle v_m|$ by spectral decomposition, define $K_m(a,b) = v_m(a,b)/c$, and recover $T(X) = \sum_m K_m X K_m^\dagger$ by linearity on matrix units. ☐

Theorem 5.45 (Prop. 2.1: TP implies partial trace condition)**.** *If T is trace-preserving, then* $\mathrm{tr}_A(\tau) = \frac{1}{D}\mathbb{1}_D$.

Proof. $(\mathrm{tr}_A(\tau))_{ij} = \mathrm{tr}(T(\Omega^{(ij)})) = \mathrm{tr}(\Omega^{(ij)})$ by trace preservation, where $\Omega^{(ij)} = \frac{1}{D}E_{ij}$ is the (i,j)-slice of $|\Omega\rangle\langle\Omega|$. Taking the trace gives $\frac{1}{D}\delta_{ij}$. ☐

Theorem 5.46 (Prop. 2.1: Hermiticity correspondence)**.** *The Choi matrix τ is Hermitian if and only if $T(B^\dagger) = T(B)^\dagger$ for all $B \in M_D(\mathbb{C})$.*

Proof. The Choi matrix entries satisfy $\tau_{(a,i),(b,j)} = \frac{1}{D}(T(E_{ij}))_{ab}$. Hermiticity of τ says $\overline{\tau_{(b,j),(a,i)}} = \tau_{(a,i),(b,j)}$, which unravels to $T(E_{ji})_{ba} = \overline{T(E_{ij})_{ab}}$, i.e., $T(E_{ij}^\dagger) = T(E_{ij})^\dagger$. By linearity this extends to all B. ☐

Theorem 5.47 (Prop. 2.1: Trace of Choi matrix for TP map)**.** *If T is trace-preserving, then* $\mathrm{tr}(\tau) = 1$.

Proof. $\mathrm{tr}(\tau) = \mathrm{tr}(\mathrm{tr}_A(\tau)) = \mathrm{tr}(\frac{1}{D}\mathbb{1}_D) = 1$. ☐

5.8 Kraus representation theorem (Wolf Thm. 2.1)

Theorem 5.48 (Thm. 2.1: TP implies Kraus normalization)**.** *If $T(X) = \sum_i K_i X K_i^\dagger$ is trace-preserving, then $\sum_i K_i^\dagger K_i = \mathbb{1}$.*

Proof. For any N, $\mathrm{tr}((\sum_i K_i^\dagger K_i)\,N) = \sum_i \mathrm{tr}(K_i^\dagger K_i N) = \sum_i \mathrm{tr}(K_i N K_i^\dagger) = \mathrm{tr}(T(N)) = \mathrm{tr}(N)$. Non-degeneracy of the trace pairing forces $\sum_i K_i^\dagger K_i = \mathbb{1}$. ☐

Theorem 5.49 (Thm. 2.1: Kraus normalization implies TP). *If* $\sum_i K_i^\dagger K_i = \mathbb{1}$*, then* $T(X) = \sum_i K_i X K_i^\dagger$ *is trace-preserving.*

Proof. $\mathrm{tr}(T(X)) = \sum_i \mathrm{tr}(K_i X K_i^\dagger) = \sum_i \mathrm{tr}(K_i^\dagger K_i X) = \mathrm{tr}((\sum_i K_i^\dagger K_i)\, X) = \mathrm{tr}(X)$. □

Theorem 5.50 (Thm. 2.1: Unitary freedom in Kraus operators). *Let* U *be a unitary* $r \times r$ *matrix* ($U^\dagger U = \mathbb{1}$) *and suppose* $K_j = \sum_\ell U_{j\ell}\, \tilde{K}_\ell$*. Then* $\{K_j\}$ *and* $\{\tilde{K}_\ell\}$ *define the same Kraus map:*

$$\sum_j K_j X K_j^\dagger = \sum_\ell \tilde{K}_\ell X \tilde{K}_\ell^\dagger \quad \forall\, X \in M_D(\mathbb{C}).$$

Proof. Substitute the combination formula, expand the triple sum over j, ℓ, ℓ', swap summation order, and use the entry-wise form of $U^\dagger U = \mathbb{1}$ to collapse to $\ell = \ell'$ diagonal terms. □

Theorem 5.51 (Thm. 2.1: Orthonormal Kraus transition is unitary). *Let* $\{K_j\}_{j=0}^{r-1}$ *and* $\{K'_j\}_{j=0}^{r-1}$ *be two Hilbert–Schmidt orthonormal Kraus families, and suppose*

$$K_j = \sum_{\ell=0}^{r-1} U_{j\ell}\, K'_\ell.$$

Then the transition matrix U *is unitary:* $U^\dagger U = \mathbb{1}_r$*.*

Proof. Expand $\mathrm{tr}(K_j^\dagger K_i)$ using the change of basis. Hilbert–Schmidt orthonormality of both Kraus families identifies the Gram matrix with the identity, yielding the matrix identity $UU^\dagger = \mathbb{1}_r$, hence also $U^\dagger U = \mathbb{1}_r$. □

Theorem 5.52 (Thm. 2.1: Dual map equality from primal map equality). *If two Kraus families satisfy* $\sum_\alpha B_\alpha X B_\alpha^\dagger = \sum_j A_j X A_j^\dagger$ *for all* X*, then* $\sum_\alpha B_\alpha^\dagger Y B_\alpha = \sum_j A_j^\dagger Y A_j$ *for all* Y*.*

Proof. Use the trace pairing: for all X, $\mathrm{tr}(X^\dagger(\sum B_\alpha^\dagger Y B_\alpha)) = \mathrm{tr}((\sum B_\alpha X^\dagger B_\alpha^\dagger)Y) = \mathrm{tr}((\sum A_j X^\dagger A_j^\dagger)Y) = \mathrm{tr}(X^\dagger(\sum A_j^\dagger Y A_j))$. Nondegeneracy of the trace pairing gives the result. □

Theorem 5.53 (Thm. 2.1: Equal Stinespring Gramians). *If two Kraus families define the same CPM, then* $\sum_\alpha B_\alpha^\dagger B_\alpha = \sum_j A_j^\dagger A_j$*.*

Proof. Specialise Theorem 5.52 at $Y = \mathbb{1}$. □

Theorem 5.54 (Thm. 2.1 item 4: Rectangular Kraus freedom (necessary direction)). *If* $\sum_\alpha B_\alpha X B_\alpha^\dagger = \sum_j A_j X A_j^\dagger$ *for all* X *and* $r_2 \le r_1$ *(where* $\{B_\alpha\}_{\alpha=0}^{r_1-1}$ *is the larger family and* $\{A_j\}_{j=0}^{r_2-1}$ *the smaller), then there exists a rectangular isometry* V ($r_1 \times r_2$, $V^\dagger V = \mathbb{1}_{r_2}$) *such that* $B_\alpha = \sum_j V_{\alpha j}\, A_j$*.*

Proof. Pad the smaller family to size r_1, establish Gram matrix equality $M_B^\dagger M_B = M_{A'}^\dagger M_{A'}$ from dual map equality, construct a partial isometry via inner product preservation on the range, and extend to a full isometry V with $V^\dagger V = \mathbb{1}_{r_2}$. □

Theorem 5.55 (Thm. 2.1 item 4: Rectangular Kraus freedom (general index variant)). *Variant of Theorem 5.54 with general finite index types* ι_1, ι_2 *in place of* $\mathrm{Fin}\, r_1, \mathrm{Fin}\, r_2$*.*

Proof. Reindex via equivalences to Fin and apply Theorem 5.54. □

5.9 Stinespring dilation (Wolf Thm. 2.2)

Definition 5.56 (Stinespring isometry). Given Kraus operators $\{K_j\}_{j=0}^{r-1}$ with $K_j \in M_D(\mathbb{C})$, the *Stinespring isometry* $V : \mathbb{C}^D \to \mathbb{C}^D \otimes \mathbb{C}^r$ is the $(D \cdot r) \times D$ matrix

$$V_{(i,j),k} = (K_j)_{ik}.$$

This implements $V = \sum_j K_j \otimes |j\rangle$.

Lemma 5.57 (Entry formula for the Stinespring isometry). *For indices $i, k \in \{0, \dots, D-1\}$ and $j \in \{0, \dots, r-1\}$,*

$$V_{(i,j),k} = (K_j)_{ik}.$$

Theorem 5.58 (Stinespring Gram identity). *The Stinespring isometry satisfies*

$$V^\dagger V = \sum_{j=0}^{r-1} K_j^\dagger K_j.$$

Theorem 5.59 (Thm. 2.2: Stinespring dual representation). *For Kraus operators $\{K_j\}$ and $A \in M_D(\mathbb{C})$,*

$$V^\dagger \,(A \otimes \mathbb{1}_r)\, V = \sum_j K_j^\dagger A\, K_j = T^*(A).$$

This is the Stinespring representation of the dual *(Heisenberg picture) map T^*.*

Proof. Compute $(V^\dagger(A \otimes \mathbb{1}_r)V)_{ab}$ by summing over the product index (i,j); the δ_{jl} from $\mathbb{1}_r$ collapses the sum over l, giving $\sum_j K_j^\dagger A K_j$. □

Theorem 5.60 (Thm. 2.2: Isometry condition iff TP). $V^\dagger V = \mathbb{1}_D$ *if and only if* $\sum_j K_j^\dagger K_j = \mathbb{1}_D$. *In particular, V is an isometry precisely when the Kraus map is trace-preserving.*

Proof. $(V^\dagger V)_{ab} = \sum_{(i,j)} \overline{V_{(i,j),a}}\, V_{(i,j),b} = \sum_j \sum_i \overline{(K_j)_{ia}}\, (K_j)_{ib} = (\sum_j K_j^\dagger K_j)_{ab}$. □

Theorem 5.61 (Thm. 2.2: Stinespring Schrödinger representation). *The Kraus map $T(\rho) = \sum_j K_j \rho K_j^\dagger$ equals the partial trace over the dilation space:*

$$T(\rho)_{ij} = \sum_{k=0}^{r-1} (V \rho V^\dagger)_{(i,k),(j,k)} = \mathrm{tr}_r(V \rho V^\dagger).$$

This is the Schrödinger-picture Stinespring representation.

Proof. Expand $(V\rho V^\dagger)_{(i,k),(j,k)}$ and sum over k to recover $\sum_k K_k \rho K_k^\dagger$. □

5.9.1 Trace-pairing expansion in transfer-matrix form

Definition 5.62 (Trace-self-dual basis). A basis $\{\sigma_i\}_i$ of $M_D(\mathbb{C})$ is *trace-self-dual* when its coordinate functionals are given by trace pairing:

$$X = \sum_i \mathrm{tr}(\sigma_i X)\, \sigma_i.$$

Definition 5.63 (Trace-pairing coefficients). For a linear map $T : M_D(\mathbb{C}) \to M_D(\mathbb{C})$ and a trace-self-dual basis $\{\sigma_i\}$, define coefficients

$$t_{ij} := \mathrm{tr}(\sigma_i\, T(\sigma_j)).$$

Theorem 5.64 (Trace-pairing expansion of a linear map). *If $\{\sigma_i\}$ is trace-self-dual, then every linear map $T : M_D(\mathbb{C}) \to M_D(\mathbb{C})$ admits the expansion*

$$T(\rho) = \sum_i \sum_j t_{ij}\ \mathrm{tr}(\sigma_j \rho)\, \sigma_i, \qquad t_{ij} = \mathrm{tr}(\sigma_i T(\sigma_j)).$$

Proof. Expand ρ and each $T(\sigma_j)$ in the chosen basis and substitute. The trace-self-dual identity replaces coordinates by traces, giving the stated double-sum formula. □

5.10 Determinant of a quantum channel (Wolf §6.1.1)

Definition 5.65 (Channel determinant). For a linear map $T : M_D(\mathbb{C}) \to M_D(\mathbb{C})$, the *channel determinant* $\det T$ is the determinant of the matrix of T with respect to the standard matrix-unit basis of $M_D(\mathbb{C})$. This is the quantity studied in [Wol12, §6.1.1].

Definition 5.66 (Unitary channel). For a unitary matrix $U \in \mathcal{U}(D)$, the *unitary channel* is $T(\rho) = U\rho U^\dagger$.

Theorem 5.67 (Thm. 6.1(1): Determinant bound for positive TP maps). *For any positive trace-preserving map $T : M_D(\mathbb{C}) \to M_D(\mathbb{C})$, $|\det T| \le 1$. This is [Wol12, Thm. 6.1(1)].*

Proof. Every eigenvalue μ of T satisfies $|\mu| \le 1$ (positivity and trace preservation force spectral radius ≤ 1). Since $\det T$ is the product of the eigenvalues (counted with algebraic multiplicity), the bound $|\det T| = \prod_i |\mu_i| \le 1$ follows by induction. □

Theorem 5.68 (Thm. 6.1(2): Determinant one iff unitary channel). *For a CPTP map $T : M_D(\mathbb{C}) \to M_D(\mathbb{C})$,*

$$|\det T| = 1 \iff \exists U \in \mathcal{U}(D), \quad T(\rho) = U\rho U^\dagger.$$

This is [Wol12, Thm. 6.1(2)].

Proof. Reverse direction: $|\det(U \cdot U^\dagger)| = |\det U|^{2D} = 1$. Forward direction: $|\det T| = 1$ together with $|\mu| \le 1$ for every eigenvalue μ (from trace preservation and positivity) forces every eigenvalue to satisfy $|\mu| = 1$. Hence T is bijective. A bijective CPTP map is a $*$-automorphism: the Heisenberg dual $T^*(Y) = \sum_i K_i^\dagger Y K_i$ is unital (since T is TP), so the Kadison–Schwarz inequality $T^*(Y^\dagger Y) \ge T^*(Y)^\dagger T^*(Y)$ becomes an equality for every Y. Hence T^*, and therefore T, preserves products. By the Skolem–Noether theorem, every $*$-automorphism of $M_D(\mathbb{C})$ has the form $T(\rho) = U\rho U^\dagger$ for some unitary U. Kraus freedom then gives $K_i = c_i U$ with $\sum_i |c_i|^2 = 1$. □

5.11 Fixed-point algebra (Wolf Thms. 6.12–6.13)

Definition 5.69 (Kraus fixed points). The *fixed-point set* of a Kraus map $E(X) = \sum_i K_i X K_i^\dagger$ is $\mathrm{Fix}(E) = \{X \in M_D(\mathbb{C}) : E(X) = X\}$.

Theorem 5.70 (Thm. 6.12: Fixed points form a $*$-subalgebra). *Let E be a unital Kraus map whose adjoint map E^* has a positive definite fixed point $\rho > 0$. Then $\mathrm{Fix}(E)$ is a $*$-subalgebra of $M_D(\mathbb{C})$. This is [Wol12, Thm. 6.12].*

Proof. Closure under addition and scalar multiplication is immediate from linearity. Closure under $\dagger$: if $E(X) = X$ then $E(X^\dagger) = X^\dagger$ because E preserves Hermitian and skew-Hermitian parts. Closure under multiplication: if $E(X) = X$ and $E(Y) = Y$, the Kadison–Schwarz gap $E(X^\dagger X) - X^\dagger X$ is PSD and has zero weighted trace against $\rho > 0$, so it vanishes. This places X in the multiplicative domain, and then $E(XY) = E(X)E(Y) = XY$. □

Remark 5.71. The multiplicative-domain step in the proof above uses the left multiplicative identity (Theorem 7.25) from Chapter 7, which appears later in the document. The logical dependency is acyclic: Chapter 7 does not use any result from §5.11.

Definition 5.72 (Adjoint fixed points)**.** The *adjoint fixed-point set* $\mathrm{Fix}(E^*) = \{X \in M_D(\mathbb{C}) : E^*(X) = X\}$, where $E^*(X) = \sum_i K_i^\dagger X K_i$.

Theorem 5.73 (Thm. 6.12 in the Heisenberg picture)**.** *If the Kraus map is trace-preserving and has a positive definite fixed point $\rho > 0$ with $E(\rho) = \rho$, then $\mathrm{Fix}(E^*)$ is a $*$-subalgebra of $M_D(\mathbb{C})$.*

Proof. Apply Theorem 5.70 to the adjoint family $\{K_i^\dagger\}$. □

Definition 5.74 (Kraus commutant)**.** The *Kraus commutant* is $\{K_i, K_i^\dagger\}' = \{X \in M_D(\mathbb{C}) : [X, K_i] = [X, K_i^\dagger] = 0\ \forall i\}$. It is a $*$-subalgebra of $M_D(\mathbb{C})$.

Theorem 5.75 (Thm. 6.13: Adjoint fixed points = Kraus commutant)**.** *Under the hypotheses of Theorem 5.73, the adjoint fixed-point $*$-subalgebra equals the Kraus commutant:*

$$\mathrm{Fix}(E^*) = \{K_i, K_i^\dagger\}'.$$

This is [Wol12, Thm. 6.13].

Proof. ($\supseteq$): if X commutes with all K_i and $K_i^\dagger$, then $E^*(X) = \sum_i K_i^\dagger X K_i = X \sum_i K_i^\dagger K_i = X$ by trace preservation. ($\subseteq$): any $*$-subalgebra inside $\mathrm{Fix}(E^*)$ lies in the Kraus commutant, because for $X \in \mathrm{Fix}(E^*)$ with $X^\dagger X \in \mathrm{Fix}(E^*)$ the Kadison–Schwarz gap vanishes, which forces $XK_i = K_i X$ and $XK_i^\dagger = K_i^\dagger X$. □

5.12 Quantum dynamical semigroups

The one-parameter semigroup theory of Wolf Chapter 7 is treated in Chapter 15. That later chapter records the exponential representation of norm-continuous semigroups, perturbation bounds, and the GKSL/Lindblad description of generators. We do not duplicate that material here.

Chapter 6

Quantum Entropy

This chapter develops the von Neumann entropy and its basic properties for finite-dimensional quantum systems.

6.1 Von Neumann entropy

Definition 6.1 (Von Neumann entropy)**.** For a Hermitian matrix $\rho \in M_D(\mathbb{C})$ with eigenvalues $\lambda_0, \dots, \lambda_{D-1}$, the *von Neumann entropy* is

$$S(\rho) = -\sum_{i=0}^{D-1} \lambda_i \log \lambda_i,$$

where $0 \log 0 := 0$.

Theorem 6.2 (Entropy is non-negative)**.** *For any density matrix ρ, $S(\rho) \geq 0$.*

Proof. Each eigenvalue λ_i of a density matrix satisfies $0 \leq \lambda_i \leq 1$, and $-x \log x \geq 0$ on $[0, 1]$. □

Lemma 6.3 (Eigenvalue sum)**.** *The eigenvalues of a density matrix sum to* 1.

Proof. Follows from $\mathrm{tr}(\rho) = \sum_i \lambda_i = 1$. □

Theorem 6.4 (Eigenvalue bound)**.** *Each eigenvalue of a density matrix lies in* $[0, 1]$.

Proof. Non-negativity comes from positive semidefiniteness. The upper bound follows because the eigenvalues are non-negative and sum to 1. □

Theorem 6.5 (Entropy upper bound)**.** *For a density matrix $\rho \in M_D(\mathbb{C})$ with $D \geq 1$,*

$$S(\rho) \leq \log D.$$

Proof. By Jensen's inequality applied to the concave function $-x \log x$, the entropy is maximized when all eigenvalues equal $1/D$. □

6.2 Tripartite partial traces

Definition 6.6 (Partial trace over A)**.** For a tripartite matrix ρ_{ABC} on $\mathbb{C}^{d_A} \otimes \mathbb{C}^{d_B} \otimes \mathbb{C}^{d_C}$, the partial trace over A is

$$(\mathrm{tr}_A\, \rho_{ABC})_{(b_1,c_1)(b_2,c_2)} = \sum_{a=0}^{d_A-1} (\rho_{ABC})_{(a,b_1,c_1)(a,b_2,c_2)}.$$

Definition 6.7 (Partial trace over C)**.** The partial trace over C:

$$(\mathrm{tr}_C\, \rho_{ABC})_{(a_1,b_1)(a_2,b_2)} = \sum_{c=0}^{d_C-1} (\rho_{ABC})_{(a_1,b_1,c)(a_2,b_2,c)}.$$

Definition 6.8 (Partial trace over AC)**.** The partial trace over A and C:

$$(\mathrm{tr}_{AC}\, \rho_{ABC})_{b_1 b_2} = \sum_{a=0}^{d_A-1} \sum_{c=0}^{d_C-1} (\rho_{ABC})_{(a,b_1,c)(a,b_2,c)}.$$

Lemma 6.9 (Partial trace preserves Hermiticity)**.** *If ρ_{ABC} is Hermitian, then $\mathrm{tr}_A(\rho_{ABC})$, $\mathrm{tr}_C(\rho_{ABC})$, and $\mathrm{tr}_{AC}(\rho_{ABC})$ are all Hermitian. The same holds for bipartite partial traces $\mathrm{tr}_A(\rho_{AB})$ and $\mathrm{tr}_B(\rho_{AB})$.*

Proof. Follows from $\overline{\rho_{ji}} = \rho_{ij}$ applied entry-wise inside the summation defining each partial trace. □

6.3 Strong subadditivity

Theorem 6.10 (Strong subadditivity)**.** *For a tripartite density matrix ρ_{ABC},*

$$S(\rho_{ABC}) + S(\rho_B) \leq S(\rho_{AB}) + S(\rho_{BC}).$$

Definition 6.11 (SSA equality)**.** A tripartite density matrix ρ_{ABC} satisfies *SSA equality* if

$$S(\rho_{ABC}) + S(\rho_B) = S(\rho_{AB}) + S(\rho_{BC}).$$

6.4 Mutual information

Definition 6.12 (Mutual information)**.** The *quantum mutual information* of a bipartite state ρ_{AB} is

$$I(A{:}B) = S(\rho_A) + S(\rho_B) - S(\rho_{AB}).$$

Theorem 6.13 (Mutual information is non-negative)**.** *For any bipartite density matrix ρ_{AB}, $I(A{:}B) \geq 0$.*

Proof. Apply strong subadditivity (Theorem 6.10) with trivial B (one-dimensional middle system). The SSA inequality $S(\rho_{ABC}) + S(\rho_B) \leq S(\rho_{AB}) + S(\rho_{BC})$ reduces to subadditivity $S(\rho_{AC}) \leq S(\rho_A) + S(\rho_C)$, which gives $I(A{:}C) \geq 0$. □

Chapter 7

Schwarz Inequalities and Multiplicative Domains

This chapter develops the Schwarz-inequality theory from Wolf's Chapter 5 [Wol12]. We begin with the Kadison–Schwarz inequality for Kraus maps, then describe the multiplicative domain and its algebraic structure. We also establish the extension to normal operators for positive subunital maps, the resulting order and spectral contractivity statements, and Wolf's standard example of a Schwarz map that is not completely positive.

7.1 Kadison–Schwarz inequality

A Kraus map is the concrete realisation of a completely positive map (Definition 5.4). The Kadison–Schwarz inequality and the multiplicative-domain characterisation are proved first for Kraus maps. The transfer map is a Kraus map by construction (Remark 5.14), so these results apply directly to transfer maps.

Definitions 7.1–7.4 below restate the Kraus-map notions from Chapter 5 for the reader's convenience.

Definition 7.1 (Kraus map)**.** Given operators $\{K_i\}_{i=0}^{d-1}$ with $K_i \in M_D(\mathbb{C})$, the *Kraus map* is $E(X) = \sum_{i=0}^{d-1} K_i X K_i^\dagger$. A linear map is completely positive (Definition 5.4) if and only if it can be written in this form.

Definition 7.2 (Adjoint Kraus map)**.** The *adjoint Kraus map* is $E^*(X) = \sum_{i=0}^{d-1} K_i^\dagger X K_i$. When the $\{K_i\}$ are the matrices of an MPS tensor A, this is the transfer map of the conjugate-transposed family $i \mapsto (A^i)^\dagger$.

Definition 7.3 (Unital Kraus map)**.** A Kraus map is *unital* if $\sum_{i=0}^{d-1} K_i K_i^\dagger = \mathbb{1}$. Equivalently, $E(\mathbb{1}) = \mathbb{1}$. In the later gauge language this is the right-canonical condition.

Definition 7.4 (Trace-preserving Kraus map)**.** A Kraus map is *trace-preserving* if $\sum_{i=0}^{d-1} K_i^\dagger K_i = \mathbb{1}$. Equivalently, the adjoint Kraus map is unital. This is the standard MPS normalization condition. In the later gauge language it is the left-canonical condition, so Kadison–Schwarz arguments are often applied to the adjoint map.

Theorem 7.5 (Kadison–Schwarz inequality)**.** *Let $E(X) = \sum_i K_i X K_i^\dagger$ be a* unital *Kraus map (i.e. $\sum_i K_i K_i^\dagger = \mathbb{1}$). Then for all $X \in M_D(\mathbb{C})$,*

$$E(X^\dagger X) \geq E(X)^\dagger E(X).$$

This is [Wol12, Eq. (5.2)].

Proof. The 2×2 block matrix $\left[\begin{smallmatrix} X^\dagger X & X^\dagger \\ X & \mathbb{1} \end{smallmatrix}\right]$ is PSD ($= vv^\dagger$ with $v = [X^\dagger, \mathbb{1}]^T$). Applying the unital CP map blockwise preserves positive semidefiniteness (each block $K_i(\cdot)K_i^\dagger$ preserves PSD, and unitality ensures the $(2,2)$-block maps to $\mathbb{1}$). The Schur complement of the $(2,2)$-block $\mathbb{1}$ gives the result. $\square$

Theorem 7.6 (Kadison–Schwarz for the adjoint channel)**.** *If $\sum_i K_i^\dagger K_i = \mathbb{1}$ (trace-preserving), then the adjoint Kraus map satisfies $E^*(X^\dagger X) \geq E^*(X)^\dagger E^*(X)$.*

Proof. The adjoint operators $\{K_i^\dagger\}$ satisfy $\sum_i K_i^\dagger (K_i^\dagger)^\dagger = \sum_i K_i^\dagger K_i = \mathbb{1}$, so they form a unital family. Apply Theorem 7.5 to $\{K_i^\dagger\}$. $\square$

7.1.1 Two-positive maps and the generalized Schwarz inequality

Definition 7.7 (k-positive map)**.** A linear map $E : M_D(\mathbb{C}) \to M_D(\mathbb{C})$ is k-positive if $E \otimes \mathbb{1}_k$ is positive on $M_D(\mathbb{C}) \otimes M_k(\mathbb{C})$.

Definition 7.8 (Two-positive map)**.** A linear map is *two-positive* if it is 2-positive.

Theorem 7.9 (Completely positive maps are k-positive)**.** *Every completely positive map is k-positive for all $k \geq 1$.*

Theorem 7.10 ($(k+1)$-positive implies k-positive)**.** *Positivity under amplification is monotone in k.*

Theorem 7.11 (Completely positive implies two-positive)**.** *Every completely positive map is two-positive.*

Theorem 7.12 (Two-positive implies positive)**.** *Every two-positive map is positive.*

Definition 7.13 (Unital linear map)**.** A linear map E is unital if $E(\mathbb{1}) = \mathbb{1}$.

Theorem 7.14 (Kadison–Schwarz for unital two-positive maps)**.** *If E is unital and two-positive, then*

$$E(X^\dagger X) \geq E(X)^\dagger E(X)$$

for all $X \in M_D(\mathbb{C})$.

Proof. Form the 2×2 block matrix $\left(\begin{smallmatrix} \mathbb{1} & X \\ X^\dagger & X^\dagger X \end{smallmatrix}\right) \geq 0$. Apply $E \otimes \mathrm{id}_2$ (positive by 2-positivity) and read off the $(2,2)$-block Schur complement: $E(X^\dagger X) - E(X)^\dagger E(X) \geq 0$. $\square$

Theorem 7.15 (Kraus-form Kadison–Schwarz as a special case)**.** *The Kraus-map Kadison–Schwarz inequality is recovered from the two-positive formulation.*

7.1.2 Douglas-type factorization lemmas

Theorem 7.16 (Range inclusion implies right factorization). *If* $\operatorname{ran}(A) \subseteq \operatorname{ran}(B)$*, then there exists* C *such that* $A = BC$*.*

Proof. In finite dimensions, $\operatorname{ran}(A) \subseteq \operatorname{ran}(B)$ implies $A = B(B^\dagger B)^\dagger B^\dagger A$, where $(B^\dagger B)^\dagger$ denotes the Moore–Penrose pseudoinverse. Set $C = (B^\dagger B)^\dagger B^\dagger A$. □

Theorem 7.17 (Vectorwise range criterion implies factorization). *If* $Av \in \operatorname{ran}(B)$ *for every vector* v*, then there exists* C *with* $A = BC$*.*

Theorem 7.18 (Hilbert–Schmidt contraction). *If a Kraus map is both unital and trace-preserving, then* $\operatorname{tr}(E(X)^\dagger E(X)) \leq \operatorname{tr}(X^\dagger X)$ *for all* $X \in M_D(\mathbb{C})$*.*

Proof. By Theorem 7.5, $E(X^\dagger X) - E(X)^\dagger E(X) \geq 0$, so $\operatorname{tr}(E(X^\dagger X)) \geq \operatorname{tr}(E(X)^\dagger E(X))$. Trace preservation gives $\operatorname{tr}(E(X^\dagger X)) = \operatorname{tr}(X^\dagger X)$. □

Lemma 7.19 (Trace pairing for Kraus map and adjoint map). *For all* $X, Y \in M_D(\mathbb{C})$*,*

$$\operatorname{tr}\!\big(X\,E(Y)\big) = \operatorname{tr}\!\big(E^*(X)\,Y\big).$$

Lemma 7.20 (Positive-definite trace test). *If* $A > 0$*,* $X \geq 0$*, and* $\operatorname{tr}(AX) = 0$*, then* $X = 0$*.*

Theorem 7.21 (Fixed-point peripheral eigenvectors satisfy KS equality). *Let* E *be unital and let* E^* *be trace-preserving with a positive definite fixed point* $\rho > 0$*. If* $E(X) = \mu X$ *with* $|\mu| = 1$*, then*

$$E(X^\dagger X) = E(X)^\dagger E(X).$$

7.2 Multiplicative domain

Theorem 7.22 (KS gap decomposition). *For a unital Kraus map* E*, the Kadison–Schwarz gap decomposes as*

$$E(X^\dagger X) - E(X)^\dagger E(X) = \sum_{i=0}^{d-1} R_i^\dagger R_i,$$

where $R_i := XK_i^\dagger - K_i^\dagger E(X)$*.*

Proof. Expand the right-hand side $\sum_i (XK_i^\dagger - K_i^\dagger E(X))^\dagger (XK_i^\dagger - K_i^\dagger E(X))$, distribute, and use unitality $\sum_i K_i K_i^\dagger = \mathbb{1}$ to identify the result with the KS gap. □

Theorem 7.23 (KS equality implies Kraus commutation). *Let* E *be a unital Kraus map. If* $E(X^\dagger X) = E(X)^\dagger E(X)$ *for some* X*, then* $XK_i^\dagger = K_i^\dagger E(X)$ *for all* i*.*

Proof. By Theorem 7.22, the KS gap is $\sum_i R_i^\dagger R_i$ with $R_i = XK_i^\dagger - K_i^\dagger E(X)$. If the gap vanishes, $\sum_i R_i^\dagger R_i = 0$. Each summand $R_i^\dagger R_i$ is PSD, so each must be zero, giving $R_i = 0$. □

Theorem 7.24 (KS equality for peripheral eigenvectors). *Let* E *be a Kraus map that is both unital and trace-preserving. If* $E(X) = \mu X$ *with* $|\mu| = 1$*, then the KS gap vanishes:* $E(X^\dagger X) = E(X)^\dagger E(X)$*.*

Proof. The gap $G := E(X^\dagger X) - E(X)^\dagger E(X)$ is PSD by Theorem 7.5. Its trace satisfies $\operatorname{tr}(G) = \operatorname{tr}(E(X^\dagger X)) - \operatorname{tr}(E(X)^\dagger E(X))$. Trace preservation gives $\operatorname{tr}(E(X^\dagger X)) = \operatorname{tr}(X^\dagger X)$. Using $E(X) = \mu X$ and $|\mu| = 1$, $\operatorname{tr}(E(X)^\dagger E(X)) = |\mu|^2 \operatorname{tr}(X^\dagger X) = \operatorname{tr}(X^\dagger X)$. So $\operatorname{tr}(G) = 0$, and a PSD matrix with zero trace must be zero. □

Theorem 7.25 (Left multiplicative identity from KS equality)**.** *Let E be a unital Kraus map. If $E(X^\dagger X) = E(X)^\dagger E(X)$, then $E(X^\dagger Y) = E(X)^\dagger E(Y)$ for all $Y \in M_D(\mathbb{C})$.*

Proof. From Theorem 7.23, $XK_i^\dagger = K_i^\dagger E(X)$ for all i. Taking conjugate transposes: $K_i X^\dagger = E(X)^\dagger K_i$. Then

$$\begin{aligned} E(X^\dagger Y) &= \sum_i K_i X^\dagger Y K_i^\dagger \\ &= \sum_i E(X)^\dagger K_i Y K_i^\dagger \\ &= E(X)^\dagger E(Y). \end{aligned}$$

□

7.2.1 Full multiplicative-domain structure

Definition 7.26 (Right and left multiplicative domains)**.** For a Kraus map E, define

$$\mathcal{A}_R(E) := \{X \in M_D(\mathbb{C}) : E(XY) = E(X)E(Y) \text{ for all } Y \in M_D(\mathbb{C})\}$$

and

$$\mathcal{A}_L(E) := \{X \in M_D(\mathbb{C}) : E(YX) = E(Y)E(X) \text{ for all } Y \in M_D(\mathbb{C})\}.$$

We follow the convention that $\mathcal{A}_R(E)$ controls right multiplication and $\mathcal{A}_L(E)$ controls left multiplication. These are the two one-sided multiplicative domains from [Wol12, Eqs. (5.11)–(5.12)].

Definition 7.27 (Full multiplicative domain)**.** The *multiplicative domain* of E is

$$\mathcal{A}(E) := \mathcal{A}_R(E) \cap \mathcal{A}_L(E).$$

Theorem 7.28 (Multiplicative-domain characterization)**.** *Let E be a unital Kraus map. Then*

$$X \in \mathcal{A}_R(E) \iff E(XX^\dagger) = E(X)E(X)^\dagger$$

and

$$X \in \mathcal{A}_L(E) \iff E(X^\dagger X) = E(X)^\dagger E(X).$$

Together these are the two one-sided characterizations from [Wol12, Thm. 5.7].

Proof. If X lies in a one-sided multiplicative domain, evaluate the defining identity at $Y = X^\dagger$. Conversely, the left-domain implication is exactly Theorem 7.25, and the right-domain implication follows by applying Theorem 7.25 to $X^\dagger$. □

Theorem 7.29 (One-sided multiplicative domains are subalgebras)**.** *For a unital Kraus map E, both $\mathcal{A}_R(E)$ and $\mathcal{A}_L(E)$ are unital subalgebras of $M_D(\mathbb{C})$.*

Proof. Closure under addition follows from linearity of E. Closure under multiplication: if $X, Y \in \mathcal{A}_R(E)$, then $E((XY)Z) = E(X)E(YZ) = E(X)E(Y)E(Z)$ for all Z, using Theorem 7.28 iteratively, and the $\mathcal{A}_L(E)$ case is analogous. Containment of $\mathbb{1}$ follows from unitality of E. □

Theorem 7.30 (Multiplicative domain is a $*$-subalgebra)**.** *For a unital Kraus map E, the full multiplicative domain $\mathcal{A}(E)$ is a $*$-subalgebra of $M_D(\mathbb{C})$. This is the algebraic content of [Wol12, Thm. 5.7].*

Proof. If $X \in \mathcal{A}(E) = \mathcal{A}_R(E) \cap \mathcal{A}_L(E)$, then $E(X^\dagger X) = E(X)^\dagger E(X)$ and $E(XX^\dagger) = E(X)E(X)^\dagger$. The characterization of Theorem 7.28 shows $X^\dagger \in \mathcal{A}(E)$. Combined with Theorem 7.29, this gives a $*$-subalgebra. □

Theorem 7.31 (Restriction to multiplicative domain is a $*$-homomorphism)**.** *If E is unital, then the restricted map $E|_{\mathcal{A}(E)} : \mathcal{A}(E) \to M_D(\mathbb{C})$ is a $*$-algebra homomorphism. This is the homomorphism conclusion of [Wol12, Thm. 5.7].*

Proof. Multiplicativity on $\mathcal{A}(E)$ is immediate from the definition. For $X \in \mathcal{A}(E)$, the Kraus commutation relation from Theorem 7.23 gives $E(X^\dagger) = E(X)^\dagger$, so the restriction preserves adjoints. □

7.2.2 Schwarz inequality for normal operators

Theorem 7.32 (CP Schwarz inequality for normal operators)**.** *Let $E^*(X) = \sum_i K_i^\dagger X K_i$ be the adjoint Kraus map of a trace-preserving Kraus family, and let $A \in M_D(\mathbb{C})$ be normal. Then the Schwarz gap*

$$E^*(A^\dagger A) - E^*(A^\dagger)E^*(A)$$

is positive semidefinite. Equivalently,

$$E^*(A^\dagger)E^*(A) \le E^*(A^\dagger A).$$

This is the completely positive case of [Wol12, Prop. 5.1].

Proof. The positivity statement is exactly Theorem 7.6. The second formulation is just the corresponding Loewner-order reformulation. □

Theorem 7.33 (Prop. 5.1: Schwarz inequality for normal operators)**.** *Let $T : M_D(\mathbb{C}) \to M_D(\mathbb{C})$ be a positive linear map with $T(\mathbb{1}) \le \mathbb{1}$. If $A \in M_D(\mathbb{C})$ is normal, then*

$$T(A^\dagger A) - T(A^\dagger)T(A) \ge 0.$$

This is [Wol12, Prop. 5.1].

Proof. Since A is normal, diagonalize $A = UDU^\dagger$ with D diagonal and U unitary. Express $A^\dagger A = U|D|^2U^\dagger$ as a sum $\sum_{i,j} \bar{d}_i d_j P_i P_j$ where $P_i = U e_i e_i^\dagger U^\dagger$. Each P_i is rank-one PSD, and the images $T(P_i)$ pairwise commute because they are images of mutually orthogonal projectors under a positive map. The Schwarz gap then reduces to a block quadratic form in the images, which is shown nonneg by a simultaneous-diagonalization argument: positivity of T ensures the scalar PSD matrix $(\langle v_k, T(P_i P_j) v_k \rangle)_{i,j}$ dominates the product $(\langle v_k, T(P_i) v_k \rangle \langle v_k, T(P_j) v_k \rangle)_{i,j}$, and combining over a common eigenbasis of the $T(P_i)$ yields the result. □

Remark 7.34. The proof follows the positivity-on-abelian argument of [Wol12, Prop. 1.6]: a positive map restricted to the commutative $*$-subalgebra generated by the spectral projectors of A satisfies the Schwarz inequality directly, without invoking the Arveson extension theorem.

7.2.3 Schwarz inequality for subnormal and commuting-dominant operators

Theorem 7.35 (Thm. 5.5: Schwarz inequality for subnormal operators)**.** *Let $T : M_D(\mathbb{C}) \to M_D(\mathbb{C})$ be a positive linear map with $T(\mathbb{1}) \le \mathbb{1}$. If $A \in M_D(\mathbb{C})$ is subnormal (i.e. there exists a normal operator on a larger space whose compression to $\mathbb{C}^D$ is A), then*

$$T(A^\dagger A) - T(A^\dagger)T(A) \ge 0.$$

This is [Wol12, Thm. 5.5].

Proof. Embed A as the northwest corner of a normal operator N on $\mathbb{C}^D \oplus \mathbb{C}^E$. Extend T to a positive subunital map $\tilde{T}$ on the larger space (acting as T on the northwest block, zero elsewhere). Apply Theorem 7.33 to $\tilde{T}$ and N, then compress the resulting inequality back to $\mathbb{C}^D$. □

Theorem 7.36 (Thm. 5.6: Schwarz inequality for commuting-dominant operators)**.** *Let $T : M_D(\mathbb{C}) \to M_D(\mathbb{C})$ be a positive linear map with $T(\mathbb{1}) \le \mathbb{1}$. If $A \in M_D(\mathbb{C})$ admits a positive semidefinite* dominant *$B \ge 0$ with $A^\dagger A \le B$ and B commuting with A (in the sense $BA = AB$, $BA^\dagger = A^\dagger B$), then*

$$T(A^\dagger)T(A) \le T(B).$$

This is [Wol12, Thm. 5.6].

Proof. Let $B^{1/2}$ be the positive square root of B. Since B commutes with A and $A^\dagger$, the contraction $C = B^{-1/2}A$ (defined on the range of $B^{1/2}$) is subnormal. An ε-regularisation $B_\varepsilon = B + \varepsilon\mathbb{1}$ makes $B_\varepsilon > 0$ and the contraction C_ε globally defined. Apply Theorem 7.35 to C_ε, multiply both sides by $T(B_\varepsilon)$, use positivity and monotonicity, and take $\varepsilon \to 0$. □

Theorem 7.37 (CP variant: Kadison–Schwarz for commuting-dominant inputs)**.** *Let E^* be the adjoint of a trace-preserving Kraus map. If $A \in M_D(\mathbb{C})$ admits a positive semidefinite dominant B commuting with A, $A^\dagger$, and satisfying $A^\dagger A \le B$, then $E^*(A^\dagger)E^*(A) \le E^*(B)$.*

Proof. Apply Theorem 7.36 to the adjoint Kraus map (which is unital under the trace-preservation hypothesis). □

7.2.4 Positive maps preserve order and spectral intervals

Theorem 7.38 (Positive maps are monotone)**.** *If $T : M_D(\mathbb{C}) \to M_D(\mathbb{C})$ is positive and $A \le B$, then $T(A) \le T(B)$.*

Proof. Since $B - A \ge 0$, positivity gives $T(B - A) \ge 0$. By linearity, this is $T(B) - T(A) \ge 0$. □

Theorem 7.39 (Positive maps preserve adjoints)**.** *If $T : M_D(\mathbb{C}) \to M_D(\mathbb{C})$ is positive, then $T(A^\dagger) = T(A)^\dagger$ for all $A \in M_D(\mathbb{C})$.*

Proof. Write $A = B + iC$ with B and C Hermitian. Positive maps send Hermitian matrices to Hermitian matrices, so they preserve this decomposition and commute with taking adjoints. □

Theorem 7.40 (Order intervals are preserved)**.** *Let $T : M_D(\mathbb{C}) \to M_D(\mathbb{C})$ be positive with $T(\mathbb{1}) \le \mathbb{1}$. If $a \le 0 \le b$ and $a\mathbb{1} \le A \le b\mathbb{1}$, then*

$$a\mathbb{1} \le T(A) \le b\mathbb{1}.$$

This is the matrix form of [Wol12, Eq. (5.21)].

Proof. By monotonicity, $T(a\mathbb{1}) \le T(A) \le T(b\mathbb{1})$. For the lower bound, $T(a\mathbb{1}) = a\,T(\mathbb{1})$. Since $a \le 0$ and $T(\mathbb{1}) \le \mathbb{1}$, we have $a\,T(\mathbb{1}) \ge a\mathbb{1}$. For the upper bound, $T(b\mathbb{1}) = b\,T(\mathbb{1}) \le b\mathbb{1}$ since $b \ge 0$. Therefore $a\mathbb{1} \le T(A) \le b\mathbb{1}$. □

Theorem 7.41 (Spectral interval contractivity)**.** *Let $T : M_D(\mathbb{C}) \to M_D(\mathbb{C})$ be positive with $T(\mathbb{1}) \le \mathbb{1}$, and let $A \in M_D(\mathbb{C})$ be Hermitian. If* spec$(A) \subseteq [a, b]$ *with $a \le 0 \le b$, then*

$$\operatorname{spec}(T(A)) \subseteq [a, b].$$

This is [Wol12, Eq. (5.21)].

Proof. For Hermitian matrices, the inclusion $\operatorname{spec}(A) \subseteq [a, b]$ is equivalent to the order bounds $a\mathbb{1} \le A \le b\mathbb{1}$. Apply Theorem 7.40 and translate the resulting inequalities back into spectral bounds. □

7.2.5 Wolf Example 5.3

Example 7.42 (Wolf Example 5.3 map)**.** Consider the linear map $T : M_2(\mathbb{C}) \to M_2(\mathbb{C})$ defined by

$$T(A) = \frac{1}{2}A^T + \frac{1}{4}\operatorname{tr}(A)\mathbb{1}.$$

This is the map from [Wol12, Ex. 5.3].

Theorem 7.43 (Wolf Example 5.3 is positive)**.** *The map from Example 7.42 is positive.*

Proof. It is a nonnegative linear combination of the transpose map and the map $A \mapsto \operatorname{tr}(A)\mathbb{1}$, and both of these maps send positive semidefinite matrices to positive semidefinite matrices. □

Theorem 7.44 (Wolf Example 5.3 satisfies the Schwarz inequality)**.** *For every $A \in M_2(\mathbb{C})$,*

$$T(A^\dagger A) - T(A^\dagger)T(A) \ge 0,$$

where T is the map from Example 7.42.

Proof. Write the Schwarz gap as $F(A)$. One checks that $F(A + \lambda\mathbb{1}) = F(A)$ for every scalar λ, so it is enough to treat the case $\operatorname{tr}(A) = 0$. In that case a direct 2×2 computation shows that $F(A)$ dominates $\frac{1}{2}(A^\dagger A)^T$, hence is positive semidefinite. □

Theorem 7.45 (Wolf Example 5.3 is not completely positive)**.** *The map from Example 7.42 is not completely positive.*

Proof. Compute the Choi matrix of T and test it on the antisymmetric vector $|01\rangle - |10\rangle$. The resulting expectation value is negative, so the Choi matrix is not positive semidefinite. □

Chapter 8

Perron–Frobenius Theory for Channels and Transfer Maps

This chapter establishes the Perron–Frobenius theory used in subsequent chapters: existence, positive definiteness, and uniqueness of fixed points for injective and irreducible transfer maps, canonical gauge constructions, ergodicity of irreducible channels, the spectral-radius identification of Perron eigenvalues, and Wolf's spectral characterization of irreducibility. The conceptual source is Wolf's treatment of irreducible positive maps [Wol12, Ch. 6, especially Thms. 6.2–6.5 and Cor. 6.3] and [EHK78].

8.1 Positive definiteness

Remark 8.1. Wolf's general theory treats irreducible positive maps first ([Wol12, Thm. 6.2]), then derives non-degeneracy and positive definiteness ([Wol12, Thm. 6.3]). The injective-tensor version (Theorem 8.2) uses a shorter direct kernel argument; the irreducible CP-map version (Theorem 8.3) follows via the support-projection argument of [Wol12, Thm. 6.2].

Theorem 8.2 (PSD fixed point is positive definite)**.** *Let A be an injective MPS tensor and let $\rho \geq 0$, $\rho \neq 0$, be a fixed point of the transfer map $\mathcal{E}_A(\rho) = \rho$. Then ρ is positive definite.*

Proof. Suppose $v^\dagger \rho v = 0$ for some $v \neq 0$. Since $\rho \geq 0$, this implies $\rho v = 0$. The fixed-point equation gives $v^\dagger \rho v = \sum_i ((A^i)^\dagger v)^\dagger \rho ((A^i)^\dagger v)$. Each term is nonneg; if the sum is zero, then $\rho (A^i)^\dagger v = 0$ for all i: the kernel of ρ is invariant under all $(A^i)^\dagger$. Since the $\{A^i\}$ span $M_D(\mathbb{C})$, we have $\rho M v = 0$ for every matrix M. Given any vector w, choose M with $Mv = w$. Then $\rho w = 0$ for every w, so $\rho = 0$, contradicting $\rho \neq 0$. In [Wol12, Thm. 6.3, item 2], the same conclusion is reached via the growth condition $(\mathbb{1} + T)^{d-1}(X) > 0$ of [Wol12, Thm. 6.2, condition (2)]; here the direct kernel argument is shorter under injectivity. $\square$

Theorem 8.3 (PSD fixed point is PD — irreducible version)**.** *If $\mathcal{E}_A$ is irreducible and $\rho \geq 0$, $\rho \neq 0$ is a fixed point, then ρ is positive definite.*

Proof. Suppose ρ is not positive definite. Then ρ has a nontrivial kernel; let Q be the orthogonal projection onto the support of ρ (the complement of the kernel). A direct computation using the fixed-point equation $\mathcal{E}_A(\rho) = \rho$ shows that Q is an invariant projection for $\mathcal{E}_A$: it satisfies $Q\,\mathcal{E}_A(QXQ)\,Q = \mathcal{E}_A(QXQ)$ for all X. By irreducibility of $\mathcal{E}_A$ (Definition 5.12), every invariant projection is 0 or $\mathbb{1}$. If $Q = 0$ then $\rho = 0$, contradicting $\rho \neq 0$. If $Q = \mathbb{1}$ then ρ is positive definite, contradicting our assumption. Hence ρ must be positive definite. This is the support-projection direction of [Wol12, Thm. 6.2, condition (1)]. $\square$

Theorem 8.4 (Orthogonal trace condition)**.** *Let E be an irreducible completely positive map on $M_D(\mathbb{C})$, and let $A, B \geq 0$ be nonzero positive semidefinite matrices with $\operatorname{tr}(BA) = 0$. Then there exists t with $1 \leq t \leq D-1$ such that*

$$\operatorname{tr}(B \cdot E^t(A)) > 0.$$

Proof. The growth theorem for irreducible CP maps ([Wol12, Thm. 6.2, item 2]) gives $(\mathbb{1} + E)^{D-1}(A) > 0$ (positive definite). Since $B \geq 0$ is nonzero, $\operatorname{tr}(B \cdot (\mathbb{1} + E)^{D-1}(A)) > 0$. Expanding by the binomial theorem:

$$\operatorname{tr}\Big(B \cdot \sum_{k=0}^{D-1} \tbinom{D-1}{k} E^k(A)\Big) > 0.$$

Each term $\operatorname{tr}(B \cdot E^k(A)) \geq 0$ since both factors are PSD. The $k = 0$ term vanishes by hypothesis ($\operatorname{tr}(BA) = 0$). Hence at least one $k \in \{1, \dots, D-1\}$ contributes a strictly positive term. This is [Wol12, Thm. 6.2, item 4]: item (2) (growth condition) implies item (4) by a binomial expansion and PSD positivity argument. $\square$

8.2 Uniqueness

Theorem 8.5 (Uniqueness of PSD fixed point)**.** *Let A be an injective MPS tensor. If $\rho, \sigma \geq 0$ are both nonzero fixed points of $\mathcal{E}_A$, then $\sigma = c\rho$ for some $c \in \mathbb{C}$.*

Proof. Both ρ and σ are positive definite by Theorem 8.2. Write $\rho = SS^\dagger$ via a spectral square root and set $H = (S^\dagger)^{-1}\sigma S^{-1}$, which is Hermitian and positive definite. Let c_0 be the minimal eigenvalue of H. Then $\tau := \sigma - c_0\rho$ is a PSD fixed point which is *not* positive definite (it has a zero eigenvalue by construction of c_0). By Theorem 8.2, τ must be zero, giving $\sigma = c_0\rho$. **Relation to Wolf:** In [Wol12, Thm. 6.3, item 2], uniqueness is proved via the growth condition of [Wol12, Thm. 6.2]. The same minimal-eigenvalue argument applies here, using the injective-tensor positive-definiteness (Theorem 8.2) in place of the growth condition. $\square$

Remark 8.6. The proof produces a positive real scalar $c_0 > 0$: it is the minimal eigenvalue of the positive definite Hermitian matrix $H = (S^\dagger)^{-1}\sigma S^{-1}$.

Theorem 8.7 (Uniqueness of positive eigenvalue for irreducible CP maps)**.** *Let $D \geq 1$ and let E be an irreducible completely positive map on $M_D(\mathbb{C})$. Suppose $\rho, \sigma \geq 0$ are nonzero positive semidefinite matrices satisfying $E(\rho) = r_1\rho$ and $E(\sigma) = r_2\sigma$ for real scalars $r_1, r_2 > 0$. Then $r_1 = r_2$.*

Proof. Extract Kraus operators K for E. Irreducibility of E passes to the adjoint transfer map $E^\dagger(\,\cdot\,) = \sum_i K_i^\dagger\,(\,\cdot\,)\,K_i$, which is a positive CP map. The Perron–Frobenius existence theorem (Theorem 8.15) together with the irreducible PSD-to-PosDef upgrade (Theorem 8.3) gives a positive definite eigenvector $\tau > 0$ with eigenvalue $t > 0$: $E^\dagger(\tau) = t\,\tau$. For any nonzero PSD X with $E(X) = sX$, the trace pairing identity yields

$$s \cdot \operatorname{tr}(\tau X) = \operatorname{tr}(\tau \cdot E(X)) = \operatorname{tr}(E^\dagger(\tau) \cdot X) = t \cdot \operatorname{tr}(\tau X).$$

Since $\tau > 0$ and $X \geq 0$, $X \neq 0$, the scalar $\operatorname{tr}(\tau X) > 0$, so $s = t$. Applying this to both (ρ, r_1) and (σ, r_2) gives $r_1 = t = r_2$. **Relation to Wolf:** This is [Wol12, Thm. 6.3, item 3], establishing non-degeneracy of the spectral-radius eigenvalue for irreducible CP maps. The proof follows Wolf's dual-map trace argument. $\square$

8.3 Existence and the Perron–Frobenius theorem

Theorem 8.8 (Existence of PSD fixed point)**.** *Assume $D \geq 1$. Let A be an MPS tensor with $\sum_i (A^i)^\dagger A^i = \mathbb{1}$ (so that $\mathcal{E}_A$ is trace-preserving). Then there exists $\rho \geq 0$, $\rho \neq 0$, with $\mathcal{E}_A(\rho) = \rho$.*

Proof. Under the normalization $\sum_i (A^i)^\dagger A^i = \mathbb{1}$, the transfer map is a quantum channel (Theorem 5.15). The Cesàro fixed-point theorem (Theorem 5.19) then provides the fixed point; injectivity is not needed for this step. □

Theorem 8.9 (Quantum Perron–Frobenius)**.** *Assume $D \geq 1$. Let A be an injective MPS tensor with $\sum_i (A^i)^\dagger A^i = \mathbb{1}$. Then the transfer map $\mathcal{E}_A$ has a unique PSD fixed point ρ (up to scaling), and ρ is positive definite.*

Proof. Combine Theorems 8.8, 8.2, and 8.5. □

8.4 Right- and left-canonical gauges

Theorem 8.10 (Right-canonical (unital) gauge)**.** *Let $\mathcal{E}_A(\rho) = \rho$ and let S be an invertible matrix with $SS^\dagger = \rho$. Define the gauged operators $A'^i = S^{-1} A^i S$. Then $\sum_i A'^i (A'^i)^\dagger = \mathbb{1}$, i.e. the gauged Kraus map is unital. This is the right-canonical normalization used in subsequent chapters.*

Proof. Expand each summand: $(S^{-1}A^iS)(S^{-1}A^iS)^\dagger = S^{-1}A^i \underbrace{SS^\dagger}_{\rho} (A^i)^\dagger (S^\dagger)^{-1}$. Summing over i and using $\sum_i A^i \rho (A^i)^\dagger = \rho$ gives $S^{-1}\rho(S^\dagger)^{-1} = S^{-1}SS^\dagger(S^\dagger)^{-1} = \mathbb{1}$. □

Theorem 8.11 (Left-canonical (trace-preserving) gauge)**.** *Let σ be a fixed point of the adjoint transfer map $\sum_i (A^i)^\dagger \sigma A^i = \sigma$, and let S be invertible with $S^\dagger S = \sigma$. Define $A'^i = SA^iS^{-1}$. Then $\sum_i (A'^i)^\dagger A'^i = \mathbb{1}$, i.e. the gauged Kraus map is trace-preserving. This is the left-canonical normalization used in subsequent chapters.*

Proof. Expand each summand: $(SA^iS^{-1})^\dagger(SA^iS^{-1}) = (S^\dagger)^{-1}(A^i)^\dagger \underbrace{S^\dagger S}_{\sigma} A^iS^{-1}$. Summing over i and using $\sum_i (A^i)^\dagger \sigma A^i = \sigma$ gives $(S^\dagger)^{-1}\sigma S^{-1} = (S^\dagger)^{-1}S^\dagger SS^{-1} = \mathbb{1}$. □

Remark 8.12. The right-canonical gauge of Theorem 8.10 and the left-canonical gauge of Theorem 8.11 are generally different similarity transforms: one is built from a fixed point of $\mathcal{E}_A$, the other from a fixed point of the adjoint transfer map. This section does not claim that a single gauge makes the transfer map simultaneously unital and trace-preserving.

8.5 Similarity preserves irreducibility

Lemma 8.13 (Similarity transform preserves irreducibility)**.** *Let E be an irreducible completely positive map on $M_D(\mathbb{C})$, let $C \in M_D(\mathbb{C})$ be invertible ($\det C \neq 0$), and let $c > 0$. Define the similarity-transformed map*

$$E'(X) \;=\; c \cdot C^{-1}\, E\!\left(C\,X\,C^\dagger\right)(C^\dagger)^{-1}.$$

Then E' is also irreducible.

Proof. Suppose Q is an invariant projection for E': $Q \neq 0$, $Q \neq \mathbb{1}$, and $(\mathbb{1} - Q)E'(QXQ)Q = 0$ for all X. Setting $R = CQC^\dagger$, the support projection P of R is an invariant projection for E. Irreducibility of E forces $P = 0$ or $P = \mathbb{1}$. The invertibility of C then forces $Q = 0$ or $Q = \mathbb{1}$, a contradiction. This is the full similarity version of [Wol12, Prop. 6.6]; the scalar case ($C = \mathbb{1}$) gives the scaling result stated next. ☐

Theorem 8.14 (Scaling preserves irreducibility)**.** *If E is an irreducible map and $c \neq 0$, then cE is also irreducible. This is the scalar special case ($C = \mathbb{1}$) of [Wol12, Prop. 6.6].*

Proof. An invariant projection for cE is an invariant projection for E (scale out the nonzero constant from the commutation relation). ☐

8.6 Perron–Frobenius eigenvector existence

This section establishes the existence of a positive definite eigenvector for the adjoint transfer map of an irreducible MPS tensor, and uses it to construct a TP-gauge normalization. The PSD-eigenvector step corresponds to [Wol12, Thm. 6.5] (general positive maps) and [EHK78]; the upgrade to positive definiteness uses the irreducible-map theory of [Wol12, Thm. 6.3]. The TP-gauge application follows [CPGSV17a, Appendix A].

Theorem 8.15 (PSD eigenvector for positive maps)**.** *Let $E : M_D(\mathbb{C}) \to M_D(\mathbb{C})$ be a positive linear map with $D > 0$, and assume that $E(\sigma) \neq 0$ for every nonzero PSD matrix σ. Then there exist a nonzero PSD matrix ρ and a positive real $r > 0$ such that $E(\rho) = r\rho$.*

This is the finite-dimensional Perron–Frobenius theorem for positive maps on matrix algebras ([Wol12, Thm. 6.5]; see also [EHK78]). The proof applies Brouwer's fixed-point theorem to the normalization map $\rho \mapsto E(\rho)/\operatorname{tr}(E(\rho))$ on the compact convex set of density matrices.

Theorem 8.16 (Adjoint transfer map is nonzero)**.** *If some Kraus operator $A^i \neq 0$, then the adjoint transfer map $\mathcal{E}_A^\dagger(X) = \sum_i (A^i)^\dagger X A^i$ is nonzero.*

Proof. If $\mathcal{E}_A^\dagger = 0$ then $\mathcal{E}_A^\dagger(\mathbb{1}) = 0$, so $\sum_i (A^i)^\dagger A^i = 0$. Since each summand $(A^i)^\dagger A^i$ is PSD, each $(A^i)^\dagger = 0$ and hence each $A^i = 0$. ☐

Theorem 8.17 (PosDef adjoint eigenvector for irreducible tensors)**.** *Let A be an irreducible MPS tensor with $D > 0$ and some $A^i \neq 0$. Then there exist a positive definite matrix σ and a positive real $r > 0$ such that $\mathcal{E}_A^\dagger(\sigma) = r\sigma$.*

This combines the general PF existence ([Wol12, Thm. 6.5]) with the irreducible upgrade to positive definiteness ([Wol12, Thm. 6.3, item 2]); the application to the adjoint transfer map follows [CPGSV17a, Appendix A].

Proof. By Theorem 8.16, $\mathcal{E}_A^\dagger \neq 0$. Since $\mathcal{E}_A^\dagger(X) = \sum_i (A^i)^\dagger X A^i$ is a Kraus map, it is positive. Irreducibility of A transfers to the adjoint transfer map. More precisely: if P is invariant for $\mathcal{E}_A$ (i.e. $(\mathbb{1} - P)K_i P = 0$ for all Kraus operators K_i), then taking adjoints gives $PK_i^\dagger(\mathbb{1} - P) = 0$, which means $\mathbb{1} - P$ is invariant for $\mathcal{E}_A^\dagger$. Hence $\mathcal{E}_A$ irreducible $\iff$ $\mathcal{E}_A^\dagger$ irreducible, and together with $\mathcal{E}_A^\dagger \neq 0$ this implies $\mathcal{E}_A^\dagger(\tau) \neq 0$ for every nonzero PSD matrix τ. Hence Theorem 8.15 applies to $\mathcal{E}_A^\dagger$, giving $\sigma_0 \geq 0$ (PSD, nonzero) and $r > 0$ with $\mathcal{E}_A^\dagger(\sigma_0) = r\sigma_0$. Rescale: set $T^i = r^{-1/2}(A^i)^\dagger$. Then $\mathcal{E}_T(\sigma_0) = \sigma_0$, and $\mathcal{E}_T$ is irreducible (by Theorem 8.14, since irreducibility of $\mathcal{E}_A$ transfers to the adjoint and is preserved under scaling). Since $\mathcal{E}_T$ is irreducible with a PSD fixed point, the PSD-to-PosDef upgrade (Theorem 8.3) gives σ_0 positive definite. ☐

Theorem 8.18 (TP-gauge existence for irreducible tensors)**.** *Let A be an irreducible MPS tensor with $D > 0$ and some $A^i \neq 0$. Then there exist a positive real r, a positive definite matrix σ, and a gauge-equivalent tensor B such that*

$$B^i = \sigma^{1/2} \cdot (r^{-1/2} A^i) \cdot \sigma^{-1/2}$$

is trace-preserving: $\sum_i (B^i)^\dagger B^i = \mathbb{1}$.

This is the "spectral rescaling + TP gauge" step of [CPGSV17a, Appendix A]. The similarity is generally non-unitary.

Proof. By Theorem 8.17, obtain σ positive definite and $r > 0$ with $\mathcal{E}_A^\dagger(\sigma) = r\sigma$. Set $c = r^{-1/2}$ and $A'^i = cA^i$. Then $\mathcal{E}_{A'}^\dagger(\sigma) = \sigma$. By the TP gauge construction (Theorem 8.11), $B^i = \sigma^{1/2} A'^i \sigma^{-1/2}$ satisfies $\sum_i (B^i)^\dagger B^i = \mathbb{1}$. ☐

8.7 Exponential positivity for irreducible CP maps

Theorem 8.19 (Exponential truncation is positive definite)**.** *Let E be an irreducible completely positive map on $M_D(\mathbb{C})$, let $A \geq 0$ be nonzero, and let $t > 0$. Then the finite exponential truncation*

$$\sum_{k=0}^{D-1} \frac{t^k}{k!} E^k(A)$$

is positive definite. This is the finite-sum core of [Wol12, Thm. 6.2, item (3)].

Proof. If a nonzero vector v annihilated this quadratic form, then every term $E^k(A)$ with $0 \leq k \leq D-1$ would annihilate v. The growth theorem for irreducible CP maps (that is, $(\mathbb{1}+E)^{D-1}(A) > 0$ for irreducible E and nonzero PSD A, as established in the proof of Theorem 8.4) then forces $(\mathbb{1} + E)^{D-1}(A)$ to annihilate v, contradicting its positive definiteness. ☐

Theorem 8.20 (Exponential positivity for irreducible CP maps)**.** *Let E be an irreducible completely positive map on $M_D(\mathbb{C})$, let $A \geq 0$ be nonzero, and let $t > 0$. Then*

$$\exp(tE)(A) = \sum_{k=0}^{\infty} \frac{t^k}{k!} E^k(A)$$

is positive definite. This is [Wol12, Thm. 6.2, item (3)].

Proof. By Theorem 8.19, the first D terms already form a positive definite matrix. Every remaining term is positive semidefinite, so the full exponential series is the sum of a positive definite matrix and a positive semidefinite tail. ☐

8.8 Ergodicity of irreducible channels

Theorem 8.21 (Unique density-matrix fixed point for an irreducible channel)**.** *Let E be an irreducible quantum channel on $M_D(\mathbb{C})$ with $D > 0$. Then there exists a unique density matrix σ such that $E(\sigma) = \sigma$. Moreover σ is positive definite. This is the qualitative form of [Wol12, Cor. 6.3].*

Proof. Every Cesàro mean of a density matrix stays inside the compact convex set of density matrices, so a subsequence converges. Its limit is a fixed point by the telescoping argument, positive definiteness follows from irreducibility, and uniqueness comes from the uniqueness of positive semidefinite Perron eigenvectors at eigenvalue 1. ☐

Theorem 8.22 (Cesàro convergence for irreducible channels)**.** *Let E be an irreducible quantum channel on $M_D(\mathbb{C})$ with $D > 0$, and let ρ be any density matrix. Then the Cesàro means*

$$\frac{1}{N}\sum_{n=0}^{N-1} E^n(\rho)$$

converge to the unique positive definite density-matrix fixed point of E. This is [Wol12, Cor. 6.3].

Proof. Every subsequential limit of the Cesàro means is again a density-matrix fixed point. By Theorem 8.21, there is only one such fixed point, so every convergent subsequence has the same limit. Compactness then forces the whole sequence to converge to that limit. □

8.9 Spectral radius at the Perron eigenvalue

Theorem 8.23 (Spectral radius is the Perron eigenvalue)**.** *Let E be an irreducible completely positive map on $M_D(\mathbb{C})$. Suppose $\rho > 0$ and $E(\rho) = r\rho$ with $r > 0$. Then the spectral radius of E is r. This is [Wol12, Thm. 6.3, item (4)].*

Proof. Choose a positive definite eigenvector for the adjoint map with the same eigenvalue, gauge by its square root, and rescale by r^{-1}. The resulting map is trace-preserving and has a positive definite fixed point, so 1 is an eigenvalue. The spectral radius of a trace-preserving map F satisfies $\rho(F) \leq 1$, since $\mathrm{tr}(F^n(X)) = \mathrm{tr}(X)$ for all n bounds the growth of iterates. Hence its spectral radius is 1. Undoing the similarity and scalar rescaling recovers the spectral radius of E. □

Theorem 8.24 (Real spectral-radius identity)**.** *Under the hypotheses of Theorem 8.23, the real-valued spectral radius of E is also equal to r.*

Proof. This is the real-valued reformulation of Theorem 8.23. □

8.10 Spectral characterization of irreducibility

Definition 8.25 (Wolf spectral properties)**.** A completely positive map E on $M_D(\mathbb{C})$ has the *spectral properties* of [Wol12, Thm. 6.4] if there exist a positive real number r, a positive definite right eigenvector ρ, and a positive definite left eigenvector σ for the adjoint map such that $E(\rho) = r\rho$, $E^\dagger(\sigma) = r\sigma$, every positive semidefinite right eigenvector for eigenvalue r is a scalar multiple of ρ, and the spectral radius of E is equal to r.

Theorem 8.26 (Irreducibility gives the spectral properties)**.** *Every nonzero irreducible completely positive map on $M_D(\mathbb{C})$ has the spectral properties of Definition 8.25.*

Proof. Combine Perron–Frobenius existence for the map and its adjoint with the uniqueness theorem for positive semidefinite Perron eigenvectors and the spectral-radius identity of Theorem 8.23. □

Theorem 8.27 (Spectral properties imply irreducibility)**.** *If a completely positive map on $M_D(\mathbb{C})$ has the spectral properties of Definition 8.25, then it is irreducible.*

Proof. Gauge by the positive definite left eigenvector and rescale by the Perron eigenvalue to obtain a trace-preserving map with a positive definite fixed point. A nontrivial invariant projection would then produce, by Cesàro averaging in its corner algebra, a second positive semidefinite fixed point not proportional to the Perron vector, contradicting the uniqueness clause in Definition 8.25. □

Theorem 8.28 (Irreducibility iff Wolf's spectral properties)**.** *Let E be a nonzero completely positive map on $M_D(\mathbb{C})$. Then E is irreducible if and only if it has the spectral properties of Definition 8.25. This is [Wol12, Thm. 6.4].*

Proof. Combine Theorems 8.26 and 8.27. $\square$

Chapter 9

Spectral Gap and Block Separation

This chapter proves that the MPV overlaps between distinct (non-gauge-phase-equivalent) MPS blocks decay to zero as system size grows. It also proves the corresponding self-overlap convergence to 1 under the complementary spectral-gap hypothesis that encodes primitivity in the cases of interest. The argument is based on a spectral analysis of the *mixed transfer operator*, following [PGVWC07].

The algebraic mixed-transfer identities in the first two sections do not use normalization. Starting with the spectral-radius estimates, we assume the trace-preserving normalization

$$\sum_i (A^i)^\dagger A^i = \mathbb{1}.$$

9.1 Mixed transfer operator

Definition 9.1 (Mixed transfer operator)**.** The *mixed transfer operator* (or *cross transfer operator*) for two MPS tensors A and B of the same bond dimension D is the linear map $F_{AB} : M_D(\mathbb{C}) \to M_D(\mathbb{C})$ defined by

$$F_{AB}(X) = \sum_{i=0}^{d-1} A^i X (B^i)^\dagger.$$

Definition 9.2 (Rectangular mixed transfer operator)**.** For MPS tensors A of bond dimension D_1 and B of bond dimension D_2 (possibly $D_1 \neq D_2$), the *rectangular mixed transfer operator* is $F_{AB}^{\mathrm{rect}} : M_{D_1 \times D_2}(\mathbb{C}) \to M_{D_1 \times D_2}(\mathbb{C})$ defined by

$$F_{AB}^{\mathrm{rect}}(X) = \sum_{i=0}^{d-1} A^i X (B^i)^\dagger.$$

Remark 9.3 (Self-transfer)*.* Setting $B = A$ recovers the ordinary transfer map: $F_{AA} = \mathcal{E}_A$.

Theorem 9.4 (Iterated mixed transfer)**.** *For all $X \in M_D(\mathbb{C})$ and $N \geq 0$,*

$$F_{AB}^N(X) = \sum_{\sigma \in \{0,\dots,d-1\}^N} A^\sigma X (B^\sigma)^\dagger,$$

where $A^\sigma = A^{\sigma_1} \cdots A^{\sigma_N}$ denotes the word evaluation (Definition 2.2).

Proof. Induction on N: the base case is $F^0(X) = X$; the inductive step applies F_{AB} to each summand and re-indexes via $\{0, \dots, d-1\}^{N+1} \cong \{0, \dots, d-1\} \times \{0, \dots, d-1\}^N$. $\square$

Theorem 9.5 (Matrix trace of iterated mixed transfer at the identity)**.** *For $N \geq 0$,*

$$\mathrm{tr}(F_{AB}^N(\mathbb{1})) = \sum_\sigma \mathrm{tr}(A^\sigma (B^\sigma)^\dagger).$$

This is the matrix trace of F_{AB}^N evaluated at the identity matrix. It should not *be confused with the operator trace $\mathrm{Tr}(F_{AB}^N)$, which equals the MPV overlap (Theorem 9.7).*

Proof. Apply the matrix trace to each summand in Theorem 9.4 with $X = \mathbb{1}$: $\mathrm{tr}(F_{AB}^N(\mathbb{1})) = \sum_\sigma \mathrm{tr}(A^\sigma \cdot \mathbb{1} \cdot (B^\sigma)^\dagger) = \sum_\sigma \mathrm{tr}(A^\sigma (B^\sigma)^\dagger)$. $\square$

9.2 MPV overlap as transfer trace

Recall from Definition 2.34 that

$$O_{AB}(N) = \sum_{\sigma \in \{0,\dots,d-1\}^N} V^{(N)}(A)_\sigma \overline{V^{(N)}(B)_\sigma}.$$

This section rewrites that scalar as the operator trace of the mixed transfer power.

Lemma 9.6 (Trace expansion over matrix units)**.** *For any linear endomorphism T on $M_D(\mathbb{C})$, the operator trace can be expanded as*

$$\mathrm{Tr}(T) = \sum_{p,q=0}^{D-1} (T(E_{pq}))_{pq},$$

where E_{pq} is the matrix with 1 in position (p, q) and 0 elsewhere.

Proof. The operator trace $\mathrm{Tr}(T)$ equals the sum of $\mathrm{tr}(T(E_{pq}) \cdot E_{qp})$ over all (p, q), and $\mathrm{tr}(M \cdot E_{qp}) = M_{pq}$. $\square$

Theorem 9.7 (Overlap equals transfer trace)**.** *For tensors A, B of bond dimension $D \geq 1$,*

$$O_{AB}(N) = \mathrm{Tr}(F_{AB}^N).$$

Proof. Expand the operator trace using Lemma 9.6 as a sum over matrix units E_{pq}. Apply Theorem 9.4 to evaluate $F_{AB}^N(E_{pq})$, extract the (p, q)-entry, and reassemble. The resulting double sum over (p, q) and σ equals $\sum_\sigma V^{(N)}(A)_\sigma \overline{V^{(N)}(B)_\sigma} = O_{AB}(N)$. $\square$

Theorem 9.8 (Rectangular overlap as transfer trace)**.** *For tensors A of bond dimension $D_1 \geq 1$ and B of bond dimension $D_2 \geq 1$,*

$$O_{AB}(N) = \mathrm{Tr}((F_{AB}^{\mathrm{rect}})^N).$$

Proof. Expand the operator trace over the rectangular matrix-unit basis. Then apply the rectangular iterated-transfer formula and sum over the matrix-unit indices exactly as in Theorem 9.7. $\square$

Lemma 9.9 (Word trace as entrywise inner product)**.** *For a word $\sigma \in \{0, \dots, d-1\}^N$,*

$$\mathrm{tr}(A^\sigma (B^\sigma)^\dagger) = \sum_{j,k=0}^{D-1} (A^\sigma)_{jk} \overline{(B^\sigma)_{jk}}.$$

Proof. Expand the matrix trace and the (j, j)-entry of $A^\sigma (B^\sigma)^\dagger$. $\square$

9.3 Frobenius norm estimates

The spectral-gap proofs use the Frobenius (Hilbert–Schmidt) norm as the natural norm for the finite-dimensional matrix algebra. We write $\|\cdot\|_F$ for this norm; it coincides with, and we use it interchangeably with, the Hilbert–Schmidt norm $\|\cdot\|_{\mathrm{HS}}$ on finite-dimensional spaces. We record the squared version and an isometric embedding into Euclidean space.

Definition 9.10 (Frobenius norm squared)**.** For a (possibly rectangular) matrix $X \in M_{m\times n}(\mathbb{C})$, the *Frobenius norm squared* is

$$\|X\|_F^2 := \sum_{i=0}^{m-1}\sum_{j=0}^{n-1} |X_{ij}|^2.$$

Lemma 9.11 (Trace formula for Frobenius norm)**.** $\|X\|_F^2 = \operatorname{Re}\operatorname{tr}(X^\dagger X)$.

Proof. Expand the matrix product and trace entry by entry. □

Definition 9.12 (Euclidean-space embedding)**.** The map $\mathrm{vec} : M_{m\times n}(\mathbb{C}) \to \mathbb{C}^{mn}$ that flattens matrix entries is an isometric embedding with respect to the Frobenius norm: $\|\mathrm{vec}(X)\|^2 = \|X\|_F^2$.

Lemma 9.13 (Norm of embedded matrix)**.** $\|\mathrm{vec}(X)\|^2 = \|X\|_F^2$.

Proof. Expand both sides as sums over entries and use $|z|^2 = \bar{z}z$. □

9.4 Eigenvalue bound and spectral gap

The eigenvalue bound comes from a uniform Frobenius-norm estimate obtained directly from the trace-preserving normalization. For the rigidity theorem, one gauges A and B separately using positive fixed points of their transfer maps to obtain unital Kraus families, and then applies the Kadison–Schwarz equality argument to a block embedding of a mixed-transfer eigenvector.

Theorem 9.14 (Eigenvalue bound)**.** *Let A, B be normalized MPS tensors. Every eigenvalue μ of F_{AB} satisfies $|\mu| \le 1$.*

Proof. Expand $F_{AB}^n(X)$ as a sum over words and use Cauchy–Schwarz, together with $\sum_i (A^i)^\dagger A^i = \sum_i (B^i)^\dagger B^i = \mathbb{1}$, to obtain a uniform Frobenius-norm bound $\|F_{AB}^n(X)\|_{\mathrm{HS}} \le C_D\|X\|_{\mathrm{HS}}$ for all n, where C_D depends only on D. If $F_{AB}(X) = \mu X$ and $X \ne 0$, then $|\mu|^n\|X\|_{\mathrm{HS}} = \|F_{AB}^n(X)\|_{\mathrm{HS}} \le C_D\|X\|_{\mathrm{HS}}$ for every n. Hence $|\mu| \le 1$.

The argument uses only the trace-preserving normalization and does not appeal to Perron–Frobenius theory. □

Theorem 9.15 (Spectral radius bound)**.** $\rho(F_{AB}) \le 1$ *for normalized tensors.*

Proof. The spectral radius is the supremum of $|\mu|$ over all eigenvalues, and each satisfies $|\mu| \le 1$ by Theorem 9.14. □

Theorem 9.16 (Spectral radius ≥ 1 implies gauge-phase equivalence)**.** *Let A, B be injective normalized MPS tensors. If $\rho(F_{AB}) \ge 1$, then A and B are gauge-phase equivalent.*

Proof. Since $\rho(F_{AB}) \le 1$ (Theorem 9.15) and $\rho(F_{AB}) \ge 1$ by hypothesis, there exists an eigenvalue μ with $|\mu| = 1$ and eigenvector $X \ne 0$: $F_{AB}(X) = \mu X$. The proof proceeds in five steps.

Step 1 (Separate gauging). Choose positive definite fixed points ρ_A, ρ_B of the transfer maps $\mathcal{E}_A, \mathcal{E}_B$ (Theorem 8.9) and gauge each tensor separately to a unital Kraus family (Theorem 8.10):

set $A'^i = \rho_A^{-1/2} A^i \rho_A^{1/2}$ and similarly for B. Let $X' = \rho_A^{-1/2} X \rho_B^{-1/2}$ denote the correspondingly gauged eigenvector.

Step 2 (Block Kraus family and off-diagonal embedding). Form the block Kraus operators $C^i = \left(\begin{smallmatrix} A'^i & 0 \\ 0 & B'^i \end{smallmatrix}\right)$. These constitute a unital Kraus family on $M_{2D}(\mathbb{C})$. The off-diagonal matrix $Y = \left(\begin{smallmatrix} 0 & X' \\ 0 & 0 \end{smallmatrix}\right)$ satisfies $E_C(Y) = \mu Y$.

Step 3 (KS equality and Kraus commutation). Because $|\mu| = 1$ and E_C is unital and trace-preserving, the Kadison–Schwarz equality for peripheral eigenvectors (Theorem 7.24) gives $E_C(Y^\dagger Y) = E_C(Y)^\dagger E_C(Y)$. By the Kraus commutation relation (Theorem 7.23), Y commutes with every block Kraus operator: $C^i Y = Y C^i$ for all i.

Step 4 (Intertwining identity and invertibility). Expanding the commutation relation block by block gives $A'^i X' = X' B'^i$ for every i. Since $\{A'^i\}$ span $M_D(\mathbb{C})$ (injectivity of A), X' is a nonzero intertwiner from $M_D(\mathbb{C})$ to $M_D(\mathbb{C})$. The left multiplicative identity (Theorem 7.25) gives $E_C(Y^\dagger Y) = E_C(Y^\dagger) E_C(Y)$, so $X'^\dagger X'$ is a positive semi-definite fixed point of $\mathcal{E}_{B'}$, positive definite by injectivity. Hence X' is invertible.

Step 5 (Skolem–Noether conclusion). The map $M \mapsto X' M (X')^{-1}$ is a $\mathbb{C}$-algebra automorphism of $M_D(\mathbb{C})$ sending B'^i to A'^i. By Skolem–Noether (Theorem 4.9) this is an inner automorphism. Undoing the separate gauges gives $B^i = \bar{\mu}\, X\, A^i\, X^{-1}$ for all i, establishing gauge-phase equivalence (Definition 2.16).

The crucial point is that one gauges the individual transfer maps to unital form; one does not require the mixed transfer map itself to be simultaneously unital and trace-preserving. ☐

Remark 9.17 (Two gauge conventions). The right-canonical gauge used here, $A'^i = \rho^{-1/2} A^i \rho^{1/2}$, makes $\{A'^i\}$ a *unital* Kraus family ($\sum_i A'^i (A'^i)^\dagger = \mathbb{1}$). A different convention, $\tilde{A}^i = \rho^{1/2} A^i \rho^{-1/2}$, makes the family *trace-preserving* ($\sum_i (\tilde{A}^i)^\dagger \tilde{A}^i = \mathbb{1}$). The Lean codebase implements both: the unital version via direct matrix square-root inversion (used in the spectral-gap proof), and the trace-preserving version (used in the canonical-form reduction).

Theorem 9.18 (Strict spectral gap). *Let A, B be injective normalized MPS tensors that are* not *gauge-phase equivalent. Then $\rho(F_{AB}) < 1$.*

Proof. $\rho(F_{AB}) \le 1$ by Theorem 9.15. If $\rho(F_{AB}) = 1$, Theorem 9.16 gives gauge-phase equivalence, contradicting the hypothesis. ☐

9.5 Spectral gap — rectangular case

Theorem 9.19 (Rectangular spectral gap). *Let A be an injective normalized tensor of bond dimension D_1 and B an injective normalized tensor of bond dimension D_2, with $D_1 \neq D_2$. Then $\rho(F_{AB}^{\mathrm{rect}}) < 1$.*

Proof. The same word-expansion argument as in Theorem 9.14 gives $|\mu| \le 1$ for every eigenvalue of F_{AB}^{rect}. If $\rho(F_{AB}^{\mathrm{rect}}) = 1$, choose a modulus-one eigenvector $X \in M_{D_1 \times D_2}(\mathbb{C}) \setminus \{0\}$.

Gauge A and B separately to unital Kraus families and apply the block-embedding Kadison–Schwarz argument as in the square case (Theorem 9.16, Steps 2–3). The Kraus commutation relation gives $A'^i X' = X' B'^i$ for all i. Now consider $\ker(X') \subseteq \mathbb{C}^{D_2}$. If $v \in \ker(X')$, then $X' B'^i v = A'^i X' v = 0$, so $B'^i v \in \ker(X')$ for every i. Since B is injective, $\{B'^i\}$ spans $M_{D_2}(\mathbb{C})$, so $\ker(X')$ is invariant under every $D_2 \times D_2$ matrix and is therefore either $\{0\}$ or $\mathbb{C}^{D_2}$. Since $X' \neq 0$, we have $\ker(X') = \{0\}$, so X' is injective as a linear map $\mathbb{C}^{D_2} \to \mathbb{C}^{D_1}$ and $D_2 \le D_1$. Applying the same argument to $X'^\dagger$ (using the commutation relation for the adjoint block family) gives $D_1 \le D_2$. Hence $D_1 = D_2$, contradicting the hypothesis. ☐

Theorem 9.20 (Rectangular overlap decay)**.** *If $D_1 \neq D_2$ and both tensors are injective and normalized, then $O_{AB}(N) \to 0$ as $N \to \infty$.*

Proof. The spectral gap $\rho(F_{AB}^{\mathrm{rect}}) < 1$ (Theorem 9.19) implies $(F_{AB}^{\mathrm{rect}})^N \to 0$, so the operator trace converges to 0. By Theorem 9.8, $O_{AB}(N) = \mathrm{Tr}((F_{AB}^{\mathrm{rect}})^N) \to 0$. □

9.6 MPV overlap decay

Theorem 9.21 (Transfer powers converge to zero)**.** *Let A, B be injective normalized tensors that are not gauge-phase equivalent. Then $F_{AB}^n(X) \to 0$ for every $X \in M_D(\mathbb{C})$.*

Proof. The spectral gap $\rho(F_{AB}) < 1$ (Theorem 9.18) implies $\|F_{AB}^n\|_{\mathrm{op}} \to 0$ by the Gelfand formula, so $F_{AB}^n(X) \to 0$ for every X. □

Theorem 9.22 (Overlap decay)**.** *Let A, B be injective normalized tensors of the same bond dimension that are not gauge-phase equivalent. Then $O_{AB}(N) \to 0$ as $N \to \infty$.*

Proof. By Theorem 9.7, $O_{AB}(N) = \mathrm{Tr}(F_{AB}^N)$. Expand the operator trace over matrix units (Lemma 9.6): $\mathrm{Tr}(F_{AB}^N) = \sum_{p,q}(F_{AB}^N(E_{pq}))_{pq}$. By Theorem 9.21, each $F_{AB}^N(E_{pq}) \to 0$ entry-wise. Since the sum is finite, $O_{AB}(N) \to 0$. □

9.7 Block separation

These asymptotic facts are the ingredients used later in the canonical-form separation arguments.

Remark 9.23 (Cross-correlation decay). If A and B are injective normalized tensors that are not gauge-phase equivalent, then for every $X \in M_D(\mathbb{C})$ one has $\mathrm{tr}(F_{AB}^N(X)) \to 0$ as $N \to \infty$. This is the scalar trace consequence of the operator convergence $F_{AB}^N(X) \to 0$ from Theorem 9.21.

Remark 9.24 (Self-correlation persists). If ρ is a fixed point of $\mathcal{E}_A$, then $\mathrm{tr}(\mathcal{E}_A^N(\rho)) = \mathrm{tr}(\rho)$ for all $N \geq 0$. In particular this applies to the distinguished QPF fixed point from Theorem 8.9.

9.8 Spectral gap under irreducible-TP hypotheses

The preceding spectral-gap results require *injectivity* of both tensors. The following variants weaken this to *irreducibility* combined with trace-preserving normalization ($\sum_i (A^i)^\dagger A^i = \mathbb{1}$), following Cirac et al. [CPGSV17b]. The argument replaces the Kraus-commutation step with a Cauchy–Schwarz rigidity argument for the Perron–Frobenius fixed point.

Theorem 9.25 (Gauge-equivalence from irreducible-TP spectral radius)**.** *Let A, B be irreducible trace-preserving normalized MPS tensors of bond dimension D. If $\rho(F_{AB}) \geq 1$, then A and B are gauge-phase equivalent.*

Proof. Choose a modulus-one eigenvector as in Theorem 9.16. The proof follows the five-step structure of Theorem 9.16, with one key adaptation. Steps 1–3 (separate gauging, block Kraus family, KS equality and Kraus commutation) proceed identically: the gauging uses the unique positive definite fixed points guaranteed by irreducibility (Theorem 8.9) and the right-canonical gauge (Theorem 8.10). The change is in Step 4: the Kadison–Schwarz equality gives $X'^\dagger X'$ as a nonzero positive semidefinite fixed point of $\mathcal{E}_{B'}$. Under injectivity this is immediately positive definite (the fixed point is unique up to scaling). Under the weaker hypothesis of irreducibility, Theorem 8.3 upgrades it to positive definite, giving invertibility of X'. The intertwining identity

$A'^i X' = X' B'^i$ follows from Kraus commutation alone and does not require the Kraus operators to span $M_D(\mathbb{C})$. Step 5 (Skolem–Noether conclusion) is unchanged. □

Theorem 9.26 (Strict spectral gap (irreducible-TP)). *Let A, B be irreducible trace-preserving normalized MPS tensors of the same bond dimension that are not gauge-phase equivalent. Then $\rho(F_{AB}) < 1$.*

Proof. Contrapositive of Theorem 9.25, combined with $\rho(F_{AB}) \le 1$ (Theorem 9.15). □

Theorem 9.27 (Transfer power decay (irreducible-TP)). *Under the hypotheses of Theorem 9.26, $F_{AB}^n(X) \to 0$ for every $X \in M_D(\mathbb{C})$.*

Proof. By the Gelfand formula, $\rho(F_{AB}) < 1$ implies $\|F_{AB}^n\|_{\mathrm{op}} \to 0$. □

Theorem 9.28 (Overlap decay (irreducible-TP)). *Let A, B be irreducible trace-preserving normalized tensors of the same bond dimension that are not gauge-phase equivalent. Then $O_{AB}(N) \to 0$ as $N \to \infty$.*

Proof. Combine the spectral gap (Theorem 9.26) with the overlap–trace identity (Theorem 9.7). □

Theorem 9.29 (Rectangular spectral gap (irreducible-TP)). *Let A (bond dimension D_1) and B (bond dimension D_2) be irreducible trace-preserving normalized tensors with $D_1 \neq D_2$. Then $\rho(F_{AB}^{\mathrm{rect}}) < 1$.*

Proof. Assume for contradiction that $\rho = 1$. Choose a modulus-one eigenvector $X \in M_{D_1 \times D_2}(\mathbb{C})$. The Cauchy–Schwarz rigidity argument produces an intertwiner X' with $A'^i X' = X' B'^i$ for all i. Since $\ker(X')$ is invariant under all B'^i, irreducibility forces $\ker(X') = \{0\}$; similarly for $X'^\dagger$. This gives $D_1 = D_2$, contradicting the hypothesis. □

Theorem 9.30 (Rectangular overlap decay (irreducible-TP)). *If $D_1 \neq D_2$ and both tensors are irreducible and trace-preserving normalized, then $O_{AB}(N) \to 0$ as $N \to \infty$.*

Proof. Combine Theorem 9.29 with Theorem 9.8. □

9.9 Conditional expectation from a faithful fixed point (Wolf Thm. 6.15)

Definition 9.31 (Conditional expectation). Given a $*$-subalgebra $S \subseteq M_D(\mathbb{C})$, a linear map $P : M_D(\mathbb{C}) \to M_D(\mathbb{C})$ is a conditional expectation onto S if it is idempotent, unital, has range contained in S, and satisfies $P(X) = X$ for all $X \in S$.

Definition 9.32 (Scalar conditional expectation). For $\sigma \ge 0$ with $\mathrm{tr}(\sigma) \neq 0$, define

$$E_\sigma(X) := \frac{\mathrm{tr}(\sigma X)}{\mathrm{tr}(\sigma)} \mathbb{1}.$$

Lemma 9.33 (Evaluation formula). $E_\sigma(X) = \dfrac{\mathrm{tr}(\sigma X)}{\mathrm{tr}(\sigma)} \mathbb{1}$.

Theorem 9.34 (Unitality). $E_\sigma(\mathbb{1}) = \mathbb{1}$.

Theorem 9.35 (Idempotence). $E_\sigma(E_\sigma(X)) = E_\sigma(X)$ *for all* X.

Lemma 9.36 (Range is scalar)**.** *For every X, there exists $c \in \mathbb{C}$ with $E_\sigma(X) = c\,\mathbb{1}$.*

Lemma 9.37 (Fixes scalar operators)**.** *For $c \in \mathbb{C}$, $E_\sigma(c\,\mathbb{1}) = c\,\mathbb{1}$.*

Theorem 9.38 (Absorbs the adjoint map)**.** *If $T(\sigma) = \sigma$, then $E_\sigma \circ T^* = E_\sigma$.*

Theorem 9.39 (Adjoint map absorbs scalar projection)**.** *If T is trace-preserving, then $T^* \circ E_\sigma = E_\sigma$.*

Lemma 9.40 (Image lies in adjoint fixed points)**.** *For all X, $E_\sigma(X)$ is fixed by T^*.*

Theorem 9.41 (Scalar map is a conditional expectation)**.** *Let K be trace-preserving, let $\rho > 0$ be a positive-definite fixed point of the associated channel $T(\rho) = \sum_i K_i \rho K_i^\dagger$ (so $T(\rho) = \rho$), and assume every adjoint fixed point of T^* is a scalar multiple of $\mathbb{1}$. Then E_ρ is a conditional expectation onto the adjoint fixed-point $*$-subalgebra.*

9.10 Stationary support (Wolf §6.4)

This section records the support-projection theory behind stationary states, following [Wol12, §6.4].

Theorem 9.42 (Support projection invariance (Wolf Lemma 6.4))**.** *Let E be a quantum channel and $\rho \geq 0$ a positive semidefinite fixed point ($E(\rho) = \rho$). Let P denote the support projection of ρ. Then*

$$P\,E(PXP)\,P = E(PXP)$$

for all $X \in M_D(\mathbb{C})$.

Proof. Write E in Kraus form $E(X) = \sum_i K_i X K_i^\dagger$. From $E(\rho) = \rho$ and $\rho \geq 0$ one deduces $(\mathbb{1} - P)K_i P = 0$ for all i, i.e. each Kraus operator maps the range of P into itself. The identity then follows by expanding the Kraus sum and using $K_i P = P K_i P$. □

Definition 9.43 (Stationary state)**.** For an irreducible channel E on $M_D(\mathbb{C})$ (with $D \geq 1$), the *stationary state* is the unique density-matrix fixed point ρ_∞ satisfying $E(\rho_\infty) = \rho_\infty$, $\rho_\infty > 0$, and $\mathrm{tr}(\rho_\infty) = 1$. Uniqueness follows from the quantum Perron–Frobenius theorem (Theorem 8.9).

Definition 9.44 (Stationary support)**.** The *stationary support* of an irreducible channel E is the support projection of the stationary state (Definition 9.43).

Theorem 9.45 (Prop. 6.9: stationary support is full)**.** *For an irreducible channel E, the stationary support equals the identity:* $\mathrm{supp}(\rho_\infty) = \mathbb{1}$.

Proof. The support projection P of ρ_∞ is an orthogonal projection satisfying $PE(PXP)P = E(PXP)$ for all X (Theorem 9.42). By irreducibility, $P = 0$ or $P = \mathbb{1}$. Since $\rho_\infty \neq 0$, we have $P \neq 0$, so $P = \mathbb{1}$. □

Remark 9.46 (Props. 6.10–6.11: non-trivial formulations). The non-trivial formulations needed here are:

- **Prop. 6.10** (converse of Prop. 6.9): if the support projection of *every* PSD fixed point is full, then the channel is irreducible.
- **Prop. 6.11**: minimality of the stationary support among invariant projections, without assuming irreducibility.

These statements are deferred in the present chapter.

9.11 Wedderburn decomposition of the fixed-point algebra (Wolf Thm. 6.14)

Theorem 9.47 (Fixed-point algebra is semisimple)**.** *The adjoint-fixed-point $*$-subalgebra of a trace-preserving Kraus map on $M_D(\mathbb{C})$ is a semisimple ring.*

Theorem 9.48 (Abstract Wedderburn–Artin decomposition (Wolf Thm. 6.14))**.** *There exist $n \in \mathbb{N}$ and dimensions $d_1, \dots, d_n \geq 1$ such that the adjoint-fixed-point $*$-subalgebra is $\mathbb{C}$-algebra isomorphic to $\prod_{k=1}^{n} M_{d_k}(\mathbb{C})$.*

Theorem 9.49 (Fixed-point algebra admits Wedderburn block decomposition)**.** *The adjoint-fixed-point $*$-subalgebra admits a concrete Wedderburn block decomposition inside the ambient matrix algebra (Wolf Eq. 1.39).*

9.12 Additional Wolf Chapter 6 equivalences

This section records several equivalences from Wolf's Chapter 6 that are used later in the blueprint.

Theorem 9.50 (Wolf Thm. 6.2(3): exponential positivity condition)**.** *Let E be an irreducible completely positive map, $A \geq 0$ with $A \neq 0$, and $t > 0$. Then* $\exp(tE)(A)$ *is positive definite. Conversely, if* $\exp(tE)(A)$ *is positive definite for every such t, A, then E is irreducible (full equivalence).*

Theorem 9.51 (Wolf Thm. 6.8: Kraus-span equivalence)**.** *Under normalization, primitivity in Wolf's sense is equivalent to eventual full Kraus rank.*

Theorem 9.52 (Wolf Thm. 6.8: combined equivalence)**.** *Under normalization, primitivity in Wolf's sense is equivalent to the conjunction of eventual full Kraus rank, normality, and strong irreducibility.*

Remark 9.53 (Wolf Thm. 6.8: pairwise equivalences). These are the pairwise equivalences appearing in Wolf's Theorem 6.8.

Theorem 9.54 (Wolf Thm. 6.15 (scalar fixed-point algebra case))**.** *In the scalar fixed-point algebra setting, the weighted map $X \mapsto \frac{\mathrm{tr}(\rho X)}{\mathrm{tr}(\rho)}\mathbb{1}$ is a conditional expectation onto the adjoint fixed-point $*$-subalgebra.*

9.13 Peripheral eigenvalue group structure (Wolf Thm. 6.6)

Lemma 9.55 (Peripheral eigenvectors are invertible)**.** *Let E be an irreducible unital Kraus map on $M_D(\mathbb{C})$ with a positive-definite adjoint fixed point. If $X \neq 0$ satisfies $E(X) = \mu X$ with $|\mu| = 1$, then X is invertible.*

Proof. The KS equality gives $E(X^\dagger X) = X^\dagger X$, so $X^\dagger X$ is a nonzero PSD fixed point. Irreducibility upgrades it to positive definite, hence invertible, so $\det(X) \neq 0$. ☐

Lemma 9.56 (Peripheral eigenvalues: product closure)**.** *Let E be an irreducible unital Kraus map on $M_D(\mathbb{C})$ with a positive-definite adjoint fixed point. If μ, ν are peripheral eigenvalues of E, then so is $\mu\nu$.*

Proof. Take eigenvectors X, Y for μ, ν. The KS equality at X puts X in the multiplicative domain, so $E(YX) = E(Y)E(X) = (\nu\mu)(YX)$. Since X, Y are units (Lemma 9.55), $YX \neq 0$ and $|\mu\nu| = 1$. $\square$

Theorem 9.57 (Peripheral eigenvalues: cyclic structure). *Let E be an irreducible unital Schwarz map on $M_D(\mathbb{C})$ with a positive-definite adjoint fixed point (trace-preservation is* not *required). Then there exist $m \geq 1$ and a primitive m-th root of unity γ such that the peripheral eigenvalues of E are exactly $\{1, \gamma, \gamma^2, \dots, \gamma^{m-1}\}$.*

Proof. The peripheral eigenvalues form a finite subset of $\mathbb{C}$, contain 1, and are closed under multiplication and inversion, so they form a finite subgroup of $\mathbb{C}^\times$. Any finite subgroup of $\mathbb{C}^\times$ is cyclic. The generator γ has order $m = |\text{peripheral spectrum}|$. $\square$

Theorem 9.58 (Peripheral eigenvalues form a cyclic group). *Let E be an irreducible unital trace-preserving Schwarz map on $M_D(\mathbb{C})$ with a positive-definite adjoint fixed point. Then there exist $m \geq 1$ and a primitive m-th root of unity γ such that $m \mid D$ and the peripheral eigenvalues of E are exactly $\{1, \gamma, \gamma^2, \dots, \gamma^{m-1}\}$.*

Proof. Apply Theorem 9.57 for the cyclic structure. Divisibility $m \mid D$ follows from the cyclic decomposition: m mutually orthogonal projections summing to $\mathbb{1}$ each have equal trace (by trace preservation), so $m \cdot \mathrm{tr}(P_0) = D$ with $\mathrm{tr}(P_0) \in \mathbb{N}$. $\square$

Theorem 9.59 (Period divides bond dimension). *Let E be as in Theorem 9.58, and let γ be a primitive m-th root of unity generating the peripheral eigenvalues. Then $m \mid D$.*

Proof. Follows from the cyclic projection decomposition and trace preservation, as in the final step of Theorem 9.58. $\square$

9.13.1 Cyclic decomposition

Definition 9.60 (Corner preservation). For an orthogonal projection P and linear map E, the corner $PM_D(\mathbb{C})P$ is preserved if $E(PXP) = PE(PXP)P$ for all X.

Definition 9.61 (Corner subspace). The corner subspace associated with P is $\{PXP : X \in M_D(\mathbb{C})\}$.

Definition 9.62 (Restricted corner map). The restriction of E to an invariant corner $PM_D(\mathbb{C})P$.

Definition 9.63 (Irreducible restriction on a corner). Irreducibility of the corner-restricted map.

Theorem 9.64 (Peripheral unitary eigenvector). *For an irreducible unital Schwarz map with faithful adjoint fixed point, each peripheral eigenvalue admits a unitary eigenvector.*

Theorem 9.65 (Powers on a peripheral-unitary orbit). *If $E(U) = \mu U$ with $|\mu| = 1$ and U unitary, then $E(U^n) = \mu^n U^n$ for all n.*

Theorem 9.66 (Normalized peripheral unitary). *Peripheral unitary eigenvectors can be normalized compatibly with the faithful fixed-point state.*

Theorem 9.67 (Cyclic projections from a peripheral unitary). *A primitive m-th peripheral root yields m orthogonal projections summing to $\mathbb{1}$ and cyclically permuted by the channel.*

Theorem 9.68 (Cyclic decomposition for irreducible Schwarz maps). *Under irreducibility and faithful fixed point hypotheses, there is a full cyclic projection decomposition realizing the peripheral period.*

Theorem 9.69 (Power maps preserve each cyclic sector)**.** *Appropriate powers of the channel preserve each sector corner.*

Theorem 9.70 (Irreducibility of restricted sector dynamics)**.** *Each sector restriction is irreducible.*

Theorem 9.71 (Primitivity of restricted sector dynamics)**.** *Each sector restriction is primitive.*

Theorem 9.72 (Corner invariance at the period)**.** *The channel raised to the period preserves every cyclic corner.*

9.13.2 Peripheral group structure

Theorem 9.73 (Peripheral product closure)**.** *The product of two peripheral eigenvalues is peripheral.*

Theorem 9.74 (Peripheral inverse closure)**.** *The inverse of a peripheral eigenvalue is peripheral.*

Theorem 9.75 (Peripheral cyclic group with period divisibility)**.** *The peripheral eigenvalues form a finite cyclic group whose order divides the dimension.*

Theorem 9.76 (Peripheral eigenvalue multiplicity one)**.** *Each peripheral eigenvalue has algebraic multiplicity one.*

Proof. Given a peripheral unitary eigenvector U for γ, the multiplicative-domain identity gives $E(XU^\dagger) = XU^\dagger$ for any γ-eigenvector X. By the scalar fixed-point lemma for irreducible unital maps, $XU^\dagger = c \cdot I$, hence $X = c \cdot U$ and the eigenspace is one-dimensional. □

9.14 Primitive overlap convergence (spectral-gap form)

The theorems in this section are stated directly in terms of the complementary map $E - P$. This is the analytic form needed for the later overlap argument. For primitive normalized transfer maps, Chapter 5 supplies this spectral-gap input under explicit hypotheses.

Theorem 9.77 (Complementary spectral gap implies trace convergence to 1)**.** *Let E be trace-preserving with fixed point ρ ($\mathrm{tr}(\rho) \neq 0$) and let P be the fixed-point projection (Definition 5.20). If $\rho_{\mathrm{spec}}(E - P) < 1$, then $\mathrm{Tr}(E^n) \to 1$ as $n \to \infty$.*

Proof. By the power decomposition (Theorem 5.21), $E^n = P + (E - P)^n$ for $n \geq 1$. Since $\rho_{\mathrm{spec}}(E - P) < 1$, the Gelfand formula gives $(E - P)^n \to 0$, so $\mathrm{Tr}(E^n) \to \mathrm{Tr}(P)$. The operator trace of P equals 1: since $P(X) = \frac{\mathrm{tr}(X)}{\mathrm{tr}(\rho)}\rho$, we have $\mathrm{Tr}(P) = \frac{\mathrm{tr}(\rho)}{\mathrm{tr}(\rho)} = 1$. □

Theorem 9.78 (Self-overlap convergence from a complementary spectral gap)**.** *Let A be a normalized MPS tensor with a nonzero PSD fixed point ρ of the transfer map. If $\rho_{\mathrm{spec}}(\mathcal{E}_A - P) < 1$ (where P is the fixed-point projection), then $O_{AA}(N) \to 1$ as $N \to \infty$.*

Proof. By Theorem 9.7 (with $A = B$), $O_{AA}(N) = \mathrm{Tr}(\mathcal{E}_A^N)$. By Theorem 9.77, $\mathrm{Tr}(\mathcal{E}_A^N) \to 1$. □

Chapter 10

Wielandt Bound

This chapter proves a quantum analog of Wielandt's inequality: for a normal MPS tensor with bond dimension D, the cumulative span of matrix products stabilises after at most D^2 steps. The results follow [SPGWC10].

10.1 Cumulative span

Definition 10.1 (Word span)**.** The *word span* at length n is the subspace

$$S_n(A) = \operatorname{span}_{\mathbb{C}}\{A^w : |w| = n\} \subseteq M_D(\mathbb{C}).$$

This is [SPGWC10, §II].

Definition 10.2 (Cumulative span)**.** The *cumulative span* at level n is

$$T_n(A) = S_0(A) + S_1(A) + \cdots + S_n(A).$$

Lemma 10.3 (Monotonicity of cumulative span)**.** *$T_n(A) \le T_{n+1}(A)$ for all n.*

Proof. T_{n+1} includes all words of length $\le n+1$, which contains all words of length $\le n$. ☐

Lemma 10.4 (Dimension bound)**.** *$\dim T_n(A) \le D^2$ for all n.*

Proof. $T_n(A) \subseteq M_D(\mathbb{C})$ and $\dim M_D(\mathbb{C}) = D^2$. ☐

Theorem 10.5 (Cumulative span stabilisation)**.** *If $T_n(A) = T_{n+1}(A)$, then $T_m(A) = T_n(A)$ for all $m \ge n$.*

Proof. If $T_n = T_{n+1}$, then $S_{n+1} \subseteq T_n$. Left-multiplying any element of S_{n+1} by A^i gives an element of S_{n+2}, so $S_{n+2} \subseteq T_{n+1} = T_n$. By induction, $S_m \subseteq T_n$ for all $m > n$, hence $T_m = T_n$. ☐

Theorem 10.6 (Dimension growth)**.** *If $T_n(A) \ne T_{n+1}(A)$, then $\dim T_{n+1}(A) > \dim T_n(A)$.*

Proof. $T_n \le T_{n+1}$ (Lemma 10.3) and $T_n \ne T_{n+1}$ imply strict submodule inclusion, hence strict rank increase. ☐

Lemma 10.7 (Word span characterises block injectivity)**.** *$S_L(A) = M_D(\mathbb{C})$ if and only if A is L-block injective.*

Proof. Both conditions assert that the span of all products of length L equals $M_D(\mathbb{C})$. ☐

10.2 Fitting decomposition

Definition 10.8 (Fitting decomposition)**.** A *Fitting decomposition* of a linear endomorphism $f : V \to V$ on a finite-dimensional vector space over an algebraically closed field consists of:

1. f is nilpotent on the generalized 0-eigenspace,
2. f is invertible on each generalized μ-eigenspace for $\mu \neq 0$,
3. the generalized eigenspaces span V,
4. the generalized eigenspaces are linearly independent.

Theorem 10.9 (Fitting decomposition exists)**.** *Every linear endomorphism on a finite-dimensional vector space over an algebraically closed field admits a Fitting decomposition.*

Proof. Nilpotency on the zero generalized eigenspace follows from the definition of generalized eigenspaces. Invertibility on nonzero generalized eigenspaces follows because $f - \mu$ is nilpotent there, so $f = \mu(1 - (1 - f/\mu))$ is invertible. Spanning and independence are standard results for generalized eigenspaces over algebraically closed fields. In [SPGWC10] and [Wol12, Lemma 6.3(b)], the Jordan normal form is used directly. The argument is phrased in terms of generalized eigenspaces instead. □

Theorem 10.10 (Nilpotent power bound)**.** *If f is nilpotent on a space of dimension n, then $f^n = 0$.*

Proof. A nilpotent endomorphism on an n-dimensional space has nilpotency index $\leq n$. □

10.3 Nonzero trace product

Theorem 10.11 (Cumulative span reaches $M_D(\mathbb{C})$ — explicit bound)**.** *If A is normal, then there exists $n \leq D^2$ with $T_n(A) = M_D(\mathbb{C})$.*

Proof. By Theorem 10.6, if $T_n \neq T_{n+1}$ then $\dim T_{n+1} > \dim T_n$. Since $\dim T_n \leq D^2$ (Lemma 10.4), after at most D^2 steps either $T_n = M_D(\mathbb{C})$ or T_n has stabilised. Stabilisation before reaching $M_D(\mathbb{C})$ contradicts normality (which requires $S_{L_0} = M_D(\mathbb{C})$ for some L_0, hence $T_{L_0} = M_D(\mathbb{C})$). □

Theorem 10.12 (Cumulative span reaches $M_D(\mathbb{C})$)**.** *If A is normal, then $T_{D^2}(A) = M_D(\mathbb{C})$.*

Proof. Take n from Theorem 10.11; since $n \leq D^2$ and T is monotone, $T_{D^2} = M_D(\mathbb{C})$. □

Theorem 10.13 (Nonzero trace product)**.** *If A is normal, then there exists a word w of length $\leq D^2$ such that $\operatorname{tr}(A^w) \neq 0$. This is [SPGWC10, Lemma 1]. For the trace argument below, the nontrivial case is $D \geq 1$; when $D = 0$ the normality hypothesis is vacuous.*

Proof. By Theorem 10.12, $\mathbb{1} \in T_{D^2}(A)$, so $\mathbb{1}$ is a linear combination of word products of length $\leq D^2$. In the nontrivial case $D \geq 1$, at least one has nonzero trace (since $\operatorname{tr}(\mathbb{1}) = D \neq 0$). □

Theorem 10.14 (Cumulative span reaches $M_D(\mathbb{C})$ — sharp bound)**.** *If A is normal, then $T_{D^2-d'+1}(A) = M_D(\mathbb{C})$, where $d' = \dim S_1(A) = \operatorname{kr}(A)$. This is the sharp form of [SPGWC10, Lemma 1]: the cumulative span already reaches $M_D(\mathbb{C})$ by index $D^2 - d' + 1 \leq D^2$.*

Proof. The key observation is that if $T_n(A) \neq T_{n+1}(A)$, the dimension strictly increases. Since $S_1(A) \subseteq T_1(A)$ and $\dim S_1(A) = d'$, we have $\dim T_1(A) \geq d'$. After at most $D^2 - d'$ additional strict-growth steps the dimension reaches $D^2 = \dim M_D(\mathbb{C})$. Stabilisation before reaching $M_D(\mathbb{C})$ contradicts normality. Hence $T_{D^2-d'+1}(A) = M_D(\mathbb{C})$. □

Theorem 10.15 (Sharp nonzero trace product)**.** *If A is normal, then there exists a word w of length $\leq D^2 - \mathrm{kr}(A) + 1$ such that $\mathrm{tr}(A^w) \neq 0$. This is [SPGWC10, Lemma 1] with the sharp length bound.*

Proof. By Theorem 10.14, the identity $\mathbb{1}$ lies in $T_{D^2-d'+1}(A)$, hence is a linear combination of word products of length $\leq D^2 - d' + 1$. At least one such product has nonzero trace. □

Theorem 10.16 (Sharp nonzero trace product — positive length)**.** *If A is normal and $D \geq 2$, then there exists a word w of* positive *length $1 \leq |w| \leq D^2 - \mathrm{kr}(A) + 1$ such that $\mathrm{tr}(A^w) \neq 0$. This strengthens Theorem 10.15 by requiring $|w| \geq 1$, which is needed for the blocking argument in Theorem 10.28.*

Proof. Define the *positive-level* cumulative span $V_n = \mathrm{span}\{A^w : 1 \leq |w| \leq n\}$. For $D \geq 2$, the span $S_0(A) = \mathrm{span}\{\mathbb{1}\}$ has dimension 1, so $V_1 \supseteq S_1(A)$ has dimension $\geq d' \geq 1$. The same stabilization-or-strict-growth dichotomy shows $V_{D^2-d'+1} = M_D(\mathbb{C})$: if V_n stabilises and is not $M_D(\mathbb{C})$, then V_n is closed under left-multiplication by each A^i, so $S_N(A) \subseteq V_n \neq M_D(\mathbb{C})$ for all $N \geq 1$, contradicting normality. Since $V_{D^2-d'+1} = M_D(\mathbb{C})$, the identity $\mathbf{1}$ lies in $V_{D^2-d'+1}$ and is therefore a linear combination of word products of positive length. At least one such product has nonzero trace (since $\mathrm{tr}(\mathbf{1}) = D \geq 2 \neq 0$). □

10.4 Eigenvector extraction

Theorem 10.17 (Nonzero trace implies eigenvector)**.** *If $M \in M_D(\mathbb{C})$ has $\mathrm{tr}(M) \neq 0$, then M has a nonzero eigenvalue $\mu \neq 0$ and a corresponding eigenvector $\varphi \neq 0$ with $M\varphi = \mu\varphi$.*

Proof. A matrix with nonzero trace has a nonzero eigenvalue (since trace = sum of eigenvalues counted with multiplicity). The Fitting decomposition (Theorem 10.9) ensures that the corresponding generalized eigenspace is nontrivial. On that generalized eigenspace, $M - \mu\mathbb{1}$ is nilpotent, so its kernel is nontrivial. Any nonzero vector in $\ker(M - \mu\mathbb{1})$ is therefore a genuine eigenvector with eigenvalue μ. □

Theorem 10.18 (Eigenvalue extraction)**.** *If A is normal, then there exist a word w_0 with $|w_0| \leq D^2$, a nonzero scalar $\mu \neq 0$, and a nonzero vector $\varphi \neq 0$ such that $A^{w_0}\varphi = \mu\varphi$.*

Proof. By Theorem 10.13, there is a word w_0 with $\mathrm{tr}(A^{w_0}) \neq 0$ and $|w_0| \leq D^2$. By Theorem 10.17, A^{w_0} has a nonzero eigenvector. □

10.5 Eigenvector spreading

Definition 10.19 (Cumulative vector span)**.** Given an MPS tensor A and a vector $\varphi \in \mathbb{C}^D$, the *cumulative vector span* is

$$K_n(A, \varphi) = \mathrm{span}_{\mathbb{C}}\{A^w\varphi : |w| \leq n\} \subseteq \mathbb{C}^D.$$

This is [SPGWC10, proof of Lemma 2(a)].

Theorem 10.20 (Vector span stabilisation)**.** *If $K_n(A, \varphi) = K_{n+1}(A, \varphi)$, then $K_m(A, \varphi) = K_n(A, \varphi)$ for all $m \geq n$.*

Proof. Same argument as Theorem 10.5: left-multiplication by A^i cannot escape the stabilised span. $\square$

Lemma 10.21 (Vector span from matrix span)**.** *If $T_N(A) = M_D(\mathbb{C})$ and $\varphi \neq 0$, then $K_N(A, \varphi) = \mathbb{C}^D$.*

Proof. Every matrix $M \in M_D(\mathbb{C})$ is in $T_N(A)$, so in particular every rank-one matrix sending φ to any target vector v is in T_N. Applying such matrices to φ recovers all of $\mathbb{C}^D$. $\square$

Theorem 10.22 (Eigenvector spreading)**.** *Let A be a normal MPS tensor and let $A^{i_0}\varphi = \mu\varphi$ with $\mu \neq 0$ and $\varphi \neq 0$. Then $K_{D-1}(A, \varphi) = \mathbb{C}^D$. This is [SPGWC10, Lemma 2(a)].*

Proof. The vector span K_n is monotone and $\dim K_n \leq D$. If $K_n = K_{n+1}$ for some $n < D - 1$, the span stabilises (Theorem 10.20). But $T_{D^2}(A) = M_D(\mathbb{C})$ by Theorem 10.12, so Lemma 10.21 gives $K_{D^2}(A, \varphi) = \mathbb{C}^D$, contradicting early stabilisation. Hence $\dim K_n$ grows by ≥ 1 at each step, reaching D by step $D - 1$. $\square$

10.6 Proof of the cumulative Wielandt bound

Definition 10.23 (Wielandt analysis)**.** A *Wielandt analysis* for a normal MPS tensor A consists of: a word w_0 of length $\leq D^2$, a nonzero eigenvalue μ, and a nonzero eigenvector φ of the matrix A^{w_0}, such that $A^{w_0}\varphi = \mu\varphi$.

Theorem 10.24 (Wielandt analysis exists)**.** *Every normal MPS tensor admits a Wielandt analysis.*

Proof. Immediate from Theorem 10.18. $\square$

Theorem 10.25 (Wielandt chain)**.** *Let A be a normal MPS tensor with bond dimension $D \geq 1$. Then the following hold simultaneously:*

1. *$T_{D^2}(A) = M_D(\mathbb{C})$ (cumulative span stabilises);*
2. *there exists a word w_0 with $|w_0| \leq D^2$ and $\operatorname{tr}(A^{w_0}) \neq 0$;*
3. *there exist $w_0, \mu \neq 0, \varphi \neq 0$ with $|w_0| \leq D^2$ and $A^{w_0}\varphi = \mu\,\varphi$;*
4. *for every $\varphi \neq 0$, $K_{D^2}(A, \varphi) = \mathbb{C}^D$.*

Proof. Items (1)–(3) follow from earlier results (Theorems 10.12, 10.24). Item (4) combines these with the eigenvector spreading bound (Theorem 10.22). $\square$

Theorem 10.26 (Wielandt bound)**.** *If A is a normal MPS tensor with bond dimension D, then $T_{D^2}(A) = M_D(\mathbb{C})$. In particular, the cumulative span stabilises at index at most D^2. This is the cumulative-span part of [SPGWC10, Theorem 1]. The paper's theorem is stronger: via Lemma 2(b) it upgrades this to a fixed-length conclusion $S_N(A) = M_D(\mathbb{C})$, proved below in Theorem 10.40.*

Proof. Immediate from Theorem 10.12. $\square$

Remark 10.27 (Explicit blocking-length estimates). Theorem 10.26 gives the D^2 cumulative-span bound. The full-rank index bound $\iota(A) \leq (D^2 - d + 1) \cdot D^2$ (Theorem 1 case (1) of [SPGWC10]) is proved in Theorem 10.28 below. The sharp blocking-length estimates for injectivity (D^4) and bi-canonical form ($3D^5$) from [SPGWC10] are not proved here.

Theorem 10.28 (Wielandt's inequality — general bound)**.** *Let A be a normalized MPS tensor ($\sum_i (A^i)^\dagger A^i = \mathbb{1}$) that is primitive in the paper sense. Then*

$$\iota(A) \;\leq\; (D^2 - \mathrm{kr}(A) + 1) \cdot D^2,$$

where $\iota(A) = \min\{n : S_n(A) = M_D(\mathbb{C})\}$ is the full-Kraus-rank index. This is [SPGWC10, Theorem 1, case (1)].

Proof.

1. $D = 1$*:* In this case $\iota(A) = 0$, so the bound holds trivially.
2. $D \geq 2$*:* By Theorem 10.16, there exists a *positive-length* word w with $|w| \leq D^2 - d' + 1$ and $\mathrm{tr}(A^w) \neq 0$, where $d' = \mathrm{kr}(A)$. The matrix A^w has nonzero trace, hence a nonzero eigenvalue μ and eigenvector φ (Theorem 10.17). Set $n = |w|$ and let $B = A^{[n]}$ be the blocked tensor. Then B is normal (Theorem 10.35), and the encoding of w as a single block index gives $B^{i_0} = A^w$.
   (a) If A^w is invertible (invertible case), $S_{D^2}(B) = M_D(\mathbb{C})$, whence $S_{D^2 \cdot n}(A) = M_D(\mathbb{C})$.
   (b) If A^w is non-invertible (non-invertible case), the eigenvector φ spans a proper subspace, and the Wielandt chain argument gives $S_{D^2}(B) = M_D(\mathbb{C})$, whence $S_{D^2 \cdot n}(A) = M_D(\mathbb{C})$.

   In the invertible case, the invertible Kraus operator B^{i_0} allows direct dimension growth, giving $S_{D^2 - k_B + 1}(B) = M_D(\mathbb{C})$ where $k_B = \mathrm{kr}(B)$. In the non-invertible case, the bound $S_{D^2}(B) = M_D(\mathbb{C})$ follows from tracking indices through the rank-one extraction argument of [SPGWC10, Lemma 2(b)]; Theorem 10.40 only gives the qualitative conclusion $\exists N : S_N(B) = M_D(\mathbb{C})$ without this explicit bound. In both cases $\iota(A) \leq D^2 \cdot n \leq D^2 \cdot (D^2 - d' + 1)$.

□

10.7 Lemma 2(b): fixed-length matrix spanning

The preceding sections establish the cumulative-spanning part of the Lemma 2(a) argument from [SPGWC10]: eigenvector spreading gives a cumulative vector span $K_{D-1}(A, \varphi) = \mathbb{C}^D$. The next stage is to pass to a span at one fixed length, namely to find n_0 such that $S_{n_0}(A) = M_D(\mathbb{C})$ (i.e. word products of *exactly* one length span all matrices).

Definition 10.29 (Fixed-length vector span)**.** The *fixed-length vector span* at length n is

$$H_n(A, \varphi) = \mathrm{span}_{\mathbb{C}}\{A^w \varphi : |w| = n\} \subseteq \mathbb{C}^D.$$

This is $S_n(A)|\varphi\rangle$ in the notation of [SPGWC10].

Lemma 10.30 (Word span products)**.** $S_m(A) \cdot S_n(A) \subseteq S_{m+n}(A)$*: the product of word spans at lengths m and n is contained in the word span at length $m + n$.*

Proof. Every product $A^{w_1}A^{w_2}$ with $|w_1| = m$ and $|w_2| = n$ equals $A^{w_1w_2}$ with $|w_1w_2| = m + n$. $\square$

Lemma 10.31 (Eigenvector padding). *Suppose $A^{i_0}\varphi = \mu\varphi$ with $\mu \neq 0$. Then for all n,*

$$K_n(A, \varphi) = H_n(A, \varphi).$$

In particular, if $K_n(A, \varphi) = \mathbb{C}^D$ then already $H_n(A, \varphi) = \mathbb{C}^D$.

Proof. Any word w with $|w| \leq n$ can be padded to exact length n by appending $n - |w|$ copies of the index i_0. The eigenvector relation gives $A^{w \cdot i_0^k}\varphi = \mu^k \cdot A^w\varphi$, so the padded vector is a nonzero scalar multiple of the original. Hence every generator of K_n lies in H_n. The reverse inclusion $H_n \leq K_n$ is immediate. $\square$

Theorem 10.32 (Fixed-length vector spanning implies matrix spanning). *Assume:*

1. *$H_n(A, \varphi) = \mathbb{C}^D$ (length-n word products applied to φ span all of $\mathbb{C}^D$), and*
2. *for each standard basis vector e_j, the rank-one operator $|\varphi\rangle\langle e_j|$ lies in $S_m(A)$.*

Then $S_{n+m}(A) = M_D(\mathbb{C})$.

Proof. Fix a matrix unit E_{ij}. By hypothesis (1), there exists $M_i \in S_n(A)$ with $M_i\varphi = e_i$. By hypothesis (2), $|\varphi\rangle\langle e_j| \in S_m(A)$. Their product satisfies

$$M_i \cdot |\varphi\rangle\langle e_j| = |M_i\varphi\rangle\langle e_j| = |e_i\rangle\langle e_j| = E_{ij},$$

and $E_{ij} \in S_{n+m}(A)$ by Lemma 10.30. Since the matrix units span $M_D(\mathbb{C})$, $S_{n+m}(A) = M_D(\mathbb{C})$. $\square$

Lemma 10.33 (Blocking transfer). *If the blocked tensor $A^{[L]}$ satisfies $S_n(A^{[L]}) = M_D(\mathbb{C})$, then $S_{nL}(A) = M_D(\mathbb{C})$.*

Proof. Each length-n word in the blocked alphabet $\{0, \dots, d^L - 1\}$ decodes to a length-nL word in the original alphabet $\{0, \dots, d - 1\}$, so $S_n(A^{[L]}) \subseteq S_{nL}(A)$. $\square$

Remark 10.34 (Relation with Lemma 2(b)). In [SPGWC10], Lemma 2(b) combines a rank-one extraction for fixed-length spans with a product argument turning fixed-length vector spanning into fixed-length matrix spanning. Theorem 10.32 is the second step. Theorem 10.39 supplies the rank-one input in blocked form, and Theorem 10.40 then gives a blocked, fixed-length word-span saturation result.

10.8 Blocked tensor and rectangular span

This section extends the Lemma 2(b) argument to the *blocked tensor* $A^{[L]}$ and develops the "rectangular span" used to produce rank-one elements.

Theorem 10.35 (Blocking preserves normality). *If A is normal and $L > 0$, then the blocked tensor $A^{[L]}$ is also normal.*

Proof. Choose N with $S_N(A) = M_D(\mathbb{C})$. Then $\mathbb{1} \in S_N(A)$. For any $M \in S_N(A)$, write $\mathbb{1} = \sum_r c_r A^{w_r}$ with each $|w_r| = N$. Then

$$M = M \cdot \mathbb{1} = \sum_r c_r\, M A^{w_r} \in S_N(A) \cdot S_N(A) \subseteq S_{2N}(A)$$

by Lemma 10.30. Iterating this inclusion gives $S_N(A) \subseteq S_{kN}(A)$ for every $k \geq 1$; taking $k = L$ yields $M_D(\mathbb{C}) = S_N(A) \subseteq S_{NL}(A)$. By Lemma 10.36, $S_{NL}(A) \subseteq S_N(A^{[L]})$, so $S_N(A^{[L]}) = M_D(\mathbb{C})$. □

Lemma 10.36 (Chunking: word span embeds into blocked word span)**.** *$S_{nL}(A) \subseteq S_n(A^{[L]})$ for all n, L.*

Proof. Every length-nL word in the original alphabet can be split into n consecutive chunks of length L, each of which is a single blocked Kraus operator. By induction on n, using the word span product lemma (Lemma 10.30). □

Theorem 10.37 (Word eigenvector gives blocked eigenvector)**.** *If $A^{w_0}\varphi = \mu\varphi$ for a word w_0 of length L, then $(A^{[L]})^{j_0}\varphi = \mu\varphi$, where j_0 is the encoded blocked index corresponding to w_0.*

Proof. By definition, $(A^{[L]})^{j_0} = A^{w_0}$. □

Theorem 10.38 (Blocked fixed-length matrix spanning)**.** *Suppose A is normal and $L > 0$, and we have:*

1. *a word w_0 of length L with eigenvector $A^{w_0}\varphi = \mu\varphi$ ($\mu \neq 0$, $\varphi \neq 0$),*
2. *a word τ_0 of length L with transpose eigenvector $(A^{\tau_0})^T\psi = \nu\psi$ ($\nu \neq 0$, $\psi \neq 0$),*
3. *a rank-one element $\varphi\psi^T \in S_m(A^{[L]})$ for some m.*

Then there exists N such that $S_N(A) = M_D(\mathbb{C})$.
Remark. *The proof gives the explicit bound $N = ((D-1) + m + (D-1)) \cdot L$; the statement above records only the existential conclusion.*

Proof. Apply Theorem 10.32 to the blocked tensor $A^{[L]}$, which is normal by Theorem 10.35. The eigenvector and transpose eigenvector are transferred to single-index eigenvectors of $A^{[L]}$ by Theorem 10.37. Transfer back to the original tensor via Lemma 10.33. □

Theorem 10.39 (Rank-one extraction for blocked tensor)**.** *Suppose A is normal and $N_0 \geq 1$ with $S_{N_0}(A) = M_D(\mathbb{C})$. Then there exist words σ_0, τ_0 of length N_0, nonzero vectors φ, ψ, nonzero scalars μ, ν, and $m \in \mathbb{N}$ such that:*

1. *$A^{\sigma_0}\varphi = \mu\varphi$,*
2. *$(A^{\tau_0})^T\psi = \nu\psi$,*
3. *$\varphi\psi^T \in S_m(A^{[N_0]})$.*

Proof. Set $B = A^{[N_0]}$. Since $S_{N_0}(A) = M_D(\mathbb{C})$, we have $S_1(B) = M_D(\mathbb{C})$ by definition of the blocked tensor. Hence $\mathbb{1} \in S_1(B)$, so in the nontrivial case $D \geq 1$ not all blocked generators can have trace zero. Choose j_0 with $\mathrm{tr}(B^{j_0}) \neq 0$. By Theorem 10.17, B^{j_0} has a nonzero eigenvector φ with eigenvalue $\mu \neq 0$. Writing j_0 as the blocked index corresponding to a word σ_0 of length N_0 gives $A^{\sigma_0}\varphi = \mu\varphi$. Applying the same argument to the transpose blocked tensor gives a word τ_0 of length N_0, a nonzero vector ψ, and $\nu \neq 0$ with $(A^{\tau_0})^T\psi = \nu\psi$. Since A is normal and $N_0 \geq 1$, the blocked tensor B is normal by Theorem 10.35. Therefore there exists M with $S_M(B) = M_D(\mathbb{C})$. In particular $\varphi\psi^T \in S_M(B)$. Taking $m = M$ proves (3). □

Theorem 10.40 (Fixed-length word-span saturation via blocking). *If A is a normal MPS tensor with bond dimension D, then there exists N such that $S_N(A) = M_D(\mathbb{C})$. This gives the fixed-length spanning conclusion corresponding to [SPGWC10, Lemma 2(b)].*

Proof. By normality, some N_0 has $S_{N_0}(A) = M_D(\mathbb{C})$. If $N_0 = 0$ we are done. Otherwise, Theorem 10.39 produces eigenvectors and a rank-one element $\varphi\psi^T \in S_m(A^{[N_0]})$. Theorem 10.38 then gives $S_{((D-1)+m+(D-1))\cdot N_0}(A) = M_D(\mathbb{C})$. □

Remark 10.41 (Blocking argument for fixed-length spanning). Theorem 10.40 is obtained by first extracting a rank-one operator in a blocked tensor (Theorem 10.39) and then applying the fixed-length matrix-spanning theorem (Theorem 10.38). Thus the fixed-length span of A is reached through a blocked auxiliary tensor rather than by a direct unblocked argument.

10.9 Primitive MPS tensors

Definition 10.42 (Primitive MPS tensor). Assume $D > 0$. A tensor A together with a matrix $\rho \in M_D(\mathbb{C})$ is *primitive* if

$$\sum_i (A^i)^\dagger A^i = \mathbb{1}, \qquad \rho \geq 0, \qquad \rho \neq 0, \qquad \mathcal{E}_A(\rho) = \rho,$$

and, if P is the fixed-point projection associated to ρ, then $\rho_{\text{spec}}(\mathcal{E}_A - P) < 1$. When the choice of ρ is irrelevant, we simply say that A is a primitive MPS tensor. This is weaker than the paper's notion, which requires $\rho > 0$; see Remark 10.48. The stronger positive-definite hypothesis is imposed separately in Theorems 10.49 and 10.60.

Theorem 10.43 (Primitive overlap convergence). *If A is a primitive MPS tensor, then $O_{AA}(N) \to 1$.*

Proof. Immediate from the self-overlap convergence under a complementary spectral gap (Theorem 9.78). □

10.10 Primitivity and normality

Throughout this section, primitivity means the condition of Definition 10.42: the tensor is in TP gauge, it has a nonzero PSD fixed point, and the complementary map has spectral radius < 1.

Theorem 10.44 (Primitive MPS tensors have nonzero word products). *If A is a primitive MPS tensor, then for every $n \geq 0$ there exists a word w of length* exactly *n with $A^w \neq 0$.*

Proof. The PSD fixed point ρ of $\mathcal{E}_A$ satisfies $\mathcal{E}_A^n(\rho) = \rho \neq 0$. Since $\mathcal{E}_A^n(\rho) = \sum_{|w|=n} A^w \rho (A^w)^\dagger$, at least one summand is nonzero, hence some $A^w \neq 0$. □

Theorem 10.45 (Iterated transfer does not vanish). *If A is a primitive MPS tensor, then $\mathcal{E}_A^n \neq 0$ for all $n \geq 0$.*

Proof. By Theorem 10.44, there exists a word w of length n with $A^w \neq 0$, so $\mathcal{E}_A^n(\rho)$ has a nonzero summand. □

10.10.1 From primitivity to normality

The primitivity condition of Definition 10.42 combines TP normalization, a nonzero PSD fixed point, and a complementary spectral gap. It is connected to normality through the following results.

Theorem 10.46 (Fixed-point uniqueness from spectral gap)**.** *If A is a primitive MPS tensor with PSD fixed point ρ, then every fixed point σ of $\mathcal{E}_A$ is proportional to ρ: $\sigma = \frac{\operatorname{tr}(\sigma)}{\operatorname{tr}(\rho)}\rho$.*

Proof. Set $\sigma' = \sigma - \frac{\operatorname{tr}(\sigma)}{\operatorname{tr}(\rho)}\rho$. Then $\operatorname{tr}(\sigma') = 0$ and $(E - P_\rho)(\sigma') = \sigma'$. If $\sigma' \neq 0$, then 1 is an eigenvalue of $E - P_\rho$, contradicting the spectral gap $\rho_{\text{spec}}(E - P_\rho) < 1$. □

Theorem 10.47 (Complement powers tend to zero)**.** *If A is a primitive MPS tensor with PSD fixed point ρ, then $(\mathcal{E}_A - P_\rho)^n \to 0$ in operator norm.*

Proof. The spectral radius of $\mathcal{E}_A - P_\rho$ is strictly less than 1 by the spectral-gap hypothesis. Apply the general result that powers of an operator with spectral radius < 1 tend to zero. □

Remark 10.48 (Primitivity: spectral gap vs. positive definiteness). The primitivity condition used in §10.10.1 has three parts: TP normalization $\sum_i (A^i)^\dagger A^i = \mathbb{1}$, a nonzero PSD fixed point ρ, and the spectral gap $\rho_{\text{spec}}(\mathcal{E}_A - P_\rho) < 1$. The notion of "primitive" in [SPGWC10, Proposition 3] additionally requires ρ to be positive definite (full rank). Without positive definiteness, the spectral-gap condition alone does *not* imply irreducibility or normality. A counterexample is $D = 2$, $\rho = \operatorname{diag}(1, 0)$.

With the additional hypothesis $\rho > 0$, Theorem 10.49 gives irreducibility. Together with Burnside's theorem (Chapter 11, §11.4.1) and Theorem 10.58, one obtains

$$\text{primitive} + \rho \text{ PosDef} \;\Rightarrow\; \text{irreducible} \;\Rightarrow\; \operatorname{alg}(A) = M_D(\mathbb{C}) \;\Rightarrow\; \text{normal}.$$

The last step uses the aperiodicity condition $\mathbb{1} \in S_1(A)$ from Remark 10.53. After restoring the positive-definite hypothesis, [SPGWC10, Proposition 3] identifies the paper's primitive/strongly irreducible condition with eventual full Kraus rank, i.e. with our notion of normality.

10.10.2 Primitive implies irreducible

Theorem 10.49 (Primitive with PosDef fixed point implies irreducible tensor)**.** *If A is a primitive MPS tensor with PSD fixed point ρ and ρ is positive definite, then A is an irreducible tensor. This is part of [SPGWC10, Proposition 3] (see also [CPGSV17a, §2.3]).*

Proof. By contrapositive. Assume A has an invariant projection P ($P \neq 0$, $P \neq \mathbb{1}$, $(\mathbb{1}-P)A^iP = 0$ for all i). Set $\sigma = P\rho P$.

1. *Invariance of σ under $\mathcal{E}_A^n$:* the projection relation $(\mathbb{1}-P)A^iP = 0$ implies $(\mathbb{1}-P)\cdot\mathcal{E}_A^n(\sigma) = 0$ for all n (by induction on n).

2. *Convergence:* By Theorem 10.47, $\mathcal{E}_A^n(\sigma) \to \frac{\operatorname{tr}(\sigma)}{\operatorname{tr}(\rho)}\rho$. Hence $(\mathbb{1} - P) \cdot \frac{\operatorname{tr}(\sigma)}{\operatorname{tr}(\rho)}\rho = 0$.

3. *Nonzero scalar:* Since $P \neq 0$ and ρ is positive definite, $\operatorname{tr}(P\rho P) > 0$, so $\frac{\operatorname{tr}(\sigma)}{\operatorname{tr}(\rho)} \neq 0$. Therefore $(\mathbb{1} - P)\rho = 0$.

4. *Contradiction:* Since ρ is positive definite, it is a unit in $M_D(\mathbb{C})$. From $(\mathbb{1} - P)\rho = 0$ we get $\mathbb{1} - P = 0$, i.e. $P = \mathbb{1}$, contradicting $P \neq \mathbb{1}$.

□

10.10.3 From primitivity to strong irreducibility and normality

The results in this subsection remove the aperiodicity assumption from the normality conclusions. They connect the spectral-gap primitivity of Definition 10.42 directly to normality (eventually full Kraus rank), using the convergence of the transfer-map iterates to the projection onto the fixed-point subspace.

Theorem 10.50 (Primitive MPS with PosDef fixed point has irreducible transfer map)**.** *If A is a primitive MPS tensor with positive-definite fixed point ρ, then the transfer map $\mathcal{E}_A$ is irreducible.*

The proof uses fixed-point uniqueness (Theorem 10.46): every PSD fixed point is proportional to ρ, so Wolf's criterion for irreducibility (positive-definite fixed point implies irreducibility) applies directly.

Proof. By Theorem 10.46, if $\sigma \geq 0$ and $\mathcal{E}_A(\sigma) = \sigma$, then $\sigma = \frac{\operatorname{tr}\sigma}{\operatorname{tr}\rho}\rho$. Since $\rho > 0$, Wolf's uniqueness criterion for irreducibility applies. □

Theorem 10.51 (Primitive MPS with PosDef fixed point is strongly irreducible)**.** *If A is a primitive MPS tensor with positive-definite fixed point ρ, then A is strongly irreducible in the sense of [SPGWC10, Proposition 3(c)]: $\mathcal{E}_A$ is irreducible, the fixed point ρ is positive definite, and the peripheral spectrum of $\mathcal{E}_A$ is $\{1\}$.*

Proof. Combine Theorem 10.50 (irreducibility) with the peripheral-spectrum primitivity from the spectral-gap hypothesis (Theorem 10.43). □

Theorem 10.52 (Primitive MPS with PosDef fixed point is normal)**.** *If A is a primitive MPS tensor with positive-definite fixed point ρ, then A is normal (has eventually full Kraus rank). No aperiodicity assumption is needed.*

This is the key connection between the spectral-gap primitivity of Definition 10.42 and normality, filling the gap that the aperiodicity-based argument in §10.11 leaves open.

Proof. By Theorem 10.51, A is strongly irreducible. The implication from strong irreducibility to normality proceeds as follows: strong irreducibility gives peripheral spectrum $\{1\}$, hence the aperiodicity condition $\mathbb{1} \in S_1(A)$. Combined with irreducibility, Theorem 10.59 then yields normality. □

10.11 Cumulative span to word span

Cumulative spanning ($T_N(A) = M_D(\mathbb{C})$, the union of word spans of all lengths $\leq N$) is weaker than normality ($S_L(A) = M_D(\mathbb{C})$ for a single length L). The gap is closed by an aperiodicity hypothesis.

Remark 10.53 (Aperiodicity condition). We say A satisfies the *aperiodicity condition* if $\mathbb{1} \in S_1(A)$, i.e. the identity matrix lies in the span of the Kraus operators $\{A^i\}$. This is an additional hypothesis. The one-sided TP normalization $\sum_i (A^i)^\dagger A^i = \mathbb{1}$ does *not* imply $\mathbb{1} \in S_1(A)$; it only gives a quadratic identity among the Kraus operators. For the paper-style primitive condition (peripheral-spectrum primitivity together with a positive-definite fixed point), one expects aperiodicity to follow from spectral analysis of the transfer map. That implication is not used in the present Wielandt argument, so the condition is kept explicit.

Remark 10.54 (Counterexample: cumulative spanning does not imply normality). In $M_2(\mathbb{C})$, let $A^0 = E_{12}$ and $A^1 = E_{21}$ (the off-diagonal matrix units). Then $\operatorname{alg}(A) = M_2(\mathbb{C})$ and $T_2(A) = M_2(\mathbb{C})$, but the word spans alternate: $S_n(A)$ contains only off-diagonal matrices for odd n and only diagonal matrices for even n. Hence $S_n(A) \neq M_2(\mathbb{C})$ for every n, and A is not normal. The aperiodicity condition $\mathbb{1} \in S_1(A)$ fails: $S_1(A) = \operatorname{span}_{\mathbb{C}}\{E_{12}, E_{21}\}$.

Theorem 10.55 (Word span monotonicity from aperiodicity)**.** *If $\mathbb{1} \in S_1(A)$, then $S_n(A) \leq S_{n+1}(A)$ for all n.*

Proof. For any $M \in S_n(A)$, $M = M \cdot \mathbb{1} \in S_n(A) \cdot S_1(A) \subseteq S_{n+1}(A)$ by Lemma 10.30. $\square$

Theorem 10.56 (Cumulative span equals word span under aperiodicity)**.** *If $\mathbb{1} \in S_1(A)$, then $T_n(A) = S_n(A)$ for all n.*

Proof. By Theorem 10.55, the word spans are monotone: $S_0 \leq S_1 \leq \cdots$. The supremum $T_n = S_0 + \cdots + S_n$ therefore equals the largest term S_n. $\square$

Theorem 10.57 (Cumulative spanning plus aperiodicity implies normality)**.** *If $T_N(A) = M_D(\mathbb{C})$ and $\mathbb{1} \in S_1(A)$, then A is normal.*

Proof. By Theorem 10.56, $S_N(A) = T_N(A) = M_D(\mathbb{C})$. $\square$

Theorem 10.58 (Algebra span plus aperiodicity implies normality)**.** *If $\mathrm{alg}(A) = M_D(\mathbb{C})$ and $\mathbb{1} \in S_1(A)$, then A is normal.*

Proof. Since $\mathrm{alg}(A) = M_D(\mathbb{C})$, there exists N with $T_N(A) = M_D(\mathbb{C})$ (by the ascending chain condition on finite-dimensional subspaces). Apply Theorem 10.57. $\square$

Theorem 10.59 (Irreducible tensor plus aperiodicity implies normality)**.** *If A is an irreducible tensor, $D > 0$, and $\mathbb{1} \in S_1(A)$, then A is normal. This assembles the full chain:*

$$\textit{irreducible tensor} \xrightarrow{11.20} \textit{irreducible action} \xrightarrow{\textit{Burnside}} \mathrm{alg}(A) = M_D(\mathbb{C}) \xrightarrow{10.58} \textit{normal.}$$

This corresponds to the irreducibility-to-normality direction of [SPGWC10, Proposition 3]. For the Burnside step, we appeal forward to Theorems 11.20 and 11.21 in Chapter 11; this is a presentation-level forward dependency, but the mathematical dependency is acyclic.

Proof. By Theorem 11.20, A acts irreducibly on $\mathbb{C}^D$. By Theorem 11.21, $\mathrm{alg}(A) = M_D(\mathbb{C})$. By Theorem 10.58 with the aperiodicity hypothesis, A is normal. $\square$

Theorem 10.60 (Primitive + positive definite fixed point imply normality)**.** *If A is a primitive MPS tensor with fixed point ρ and ρ is positive definite, then A is normal. No aperiodicity assumption is needed.*

Proof. By Theorem 10.51, A is strongly irreducible. In particular, A is irreducible and the peripheral spectrum of $\mathcal{E}_A$ is $\{1\}$, so A is aperiodic. Theorem 10.59 then gives normality, hence eventually full Kraus rank. $\square$

Remark 10.61 (Normal canonical form avoids the aperiodicity step). The aperiodicity results in §10.11 are needed only for the block-injective argument through Definition 2.22, where one upgrades cumulative spanning to eventual block injectivity. The normal-canonical theory used later in Chapters 11–13 does not pass through this step: it works directly with irreducible TP-normalized blocks whose blocked transfer maps are primitive and, once the corresponding weighted block data are available, places them in normal canonical form in the sense of [CPGSV17a]. There are analogous cumulative-span lemmas for blocked irreducible tensors and eigenvector-spreading arguments, but they are auxiliary here and we do not list them separately.

Chapter 11

Canonical Form Reduction

This chapter reduces a general MPS tensor to block-diagonal data via iterated invariant-subspace decomposition. Two reduction schemes are relevant. The block-injective reduction aims at injective TP-normalized blocks and the canonical form conditions of Chapter 12. The normal-canonical reduction uses a weighted family of irreducible TP-normalized blocks with primitive transfer maps and pairwise distinct nonzero weight moduli, in the sense of [CPGSV17a]. The Perron–Frobenius eigenvector existence and TP-gauge construction are in Chapter 8; this chapter uses them to produce left-canonical normalization data and the normal canonical form data. The reduction draws on [PGVWC07], [CPGSV21], and [Wol12, Ch. 6].

11.1 Block-diagonal construction

Definition 11.1 (Block-diagonal tensor from blocks)**.** Recall the block-diagonal tensor of Definition 2.26. Given r blocks with bond dimensions $(D_k)_{k=1}^r$, per-block MPS tensors $\{A_k^i\}$, and scaling factors $(\mu_k)_{k=1}^r$, we write $\bigoplus_{k=1}^r \mu_k A_k$ for the tensor of total bond dimension $D = \sum_k D_k$ whose Kraus operators are $A^i = \bigoplus_{k=1}^r \mu_k A_k^i$, obtained by placing the scaled block Kraus operators on the diagonal.

Theorem 11.2 (Per-block single-block FT)**.** *If all block tensors A_k are injective and the per-block same-MPV condition holds (A_k and B_k generate the same MPV family for each k), then each pair (A_k, B_k) is gauge equivalent.*

Proof. Apply the single-block fundamental theorem (Theorem 4.11) to each block independently. $\square$

Theorem 11.3 (Per-block same MPV implies global same MPV)**.** *If A_k and B_k generate the same MPV family for each block k, then the block-diagonal tensors $\bigoplus_k \mu_k A_k$ and $\bigoplus_k \mu_k B_k$ also generate the same MPV family.*

Proof. By the MPV decomposition (Theorem 2.27), each MPV is $\sum_k \mu_k^N V^{(N)}(A_k)_\sigma$. Per-block equality of MPV coefficients gives global equality. $\square$

Theorem 11.4 (Global multi-block fundamental theorem)**.** *If each block A_k is injective and each pair (A_k, B_k) generates the same MPV family, then the block-diagonal tensors $\bigoplus_k \mu_k A_k$ and $\bigoplus_k \mu_k B_k$ are gauge equivalent.*

Proof. Apply the per-block single-block fundamental theorem (Theorem 11.2) to obtain blockwise gauge equivalences from injectivity and equality of MPV families, then combine them into a global gauge for the block-diagonal tensors. □

11.2 Transfer map normalization

Theorem 11.5 (Scaling preserves injectivity)**.** *If A is injective and $\zeta \neq 0$, then ζA is injective.*

Proof. Scaling by a nonzero scalar preserves linear independence and hence spanning. □

Theorem 11.6 (Transfer map under scaling)**.** $\mathcal{E}_{\zeta A}(X) = |\zeta|^2 \mathcal{E}_A(X)$.

Proof. Each summand: $(\zeta A^i)X(\zeta A^i)^\dagger = |\zeta|^2 A^i X (A^i)^\dagger$. □

Theorem 11.7 (Phase-scaling preserves normalization)**.** *If $\sum_i (A^i)^\dagger A^i = \mathbb{1}$ and $|\zeta| = 1$, then $\sum_i (\zeta A^i)^\dagger (\zeta A^i) = \mathbb{1}$.*

Proof. $(\zeta A^i)^\dagger (\zeta A^i) = \bar{\zeta}\zeta (A^i)^\dagger A^i = (A^i)^\dagger A^i$, since $\bar{\zeta}\zeta = |\zeta|^2 = 1$. □

Theorem 11.8 (MPV under block normalization)**.** *For any nonzero scaling factors μ_k, set $\eta_k := \mu_k / |\mu_k|$ and $B_k := |\mu_k| A_k$. Then $\bigoplus_k \mu_k A_k$ and $\bigoplus_k \eta_k B_k$ generate the same MPV family. Equivalently, the k-th block contribution factors as*

$$\mu_k^N V^{(N)}(A_k)_\sigma = \eta_k^N |\mu_k|^N V^{(N)}(A_k)_\sigma,$$

so the modulus contributes only through $|\mu_k|^N$, while the remaining factor η_k is a unit-modulus phase.

Proof. By the MPV decomposition (Theorem 2.27), the k-th contribution of $\bigoplus_k \mu_k A_k$ is $\mu_k^N V^{(N)}(A_k)_\sigma$. For $B_k = |\mu_k| A_k$, Lemma 2.15 gives $V^{(N)}(B_k)_\sigma = |\mu_k|^N V^{(N)}(A_k)_\sigma$. Hence $\eta_k^N V^{(N)}(B_k)_\sigma = \eta_k^N |\mu_k|^N V^{(N)}(A_k)_\sigma = \mu_k^N V^{(N)}(A_k)_\sigma$ for every block, so the total MPV families agree. □

11.3 Invariant subspace decomposition

Theorem 11.9 (Unitary conjugation preserves the MPV family)**.** *If U is a unitary matrix, then A and $U^\dagger A U$ generate the same MPV family.*

Proof. By cyclicity of the trace: $\mathrm{tr}((U^\dagger A^{i_1} U) \cdots (U^\dagger A^{i_N} U)) = \mathrm{tr}(A^{i_1} \cdots A^{i_N})$. □

Definition 11.10 (Support projection)**.** For a positive semi-definite matrix $\rho \geq 0$, the *support projection* P is the orthogonal projection onto the range of ρ. It is constructed via the spectral decomposition: if $\rho = U \operatorname{diag}(\lambda_1, \ldots, \lambda_D) U^\dagger$, then $P = U \operatorname{diag}(\mathbf{1}_{\lambda_1 > 0}, \ldots, \mathbf{1}_{\lambda_D > 0}) U^\dagger$, where $\mathbf{1}_{\lambda_j > 0}$ is 1 if $\lambda_j > 0$ and 0 otherwise.

Definition 11.11 (Invariant projection predicate)**.** An MPS tensor A *has an invariant projection* if there exists an orthogonal projection P with $P \neq 0$, $P \neq \mathbb{1}$, and $(\mathbb{1} - P) A^i P = 0$ for all i. A tensor is *irreducible* if it has no nontrivial invariant projection (Definition 11.14).

Theorem 11.12 (PSD fixed point gives invariant projection)**.** *Let $\rho \geq 0$ be a fixed point of $\mathcal{E}_A$, and let P be the support projection of ρ. Then for all i, $(\mathbb{1} - P) A^i P = 0$: the range of P is invariant under all A^i.*

Proof. From $\mathcal{E}_A(\rho) = \rho$ and positivity of each summand $A^i\rho(A^i)^\dagger$, $(\mathbb{1} - P)A^i\rho(A^i)^\dagger(\mathbb{1} - P) = 0$ for each i. Since ρ is supported on the range of P, this forces $(\mathbb{1} - P)A^iP = 0$.

This is the finite-dimensional version of [Wol12, Prop. 6.10] ("Stationary subspaces": the support projection of any stationary state is sub-harmonic, i.e. invariant under the channel). The equivalence between irreducibility and absence of nontrivial invariant projections is [Wol12, Thm. 6.2]; see also [CPGSV21, §IV.A.1]. □

Theorem 11.13 (Two-block decomposition)**.** *If the transfer map has a PSD fixed point ρ that is* not *positive definite (i.e. ρ has a nontrivial kernel), then there exist MPS tensors A_1, A_2 of smaller bond dimensions such that the two-block tensor $A_1 \oplus A_2$ generates the same MPV family as A: $\mathcal{V}(A) = \mathcal{V}(A_1 \oplus A_2)$.*

Proof. The support projection P of ρ is neither 0 nor $\mathbb{1}$ (since $\rho \neq 0$ and ρ is not PD). Theorem 11.12 gives $(\mathbb{1}-P)A^iP = 0$ for all i. A unitary change of basis that block-diagonalizes P puts A^i in upper-triangular block form, with diagonal blocks B^i_{11} and B^i_{22}. Theorem 11.9 ensures the MPV is preserved by the conjugation. The trace of the upper-triangular product equals the sum of the diagonal-block traces, so the block-diagonal tensor obtained by discarding the off-diagonal blocks has the same MPV coefficients as the conjugated tensor, and hence as A. □

11.4 Iterated reduction

Definition 11.14 (Irreducible tensor)**.** An MPS tensor A is *irreducible* if it has no nontrivial invariant projection.

Theorem 11.15 (Irreducible block decomposition)**.** *Every MPS tensor A admits a block-diagonal tensor $\bigoplus_{k=1}^r A_k$ where each A_k is irreducible and $\mathcal{V}(A) = \mathcal{V}(\bigoplus_k A_k)$.*

Proof. By strong induction on the bond dimension D. If A is already irreducible, take $r = 1$. Otherwise, by Definition 11.14, A has a nontrivial invariant projection P. Diagonalizing P yields the same two-block upper-triangular splitting used in Theorem 11.13; the block-diagonal tensor obtained by discarding the off-diagonal part has the same MPV family as A, and the two diagonal blocks have strictly smaller bond dimensions. Iterating this splitting produces a block-diagonal tensor $\bigoplus_k A_k$ whose blocks are irreducible. Apply the induction hypothesis to each block and concatenate. (If one prefers a positive-eigenvector argument without assuming TP normalization, the correct replacement for the channel-specific Cesàro theorem is Theorem 8.15; the induction itself needs only the invariant projection.)

Both [PGVWC07] (Theorem 4, "TI canonical form") and [CPGSV21] (§IV.A.1) obtain the block decomposition by the same invariant-projection splitting used here. An alternative derivation, developed in [Wol12, Thm. 6.14], characterises the fixed-point set of a quantum channel as a $*$-algebra and then applies the Wedderburn–Artin decomposition ([Wol12, Eq. (1.39)]) to read off the block structure directly. The block decomposition is obtained by iterating the invariant-projection splitting and using strong induction on D to ensure termination. □

Theorem 11.16 (Unitary diagonal positive fixed point in TP gauge)**.** *Let A be an irreducible MPS tensor satisfying $\sum_i (A^i)^\dagger A^i = \mathbb{1}$, and assume $D > 0$. Then there exist a unitary U and a diagonal positive definite matrix Λ such that, for $B^i := U^\dagger A^i U$,*

$$\sum_i (B^i)^\dagger B^i = \mathbb{1} \qquad \textit{and} \qquad \mathcal{E}_B(\Lambda) = \Lambda.$$

Proof. The TP hypothesis makes $\mathcal{E}_A$ a channel, so it has a nonzero PSD fixed point. The irreducible tensor condition implies that $\mathcal{E}_A$ is irreducible as a CP map, and Theorem 8.3 upgrades the fixed point to a positive definite one. The spectral theorem diagonalizes this matrix by a unitary conjugation, and unitary conjugation preserves the TP normalization. $\square$

11.4.1 From irreducibility to full spanning via Burnside's theorem

The irreducibility condition (Definition 11.14) is formulated in terms of invariant *projections*. For Burnside-style arguments it is more natural to work with invariant *subspaces* of $\mathbb{C}^D$ under the Kraus operators.

Definition 11.17 (Algebra span)**.** For an MPS tensor A, let $\mathrm{alg}(A)$ be the unital $\mathbb{C}$-subalgebra of $M_D(\mathbb{C})$ generated by the matrices $\{A^i\}$:

$$\mathrm{alg}(A) = \mathbb{C}\langle A^i : i = 0, \dots, d{-}1\rangle \subseteq M_D(\mathbb{C}).$$

Definition 11.18 (Invariant submodule)**.** A subspace $W \leq \mathbb{C}^D$ is A-*invariant* if $A^i W \subseteq W$ for every i.

Definition 11.19 (Irreducible action)**.** The Kraus operators of A act *irreducibly* on $\mathbb{C}^D$ if every A-invariant subspace is trivial, i.e. equal to 0 or to $\mathbb{C}^D$.

Theorem 11.20 (Irreducible tensor implies irreducible action)**.** *If A is irreducible as a tensor (Definition 11.14), then the Kraus operators act irreducibly on $\mathbb{C}^D$ (Definition 11.19).*

Proof. Let $W \leq \mathbb{C}^D$ be an A-invariant subspace. In finite dimension the orthogonal projection P onto W exists. Invariance of W implies $(\mathbb{1} - P)A^i P = 0$ for all i. Thus P is a nontrivial invariant projection unless $W = 0$ or $W = \mathbb{C}^D$, contradicting irreducibility of the tensor. $\square$

Theorem 11.21 (Burnside's theorem for Kraus algebras)**.** *Assume $D > 0$. If A acts irreducibly on $\mathbb{C}^D$, then the generated unital subalgebra is the full matrix algebra:*

$$\mathrm{alg}(A) = M_D(\mathbb{C}).$$

This is the complex finite-dimensional case of Burnside's theorem (equivalently, Jacobson's density theorem); see [Jac09].

Proof. Transport $\mathrm{alg}(A)$ along the algebra equivalence $M_D(\mathbb{C}) \simeq_{\mathbb{C}} \mathrm{End}_{\mathbb{C}}(\mathbb{C}^D)$. The irreducible-action hypothesis makes $\mathbb{C}^D$ a simple module for this algebra. Over the algebraically closed field $\mathbb{C}$, Schur's lemma identifies the endomorphisms commuting with the action with scalars, and Jacobson density then forces the action map to be surjective onto all of $\mathrm{End}_{\mathbb{C}}(\mathbb{C}^D)$. $\square$

Remark 11.22. This Burnside/Jacobson-density step is a substantial external algebraic input in the chapter. We use it here as the finite-dimensional complex Burnside theorem, namely the statement that an irreducible matrix algebra action already generates the full endomorphism algebra.

Theorem 11.23 (Irreducible action implies eventual cumulative spanning)**.** *Assume $D > 0$. If A acts irreducibly on $\mathbb{C}^D$, then there exists N such that $T_N(A) = M_D(\mathbb{C})$.*

Proof. By Theorem 11.21, $\mathrm{alg}(A) = M_D(\mathbb{C})$. Every element of the algebra span is a linear combination of word products, hence belongs to $T_n(A)$ for some n. Since $M_D(\mathbb{C})$ is finite-dimensional, the ascending chain $T_0(A) \leq T_1(A) \leq \cdots$ stabilizes, so one can choose a uniform N with $T_N(A) = M_D(\mathbb{C})$. $\square$

11.5 Proportional single-block theorem

Theorem 11.24 (Proportional MPV implies gauge-phase equivalence)**.** *Let A and B be injective MPS tensors satisfying $\sum_i (A^i)^\dagger A^i = \mathbb{1}$ and $\sum_i (B^i)^\dagger B^i = \mathbb{1}$. If A and B generate proportional MPV families ($V^{(N)}(A)_\sigma = c_N V^{(N)}(B)_\sigma$ for some $c_N \in \mathbb{C}$) and $O_{AA}(N) \to 1$, $O_{BB}(N) \to 1$, then A and B are gauge-phase equivalent.*

Proof. By contradiction. If A and B are not gauge-phase equivalent, then $O_{AB}(N) \to 0$ (Theorem 9.22). From the proportionality hypothesis, $O_{AB}(N) = c_N O_{BB}(N)$. Since $O_{BB}(N) \to 1$, it follows that $c_N \to 0$. But $O_{AA}(N) = |c_N|^2 O_{BB}(N) \to 0$, contradicting $O_{AA}(N) \to 1$.

In [PGVWC07], the proportional case is handled as part of the canonical form argument. Thus the proportional case is obtained directly from the spectral gap in Chapter 9. □

Theorem 11.25 (Proportional MPV + primitive implies gauge-phase equivalence)**.** *Let A and B be MPS tensors with primitive transfer maps E_A, E_B and positive definite fixed points ρ_A, ρ_B. If A and B generate proportional MPV families, then A and B are gauge-phase equivalent. This strengthens Theorem 11.24: primitivity supplies left-canonicality, irreducibility, and overlap convergence, reducing the hypothesis count from seven to four.*

Proof. From primitivity, extract a left-canonical normalization ($\sum_i (A^i)^\dagger A^i = \mathbb{1}$), overlap convergence ($O_{AA}(N) \to 1$), and irreducibility (Theorem 10.49). Apply Theorem 11.24. □

11.6 Canonical form from peripheral primitivity

The canonical form conditions (Definition 12.9) includes an explicit hypothesis that the self-overlap $O_{A_k A_k}(N) \to 1$ for each block. The following theorem derives this from per-block peripheral-spectrum primitivity, removing it as an independent assumption.

Theorem 11.26 (Canonical form from peripheral primitivity)**.** *Let $(\mu_k, A_k)_{k=1}^r$ be a collection of block tensors and scaling factors satisfying:*

1. *each A_k is injective,*
2. *each A_k satisfies the TP normalization $\sum_i (A_k^i)^\dagger A_k^i = \mathbb{1}$,*
3. *$|\mu_1| > \cdots > |\mu_r|$ (strictly decreasing),*
4. *all $\mu_k \neq 0$,*
5. *each transfer map $\mathcal{E}_{A_k}$ is primitive, i.e. has peripheral spectrum $\{1\}$.*

Then (μ_k, A_k) satisfies the canonical form conditions (Definition 12.9).

Proof. Fix a block k and write $E := \mathcal{E}_{A_k}$. We check that each primitive block already satisfies the defining properties of the canonical form conditions. By injectivity and TP normalization, Theorem 8.9 provides a nonzero positive definite fixed point for E. To apply the spectral-gap criterion (Theorem 5.39), we check its three hypotheses:

1. E is primitive, since the peripheral spectrum is $\{1\}$ by assumption;
2. every eigenvalue of E has modulus at most 1, because $E = F_{A_k A_k}$ is a normalized mixed transfer operator and Theorem 9.14 applies;

3. any trace-zero fixed point of E is zero; this is the trace-zero uniqueness consequence extracted in Chapter 8 from the uniqueness of the PSD fixed point (Theorem 8.5).

Hence E has a spectral gap on the complement of its fixed-point projection. Theorem 9.78 then gives $O_{A_k A_k}(N) \to 1$. Since this holds for every block, the overlap clause required in Definition 12.9 follows from the other hypotheses. □

Remark 11.27. Theorem 11.26 shows that the overlap convergence hypothesis in the canonical form conditions is redundant once peripheral-spectrum primitivity is known. The argument factors through the spectral-gap formulation of Chapter 9: peripheral-spectrum primitivity is first converted to that spectral-gap condition, and the overlap limit is then deduced from it. In the paper [CPGSV21], primitivity is obtained from irreducibility plus periodicity removal by blocking to period one. Theorem 11.28 is precisely this periodicity-removal step under irreducibility and TP normalization.

Theorem 11.28 (Blocking yields a peripheral-spectrum primitive transfer map)**.** *Let A be an irreducible MPS tensor with $D > 0$. Assume that A satisfies the TP normalization $\sum_i (A^i)^\dagger A^i = \mathbb{1}$. Then there exists a blocking length $p > 0$ such that the transfer map $\mathcal{E}_{A^{[p]}}$ of the blocked tensor $A^{[p]}$ is primitive, i.e. its peripheral spectrum is $\{1\}$.*

Proof. The period-p blocking step is the diagrammatic move of grouping the chain into consecutive blocks of length p. Pass to the conjugate-transposed Kraus family $K_i := (A^i)^\dagger$. The TP normalization of A makes K unital, and irreducibility of A gives irreducibility of the associated Kraus map. The TP fixed-point theorem for $\mathcal{E}_A$ provides a positive definite fixed point for the adjoint map of K, so Theorem 5.34 shows that every peripheral eigenvalue is a root of unity. Since peripheral$(\mathcal{E}_A)$ is finite (Theorem 5.26), Lemma 5.30 gives a common exponent p such that $\mu^p = 1$ for all $\mu \in$ peripheral$(\mathcal{E}_A)$. Lemma 5.31 then implies peripheral$(\mathcal{E}_A^p) = \{1\}$. Finally, blocking corresponds to taking a power of the transfer map, so $\mathcal{E}_{A^{[p]}}$ is primitive. □

Remark 11.29. Theorem 11.28 is the "periodicity removal by blocking" step in [CPGSV17a, Appendix A] for an irreducible block already placed in TP gauge. The later existence theorems simply reuse this step.

11.7 Normal canonical form

Definition 11.30 (Normal canonical form conditions)**.** A collection of scaling factors $(\mu_k)_{k=1}^r$ and block tensors $(A_k)_{k=1}^r$ satisfies the *normal canonical form conditions* if:

1. each A_k is irreducible,
2. each A_k satisfies the TP normalization $\sum_i (A_k^i)^\dagger A_k^i = \mathbb{1}$,
3. each transfer map $\mathcal{E}_{A_k}$ is primitive (equivalently, has peripheral spectrum $\{1\}$),
4. the moduli are strictly decreasing: $|\mu_1| > |\mu_2| > \cdots > |\mu_r|$,
5. all $\mu_k \neq 0$,
6. all bond dimensions are positive.

Here "normal" is used in the spectral sense of [CPGSV17a]: irreducible blocks whose TP-normalized transfer maps are primitive. As noted in the remark after Definition 2.22, this is equivalent to the eventual block-injectivity formulation of Definition 2.22; see [SPGWC10, Proposition 3].

Theorem 11.31 (Self-overlap is derived in normal canonical form)**.** *If (μ_k, A_k) satisfies the normal canonical form conditions, then $O_{A_k A_k}(N) \to 1$ for every block k.*

Proof. Peripheral-spectrum primitivity is part of the data. Together with irreducibility and TP normalization, it yields the spectral-gap primitive theorem needed for overlap convergence, so the self-overlap condition follows from the other hypotheses rather than appearing as a separate assumption. □

Theorem 11.32 (Modulus-one eigenvalue rigidity for irreducible TP blocks)**.** *Let A and B be irreducible MPS tensors satisfying $\sum_i (A^i)^\dagger A^i = \mathbb{1}$ and $\sum_i (B^i)^\dagger B^i = \mathbb{1}$. If the mixed transfer map has spectral radius at least 1 (equivalently, it has a peripheral eigenvalue of modulus 1), then A and B are gauge-phase equivalent.*

Proof. The argument is the same as in Theorem 9.16, but the fixed-point input comes from irreducibility rather than injectivity. Take a modulus-one eigenvector of the mixed transfer map. Because A and B are TP-normalized, each transfer map has a nonzero PSD fixed point by Theorem 8.8. Irreducibility upgrades these fixed points to positive definite ones by Theorem 8.3. Theorem 8.10 therefore gauges A and B separately to unital Kraus families. In this unital gauge, the embedded modulus-one eigenvector saturates the Kadison–Schwarz inequality, and the same equality / multiplicative-domain argument as in Theorem 9.16 produces an invertible intertwiner between the two Kraus families. That intertwiner is exactly the required gauge-phase equivalence. □

11.8 Steps toward canonical form existence

This section combines the components of the canonical-form construction following [CPGSV17a, §2.3 and Appendix A]. The approach here requires blocking to remove non-trivial periodicity. An alternative framework that handles periodic blocks without blocking is developed in [DlCCSPG17]; see Chapter 14 for a discussion of that extension.

Theorem 11.33 (Irreducible block decomposition)**.** *Every MPS tensor A generates the same MPV family as a block-diagonal tensor $\bigoplus_{k=1}^{r} A_k$ with each block irreducible.*

Proof. Apply Theorem 11.15. □

Theorem 11.34 (Left-canonical normalization for irreducible TP blocks)**.** *Let A be an irreducible MPS tensor already in TP gauge ($\sum_i (A^i)^\dagger A^i = \mathbb{1}$) with $D > 0$. Then there exist a unitary U and a diagonal positive definite matrix Λ such that:*

1. *$B^i := U^\dagger A^i U$ is TP,*
2. *$\mathcal{E}_B(\Lambda) = \Lambda$,*
3. *Λ is diagonal and positive definite,*
4. *A and B generate the same MPV family.*

This is the left-canonical normalization step (Canonical Form II) of [CPGSV17a, Appendix A]. The unitary conjugation occurs only after TP normalization has already been achieved; it is a second step inside TP gauge, not a unitary normalization of an arbitrary irreducible block.

Proof. Apply Theorem 11.16 to obtain the unitary and the diagonal PD fixed point. Unitary conjugation preserves trace-preservation and the MPV family (Theorem 11.9). □

Theorem 11.35 (Irreducible block decomposition with left-canonical normalization)**.** *For any MPS tensor A, there exist irreducible blocks $(A_k)_{k=1}^r$ that generate the same MPV family as A. Moreover, for each block k, if that block has already been placed in TP gauge and $D_k > 0$, one obtains a left-canonical normalization (unitary, diagonal PD fixed point, MPV preservation). This yields the left-canonical normalization; further reduction to normal canonical form is carried out in Chapter 9.*

Proof. Combine Theorems 11.33 and 11.34. □

Theorem 11.36 (Normal canonical form from a primitive block decomposition)**.** *Suppose A generates the same MPV family as a weighted block-diagonal tensor $\bigoplus_k \mu_k A_k$, and assume that:*

1. *each A_k is irreducible,*
2. *each A_k satisfies the TP normalization $\sum_i (A_k^i)^\dagger A_k^i = \mathbb{1}$,*
3. *each transfer map $\mathcal{E}_{A_k}$ is primitive (equivalently, has peripheral spectrum $\{1\}$),*
4. *the moduli $|\mu_k|$ are pairwise distinct,*
5. *all $\mu_k \neq 0$,*
6. *all bond dimensions are positive.*

Then there exist a blocking length $p > 0$, weights (ν_k), and blocked tensors (B_k) such that $A^{[p]}$ generates the same MPV family as $\bigoplus_k \nu_k B_k$, and (ν_k, B_k) satisfies the normal canonical form conditions.

Proof. No new blocking is needed here: the input blocks are already primitive, so one may take $p = 1$. The theorem is stated with a blocking length because earlier reduction steps may require a common blocking period; under these hypotheses that period can be chosen to be 1. Since the moduli $|\mu_k|$ are pairwise distinct, there is a reordering of the blocks for which they become strictly decreasing. After this reindexing, set $\nu_k := \mu_k$ and $B_k := A_k$ in the new order. Irreducibility, TP normalization, primitivity, nonzero weights, and positive bond dimensions are unchanged by the reordering, so the reordered family satisfies Definition 11.30 and generates the same MPV family as A. □

11.9 Zero-block separation

In the irreducible block decomposition, some blocks may have all-zero Kraus operators. Such blocks contribute to the MPV only at word length $N = 0$ (where their contribution equals their bond dimension). The following results separate the zero blocks from the "live" (nonzero) blocks.

Definition 11.37 (Zero MPS tensor)**.** The *zero MPS tensor* of physical dimension d and bond dimension D, denoted $\mathbf{0}_{d,D}$, is the tensor whose Kraus operators are all zero.

Theorem 11.38 (MPV of zero tensor)**.** *For any spin configuration σ of length N,*

$$V^{(N)}(\mathbf{0}_{d,D})_\sigma = \begin{cases} D & \text{if } N = 0, \\ 0 & \text{if } N \geq 1. \end{cases}$$

Proof. For $N = 0$, $(\mathbf{0}_{d,D})^\emptyset = \mathbb{1}_D$, so the MPV equals $\operatorname{tr}(\mathbb{1}_D) = D$. For $N \geq 1$, every word product of a zero tensor is zero. □

Theorem 11.39 (Zero-block separation)**.** *Every MPS tensor A admits a block decomposition into:*

1. a trivial block *of bond dimension D_0 with all Kraus operators zero (accumulating all all-zero irreducible blocks), and*
2. *a finite family of* nontrivial blocks $(B_k)_{k=1}^{r}$*, each irreducible, each with at least one nonzero Kraus operator, and each with positive bond dimension,*

such that

$$V^{(N)}(A)_\sigma = V^{(N)}(\mathbf{0}_{d,D_0})_\sigma + V^{(N)}(\textstyle\bigoplus_{k=1}^{r} B_k)_\sigma$$

for every spin configuration σ of length N. At $N = 0$ this reads $D = D_0 + \sum_{k=1}^{r} D_k$. For $N \geq 1$ the zero-tail term vanishes.

Proof. Apply the irreducible block decomposition (Theorem 11.15) and partition the blocks by whether each has a nonzero Kraus operator. The all-zero blocks satisfy a dimension bound via irreducibility; their total bond dimension is D_0. The MPV identity follows from the block structure. $\square$

11.10 Arbitrary-input TP-gauge reduction

Theorem 11.40 (Arbitrary-input TP-gauge reduction with zero-block separation)**.** *For any MPS tensor A, there exist:*

1. *a zero-tail bond dimension D_0,*
2. *nontrivial blocks $(B_k)_{k=1}^{r}$ with nonzero weights $(\mu_k)_{k=1}^{r}$, each block irreducible and left-canonical ($\sum_i (B_k^i)^\dagger B_k^i = \mathbb{1}$), with positive bond dimension,*

such that

$$V^{(N)}(A)_\sigma = V^{(N)}(\mathbf{0}_{d,D_0})_\sigma + V^{(N)}(\textstyle\bigoplus_{k=1}^{r} \mu_k B_k)_\sigma.$$

This is the furthest unconditional arbitrary-input reduction step available before periodicity removal.

Proof. Apply Theorem 11.39 to separate the trivial block and nontrivial blocks. Then apply the blockwise TP-gauge theorem to the nontrivial blocks. The MPV identity chains through both steps. $\square$

11.11 Blocking and primitivity reduction

Theorem 11.41 (Blocking distributes over MPV equivalence)**.** *If A generates the same MPV family as $\bigoplus_k \mu_k B_k$, then for any $p \geq 0$,*

$$A^{[p]} \sim \bigoplus_k \mu_k^p B_k^{[p]}.$$

Proof. The identity $V^{(N)}(A^{[p]})_\sigma = V^{(Np)}(A)_{\sigma_{\text{flat}}}$ holds for any tensor by definition of blocking. The MPV equivalence of A and $\bigoplus_k \mu_k B_k$ at length Np then gives the equivalence at length N with weights μ_k^p. $\square$

Theorem 11.42 (Primitive channels remain primitive under powers)**.** *If $\mathcal{E}$ has peripheral spectrum $\{1\}$ (is primitive) and $m \geq 1$, then $\mathcal{E}^m$ is also primitive.*

Proof. The peripheral spectrum of $\mathcal{E}^m$ is the set of m-th powers of that of $\mathcal{E}$. Since $\{1\}^m = \{1\}$, the result follows. $\square$

Theorem 11.43 (Blocking-period divisibility preserves primitivity)**.** *If the transfer map of $A^{[p]}$ is primitive, $p \mid q$, and $q \geq 1$, then the transfer map of $A^{[q]}$ is also primitive.*

Proof. Write $q = p \cdot m$. Then $\mathcal{E}_{A^{[q]}} = \mathcal{E}^m_{A^{[p]}}$. Apply Theorem 11.42 with $\mathcal{E} = \mathcal{E}_{A^{[p]}}$. $\square$

Theorem 11.44 (Iterated blocking multiplies periods at the transfer-map level)**.** *For any MPS tensor A and integers $m, n \geq 0$,*

$$\mathcal{E}_{(A^{[m]})^{[n]}} = \mathcal{E}_{A^{[mn]}}.$$

Proof. Rewrite the outer blocked transfer map as the n-th power of $\mathcal{E}_{A^{[m]}}$, then rewrite $\mathcal{E}_{A^{[m]}}$ as $\mathcal{E}^m_A$. This gives $(\mathcal{E}^m_A)^n$, which simplifies to $\mathcal{E}^{mn}_A$. Finally rewrite $\mathcal{E}^{mn}_A$ back as the transfer map of $A^{[mn]}$. $\square$

Theorem 11.45 (Common blocking period for a block family)**.** *Let $(B_k)_{k=1}^r$ be a finite family of MPS tensors with positive bond dimensions, each admitting a blocking period p_k making its transfer map primitive. Then $p = \operatorname{lcm}(p_1, \dots, p_r)$ is a common period: $\mathcal{E}_{B_k^{[p]}}$ is primitive for every k.*

Proof. Each $p_k \mid p$, so Theorem 11.43 applies. $\square$

11.12 Cyclic sector decomposition

Theorem 11.46 (Compression to a supported projection sector)**.** *Let A be an MPS tensor satisfying $\sum_i (A^i)^\dagger A^i = P$ for an orthogonal projection P, and suppose $PA^iP = A^i$ for every i. Then there exists a left-canonical tensor C of bond dimension $n = \operatorname{tr}(P)$ with*

$$V^{(N)}(C)_\sigma = \operatorname{tr}(P \cdot A^\sigma)$$

for every σ.

Proof. Diagonalize P via a unitary U (spectral theorem). The eigenvalues are in $\{0, 1\}$. Set $n = \operatorname{tr} P$; let C^i be the top-left $n \times n$ block of $U^\dagger A^i U$ (restricted to eigenvalue-1 indices). TP follows from the projection normalization; the MPV identity follows from trace cyclicity. $\square$

Corollary 11.47 (Compression preserves positive-length MPV coefficients)**.** *Under the hypotheses of Theorem 11.46, the compressed tensor C additionally satisfies $V^{(N)}(C)_\sigma = V^{(N)}(A)_\sigma$ for all positive chain lengths $N > 0$.*

Proof. For $N > 0$ the word $A^{i_1} \cdots A^{i_N}$ is nonempty, so P acts as the identity on the left by idempotence and the support hypothesis. The MPV identity then follows from trace cyclicity via Theorem 11.46. $\square$

Theorem 11.48 (Block decomposition from commuting projections)**.** *Let A be a left-canonical MPS tensor, and let $(P_k)_{k=1}^m$ be orthogonal projections summing to $\mathbb{1}$ that commute with every Kraus operator: $P_k A^i = A^i P_k$ for all i, k. Then there exist left-canonical blocks $(C_k)_{k=1}^m$ with unit weights such that A generates the same MPV family as $\bigoplus_{k=1}^m C_k$. Moreover, each sector satisfies the per-sector trace relation $V^{(N)}(C_k)(\sigma) = \operatorname{tr}(P_k \cdot A^{i_1} \cdots A^{i_N})$ for every word $\sigma = (i_1, \dots, i_N)$.*

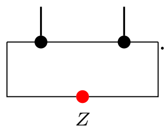
Proof. For each k, the tensor $P_k A^i$ is supported on P_k (using commutativity and $P_k^2 = P_k$). The TP condition $\sum_i (P_k A^i)^\dagger (P_k A^i) = P_k$ follows from $\sum_i (A^i)^\dagger A^i = \mathbb{1}$. Apply Theorem 11.46 to each sector. The MPV identity uses $\sum_k P_k = \mathbb{1}$ and linearity of trace. □

Theorem 11.49 (Block decomposition from adjoint-fixed projections)**.** *Let A be a left-canonical MPS tensor, and let $(P_k)_{k=1}^m$ be orthogonal projections summing to $\mathbb{1}$ that are fixed by the adjoint transfer map: $\mathcal{E}_A^\dagger(P_k) = P_k$ for every k. Then the same block decomposition conclusion holds as in Theorem 11.48, including the per-sector trace relation.*

Proof. The Kadison–Schwarz equality condition for a TP map (Theorem 7.23) implies that a projection fixed by $\mathcal{E}_A^\dagger$ commutes with every Kraus operator. Apply Theorem 11.48. □

Theorem 11.50 (Cyclic sector decomposition for irreducible TP tensors)**.** *Let A be a left-canonical MPS tensor with irreducible transfer map. Then there exist $m \geq 1$ and left-canonical blocks $(C_k)_{k=1}^m$ such that the m-blocked tensor $A^{[m]}$ decomposes as $\bigoplus_{k=1}^m C_k$ (with unit weights).*

Proof. Derive the conjugate-transposed Kraus family $K_i = (A^i)^\dagger$, its unitality and irreducibility, and a positive-definite fixed point ρ of the transfer map. Apply the cyclic structure theorem (peripheral eigenvalues form a cyclic group of order m) to obtain m cyclic spectral projections:



Here the operator Z is inserted on the closing virtual bond, while the order-m periodicity is part of the accompanying algebraic statement. Apply Theorem 11.49. □

11.12.1 Sector irreducibility helpers

Lemma 11.51 (Pairwise orthogonality from projection sum)**.** *Let $(P_k)_{k=1}^m$ be orthogonal projections summing to the identity: $\sum_k P_k = \mathbb{1}$. Then for $i \neq j$ we have $P_i P_j = 0$.*

Proof. Sandwich the sum between P_i to get $\sum_k P_i P_k P_i = P_i$. Since the i-th summand equals P_i by idempotence, all other summands vanish. Positivity ($BB^* = 0 \Rightarrow B = 0$) then gives $P_i P_k = 0$ for each $k \neq i$. □

Lemma 11.52 (Corner preservation from adjoint-fixed projection)**.** *Let A be a left-canonical MPS tensor and let P be an orthogonal projection fixed by $\mathcal{E}_A^\dagger$. Then $\mathcal{E}_A^\dagger$ preserves the corner algebra $P \cdot M_D(\mathbb{C}) \cdot P$.*

Proof. The fixed-point condition and the Kadison–Schwarz equality imply $[P, A_i] = 0$ for every Kraus operator. Thread the idempotent relation $P^2 = P$ through the corner sandwich. □

Lemma 11.53 (Orbit-sum fixed point)**.** *Let T be a linear endomorphism and Q a matrix satisfying $T^m(Q) = Q$. Then the orbit sum $\sum_{l=0}^{m-1} T^l(Q)$ is a fixed point of T.*

Proof. Telescoping: T shifts each summand $T^l(Q) \mapsto T^{l+1}(Q)$ and the periodicity condition $T^m(Q) = Q$ wraps the last term $T^{m-1}(Q) \mapsto T^m(Q) = Q$ back to the first. □

Lemma 11.54 (Orbit iterate supported on shifted sector)**.** *Let $(P_k)_{k=1}^m$ be orthogonal projections with cyclic shift $T(P_{k+1}) = P_k$, and suppose T is multiplicative on each sector corner. If Q is supported on sector P_k (i.e. $QP_k = Q = P_k Q$), then for each $0 \leq l < m$ the iterate $T^l(Q)$ is supported on the shifted sector P_{k-l}.*

Proof. Induction on l. The base case is the hypothesis $QP_k = Q = P_kQ$. For the inductive step, the multiplicative-domain identities and the cyclic shift relation $T(P_j) = P_{j-1}$ transfer the support from sector P_j to sector P_{j-1}. $\square$

Lemma 11.55 (Orbit iterate preserves orthogonal projection). *Under the same cyclic shift and multiplicative-domain hypotheses as Lemma 11.54, suppose additionally that T maps every orthogonal projection supported on a single sector to an orthogonal projection. If Q is an orthogonal projection supported on P_k, then $T^l(Q)$ is an orthogonal projection for each $0 \le l < m$.*

Proof. Induction on l. The base case is the hypothesis on Q. At each step, Lemma 11.54 guarantees that $T^n(Q)$ is supported on a single sector, so the one-step preservation hypothesis applies. $\square$

Lemma 11.56 (Orbit sum of a full sector projection). *Let $(P_k)_{k=1}^m$ be orthogonal projections with $\sum_k P_k = \mathbb{1}$ and cyclic shift $T(P_{k+1}) = P_k$. Then for every sector index k,*

$$\sum_{l=0}^{m-1} T^l(P_k) \;=\; \mathbb{1}.$$

Proof. By induction, $T^l(P_k) = P_{k-l}$ for each l. The orbit sum therefore runs over all distinct sectors, and the reindexing $l \mapsto k - l$ recovers $\sum_j P_j = \mathbb{1}$. $\square$

Theorem 11.57 (Sector irreducibility from orbit-sum lift). *Let A be a left-canonical MPS tensor with irreducible adjoint transfer map, and let $(P_k)_{k=1}^m$ be the cyclic sector projections. Suppose the orbit-sum lift hypothesis holds: for every sector k and every subprojection $Q \le P_k$ fixed by $(\mathcal{E}_A^\dagger)^m$, there exists a global projection R that is invariant under $\mathcal{E}_A^\dagger$ and faithfully reflects triviality ($Q = 0 \Leftrightarrow R = 0$, $Q = P_k \Leftrightarrow R = \mathbb{1}$). Then $(\mathcal{E}_A^\dagger)^m$ is irreducible on each corner P_k.*

Proof. Direct application of the cyclic-decomposition irreducibility theorem from spectral theory. $\square$

11.13 Reduction to primitive blocks

Theorem 11.58 (Arbitrary tensor to primitive block decomposition). *For any MPS tensor A, there exist a period $p \ge 1$, a zero-tail bond dimension D_0, nonzero weights $(\mu_k)_{k=1}^r$, and left-canonical blocks $(B_k)_{k=1}^r$ with primitive transfer maps and positive bond dimensions, such that*

$$V^{(N)}\big(A^{[p]}\big)_\sigma = V^{(N)}\big(\mathbf{0}_{d^{[p]},D_0}\big)_\sigma + V^{(N)}\big(\textstyle\bigoplus_k \mu_k B_k\big)_\sigma.$$

Proof. Apply Theorem 11.40 to get the trivial block and left-canonical nontrivial blocks. Apply Theorem 11.45 to find a common blocking period p making all transfer maps primitive. Apply Theorem 11.41. $\square$

Theorem 11.59 (TP-primitive irreducible tensor is normal). *Let A be a left-canonical MPS tensor with $D > 0$, irreducible transfer map, and peripheral spectrum $\{1\}$. Then A is normal.*

Proof. Peripheral primitivity and irreducibility yield the spectral-gap condition. The PSD fixed point is positive definite by Theorem 8.3. Apply Theorem 10.52. $\square$

Theorem 11.60 (Blocking preserves normality). *If A is normal and $p \ge 1$, then $A^{[p]}$ is normal.*

Proof. Since A is normal, there exists N with $S_N(A) = M_D(\mathbb{C})$. Then $S_{Np}(A) = M_D(\mathbb{C})$ (word products at multiples still span), and $S_N(A^{[p]}) \supseteq S_{Np}(A) = M_D(\mathbb{C})$. $\square$

11.13.1 Basis of Normal Tensors

Theorem 11.61 (Sorted reindexing by strictly decreasing weight norms)**.** *If nonzero weights* $(\mu_k)_{k=1}^r$ *have pairwise distinct norms, then there is a permutation of* $\{1, \dots, r\}$ *under which* $\|\mu_k\|$ *is strictly decreasing.*

Theorem 11.62 (Sorted decomposition preserves the represented MPS family)**.** *Under the same distinct-norm hypothesis, the sorted block family is gauge equivalent to the original decomposition and has strictly decreasing norms.*

Theorem 11.63 (Normal canonical form after sorting)**.** *If each block is irreducible, trace-preserving, primitive, and has positive dimension, then sorting by strictly decreasing weight norms produces a normal canonical form.*

Theorem 11.64 (Trivial sector decomposition for the sorted distinct-norm case)**.** *If weights are already strictly decreasing in norm, one obtains a sector decomposition with one copy per sector that preserves the represented MPS family.*

Definition 11.65 (Norm-class partition)**.** The grouping data consists of: norm classes, one representative value per class, class multiplicities, an enumeration of members in each class, and a regrouping identity recovering the full finite sum.

Definition 11.66 (Norm-class enumeration in decreasing order)**.** From any finite weight family, construct norm classes ordered by strictly decreasing norm values.

Theorem 11.67 (BNT grouping theorem)**.** *If equal-norm blocks have equal bond dimensions and represent the same MPS family, then one obtains a sector decomposition whose sector norms are strictly decreasing.*

11.13.2 Equal-Norm Blocks

Theorem 11.68 (Gauge equivalence from non-decaying cross-overlap)**.** *For trace-preserving irreducible blocks, a non-decaying cross-overlap forces equal bond dimensions and gauge equivalence up to a unit-modulus phase.*

Theorem 11.69 (BNT partition from gauge equivalence)**.** *If equal-norm blocks are related by gauge equivalence with unit-modulus phase, the phases can be absorbed into sector weights to obtain a BNT grouping with strictly decreasing sector norms.*

Theorem 11.70 (Sector decomposition from TP primitive irreducible blocks)**.** *From a trace-preserving primitive irreducible block decomposition with nonzero weights and non-decaying cross-overlaps for equal-norm blocks, one obtains a sector decomposition with strict norm ordering.*

11.13.3 Cyclic Sectors

Definition 11.71 (Left sector tensor)**.** For a projection P, the left sector tensor is obtained by left multiplication of each Kraus operator by P.

Lemma 11.72 (Commuting letters imply commuting words)**.** *If a projection commutes with every Kraus operator, it commutes with every finite word product.*

Lemma 11.73 (Left multiplication identity for sector words)**.** *If a projection commutes with all Kraus operators, then word evaluation for the left sector tensor equals left multiplication by the projection.*

Lemma 11.74 (Sector support)**.** *The left sector tensor is supported on the chosen projection.*

Theorem 11.75 (Adjoint-fixed projections commute with Kraus operators)**.** *If a projection is fixed by the adjoint transfer map of a trace-preserving tensor, then it commutes with each Kraus operator.*

Theorem 11.76 (Left-canonical primitive blocking from irreducibility)**.** *For a trace-preserving irreducible tensor, a positive blocking period yields a blocked tensor that is left-canonical and primitive.*

Theorem 11.77 (Normality from primitivity and a positive-definite fixed point)**.** *A primitive tensor admitting a positive-definite fixed point of its transfer map is normal.*

Theorem 11.78 (Zero-word evaluation for the zero tensor)**.** *Every nonempty finite word evaluates to zero when all Kraus operators are zero.*

Theorem 11.79 (Zero MPV for the zero tensor)**.** *For a nonempty spin chain, the MPS value of every spin word is zero for the zero tensor.*

Theorem 11.80 (Nonzero Kraus operator in irreducible dimension at least two)**.** *An irreducible tensor of bond dimension at least two has at least one nonzero Kraus operator.*

Theorem 11.81 (All-zero tensors are irreducible only in dimension at most one)**.** *If every Kraus operator vanishes and the tensor is irreducible, then the bond dimension is at most one.*

Theorem 11.82 (Irreducible block decomposition with trace-preserving gauge)**.** *After removing a trivial block, one obtains a block decomposition into irreducible blocks with a trace-preserving gauge.*

11.14 Open directions

Remark 11.83 (Remaining steps toward normal canonical form)*.* The arbitrary-input reduction (Theorem 11.40) produces a trivial block together with left-canonical nontrivial blocks. After a common blocking step (Theorem 11.58), the blocked blocks have primitive transfer maps.

To conclude full normal canonical form, two further ingredients are still needed: tensor irreducibility for each blocked block (blocking does not in general preserve tensor irreducibility), and pairwise distinct weight norms. The cyclic-sector decomposition (Theorems 11.48 and 11.49) provides the infrastructure for the periodic-to-aperiodic splitting.

Chapter 12

Block Permutation and Separation

This chapter establishes the algebraic structure of automorphisms of products of matrix algebras and uses it to prove that the same-MPV condition on block-diagonal tensors descends to per-block gauge equivalence. The results correspond to the canonical form uniqueness of [PGVWC07] and the block separation results of [CPGSV21]. The proof technique used here—induction on blocks via the spectral gap—is the approach adopted in this chapter.

12.1 Block permutation algebra

Theorem 12.1 (Ring isomorphisms permute block ideals)**.** *Let $\{R_k\}_{k=1}^r$ be simple rings. Every ring isomorphism $T : \prod_k R_k \to \prod_k R_k$ permutes the block ideals: there exists a permutation π such that T maps the k-th block ideal to the $\pi(k)$-th block ideal.*

Proof. Each block ideal $I_k = \{x : x_j = 0 \text{ for } j \neq k\}$ is an atom in the lattice of two-sided ideals (since R_k is simple). A ring isomorphism preserves the ideal lattice and hence permutes the atoms. Injectivity of the induced map on atoms gives the permutation π. □

Remark 12.2 (Ring level versus algebra level). Theorem 12.1 is a ring-level statement: it only uses the ideal structure of $\prod_k R_k$. Theorems 12.3 and 12.4 require an upgrade to $\mathbb{C}$-algebra automorphisms. The reason is threefold: dimension preservation compares the blocks as $\mathbb{C}$-vector spaces; Skolem–Noether applies to the resulting central simple $\mathbb{C}$-algebras; and the per-block maps extracted from T must commute with scalar multiplication. Thus the argument separates the ring-level permutation step from the later algebra-level decomposition.

Theorem 12.3 (Dimension preservation)**.** *Assume $D_k \geq 1$ for all k. If T is an algebra automorphism of $\prod_{k=1}^r M_{D_k}(\mathbb{C})$, and π is the induced permutation, then $D_{\pi(k)} = D_k$ for all k.*

Proof. The restriction of T to the k-th block ideal induces a $\mathbb{C}$-algebra isomorphism $M_{D_k}(\mathbb{C}) \cong M_{D_{\pi(k)}}(\mathbb{C})$, which forces $D_k = D_{\pi(k)}$ (by comparing dimensions as $\mathbb{C}$-vector spaces). □

Theorem 12.4 (Decomposition of automorphisms of $\prod M_{D_k}(\mathbb{C})$)**.** *Assume $D_k \geq 1$ for all k. Every $\mathbb{C}$-algebra automorphism T of $\prod_{k=1}^r M_{D_k}(\mathbb{C})$ decomposes as a block permutation π followed by per-block inner automorphisms: there exist $\pi \in S_r$ with $D_{\pi(k)} = D_k$ and invertible matrices*

$X_k \in \mathrm{GL}_{D_k}(\mathbb{C})$ such that, for every tuple $M = (M_1, \dots, M_r)$, $T(M)_k = X_k M_{\pi(k)} X_k^{-1}$, where $T(M)_k$ denotes the k-th block component of $T(M)$.

Proof. Theorem 12.1 gives the permutation π. Theorem 12.3 gives $D_{\pi(k)} = D_k$. The restriction of T to the k-th block ideal yields a $\mathbb{C}$-algebra isomorphism $M_{D_k}(\mathbb{C}) \cong M_{D_{\pi(k)}}(\mathbb{C})$; after Theorem 12.3 identifies the two matrix sizes, Skolem–Noether (Theorem 4.9) makes this restriction inner.

This is a standard result in the theory of semisimple algebras; see [CPGSV21, §IV.A] for the MPS context. □

12.2 Product algebra linear extension

Definition 12.5 (Per-block linear extension)**.** Given injective tensors A_k, B_k of bond dimension D_k with the same MPV for each block k, the *per-block linear extension* T_k is the unique linear map $M_{D_k}(\mathbb{C}) \to M_{D_k}(\mathbb{C})$ satisfying $T_k(A_k^i) = B_k^i$ for all i. It exists (and is unique) because $\{A_k^i\}$ spans $M_{D_k}(\mathbb{C})$ by injectivity.

Definition 12.6 (Product algebra automorphism)**.** The *product algebra automorphism* is the blockwise assembled map $T : \prod_k M_{D_k}(\mathbb{C}) \to \prod_k M_{D_k}(\mathbb{C})$ defined by $T(M)_k := T_k(M_k)$. Theorem 12.7 shows that each T_k is a $\mathbb{C}$-algebra homomorphism and that this assembled map is indeed a $\mathbb{C}$-algebra automorphism.

Theorem 12.7 (Per-block linear extensions assemble)**.** *Let A_k and B_k be injective tensors of bond dimension D_k for $k = 1, \dots, r$, and suppose the per-block same-MPV condition holds. Then the per-block linear extensions $T_k : A_k^i \mapsto B_k^i$ assemble into an algebra automorphism of $\prod_k M_{D_k}(\mathbb{C})$, which decomposes as a permutation π with $D_{\pi(k)} = D_k$ and per-block conjugation matrices X_k.*

Proof. Each T_k exists uniquely and is multiplicative by Theorems 4.6 and 4.7. Promotion to algebra homomorphisms (Lemma 4.10) and composition into the product gives an algebra automorphism T. Theorem 12.4 decomposes T into a permutation plus per-block inner automorphisms. Because T is assembled blockwise, it preserves each block ideal, so in this theorem the permutation is actually $\pi = \mathrm{id}$. A nontrivial block permutation only appears after block separation identifies which blocks match globally. □

Theorem 12.8 (Per-block same MPV gives global gauge equivalence)**.** *If all block tensors A_k are injective and per-block same-MPV holds, then each pair (A_k, B_k) is gauge equivalent, and the block-diagonal tensors $\bigoplus_k \mu_k A_k$ and $\bigoplus_k \mu_k B_k$ are globally gauge equivalent.*

Proof. Per-block gauge equivalence follows from the single-block FT (Theorem 4.11) applied to each block. Global gauge equivalence follows by assembling the per-block gauge matrices into a block-diagonal matrix (Theorem 11.4). □

12.3 Block separation

Recall from Definition 2.34 and Theorem 9.7 that the bilinear overlap is

$$O_{AB}(N) = \sum_{\sigma \in \{0,\dots,d-1\}^N} V^{(N)}(A)_\sigma \overline{V^{(N)}(B)_\sigma} = \mathrm{Tr}(F_{AB}^N).$$

We use this notation throughout the block-separation argument.

Definition 12.9 (Canonical form conditions)**.** A collection of scaling factors $(\mu_k)_{k=1}^r$ and block tensors $(A_k)_{k=1}^r$ satisfies the *canonical form conditions* if:

1. each A_k is injective,
2. each A_k satisfies the TP normalization $\sum_i (A_k^i)^\dagger A_k^i = \mathbb{1}$,
3. the moduli are strictly decreasing: $|\mu_1| > |\mu_2| > \cdots > |\mu_r|$,
4. all $\mu_k \neq 0$,
5. the self-overlap converges: $O_{A_k A_k}(N) \to 1$ for each k.

Only this one-sided normalization is assumed; no unital condition is assumed here. Condition (5) is a primitivity hypothesis: for a primitive channel ([Wol12, Thm. 6.7, item 3]), $T^n(\rho) \to \rho_\infty$ for every initial state, which forces the self-overlap to converge to 1. See also [Wol12, Thm. 6.8] for the completely-positive characterisation and Chapter 9 for the spectral-gap proof.

Remark 12.10 (Normalization convention)*.* The convention here differs from [PGVWC07, Thm. 4], which uses the unital gauge $\sum_i A_k^i (A_k^i)^\dagger = \mathbb{1}$. This chapter instead works in the dual TP gauge $\sum_i (A_k^i)^\dagger A_k^i = \mathbb{1}$. A positive-definite fixed point of the adjoint transfer map provides a gauge transformation between these two normalizations (Theorem 8.11).

Remark 12.11*.* In the literature [CPGSV21], condition (5) is derived from primitivity of the transfer map (peripheral spectrum $= \{1\}$). Theorem 11.26 gives this implication: per-block primitivity implies overlap convergence, so condition (5) is redundant when primitivity holds. The canonical form conditions retain condition (5) explicitly because it can be verified independently of primitivity.

Remark 12.12 (Normal-canonical companion)*.* Definition 12.9 gives the block-injective conditions. The normal-canonical analogue instead uses Definition 11.30: injectivity and the explicit self-overlap hypothesis are replaced by irreducibility, TP normalization, and primitive transfer maps. Theorem 11.31 then supplies the self-overlap limit automatically.

Theorem 12.13 (Vandermonde separation)**.** *If $\mu_1, \dots, \mu_r$ are pairwise distinct complex numbers and $\sum_{k=1}^r c_k \mu_k^N = 0$ for $N = 0, 1, \dots, r-1$, then $c_k = 0$ for all k.*

Proof. The coefficient vector c satisfies $Vc = 0$ where V is the Vandermonde matrix of the μ_k. Injectivity of μ makes V invertible. ☐

Remark 12.14*.* Theorem 12.13 is not used in the proofs given here. It is retained for potential alternative arguments; the proof of Theorem 12.5 uses BNT linear independence (Chapter 11) rather than Newton–Girard identities to match the weight multiset.

Lemma 12.15 (Block separation core)**.** *Let $((\mu_k), (A_k))$ satisfy the canonical form conditions, and let (B_k) be injective tensors satisfying the same TP normalization $\sum_i (B_k^i)^\dagger B_k^i = \mathbb{1}$. If $\sum_k \mu_k^N (V^{(N)}(A_k)_\sigma - V^{(N)}(B_k)_\sigma) = 0$ for all N, σ, then each pair (A_k, B_k) generates the same MPV family.*

Proof. By induction on the number of blocks r. In the inductive step, take overlaps with the leading block A_0; after division by μ_0^N, the contributions from $k \geq 1$ decay because $|\mu_k/\mu_0| < 1$ (strict weight ordering) and the bilinear overlaps $O_{A_0 A_k}(N)$ and $O_{A_0 B_k}(N)$ are bounded (Theorems 9.20 and 9.22 supply the relevant bounds), leaving $O_{A_0 A_0}(N) - O_{A_0 B_0}(N) \to 0$. Condition (5) gives $O_{A_0 A_0}(N) \to 1$, hence $O_{A_0 B_0}(N) \to 1$. Since $O_{A_0 B_0}(N) \to 1 \neq 0$, the bond dimensions must match: $D_{A_0} = D_{B_0}$; otherwise Theorem 9.20 would give $O_{A_0 B_0}(N) \to 0$, a contradiction. With equal bond dimensions established, the contrapositive of Theorem 9.22 forces A_0 and B_0 to be gauge-phase equivalent; the limit $O_{A_0B_0}(N) \to 1$ then makes the phase trivial, so A_0 and B_0 generate the same MPV family. Peeling off the leading block and dividing by μ_0^N, the induction hypothesis applies. □

Theorem 12.16 (Block separation from canonical form)**.** *Let $((\mu_k), (A_k))$ satisfy the canonical form conditions, and let (B_k) be injective tensors satisfying the same TP normalization $\sum_i (B_k^i)^\dagger B_k^i = \mathbb{1}$. If the block-diagonal tensors $\bigoplus_k \mu_k A_k$ and $\bigoplus_k \mu_k B_k$ generate the same MPV family, then each pair (A_k, B_k) generates the same MPV family.*

Proof. Apply Lemma 12.15 to the identity obtained from the equality of the two block-diagonal MPV families. The induction peels off the dominant block and treats the remainder as a decaying tail. Concretely, one isolates the leading term $\mu_1^N V(A_1)$ and bounds the remaining contribution by $O(|\mu_2/\mu_1|^N)$ after dividing by μ_1^N. The canonical form conditions supply injectivity, TP normalization, strict nonzero weights, and self-overlap convergence for (A_k), while the hypotheses on (B_k) supply the comparison injectivity and TP normalization. □

Theorem 12.17 (Block separation from normal canonical form)**.** *Let $((\mu_k), (A_k))$ satisfy the normal canonical form conditions, and let (B_k) be irreducible tensors of the same bond dimensions, satisfying the same TP normalization $\sum_i (B_k^i)^\dagger B_k^i = \mathbb{1}$. If the block-diagonal tensors $\bigoplus_k \mu_k A_k$ and $\bigoplus_k \mu_k B_k$ generate the same MPV family, then each pair (A_k, B_k) generates the same MPV family.*

Proof. The proof follows the same peeling argument as Theorem 12.16, with Theorem 11.32 (modulus-one eigenvalue rigidity for irreducible tensors) replacing the contrapositive of Theorem 9.22 at the identification step, since the blocks are irreducible rather than injective. Theorem 11.31 replaces the explicit self-overlap hypothesis on the leading block. Since both block families already use the same dimension function, the conclusion is direct per-block same MPV. □

Theorem 12.18 (Canonical form fundamental theorem, same-structure version)**.** *Under the hypotheses of Theorem 12.16, each pair (A_k, B_k) is gauge equivalent. Hence the block-diagonal tensors $\bigoplus_k \mu_k A_k$ and $\bigoplus_k \mu_k B_k$ are gauge equivalent.*

Proof. Theorem 12.16 gives per-block same MPV. Theorem 12.8 then converts this to per-block gauge equivalence and then to global gauge equivalence of the block-diagonal tensors. □

Remark 12.19 (Relation to the literature). The results of this chapter correspond to the canonical-form uniqueness of [PGVWC07] and the block-separation statements of [CPGSV21]. The proof technique used here—an induction on the number of blocks driven by the spectral gap of the mixed transfer operators—is the method used here.

Chapter 13

Basis of Normal Tensors

This chapter introduces the *basis of normal tensors* (BNT) notion, the canonical-form variants with BNT separation, the cross-overlap decay theorem, and the permutation-rigidity arguments used in the proof of the Fundamental Theorem. The chapter concludes with auxiliary results on coefficient convergence and Newton–Girard identities. The main references are [PGVWC07] and [CPGSV21].

13.1 Basis of normal tensors

Definition 13.1 (Basis of Normal Tensors (BNT))**.** A *basis of normal tensors* (BNT) for a total tensor A_{tot} consists of g block tensors $(A_1, \dots, A_g)$ with bond dimensions $(D_1, \dots, D_g)$ such that:

1. each A_j is normal (Definition 2.22),
2. at each system size N, the MPV of A_{tot} lies in the span of the block MPVs $\{|V^{(N)}(A_j)\rangle\}_{j=1}^{g}$,
3. for sufficiently large N, the block MPV vectors $\{|V^{(N)}(A_j)\rangle\}_{j=1}^{g}$ are linearly independent.

This is [CPGSV21, Definition 4.2].

Definition 13.2 (Canonical form with BNT separation)**.** A family is in *canonical form with BNT separation* if it satisfies Definition 12.9 and, in addition, distinct blocks of the same bond dimension are *not* gauge-phase equivalent.

Definition 13.3 (Normal canonical form with BNT separation)**.** A family is in *normal canonical form with BNT separation* if it satisfies Definition 11.30 and, in addition, distinct blocks of the same bond dimension are not gauge-phase equivalent.

Theorem 13.4 (Cross-overlap decay for distinct blocks in canonical form with BNT separation)**.** *If $((\mu_k), (A_k))$ is in canonical form with BNT separation and $j \neq k$, then $O_{A_j A_k}(N) \to 0$.*

Proof. Definition 13.2 supplies injectivity and TP normalization for each block, which are the hypotheses required by the spectral-gap theorems. If $D_j \neq D_k$, the rectangular overlap-decay theorem (Theorem 9.20) gives $O_{A_j A_k}(N) \to 0$. If $D_j = D_k$, the separation condition says that A_j and A_k are not gauge-phase equivalent, so the same-dimension overlap-decay theorem (Theorem 9.22) again gives decay to 0. $\square$

Theorem 13.5 (Canonical form with BNT separation yields a basis of normal tensors)**.** *If $((\mu_k), (A_k))$ is in canonical form with BNT separation, then the block tensors form a basis of normal tensors for the block-diagonal tensor $\bigoplus_k \mu_k A_k$.*

Proof. Each block is injective, so it is normal by taking blocking length 1. The MPV decomposition (Theorem 2.27) gives

$$|V^{(N)}(A_{\rm tot})\rangle = \sum_k \mu_k^N |V^{(N)}(A_k)\rangle,$$

so the MPV of $A_{\rm tot}$ lies in the span of the block MPVs at each system size. For eventual linear independence, consider the Gram matrix

$$G_{jk}(N) = \langle V^{(N)}(A_j)|V^{(N)}(A_k)\rangle.$$

By Lemma 2.35, $G_{jk}(N) = \overline{O_{A_j A_k}(N)}$. The self-overlap $O_{A_j A_j}(N) \to 1$ is real-valued under TP normalization, hence $G_{jj}(N) \to \overline{1} = 1$. For $j \neq k$, Theorem 13.4 gives $O_{A_j A_k}(N) \to 0$, so $G_{jk}(N) \to \overline{0} = 0$. Thus $G(N) \to I_g$, hence the Gram matrix is eventually invertible. Therefore the vectors $\{|V^{(N)}(A_j)\rangle\}_{j=1}^{g}$ are linearly independent for all sufficiently large N. □

Remark 13.6 (The normality gap for normal canonical form with BNT separation)*.* Definition 13.3 references the normal canonical form conditions (Definition 11.30), which encodes normality via the spectral characterization (primitive transfer map). Definition 13.1 requires normality via the algebraic characterization (eventual block injectivity, Definition 2.22). These are equivalent by Remark 2.20, but the equivalence must be invoked explicitly: the Wielandt theory of Chapter 10 provides the implication from primitivity to eventual block injectivity needed here.

13.2 Permutation rigidity for bases of normal tensors

Theorem 13.7 (Permutation rigidity for proportional MPV decompositions)**.** *Let $(A_j)_{j=1}^{g_A}$ and $(B_k)_{k=1}^{g_B}$ be injective families satisfying the TP normalization $\sum_i (A_j^i)^\dagger A_j^i = \mathbb{1}$ and $\sum_i (B_k^i)^\dagger B_k^i = \mathbb{1}$, whose self-overlaps converge to 1 and whose cross-overlaps within each family converge to 0. Assume moreover that there exist MPS tensors $A_{\rm tot}$ and $B_{\rm tot}$, coefficient arrays $a_{N,j}$ and $b_{N,k}$, and a sequence c_N with nonzero limits a_j^∞, b_k^∞, c^∞ such that for every N,*

$$|V^{(N)}(A_{\rm tot})\rangle = \sum_{j=1}^{g_A} a_{N,j}|V^{(N)}(A_j)\rangle, \qquad |V^{(N)}(B_{\rm tot})\rangle = \sum_{k=1}^{g_B} b_{N,k}|V^{(N)}(B_k)\rangle,$$

with $a_{N,j} \to a_j^\infty \neq 0$ and $b_{N,k} \to b_k^\infty \neq 0$, and also

$$|V^{(N)}(A_{\rm tot})\rangle = c_N|V^{(N)}(B_{\rm tot})\rangle$$

with $c_N \to c^\infty \neq 0$. Then $g_A = g_B$, and there is a permutation π of the common block index set such that $A_{\pi(k)}$ and B_k have the same bond dimension and are gauge-phase equivalent for each k. The canonical-form application of these hypotheses is described in Remark 13.8.

Proof. Substitute the two decomposition formulas into the proportionality relation and take overlaps with the block MPVs. Passing to the limit using the convergence of $a_{N,j}$, $b_{N,k}$, and c_N produces a limiting mixed-overlap matrix. The asymptotically orthonormal overlap hypotheses within each family force every row and column of that matrix to contain at most one nonzero entry, hence determine a permutation π. If one of the matched mixed overlaps came from blocks of different bond dimensions, Theorem 9.20 would force it to decay to 0; if the bond dimensions agree but the tensors are not gauge-phase equivalent, Theorem 9.22 does the same. Therefore every nonzero matched entry comes from blocks of the same bond dimension that are gauge-phase equivalent.

This is the overlap-and-decomposition form of the permutation step corresponding to [CPGSV21, Theorem 4.1]. □

Remark 13.8. In the canonical-form application, the abstract data above come from block-diagonal assemblies

$$A_{\text{tot}} = \bigoplus_{j=1}^{g_A} \mu_j A_j, \qquad B_{\text{tot}} = \bigoplus_{k=1}^{g_B} \nu_k B_k,$$

and Theorem 2.27 gives the coefficient arrays $a_{N,j} = \mu_j^N$ and $b_{N,k} = \nu_k^N$. To fit the convergence hypotheses of Theorem 13.7, one first normalizes by the dominant weight on each side: the relevant arrays are $(\mu_j/\mu_0)^N$ and $(\nu_k/\nu_0)^N$, with $(\mu_j/\mu_0)^N \to 0$ for $j \geq 1$ and $(\nu_k/\nu_0)^N \to 0$ for $k \geq 1$ by Theorem 13.10, while the overall factors μ_0^N and ν_0^N are absorbed into the proportionality constant c_N. Definition 13.2 supplies injectivity, TP normalization, and the within-family overlap limits, so the theorem isolates the abstract permutation argument from this canonical-form packaging.

Theorem 13.9 (Permutation rigidity with asymptotically orthonormal overlaps)**.** *Let $(A_j)_{j=1}^{g_A}$ and $(B_k)_{k=1}^{g_B}$ be two families of injective tensors satisfying the TP normalization $\sum_i (A_j^i)^\dagger A_j^i = \mathbb{1}$ and $\sum_i (B_k^i)^\dagger B_k^i = \mathbb{1}$, whose self-overlaps converge to 1 and whose cross-overlaps within each family converge to 0. If for every system size N,*

$$\operatorname{span}_{\mathbb{C}}\{|V^{(N)}(A_j)\rangle : 1 \leq j \leq g_A\} = \operatorname{span}_{\mathbb{C}}\{|V^{(N)}(B_k)\rangle : 1 \leq k \leq g_B\},$$

then $g_A = g_B$, and there is a permutation π of the common block index set such that A_j and $B_{\pi(j)}$ have the same bond dimension and are gauge-phase equivalent for each j.

Proof. For each family, the Gram matrix converges to the identity by Lemma 2.35 together with the self-overlap and off-diagonal decay hypotheses. Hence for all sufficiently large N both families of MPV vectors are linearly independent. At such an N, the equality of spans in the statement forces the two common subspaces to have the same dimension, so $g_A = g_B$. The span equality then gives change-of-basis matrices between the two families, and the same mixed-overlap matching argument shows that the mixed-overlap matrix has exactly one nonzero entry in each row and column. This produces the permutation π, while Theorems 9.20 and 9.22 upgrade each nonzero match to equality of bond dimensions and gauge-phase equivalence. □

13.3 Coefficient convergence

Theorem 13.10 (Coefficient ratio decay)**.** *Under the canonical form conditions of Definition 12.9, indexed so that μ_0 is the dominant weight, one has $(\mu_k/\mu_0)^N \to 0$ for every $k \geq 1$ as $N \to \infty$.*

Proof. Since $|\mu_k| < |\mu_0|$ (strict ordering of moduli), $|\mu_k/\mu_0| < 1$, so the geometric sequence converges to 0. □

Remark 13.11 (Strict-dominance regime). Theorem 13.10 applies when one modulus is strictly dominant: $|\mu_0| > |\mu_k|$ for $k \geq 1$. In the more general periodic setting of [CPGSV21, Eq. (72a)], the coefficient attached to block j can take the form $\sum_q \mu_{j,q}^N$, where the $\mu_{j,q}$ are the individual weights contributed by that block. The oscillatory case with $|\mu_{j,q}| = 1$ is treated separately in the full paper.

13.4 Newton–Girard identities

Theorem 13.12 (Newton–Girard trace recursion)**.** *If $A, B \in M_n(\mathbb{C})$ satisfy $\operatorname{tr}(A^k) = \operatorname{tr}(B^k)$ for all $k \geq 1$, then A and B have the same characteristic polynomial.*

Proof. Let e_m denote the mth elementary symmetric polynomial in the eigenvalues, with $e_0 = 1$. The Newton–Girard recursion

$$me_m = \sum_{i=1}^{m} (-1)^{i-1} e_{m-i} \operatorname{tr}(A^i)$$

expresses the coefficients of the characteristic polynomial in terms of the power-sum traces $\operatorname{tr}(A^i)$. Equal power sums for A and B therefore give equal coefficients, hence equal characteristic polynomials. □

Theorem 13.13 (Equal power sums imply equal multisets)**.** *Let $\alpha, \beta : \{1, \dots, n\} \to \mathbb{C}$. If $\sum_i \alpha_i^k = \sum_i \beta_i^k$ for all $k \geq 1$, then the multisets $\{\alpha_i\}$ and $\{\beta_i\}$ are equal.*

Proof. Construct diagonal matrices $\operatorname{diag}(\alpha)$ and $\operatorname{diag}(\beta)$. The power-sum hypothesis gives $\operatorname{tr}(\operatorname{diag}(\alpha)^k) = \operatorname{tr}(\operatorname{diag}(\beta)^k)$. By Theorem 13.12, the characteristic polynomials agree: $\prod_i (X - \alpha_i) = \prod_i (X - \beta_i)$. Extracting roots gives multiset equality. □

Remark 13.14. Theorems 13.12 and 13.13 remain relevant for the general equal-MPV Theorem 14.5, where one expects a power-sum step to recover the multiplicity-weight data. The present explicit-coefficient special case, Theorem 14.6, instead uses linear independence of the block MPV family supplied by a basis of normal tensors.

Chapter 14

Proof of the Fundamental Theorem

This chapter proves the Fundamental Theorem of Matrix Product States. We state the Fundamental Theorem in two versions: one for block-injective tensors in canonical form with BNT separation, and one for the normal-canonical variant. We then treat the equal-MPV case, including the common-block-structure specialization that deduces gauge equivalence. The results combine the canonical form reduction (Chapter 11), block separation (Chapter 12), spectral gap (Chapter 9), and the BNT permutation arguments of Chapter 13.

The theorems in Sections 14.1–14.3 follow the approach of [CPGSV17a]: they first block the tensor to remove non-trivial periodicity, then apply the aperiodic fundamental theorem. Section 14.4 outlines the stronger "irreducible form" framework of [DlCCSPG17], which gives a fundamental theorem valid for periodic tensors directly, with an extra Z-gauge degree of freedom.

14.1 Weak Fundamental Theorem

The following theorem combines the reduction to canonical form (zero-block separation, TP gauge, blocking, primitivity) with the Fundamental Theorem for normal canonical form. It is conditional on the extra hypotheses (tensor irreducibility and distinct weight norms).

Theorem 14.1 (Weak Fundamental Theorem (conditional))**.** *Let $(\mu_j^A, A_j)_{j=1}^{g_A}$ and $(\mu_k^B, B_k)_{k=1}^{g_B}$ be two families in normal canonical form, each satisfying the non-degeneracy condition that no two distinct blocks in the same family are gauge-phase equivalent. Let A_{tot} and B_{tot} be tensors with MPV decompositions*

$$V^{(N)}(A_{\mathrm{tot}})_\sigma = \sum_j a_{N,j}\, V^{(N)}(A_j)_\sigma, \quad V^{(N)}(B_{\mathrm{tot}})_\sigma = \sum_k b_{N,k}\, V^{(N)}(B_k)_\sigma,$$

with $a_{N,j} \to a_j \neq 0$ and $b_{N,k} \to b_k \neq 0$. Suppose moreover that $V^{(N)}(A_{\mathrm{tot}})_\sigma = c_N\, V^{(N)}(B_{\mathrm{tot}})_\sigma$ with $c_N \to c \neq 0$. Then $g_A = g_B$ and there is a permutation π such that $D_j^A = D_{\pi(j)}^B$ and A_j is gauge-phase equivalent to $B_{\pi(j)}$ for every j.

Proof. This is exactly the normal-canonical-form Fundamental Theorem (Theorem 14.4). The normal canonical form data provide the BNT separation and overlap convergence hypotheses; the non-degeneracy condition ensures the block matching is well-defined. □

14.2 Proportional Fundamental Theorem

Theorem 14.2 (Fundamental Theorem — proportional MPV case)**.** *Let A_{tot} and B_{tot} be MPS tensors in canonical form with BNT separation, with associated families $(\mu_j^A, A_j)_{j=1}^{g_A}$ and $(\mu_k^B, B_k)_{k=1}^{g_B}$. Suppose there exist coefficient arrays $a_{N,j}$, $b_{N,k}$ and nonzero limits a_j^∞, b_k^∞, c^∞ such that*

$$V^{(N)}(A_{\mathrm{tot}})_\sigma = \sum_{j=1}^{g_A} a_{N,j} V^{(N)}(A_j)_\sigma, \qquad V^{(N)}(B_{\mathrm{tot}})_\sigma = \sum_{k=1}^{g_B} b_{N,k} V^{(N)}(B_k)_\sigma,$$

with $a_{N,j} \to a_j^\infty \neq 0$ and $b_{N,k} \to b_k^\infty \neq 0$, and suppose also that

$$V^{(N)}(A_{\mathrm{tot}})_\sigma = c_N V^{(N)}(B_{\mathrm{tot}})_\sigma$$

for a sequence $c_N \to c^\infty \neq 0$. Then $g_A = g_B$, and there is a permutation π such that $D_j^A = D_{\pi(j)}^B$ and A_j is gauge-phase equivalent to $B_{\pi(j)}$ for each j.

Proof. Apply Theorem 13.7 to the two canonical-form decompositions with BNT separation. This yields the permutation π that reindexes the separated sectors before comparing them one by one. These hypotheses provide injectivity, TP normalization, diagonal overlap limits, and off-diagonal decay via Theorem 13.4. With the supplied decomposition identities and convergence hypotheses, Theorem 13.7 yields the equality of block counts, the permutation, and the per-block gauge-phase equivalence. □

Remark 14.3 (Coefficient arrays for canonical-form decompositions). In applications the coefficient arrays in Theorem 14.2 come from the block-diagonal decomposition itself. If

$$A_{\mathrm{tot}} = \bigoplus_{j=1}^{g_A} \mu_j^A A_j, \qquad B_{\mathrm{tot}} = \bigoplus_{k=1}^{g_B} \mu_k^B B_k,$$

then Theorem 2.27 gives

$$V^{(N)}(A_{\mathrm{tot}})_\sigma = \sum_{j=1}^{g_A} (\mu_j^A)^N V^{(N)}(A_j)_\sigma, \qquad V^{(N)}(B_{\mathrm{tot}})_\sigma = \sum_{k=1}^{g_B} (\mu_k^B)^N V^{(N)}(B_k)_\sigma.$$

Thus one takes $a_{N,j} = (\mu_j^A)^N$ and $b_{N,k} = (\mu_k^B)^N$, while the sequence c_N is the global proportionality factor from the hypothesis $V^{(N)}(A_{\mathrm{tot}}) = c_N V^{(N)}(B_{\mathrm{tot}})$. Theorem 14.2 keeps these arrays explicit because Theorem 13.7 is formulated in that abstract form.

Theorem 14.4 (Fundamental Theorem for normal canonical form in the proportional case)**.** *Let A_{tot} and B_{tot} be MPS tensors in normal canonical form with BNT separation, with associated families $(\mu_j^A, A_j)_{j=1}^{g_A}$ and $(\mu_k^B, B_k)_{k=1}^{g_B}$. Suppose there exist coefficient arrays $a_{N,j}$, $b_{N,k}$ and nonzero limits a_j^∞, b_k^∞, c^∞ such that*

$$V^{(N)}(A_{\mathrm{tot}})_\sigma = \sum_{j=1}^{g_A} a_{N,j} V^{(N)}(A_j)_\sigma, \qquad V^{(N)}(B_{\mathrm{tot}})_\sigma = \sum_{k=1}^{g_B} b_{N,k} V^{(N)}(B_k)_\sigma,$$

with $a_{N,j} \to a_j^\infty \neq 0$ and $b_{N,k} \to b_k^\infty \neq 0$, and suppose also that

$$V^{(N)}(A_{\mathrm{tot}})_\sigma = c_N V^{(N)}(B_{\mathrm{tot}})_\sigma$$

for a sequence $c_N \to c^\infty \neq 0$. Then $g_A = g_B$, and there is a permutation π such that $D_j^A = D_{\pi(j)}^B$ and A_j is gauge-phase equivalent to $B_{\pi(j)}$ for each j.

Proof. Normal canonical form with BNT separation provides irreducibility, TP normalization, and primitive self-overlap convergence via Theorem 11.31. Theorem 11.32 supplies the leading-block identification step: a nondecaying mixed overlap forces the corresponding comparison block to be gauge-phase equivalent. The same peeling argument as in the block-injective case therefore yields the block permutation and gauge-phase equivalence. Remark 13.6 explains why the argument stops at this separated normal-canonical level. □

14.3 Equal-MPV Fundamental Theorem

Theorem 14.5 (Fundamental Theorem — equal MPV case)**.** *Let A_{tot} and B_{tot} be MPS tensors in canonical form with BNT separation, with associated families $(\mu_j^A, A_j)_{j=1}^{g_A}$ and $(\mu_k^B, B_k)_{k=1}^{g_B}$. If*

$$V^{(N)}(A_{\text{tot}})_\sigma = V^{(N)}(B_{\text{tot}})_\sigma$$

for every N and σ, then the block-diagonal tensors $\bigoplus_{j=1}^{g_A} \mu_j^A A_j$ and $\bigoplus_{k=1}^{g_B} \mu_k^B B_k$ are gauge equivalent.

Proof. Begin by applying Theorem 14.2 with $c_N = 1$, which yields a permutation of the matched blocks and blockwise gauge-phase equivalences. One then compares the resulting MPV expansions from Theorem 2.27. In the general equal case the weights occur with multiplicities, so recovering the individual weights requires a power-sum argument such as Theorem 13.13. After the weights are identified, the phase factors are absorbed into them and the blockwise conjugacies combine to a global gauge equivalence. □

Theorem 14.6 (Fundamental Theorem — equal MPV case with explicit coefficients)**.** *Let A_{tot} and B_{tot} be MPS tensors in canonical form with BNT separation, with associated families $(\mu_j^A, A_j)_{j=1}^{g_A}$ and $(\mu_k^B, B_k)_{k=1}^{g_B}$. Suppose there exist coefficient arrays $a_{N,j}$, $b_{N,k}$ and nonzero limits a_j^∞, b_k^∞ such that*

$$V^{(N)}(A_{\text{tot}})_\sigma = \sum_{j=1}^{g_A} a_{N,j} V^{(N)}(A_j)_\sigma, \qquad V^{(N)}(B_{\text{tot}})_\sigma = \sum_{k=1}^{g_B} b_{N,k} V^{(N)}(B_k)_\sigma,$$

with $a_{N,j} \to a_j^\infty \neq 0$ and $b_{N,k} \to b_k^\infty \neq 0$. If

$$V^{(N)}(A_{\text{tot}})_\sigma = V^{(N)}(B_{\text{tot}})_\sigma$$

for every N and σ, then $g_A = g_B$, and there is a permutation π such that $D_j^A = D_{\pi(j)}^B$ and, after reindexing the B-blocks by π, the weighted block-diagonal tensors $\bigoplus_{j=1}^{g_A} \mu_j^A A_j$ and $\bigoplus_{j=1}^{g_A} \mu_{\pi(j)}^B B_{\pi(j)}$ are gauge equivalent.

Proof. Apply Theorem 14.2 with $c_N = 1$. This yields $g_A = g_B$, a permutation π, equal block dimensions, and for each j a gauge-phase equivalence

$$B_{\pi(j)}^i = \zeta_j X_j A_j^i X_j^{-1}.$$

By Theorem 2.27,

$$V^{(N)}(A_{\text{tot}})_\sigma = \sum_{j=1}^{g_A} (\mu_j^A)^N V^{(N)}(A_j)_\sigma,$$

and similarly for B_{tot}. Reindex the B-sum by π and use the gauge-phase identity to rewrite

$$V^{(N)}(B_{\pi(j)})_\sigma = \zeta_j^N V^{(N)}(A_j)_\sigma.$$

Then equality of MPVs implies

$$\sum_{j=1}^{g_A} \left((\mu_j^A)^N - (\mu_{\pi(j)}^B \zeta_j)^N\right) V^{(N)}(A_j)_\sigma = 0.$$

By Theorem 13.5 and Definition 13.1, the family $\{V^{(N)}(A_j)\}_{j=1}^{g_A}$ is linearly independent for all sufficiently large N, so

$$(\mu_j^A)^N = (\mu_{\pi(j)}^B \zeta_j)^N$$

for every j and all large N. Comparing two consecutive values of N and using that the weights are nonzero gives

$$\mu_j^A = \mu_{\pi(j)}^B \zeta_j.$$

Thus the phase factors can be absorbed into the weights, and the resulting blockwise conjugacies assemble to a gauge equivalence of the reindexed weighted block-diagonal tensors. □

Theorem 14.7 (Fundamental Theorem — equal MPV case, common block structure)**.** *Fix a common block structure: the same number of blocks g, the same bond dimensions $(D_k)_{k=1}^g$, and the same weights $(\mu_k)_{k=1}^g$. Let $(\mu_k, A_k)_{k=1}^g$ and $(\mu_k, B_k)_{k=1}^g$ be two families in canonical form with BNT separation on this common data. If the block-diagonal tensors $\bigoplus_k \mu_k A_k$ and $\bigoplus_k \mu_k B_k$ generate the same MPV family, then each pair (A_k, B_k) is gauge equivalent, and hence the corresponding block-diagonal tensors are gauge equivalent.*

Proof. Since both families share the canonical form data, the BNT separation condition is not needed. The canonical-form fundamental theorem (Theorem 12.18) then applies directly to this common (g, D_k, μ_k) structure, yielding per-block gauge equivalence and hence global gauge equivalence. □

14.3.1 Multiplicity structure for repeated blocks

This subsection records the multiplicity data in which repeated copies of the same basis block are kept separate rather than merged into a single weighted block.

Definition 14.8 (Sector data and sector decomposition)**.** A *sector datum* over g basis blocks specifies: multiplicities $r_j > 0$, sector weights $\mu_{j,q} \neq 0$ for $q \in \{1, \dots, r_j\}$, and the *sector coefficient* $\mathrm{coeff}(N, j) := \sum_{q=1}^{r_j} \mu_{j,q}^N$. A *sector decomposition* consists of the basis blocks $(A_j)_{j=1}^g$ together with their sector data.

Theorem 14.9 (Sector decomposition formula)**.** *For a sector decomposition P, the MPV of the associated tensor satisfies*

$$V^{(N)}(P.\mathrm{toTensor})_\sigma = \sum_{j=1}^{g} \mathrm{coeff}(N, j) \cdot V^{(N)}(A_j)_\sigma.$$

Theorem 14.10 (Eventual coefficient comparison from BNT linear independence)**.** *If two sector data S, T over the same basis produce equal total MPVs and the basis is eventually linearly independent (the BNT property), then $S.\mathrm{coeff}(N, j) = T.\mathrm{coeff}(N, j)$ for all sufficiently large N.*

Theorem 14.11 (Geometric sequence extrapolation)**.** *If two finite families of nonzero complex numbers have equal power sums* $\sum_i \alpha_i^k = \sum_j \beta_j^k$ *for all* $k > N_0$*, then equality holds for all* $k \geq 0$*. The proof uses induction on the number of terms with a telescoping argument: subtract a geometric factor to reduce the number of terms, apply the induction hypothesis, and conclude via a geometric recurrence.*

Theorem 14.12 (Weight multiset recovery from multiplicity data)**.** *Let* S *and* T *be two sector data over the same basis with the same multiplicities* ($r_j^S = r_j^T$ *for all* j)*. If the basis is eventually linearly independent and the total MPVs of* S *and* T *agree, then for each basis block* j *the sector weight multisets* $\{\mu_{j,q}^S\}$ *and* $\{\mu_{j,q}^T\}$ *are equal.*

The argument proceeds as follows:

1. *BNT LI + equal MPVs* $\Rightarrow$ *eventual coefficient equality (Theorem 14.10);*
2. *Geometric extrapolation* $\Rightarrow$ *full coefficient equality for all positive* k *(Theorem 14.11);*
3. *Newton–Girard* $\Rightarrow$ *equal weight multisets (Theorem 13.13).*

At the level of block contributions, one compares the weighted sector sums $\sum_j \operatorname{coeff}_A(N,j)\, V^{(N)}(A_j)$ *and* $\sum_j \operatorname{coeff}_B(N,j)\, V^{(N)}(B_j)$*, with the coefficients now aggregated over the multiplicity layers.*

Remark 14.13 (Remaining step in the unconditional equal-case theorem)*.* This chapter provides three complementary equal-case results:

1. Theorem 14.6: the explicit-coefficient equal-case theorem (upgrade of Theorem 14.2 with $c_N = 1$), concluding a global gauge equivalence of the reindexed weighted block-diagonal tensors.
2. Theorem 14.7: the common block-structure special case.
3. Theorem 14.12: the equal-case multiplicity corollary, which from the richer multiplicity data recovers the sector weight multisets without requiring explicit coefficient convergence hypotheses.

The unconditional equal-case theorem (Theorem 14.5, [CPGSV21, Corollary IV.5]) still requires one final global gauge-construction step: combining the weight-recovery result above with (a) the proportional FT for block matching and gauge-phase equivalences, (b) phase absorption into weights, and (c) global gauge construction via permutation and cast transport.

14.4 Periodic Fundamental Theorem (de las Cuevas–Cirac–Schuch–Pérez-García)

The theorems in the preceding sections all require that each block has a *primitive* transfer map, i.e. peripheral spectrum exactly $\{1\}$. This is achieved by blocking the original tensor by a common period. The paper [DlCCSPG17] develops a framework that avoids this blocking step, giving a fundamental theorem that applies directly to periodic tensors.

14.4.1 Irreducible form

Definition 14.14 (Periodic tensor)**.** An irreducible MPS tensor A of bond dimension D is m-*periodic* (for some $m \geq 1$) if the peripheral spectrum of its transfer map $\mathcal{E}_A$ is exactly the m-th roots of unity $\{e^{2\pi i k/m} : 0 \leq k < m\}$. A 1-periodic tensor is exactly an irreducible tensor with primitive transfer map (Definition 5.37).

Definition 14.15 (Irreducible form)**.** A tensor A is in *irreducible form* if it is block-diagonal

$$A^i = \bigoplus_{j \in J} \mu_j \, A^i_j,$$

where each block A_j is an m_j-periodic tensor, each weight $\mu_j \neq 0$, and the blocks are pairwise non-repeated (no two distinct j, j' have A_j gauge-equivalent to $A_{j'}$). This is the standard form introduced in [DlCCSPG17, §II]. It extends the normal canonical form (Definition 11.30), which requires the stronger condition that every block is 1-periodic.

Remark 14.16 (Relation to normal canonical form). Every tensor in normal canonical form (Definition 11.30) is in irreducible form with all periods equal to 1. The irreducible form is strictly more general: it accommodates blocks such as the antiferromagnet, which are 2-periodic.

The arguments in this chapter work exclusively with period-1 blocks, obtained by a blocking step. A direct treatment of the irreducible form of [DlCCSPG17] would require:

1. a definition of m-periodic irreducible tensors and their cyclic-sector decomposition (see below),
2. the off-diagonal structure of a periodic block's transfer matrix (the cyclic permutation of sectors), and
3. the notion of equivalence between periodic blocks that accounts for the Z-gauge freedom described below.

14.4.2 Common-period blocking lemmas

Definition 14.17 (LCM period for a finite block family)**.** Given a finite family of blocking periods (p_i) indexed by a finite set, their least common multiple serves as the shared blocking period.

Definition 14.18 (Common-period blocked family)**.** Given a finite family of tensors (B_i) with designated periods (p_i), block each B_i to the shared period $\mathrm{lcm}(p_i)$, so every block lives in the same physical space.

Theorem 14.19 (Common-period blocking preserves transfer-map primitivity)**.** *If each tensor B_i has a primitive transfer map after blocking by its own positive period p_i, then blocking the entire family to the common LCM period preserves primitivity of each member's transfer map.*

Proof. Each p_i divides $\mathrm{lcm}(p_i)$. Apply the divisibility monotonicity theorem for primitive blocked transfer maps to each family member separately. □

14.4.3 The Z-gauge degree of freedom

Definition 14.20 (Entrywise periodic Z-gauge ratio)**.** Given complex weights μ, ν, define the entrywise Z-gauge ratio by

$$z(\mu, \nu) := \mu/\nu.$$

Lemma 14.21 (Matched powers imply root-of-unity ratio)**.** *If $\mu^m = \nu^m$ and $\nu \neq 0$, then*

$$z(\mu, \nu)^m = 1.$$

Lemma 14.22 (Ratio rescales denominator back to numerator)**.** *If $\nu \neq 0$, then*

$$z(\mu, \nu) \, \nu = \mu.$$

Definition 14.23 (Diagonal periodic Z-gauge matrix)**.** For matched weight families $(\mu_i)_i$ and $(\nu_i)_i$ on a finite index type, define the diagonal matrix

$$Z := \operatorname{diag}\big(z(\mu_i, \nu_i)\big)_i.$$

Lemma 14.24 (Diagonal Z-gauge has finite order under matched powers)**.** *If $\mu_i^m = \nu_i^m$ and $\nu_i \neq 0$ for all i, then*

$$Z^m = \mathbb{1}.$$

Lemma 14.25 (Diagonal Z-gauge transports diagonal weights)**.** *Under $\nu_i \neq 0$ for all i,*

$$Z \operatorname{diag}(\nu) = \operatorname{diag}(\mu).$$

Definition 14.26 (Z-gauge equivalence)**.** Let A be an m-periodic irreducible tensor of bond dimension D. A *Z-gauge transformation* of A is a diagonal unitary matrix $Z \in M_D(\mathbb{C})$ satisfying:

1. Z commutes with every Kraus operator: $[A^i, Z] = 0$ for all physical indices i, and
2. $Z^m = \mathbb{1}_D$ (all diagonal entries are m-th roots of unity).

Replacing A by ZA (i.e. $A^i \mapsto ZA^i$) leaves the MPV family invariant for lengths N that are multiples of m, because the factor $\operatorname{tr}(Z^N) = \operatorname{tr}(\mathbb{1}) = D$ for $N \equiv 0 \pmod m$. Two tensors A, B are *Z-gauge equivalent* if there exists an invertible Y and a Z-gauge transformation Z for A such that $ZA^i = YB^iY^{-1}$ for all i.

Remark 14.27 (Z-gauge is trivial for period-1 blocks). When $m = 1$ the condition $Z^1 = \mathbb{1}$ forces $Z = \mathbb{1}$, so Z-gauge equivalence reduces to ordinary gauge equivalence (Definition 2.10). The Z-gauge is therefore a genuine new degree of freedom that appears only for $m > 1$.

14.4.4 Statement of the periodic Fundamental Theorem

The following is Theorem 3.8 ("equal case") of [DlCCSPG17].

Theorem 14.28 (Periodic Fundamental Theorem)**.** *Let A and B be MPS tensors in irreducible form, with bases of periodic blocks $\{A_j\}_{j \in J}$ and $\{B_k\}_{k \in K}$ respectively. Suppose that $\mathcal{V}(A) = \mathcal{V}(B)$ (equal MPV families for all lengths N).*

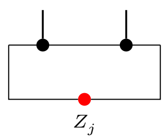
Then $|J| = |K|$ and there is a bijection $\pi : J \to K$ such that A_j and $B_{\pi(j)}$ are m_j-periodic for the same period m_j. Moreover, for each $j \in J$ there exists:

1. *an invertible matrix Y_j of size $D_j \times D_{\pi(j)}$, and*
2. *a diagonal unitary matrix Z_j of size $D_j \times D_j$ satisfying $Z_j^{m_j} = \mathbb{1}$ and $[A_j^i, Z_j] = 0$ for all i,*

such that

$$Z_j\, A_j^i = Y_j\, B_{\pi(j)}^i\, Y_j^{-1} \quad \textit{for all physical indices } i.$$

For each periodic block, the extra freedom is encoded by



rather than by an ordinary aperiodic gauge alone: the operator Z_j sits on the closing virtual bond and satisfies $Z_j^{m_j} = \mathbb{1}$. Equivalently, setting $Y = \bigoplus_j Y_j \otimes \mathbb{1}_{r_j}$ and $Z = \bigoplus_j Z_j \otimes \mathbb{1}_{r_j}$, one has $ZA^i = YB^iY^{-1}$ for all i, with $[A^i, Z] = 0$ and $\mathcal{V}(A) = \mathcal{V}(ZA)$.

Remark. *The proof is conditional on the periodic-overlap dichotomy (Proposition 3.3 of [DlCCSPG17]), which remains to be proved.*

14.4.5 Periodic overlap dichotomy (Proposition 3.3)

Definition 14.29 (Cyclic sector decomposition)**.** A family of compressed sector tensors $(C_k)_{k=0}^{m-1}$ on corner bond spaces forms a *cyclic sector decomposition* of the blocked tensor $A^{(m)}$ if there exist orthogonal projections $P_0, \dots, P_{m-1}$ on the ambient bond space such that $\sum_k P_k = \mathbb{1}$, each P_k commutes with every blocked letter $P_k A_i^{(m)} = A_i^{(m)} P_k$, and $V^{(N)}(C_k)(\sigma) = \mathrm{tr}(P_k \cdot \mathrm{evalWord}(A^{(m)}, \sigma))$ for every word σ.

Theorem 14.30 (Self-overlap along period multiples)**.** *If A is m-periodic, then the self-overlap along multiples of m converges to m (the period, viewed as a scalar).*

Theorem 14.31 (Different periods imply asymptotic orthogonality)**.** *If A and B are periodic with different periods, then $\langle V_N(A), V_N(B)\rangle \to 0$ as $N \to \infty$.*

Proof. Split on whether the bond dimensions agree. If not, dimension mismatch gives decay. If $D_1 = D_2$, suppose for contradiction that A and B are gauge-phase equivalent; Perron–Frobenius eigenvalue uniqueness (Wolf Theorem 6.3) then forces equal periods, contradicting the hypothesis. Hence the tensors are not gauge-phase equivalent, and the overlap decays. □

Lemma 14.32 (Sector-block normality for periodic tensors)**.** *For a periodic tensor equipped with a cyclic sector decomposition, each nonzero blocked sector tensor is normal.*

Theorem 14.33 (No sector match implies asymptotic orthogonality)**.** *If two periodic tensors with the same period admit no gauge-phase matching sector pair after blocking, then their overlap tends to zero.*

Lemma 14.34 (Sector-match propagation under translation)**.** *A single gauge-phase match between blocked sectors propagates to all cyclic translates of that sector pair.*

Lemma 14.35 (Per-site proportionality from blocked sector match)**.** *If all aligned blocked sectors match up to gauge and phase, then the original tensors are related by repeated-block proportionality.*

Theorem 14.36 (Sector match yields repeated-block equivalence)**.** *For periodic tensors of the same period, existence of one matching sector pair implies repeated-block equivalence.*

Theorem 14.37 (Different bond dimensions imply asymptotic orthogonality)**.** *If two periodic tensors have different bond dimensions, then $\langle V_N(A), V_N(B)\rangle \to 0$ as $N \to \infty$.*

Theorem 14.38 (Periodic overlap dichotomy)**.** *Let A and B be periodic tensors. Then at least one of the following alternatives holds:*

1. *$\langle V_N(A), V_N(B)\rangle \to 0$ as $N \to \infty$;*
2. *A and B are equivalent up to repeated blocks.*

Theorem 14.39 (Eventual linear independence of a periodic basis)**.** *For a finite family of pairwise non-equivalent periodic tensors whose periods divide a common integer p, there is N_0 such that for every $N \geq N_0$ the vectors $\{V_{pN}(A_j)\}_j$ are linearly independent.*

Remark 14.40 (Comparison with the 1606.00608 approach)*.* Theorem 14.28 and the equal-MPV Fundamental Theorem (Theorem 14.5) both characterize the gauge freedom between two tensors with equal MPV families, but they do so at different levels:

1. *Blocking approach* (Theorem 14.5, following [CPGSV17a]): Block both A and B by a common period p so that all blocks of $A^{[p]}$ and $B^{[p]}$ are primitive (1-periodic). Apply the aperiodic fundamental theorem to the blocked tensors. The conclusion is that $A^{[p]}$ and $B^{[p]}$ are gauge equivalent. The Z-gauge freedom does not appear because blocking absorbs the periodicity.

2. *Periodic approach* (Theorem 14.28, following [DlCCSPG17]): Work directly with the periodic blocks of A and B. The conclusion includes an extra Z-gauge transformation reflecting the non-trivial phase ambiguity that arises for periodic blocks. The Z matrix satisfies $Z^m = \mathbb{1}$ and commutes with all Kraus operators; it is invisible to the MPV family because it only introduces m-th roots of unity in the trace.

In this chapter only the blocking approach is developed. A direct periodic approach would require the irreducible-form and Z-gauge theory outlined in Remark 14.16.

Remark 14.41 (Proportional periodic FT). There is also a proportional-case version of the periodic Fundamental Theorem (Theorem 3.4, "proportional case", in [DlCCSPG17]): if $\mathcal{V}(A) = \lambda_N \mathcal{V}(B)$ for scalars λ_N with $\lambda_N \to \lambda \neq 0$, then the bases of periodic tensors of A and B are equivalent (up to the same gauge and Z freedom as above). This extends Theorems 14.2 and 14.4 to the periodic setting.

Chapter 15

Quantum Dynamical Semigroups

This chapter develops the theory of one-parameter semigroups of quantum channels and characterizes their generators via the GKSL (Gorini–Kossakowski–Sudarshan–Lindblad) theorem, following [Wol12, Ch. 7].

Section 15.1 establishes definitions and the basic exponential form (Proposition 15.8). Section 15.2 gives the Duhamel formula and perturbation bound. Section 15.3 develops the generator theory and proves the GKSL theorem.

15.1 Dynamical semigroups and the exponential form

Definition 15.1 (Dynamical semigroup)**.** A family of linear maps $T : \mathbb{R} \to (M_D(\mathbb{C}) \to_\ell M_D(\mathbb{C}))$ is a *dynamical semigroup* if

1. (semigroup law) $T(t+s) = T(t) \circ T(s)$ for all $t, s \geq 0$; and
2. (initial condition) $T(0) = \mathrm{id}$.

This is [Wol12, Eq. (7.1)].

Definition 15.2 (Continuous dynamical semigroup)**.** A dynamical semigroup T is *norm-continuous* if the map $t \mapsto T(t)$ is continuous in the operator norm topology. In finite dimension this coincides with strong continuity.

Definition 15.3 (Exponential semigroup)**.** Given a generator $L \in \mathrm{End}_{\mathbb{C}}(M_D(\mathbb{C}))$, the *exponential semigroup* is

$$T_t := e^{tL} = \sum_{k=0}^{\infty} \frac{(tL)^k}{k!}.$$

Theorem 15.4 (Semigroup law for the exponential)**.** *For all $t, s \in \mathbb{R}$, $e^{(t+s)L} = e^{tL} \cdot e^{sL}$.*

Proof. Since $(t \cdot L)$ commutes with $(s \cdot L)$, this follows from the exponential addition formula for commuting elements in a Banach algebra. □

Theorem 15.5 (Continuity of the exponential semigroup)**.** *The map $t \mapsto e^{tL}$ is continuous in the operator norm topology.*

Proof. The exponential is analytic (convergence radius $= \infty$), hence continuous. □

Theorem 15.6 (Derivative of the exponential semigroup). *For all $t \in \mathbb{R}$,*

$$\frac{d}{dt} e^{tL} = e^{tL} \cdot L.$$

This is [Wol12, Eq. (7.2)].

Proof. Chain rule applied to the composition $\mathbb{R} \xrightarrow{t \mapsto t \cdot L} \mathrm{End}(M_D(\mathbb{C})) \xrightarrow{\exp} \mathrm{End}(M_D(\mathbb{C}))$. $\square$

Theorem 15.7 (Generator uniqueness). *If $e^{tL} = e^{tL'}$ for all $t \geq 0$, then $L = L'$.*

Proof. Both semigroups agree on $[0, \infty)$, so their derivatives within $[0, \infty)$ at $t = 0$ agree. Since $[0, \infty)$ has unique differentials at 0, $L = L'$. $\square$

Theorem 15.8 (Prop. 7.1: Every continuous semigroup is exponential). *Every norm-continuous dynamical semigroup T on $M_D(\mathbb{C})$ is of the form*

$$T_t = e^{tL}$$

for some generator $L \in \mathrm{End}_{\mathbb{C}}(M_D(\mathbb{C}))$ and every $t \geq 0$. This is [Wol12, Prop. 7.1].

Proof. Since $T_0 = \mathbb{1}$ and T is continuous, the integral $M_\varepsilon = \int_0^\varepsilon T_s \, ds$ is invertible for small $\varepsilon > 0$. Using the semigroup property, $T_t = M_\varepsilon^{-1}(M_{t+\varepsilon} - M_t)$ is differentiable for $t \geq 0$. ODE uniqueness then gives $T_t = e^{tL}$ for all $t \geq 0$. $\square$

15.2 Perturbation theory

Theorem 15.9 (Lem. 7.1: Duhamel formula). *Let $\Delta = L' - L$. For $t \geq 0$,*

$$e^{tL'} - e^{tL} = \int_0^t e^{(t-s)L} \Delta \, e^{sL'} \, ds.$$

This is [Wol12, Lem. 7.1].

Proof. Define $f(s) = e^{(t-s)L} e^{sL'}$. Then $f'(s) = e^{(t-s)L} \Delta \, e^{sL'}$ (Theorem 15.6), and $e^{tL'} - e^{tL} = f(t) - f(0) = \int_0^t f'(s) \, ds$. $\square$

Theorem 15.10 (Cor. 7.1: Perturbation bound). *For $t \geq 0$,*

$$\left\| e^{tL'} - e^{tL} \right\| \leq t \, \|\Delta\| \cdot \sup_{s \in [0,t]} \|e^{sL}\| \cdot \sup_{s \in [0,t]} \|e^{sL'}\|,$$

where $\Delta = L' - L$. This is [Wol12, Cor. 7.1].

Proof. Apply the triangle inequality and submultiplicativity of the operator norm to the Duhamel integral (Theorem 15.9). $\square$

Lemma 15.11 (Dyson-series summability from factorial majorant). *Assume a factorial majorant of Dyson–Phillips iterates:*

$$\|D_n(t)\| \leq M \, \frac{(t \, \|L' - L\| \, M)^n}{n!}.$$

Then the Dyson series is summable in operator norm.

Proof. This is the Weierstrass M-test step, using the scalar convergence of $\sum_n a^n/n!$ and lifting via norm-bounded summability. $\square$

Theorem 15.12 (Factorial norm bound for Dyson iterates)**.** *For $s \in [0,t]$ and $M = \sup_{u\in[0,t]} \|e^{uL}\|$,*

$$\|\widetilde{T}_s^{(n)}\| \leq M\,\frac{(s\,\|\Delta\|\,M)^n}{n!}.$$

Proof. Induction on n. The base case uses the semigroup supremum bound. The inductive step bounds the integrand pointwise by $M\|\Delta\| \cdot M(u\|\Delta\|M)^n/n!$ and integrates $\int_0^s u^n\,du = s^{n+1}/(n+1)$, recovering the factorial. $\square$

Corollary 15.13 (Dyson series summability)**.** *The Dyson–Phillips series $\sum_n \widetilde{T}_t^{(n)}$ converges in operator norm for every $t \geq 0$.*

Proof. Compose the factorial norm bound (Theorem 15.12) with the Weierstrass M-test step (Lemma 15.11). $\square$

15.3 GKSL/Lindblad generators

This section characterizes generators of semigroups of completely positive and trace-preserving maps. The central result is the GKSL theorem (Theorem 15.42), which shows that such generators have the standard Lindblad form.

15.3.1 Generator decomposition and conditional complete positivity

Definition 15.14 (Generator decomposition)**.** A *generator decomposition* consists of a completely positive map $\phi : M_d(\mathbb{C}) \to M_d(\mathbb{C})$ and a matrix $\kappa \in M_d(\mathbb{C})$, defining the linear map

$$L(\rho) = \phi(\rho) - \kappa\rho - \rho\kappa^\dagger.$$

This is [Wol12, Eq. (7.14)].

Definition 15.15 (Conditional complete positivity)**.** A linear map $L : M_d(\mathbb{C}) \to M_d(\mathbb{C})$ is *conditionally completely positive* (CCP) if it admits a generator decomposition, i.e. there exist a CP map ϕ and a matrix κ such that $L(\rho) = \phi(\rho) - \kappa\rho - \rho\kappa^\dagger$. This is condition 1 of [Wol12, Prop. 7.2].

Definition 15.16 (Trace-annihilating map)**.** A linear map $L : M_d(\mathbb{C}) \to M_d(\mathbb{C})$ is *trace-annihilating* if $\operatorname{tr}(L(\rho)) = 0$ for all $\rho \in M_d(\mathbb{C})$. This is the infinitesimal version of trace preservation.

Definition 15.17 (Trace constraint)**.** A generator decomposition (ϕ, κ) satisfies the *trace constraint* if $\phi^*(\mathbb{1}) = \kappa + \kappa^\dagger$, where ϕ^* is the Hilbert–Schmidt adjoint. This is the condition in [Wol12, Eq. (7.20)].

Theorem 15.18 (Trace constraint implies trace-annihilating)**.** *If (ϕ,κ) satisfies $\phi^*(\mathbb{1}) = \kappa + \kappa^\dagger$, then $L(\rho) = \phi(\rho) - \kappa\rho - \rho\kappa^\dagger$ is trace-annihilating.*

Proof. Let $\phi(\rho) = \sum_i K_i \rho K_i^\dagger$. By the cyclic property of trace, $\operatorname{tr}(L(\rho)) = \operatorname{tr}((\sum_i K_i^\dagger K_i)\rho) - \operatorname{tr}(\kappa\rho) - \operatorname{tr}(\kappa^\dagger\rho) = \operatorname{tr}((\kappa+\kappa^\dagger)\rho) - \operatorname{tr}(\kappa\rho) - \operatorname{tr}(\kappa^\dagger\rho) = 0$. $\square$

15.3.2 The Lindblad form

Definition 15.19 (Lindblad form)**.** A *Lindblad form* consists of a Hermitian matrix $H = H^\dagger$ (the Hamiltonian) and a family of matrices $\{L_j\}_{j=0}^{r-1}$ (the Lindblad operators), defining the linear map

$$L(\rho) = i[\rho, H] + \sum_j \Big(L_j \rho L_j^\dagger - \tfrac{1}{2}\{L_j^\dagger L_j, \rho\}_+ \Big),$$

where $[A, B] = AB - BA$ and $\{A, B\}_+ = AB + BA$. This is [Wol12, Eq. (7.21)].

Theorem 15.20 (Lindblad form is trace-annihilating)**.** *Every Lindblad form defines a trace-annihilating linear map.*

Proof. The Hamiltonian part satisfies $\mathrm{tr}(i[\rho, H]) = 0$ by the cyclic property. Each dissipator term satisfies $\mathrm{tr}(L_j \rho L_j^\dagger) = \mathrm{tr}(L_j^\dagger L_j \rho) = \mathrm{tr}(\rho L_j^\dagger L_j)$, so the three terms sum to zero. □

Theorem 15.21 (Lindblad form equals generator decomposition)**.** *A Lindblad form with Hamiltonian H and operators $\{L_j\}$ defines the same linear map as the generator decomposition (ϕ, κ) with $\phi(\rho) = \sum_j L_j \rho L_j^\dagger$ and $\kappa = iH + \frac{1}{2}\sum_j L_j^\dagger L_j$. This is [Wol12, Eq. (7.24)].*

Proof. Compute $\kappa^\dagger = -iH + \frac{1}{2}\sum_j L_j^\dagger L_j$ using $H^\dagger = H$ and $(A^\dagger A)^\dagger = A^\dagger A$. Setting $S = \sum_j L_j^\dagger L_j$, expand $\phi(\rho) - \kappa\rho - \rho\kappa^\dagger = \sum_j L_j \rho L_j^\dagger - iH\rho - \frac{1}{2}S\rho + i\rho H - \frac{1}{2}\rho S$, which is $i[\rho, H] + \sum_j (L_j \rho L_j^\dagger - \frac{1}{2}L_j^\dagger L_j \rho - \frac{1}{2}\rho L_j^\dagger L_j)$. □

Theorem 15.22 (Lindblad form is CCP)**.** *Every Lindblad form defines a conditionally completely positive map.*

Proof. By Theorem 15.21, the Lindblad form equals a generator decomposition with ϕ CP. □

Definition 15.23 (Commutator form of Lindblad equation)**.** The *commutator form* of the Lindblad equation writes the generator as

$$L(\rho) = i[\rho, H] + \tfrac{1}{2}\sum_j ([L_j,\, \rho L_j^\dagger] + [L_j\rho,\, L_j^\dagger]),$$

where $[A, B] = AB - BA$. This is [Wol12, Eq. (7.22)].

Theorem 15.24 (Commutator form equals standard Lindblad form)**.** *The commutator form (Definition 15.23) defines the same linear map as the standard Lindblad form (Definition 15.19).*

Proof. Expanding the commutator brackets $[L_j, \rho L_j^\dagger]$ and $[L_j\rho, L_j^\dagger]$ yields the standard dissipator terms. □

15.3.3 Prop. 7.2: Characterization of conditional complete positivity

Theorem 15.25 (Prop. 7.2, direction 1→2)**.** *If L is CCP and $P = \mathbb{1} - |\Omega\rangle\langle\Omega|$, then*

$$P((L \otimes \mathrm{id})(|\Omega\rangle\langle\Omega|))P \geq 0.$$

This is [Wol12, Prop. 7.2].

Proof. Write $L(\rho) = \phi(\rho) - \kappa\rho - \rho\kappa^\dagger$. The left- and right-multiplication terms are annihilated by P, so the projected Choi matrix reduces to the projected Choi matrix of ϕ, which is positive. □

Theorem 15.26 (Prop. 7.2, direction 2→1)**.** *If L is Hermiticity-preserving and $P((L\otimes\mathrm{id})(|\Omega\rangle\langle\Omega|))P \geq 0$ where $P = \mathbb{1} - |\Omega\rangle\langle\Omega|$, then L is CCP. This is [Wol12, Prop. 7.2].*

Proof. Decompose the Hermitian Choi matrix as $\tau_L = Q - |\psi\rangle\langle\Omega| - |\Omega\rangle\langle\psi|$ with $Q \geq 0$ supported on $|\Omega\rangle^{\perp}$. The Choi–Jamiolkowski isomorphism applied to Q gives the CP map ϕ, and $(\kappa\otimes\mathbb{1})|\Omega\rangle = |\psi\rangle$ defines κ. □

15.3.4 Prop. 7.3: CP semigroups and CCP generators

Theorem 15.27 (Prop. 7.3: CP semigroup implies CCP generator)**.** *If e^{tL} is CP for all $t \geq 0$, then L is CCP. This is [Wol12, Prop. 7.3].*

Proof. From $(e^{tL} \otimes \mathrm{id})(|\Omega\rangle\langle\Omega|) \geq 0$, expand at infinitesimal t and project onto P to get $P(L \otimes \mathrm{id})(|\Omega\rangle\langle\Omega|)P \geq 0$. Then Theorem 15.26 gives CCP. □

Theorem 15.28 (Prop. 7.3: CCP generator implies CP semigroup)**.** *If L is CCP, then e^{tL} is CP for all $t \geq 0$. This is [Wol12, Prop. 7.3].*

Proof. Approximate e^{tL} by the completely positive Euler steps

$$\rho \mapsto (\mathbb{1} - h\kappa)\rho(\mathbb{1} - h\kappa)^{\dagger} + h\phi(\rho), \qquad h = \frac{t}{n+1}.$$

Each step is CP, hence so are its powers. These powers converge in operator norm to e^{tL}, and the cone of CP maps is closed. □

Theorem 15.29 (Prop. 7.3: CP semigroup iff CCP generator)**.** *e^{tL} is CP for all $t \geq 0$ if and only if L is CCP.*

Proof. Combine Theorems 15.27 and 15.28. □

15.3.5 Prop. 7.4: Freedom in generator representation

Theorem 15.30 (Traceless Kraus operators exist)**.** *Assume $d > 0$. Given any Kraus operators $\{K_j\}$, there exist $c_j \in \mathbb{C}$ such that $K_j' = K_j + c_j\mathbb{1}$ is traceless. This is part of [Wol12, Prop. 7.4].*

Proof. Set $c_j = -\mathrm{tr}(K_j)/d$. □

Theorem 15.31 (Shift invariance of generators)**.** *If $K_j' = K_j + c_j\mathbb{1}$ and κ' is adjusted per [Wol12, Eq. (7.19)], then (ϕ', κ') and (ϕ, κ) define the same generator.*

Proof. Direct algebraic expansion. □

15.3.6 Prop. 7.4 item 2: Uniqueness of the traceless Lindblad form

Definition 15.32 (Traceless Kraus operators)**.** A Lindblad form has *traceless Kraus operators* if $\mathrm{tr}(L_j) = 0$ for every j.

Theorem 15.33 (Prop. 7.4, item 2: CP-part uniqueness)**.** *If two Lindblad forms F, F' induce the same generator and both have traceless Kraus operators, then their CP parts agree: $\phi = \phi'$. This is part of [Wol12, Prop. 7.4, item 2].*

Proof. Compare projected Choi matrices and use Choi injectivity. □

Theorem 15.34 (Prop. 7.4, item 2: drift uniqueness modulo phase)**.** *If two Lindblad forms F, F' induce the same generator and both have traceless Kraus operators, then their drift matrices differ by an imaginary scalar: $\kappa' = \kappa + i\lambda \mathbb{1}$ for some $\lambda \in \mathbb{R}$. This is part of [Wol12, Prop. 7.4, item 2].*

Proof. After identifying $\phi = \phi'$ (Theorem 15.33), the generator equality forces $\Delta\kappa \cdot \rho + \rho \cdot (\Delta\kappa)^\dagger = 0$ for all ρ. Setting $\rho = \mathbb{1}$ gives $\Delta\kappa + (\Delta\kappa)^\dagger = 0$ (skew-Hermiticity), which converts the equation to $[\Delta\kappa, \rho] = 0$ for all ρ. The scalar commutant lemma gives $\Delta\kappa = c \cdot \mathbb{1}$, and skew-Hermiticity forces $c = i\lambda$ for some $\lambda \in \mathbb{R}$. □

Theorem 15.35 (Prop. 7.4, item 2: combined uniqueness)**.** *If two Lindblad forms F, F' induce the same generator and both have traceless Kraus operators, then $\phi = \phi'$ and $\kappa' = \kappa + i\lambda \mathbb{1}$ for some $\lambda \in \mathbb{R}$.*

Proof. Combine Theorems 15.33 and 15.34. □

15.3.7 Trace-annihilation and trace preservation

Theorem 15.36 (Trace-annihilating generators yield trace-preserving semigroups)**.** *If L is trace-annihilating, then e^{tL} is trace-preserving for every $t \in \mathbb{R}$.*

Proof. For fixed ρ, the function $f(t) = \mathrm{tr}(e^{tL}(\rho))$ has derivative $f'(t) = \mathrm{tr}(L(e^{tL}(\rho))) = 0$, so f is constant and $\mathrm{tr}(e^{tL}(\rho)) = f(t) = f(0) = \mathrm{tr}(\rho)$. □

Theorem 15.37 (Trace-preserving semigroups have trace-annihilating generators)**.** *If e^{tL} is trace-preserving for every $t \geq 0$, then L is trace-annihilating.*

Proof. Differentiate the identity $\mathrm{tr}(e^{tL}(\rho)) = \mathrm{tr}(\rho)$ at $t = 0$ from the right. □

15.3.8 Thm. 7.1: The GKSL/Lindblad theorem

Definition 15.38 (GKSL generator)**.** A linear map $L : M_d(\mathbb{C}) \to M_d(\mathbb{C})$ is a *GKSL generator* if e^{tL} is a quantum channel (CPTP) for all $t \geq 0$.

Theorem 15.39 (Thm. 7.1, Form (i))**.** *If $L(\rho) = \phi(\rho) - \kappa\rho - \rho\kappa^\dagger$ with ϕ CP and $\phi^*(\mathbb{1}) = \kappa + \kappa^\dagger$, then L is a GKSL generator. This is [Wol12, Thm. 7.1, Eq. (7.20)].*

Proof. CP follows from Theorem 15.28. The trace constraint implies trace-annihilation by Theorem 15.18, so Theorem 15.36 gives trace preservation. □

Theorem 15.40 (Thm. 7.1, Form (i) conversely)**.** *If L is a GKSL generator, then there exist a CP map ϕ and a matrix κ such that*

$$L(\rho) = \phi(\rho) - \kappa\rho - \rho\kappa^\dagger$$

and $\phi^*(\mathbb{1}) = \kappa + \kappa^\dagger$.

Proof. Apply Theorem 15.27 to obtain a CCP decomposition. Then apply Theorem 15.37 to the trace-preserving part of the semigroup and rewrite the resulting trace-annihilation condition as $\phi^*(\mathbb{1}) = \kappa + \kappa^\dagger$. □

Theorem 15.41 (Thm. 7.1: GKSL iff CCP + trace-annihilating)**.** *L is a GKSL generator if and only if L is CCP and trace-annihilating.*

Proof. Forward: apply Theorem 15.27 to the CP part and Theorem 15.37 to the TP part. Reverse: combine Theorem 15.28 with Theorem 15.36. □

Theorem 15.42 (Thm. 7.1: GKSL iff Lindblad form). *L is a GKSL generator if and only if*

$$L(\rho) = i[\rho, H] + \sum_j \Big(L_j \rho L_j^\dagger - \tfrac{1}{2}\{L_j^\dagger L_j, \rho\}_+ \Big)$$

with $H = H^\dagger$. This is [Wol12, Thm. 7.1].

Proof. Forward: apply Theorem 15.40 to write $L(\rho) = \phi(\rho) - \kappa\rho - \rho\kappa^\dagger$ with trace constraint. Then choose traceless Kraus operators by Theorem 15.30 and rewrite κ as $iH + \frac{1}{2}\phi^*(\mathbb{1})$; Theorem 15.21 gives the Lindblad formula. Reverse: Theorems 15.22 and 15.20 show that every Lindblad form satisfies the right-hand side of Theorem 15.41. □

15.3.9 Kossakowski matrix form

Definition 15.43 (Kossakowski form). A *Kossakowski form* consists of $H = H^\dagger$, a finite family of matrices $\{F_k\}_{k=0}^{n-1}$, and a positive semidefinite matrix $C \in M_n(\mathbb{C})$, defining

$$L(\rho) = i[\rho, H] + \tfrac{1}{2} \sum_{k,l} C_{lk}([F_k, \rho F_l^\dagger] + [F_k \rho, F_l^\dagger]).$$

In Wolf's formula one takes $\{F_k\}$ to be a basis of traceless matrices; here we keep only the algebraic data needed for the conversion to Lindblad form.

Theorem 15.44 (Kossakowski form iff Lindblad form). *A linear map admits a Kossakowski form iff it admits a Lindblad form.*

Proof. Forward: diagonalize $C = B^\dagger B$ and set $L_j = \sum_k B_{jk} F_k$. Reverse: keep the same Hamiltonian and family $F_k = L_k$, and take $C = \mathbb{1}$. □

15.4 Primitivity and irreducibility of QDS

15.4.1 Auxiliary spectral and semigroup lemmas

Theorem 15.45 (Semigroup power identity). *For $t \geq 0$ and $n \in \mathbb{N}$, one has $T_{nt} = (T_t)^n$.*

Theorem 15.46 (Eigenvector propagation under powers). *If $E(v) = \mu v$, then $E^n(v) = \mu^n v$ for all $n \in \mathbb{N}$.*

Lemma 15.47 (Density matrices are nonzero). *Every density matrix is nonzero.*

Theorem 15.48 (Compression invariance is stable under powers). *If a corner compression is invariant under E, it remains invariant under every power E^n.*

Theorem 15.49 (Generator eigenvectors along the exponential semigroup). *If $L(X) = \mu X$, then $e^{tL}(X) = e^{t\mu} X$ for all $t \in \mathbb{R}$.*

Theorem 15.50 (Spectral mapping for semigroup generators). *If μ is in the spectrum of L, then $e^{t\mu}$ is in the spectrum of e^{tL}.*

Theorem 15.51 (Root-of-unity rigidity for phases). *If $e^{it\theta}$ is a root of unity for every $t > 0$, then $\theta = 0$.*

Theorem 15.52 (Peripheral generator eigenvalues are purely imaginary). *Peripheral spectral points of the generator have vanishing real part.*

Theorem 15.53 (Peripheral cardinality bound)**.** *The number of peripheral eigenvalues is bounded by* $\dim(M_D(\mathbb{C}))$*.*

Theorem 15.54 (Bounded order under peripheral power-closure)**.** *If all powers of a peripheral eigenvalue remain peripheral, then some positive power not exceeding* $\dim(M_D(\mathbb{C}))$ *equals* 1*.*

Theorem 15.55 (Peripheral powers stay peripheral under irreducibility)**.** *For an irreducible channel with faithful fixed point, every power of a peripheral eigenvalue is again peripheral.*

Theorem 15.56 (Continuous-linear-map power evaluation)**.** *Evaluating powers of a linear map agrees with powers in its continuous linear realization.*

Theorem 15.57 (Spectral-radius criterion from eigenvalue bounds)**.** *If all eigenvalues satisfy* $|\lambda| < 1$*, then the spectral radius is* < 1*.*

Remark 15.58 (Semigroup irreducible/primitive equivalence). The semigroup analogue of the irreducible/primitive equivalence is now fully proved: starting from one irreducible time slice, the fraction-slice reduction constructs a primitive slice, and the result lifts to the full semigroup (Theorems 15.65 and 15.66).

Remark 15.59 (Fraction-slice reduction for Proposition 7.5). Starting from one irreducible time t_0, we construct a positive slice $u = t_0/N$ with $T_{t_0} = (T_u)^N$, prove peripheral eigenvalue collapse at time u, and deduce primitivity of T_u.

Theorem 15.60 (Fraction-slice eigenvector has nonzero trace)**.** *Under the fraction-slice hypotheses* $T_{t_0} = (T_u)^{(\dim M_D(\mathbb{C}))!}$*, if* μ *is a peripheral eigenvalue of* T_u*, then* T_u *admits a* μ*-eigenvector with nonzero trace.*

Theorem 15.61 (Fraction-slice peripheral collapse)**.** *Under the same hypotheses, every peripheral eigenvalue of* T_u *equals* 1*.*

Theorem 15.62 (Fraction-slice primitivity)**.** *Under the same hypotheses,* T_u *is primitive.*

Theorem 15.63 (Primitive-slice fixed points are trace-scalars)**.** *If* T_u *is primitive and* σ *is the common fixed density, then every fixed point of any positive-time slice* T_s *has the form* $\tau = \mathrm{tr}(\tau)\,\sigma$*.*

Theorem 15.64 (Existence of a primitive fraction slice)**.** *If* T_{t_0} *is irreducible for some* $t_0 > 0$*, then there exists a fraction slice* $u \in (0, t_0]$ *such that* T_u *is primitive.*

Theorem 15.65 (Prop. 7.5: Irreducible semigroup implies primitive)**.** *Let* $T_t = e^{tL}$ *be a quantum dynamical semigroup. If* T_{t_0} *is irreducible for some* $t_0 > 0$*, then* T_t *is primitive (and irreducible) for every* $t > 0$*. This is [Wol12, Prop. 7.5].*

Proof. The unique faithful density fixed point σ of T_{t_0} is also fixed by all T_u (semigroup commutativity), so by uniqueness it is the common fixed point. Irreducibility of each T_t then follows from uniqueness of the positive definite fixed point and the spectral-gap argument. Primitivity is then the one-dimensional peripheral-eigenvalue characterisation, using that T_t irreducible implies all peripheral eigenvalues are roots of unity, and the faithful fixed point forces those eigenvalues to equal 1. □

Theorem 15.66 (Prop. 7.5: Irreducibility iff primitivity for QDS)**.** *For a quantum dynamical semigroup* $T_t = e^{tL}$*:*

$$(\exists\, t_0 > 0,\ T_{t_0}\ \textit{irreducible}) \iff (\forall\, t > 0,\ T_t\ \textit{primitive and irreducible}).$$

This is [Wol12, Prop. 7.5].

Proof. Forward: apply Theorem 15.65. Reverse: take $t_0 = 1$ (or any positive time); primitivity implies irreducibility by definition. □

15.5 Kernel of the adjoint Liouvillian

Theorem 15.67 (Thm. 7.2: Adjoint kernel equals commutant (faithful state)). *Let L be a GKSL generator in Lindblad form with Hamiltonian H and Lindblad operators $\{L_j\}$. Define the adjoint generator L^* by $\mathrm{tr}(\rho\, L^*(A)) = \mathrm{tr}(L(\rho)\, A)$. If L has a faithful stationary state $\rho_0 > 0$ with $L(\rho_0) = 0$, then for any A:*

$$L^*(A) = 0 \implies [A, H] = [A, L_j] = [A, L_j^\dagger] = 0 \ \forall j.$$

That is, $\ker(L^*) \subseteq \{H, L_j, L_j^\dagger\}'$*. This is [Wol12, Thm. 7.2].*

Proof. From $L^*(A) = 0$ and $L^*(A^\dagger) = 0$, compute $L^*(A^\dagger A) = \sum_j [A, L_j]^\dagger [A, L_j]$. Faithfulness of ρ_0 and $\mathrm{tr}(\rho_0 L^*(A^\dagger A)) = 0$ force $[A, L_j] = 0$ for all j. Similarly $[A, L_j^\dagger] = 0$, and then $L^*(A) = 0$ directly gives $[A, H] = 0$. □

Theorem 15.68 (Thm. 7.2: Adjoint kernel equals commutant (full equivalence)). *Under the same hypotheses as Theorem 15.67,*

$$\ker(L^*) = \{H, L_j, L_j^\dagger\}',$$

i.e. the adjoint kernel equals the commutant of $\{H, L_j, L_j^\dagger\}$. This is [Wol12, Thm. 7.2].

Proof. The inclusion $\supseteq$ is the easy direction (commutant elements are in the kernel by a direct calculation). The reverse $\subseteq$ is Theorem 15.67. □

15.6 Reducibility of quantum dynamical semigroups

The four equivalent reducibility conditions of [Wol12, Prop. 7.6] are as follows.

Definition 15.69 (Rank-deficient fixed density). Condition (1): there exists a density matrix ρ_0 with nontrivial kernel such that $e^{tL}(\rho_0) = \rho_0$ for all $t \geq 0$.

Definition 15.70 (Rank-deficient kernel element). Condition (2): there exists a density matrix ρ_0 with nontrivial kernel satisfying $L(\rho_0) = 0$.

Definition 15.71 (Invariant compression). Condition (3): there exists a nontrivial orthogonal projector P such that $e^{tL}(PM_D(\mathbb{C})P) \subseteq PM_D(\mathbb{C})P$ for all $t \geq 0$.

Definition 15.72 (Block-upper-triangular Lindblad form). Condition (4): there exists a nontrivial projector P and a Lindblad form $(H, \{L_j\})$ for L such that $(1 - P)L_j P = 0$ and $(1 - P)\kappa P = 0$ for all j, where $\kappa = iH + \frac{1}{2}\sum_j L_j^\dagger L_j$.

Theorem 15.73 (Prop. 7.6, (1)↔(2)). *A rank-deficient density matrix is a fixed point of e^{tL} for all $t \geq 0$ if and only if it lies in $\ker L$. This is [Wol12, Prop. 7.6, (1)↔(2)].*

Proof. The equivalence $L(\rho) = 0 \iff e^{tL}(\rho) = \rho$ for all $t \geq 0$ follows from the generator–semigroup correspondence: differentiate at $t = 0$ for the forward direction, and exponentiate for the reverse. □

Theorem 15.74 (Prop. 7.6, (1)→(3)). *If L is a GKSL generator and there exists a rank-deficient fixed density matrix, then the semigroup preserves a nontrivial compression. This is [Wol12, Prop. 7.6, (1)→(3)].*

Proof. Let ρ_0 be a rank-deficient fixed density. Let P be the support projection of ρ_0. Since ρ_0 is rank-deficient, $P \neq \mathbb{1}$; since $\rho_0 \neq 0$, $P \neq 0$. Channel positivity and trace preservation force e^{tL} to preserve $PM_D(\mathbb{C})P$. □

Theorem 15.75 (Prop. 7.6, (3)↔(4))**.** *For a GKSL generator L, the semigroup preserves a nontrivial compression if and only if the Lindblad data can be chosen block-upper-triangular. This is [Wol12, Prop. 7.6, (3)↔(4)].*

Proof. (4)→(3): block-upper-triangular Lindblad operators immediately imply that L preserves the compression $PM_D(\mathbb{C})P$, and hence so does e^{tL}. (3)→(4): the semigroup preserving $PM_D(\mathbb{C})P$ implies (by differentiation) that L preserves $PM_D(\mathbb{C})P$; the vanishing of $(1-P)L(PXP)$ for all X forces $(1-P)L_jP = 0$ and $(1-P)\kappa P = 0$ via the sum-of-squares identity. □

Theorem 15.76 (Prop. 7.6, (4)→(2))**.** *If the Lindblad data of a GKSL generator are block-upper-triangular with respect to some nontrivial projector P, then there exists a density matrix ρ_0 with nontrivial kernel satisfying $L(\rho_0) = 0$. This is [Wol12, Prop. 7.6, (4)→(2)].*

Proof. The block-upper-triangular structure implies L preserves the compression $PM_d(\mathbb{C})P$. Therefore the semigroup e^{tL} also preserves $PM_d(\mathbb{C})P$ for all $t \geq 0$. By the Brouwer fixed-point theorem, $e^{(1/m)L}$ has a fixed density matrix ρ_m supported in $PM_d(\mathbb{C})P$ for each m. Passing to a subsequential limit gives a density matrix ρ_0 with $P\rho_0P = \rho_0$ and $L(\rho_0) = 0$. Since $P \neq \mathbb{1}$, the matrix ρ_0 has nontrivial kernel. □

Theorem 15.77 (Prop. 7.6: Full equivalence)**.** *For a GKSL generator $L : M_D(\mathbb{C}) \to M_D(\mathbb{C})$, the four conditions*

1. *rank-deficient fixed density,*
2. *rank-deficient kernel element,*
3. *invariant compression, and*
4. *block-upper-triangular Lindblad form*

are equivalent. This is [Wol12, Prop. 7.6].

Proof. The cycle (1)↔(2), (1)→(3), (3)↔(4), and (4)→(2) closes the equivalence chain. □

15.6.1 Cor. 7.2: Sufficient conditions for non-reducibility

Definition 15.78 (Lindblad span)**.** The $\mathbb{C}$-linear span of the Lindblad operators $\{L_j\}$.

Definition 15.79 (Hermitian-closed Lindblad span)**.** The span of the Lindblad operators is closed under Hermitian conjugation, i.e. it forms a $*$-subspace of $M_d(\mathbb{C})$.

Definition 15.80 (Trivial Lindblad commutant)**.** The commutant of the Lindblad family $\{L_j\}'$ equals $\mathbb{C} \cdot \mathbb{1}$, i.e. the Lindblad operators act irreducibly on $M_d(\mathbb{C})$.

Definition 15.81 (Kossakowski rank)**.** The minimal number of Lindblad operators across all GKSL representations of L, which equals the rank of the Kossakowski matrix in the orthonormal-basis representation.

Theorem 15.82 (κ block vanishing from generator compression)**.** *Let $(H, \{L_j\})$ be a Lindblad form with $\kappa = iH + \frac{1}{2}\sum_j L_j^\dagger L_j$, and let P be an orthogonal projection such that the generator preserves the compression $PM_d(\mathbb{C})P$. If $(1-P)L_jP = 0$ for every j, then $(1-P)\kappa P = 0$.*

Proof. From $L = \varphi - \kappa(\cdot) - (\cdot)\kappa^\dagger$ and $(1-P)L(P) = 0$: the $(1-P)\varphi(P)$ term vanishes because each $(1-P)L_jP = 0$, and the $(1-P)(P\kappa^\dagger)$ term vanishes because $(1-P)P = 0$. What remains is $(1-P)\kappa P = 0$. □

Theorem 15.83 (Full algebra generation forbids block-upper-triangular form)**.** *Let $(H, \{L_j\})$ be a Lindblad form. If the algebra generated by $\{L_j\}$ and κ is the entire matrix algebra $M_d(\mathbb{C})$, then the Lindblad data do not admit a block-upper-triangular decomposition.*

Proof. Assume for contradiction that a block-upper-triangular decomposition exists with nontrivial projection P. By Theorem 15.82, $(1-P)\kappa P = 0$. Induction on elements of the subalgebra shows $(1-P)AP = 0$ for every generator. Since the generated algebra is all of $M_d(\mathbb{C})$, every matrix satisfies $(1-P)AP = 0$, forcing $P = 0$ or $P = \mathbb{1}$ — a contradiction. □

Theorem 15.84 (Hermitian span + trivial commutant forbids block-upper-triangular form)**.** *If the Lindblad span is closed under Hermitian conjugation and its commutant is $\mathbb{C} \cdot \mathbb{1}$, then the Lindblad data do not admit a block-upper-triangular decomposition.*

Proof. Assume for contradiction that a block-upper-triangular decomposition exists with nontrivial projection P. The block vanishing $(1-P)L_jP = 0$ and Hermitian closure give $PL_j(1-P) = 0$ as well, so $[P, L_j] = 0$ for all j. But then P lies in the commutant of the Lindblad span, contradicting triviality. □

Theorem 15.85 (Cor. 7.2(1): full algebra generation implies non-reducibility)**.** *If the algebra generated by $\{L_j\}$ and κ is the entire matrix algebra $M_d(\mathbb{C})$, then the QDS is not reducible. This is [Wol12, Cor. 7.2, condition 1].*

Proof. By Theorem 15.83, no block-upper-triangular decomposition exists. By the equivalence of Theorem 15.75, this rules out reducibility. □

Theorem 15.86 (Cor. 7.2(2): Hermitian span + trivial commutant implies non-reducibility)**.** *If the Lindblad span is Hermitian-closed and its commutant is trivial, then the QDS is not reducible. This is [Wol12, Cor. 7.2, condition 2].*

Proof. By Theorem 15.84, no block-upper-triangular decomposition exists. The equivalence of Theorem 15.75 then rules out reducibility. □

Theorem 15.87 (Cor. 7.2(3): large Kossakowski rank implies non-reducibility)**.** *If* $\mathrm{rank}(C) > d^2 - d$*, then the QDS is not reducible. This is [Wol12, Cor. 7.2, condition 3].*

Proof. Rewrite the hypothesis as $\mathrm{rank}(C) + d \geq d^2 + 1$ and apply the formal result that this large-rank condition rules out any nontrivial block-upper-triangular Lindblad form. Hence the generator is not reducible. □

Chapter 16

The Algebraic Fundamental Theorem

This chapter develops the one-dimensional MPS results of [MGRPG+18]: the Fundamental Theorem for injective and normal MPS on a finite periodic chain of fixed length. The argument is purely algebraic, working directly with $M_D(\mathbb{C})$. Two preparatory lemmas—the virtual bond gauge (Lemma 16.11) and the proportionality lemma (Lemma 16.12)—carry the entire argument, yielding the main theorem and its corollaries for translation-invariant chains, non-TI descriptions of TI states, and normal MPS.

Throughout, D is the bond dimension and d is the physical dimension with $d \geq D^2$. The extensions to injective and normal PEPS on arbitrary graphs and the improved system-size bound ($2L+1$ in place of $3L$) are left for future work. The chain data (Definition 4.12), including same chain state, injectivity, and cyclic gauge equivalence, was introduced in Chapter 4.

16.1 One-sided inverse and decomposition

If A is injective, the linear combination map $\Phi_A(c) := \sum_i c_i A^i : \mathbb{C}^d \to M_D(\mathbb{C})$ is surjective and admits a right inverse.

Lemma 16.1 (Existence of a right inverse)**.** *If A is injective, then there exists a $\mathbb{C}$-linear map $\Psi : M_D(\mathbb{C}) \to \mathbb{C}^d$ satisfying $\Phi_A \circ \Psi = \mathrm{id}_{M_D(\mathbb{C})}$.*

Proof. Surjectivity of Φ_A yields a linear section by choice. □

Definition 16.2 (Decomposition map)**.** Let A be an injective MPS tensor. The *decomposition map* $\Psi_A : M_D(\mathbb{C}) \to \mathbb{C}^d$ is a fixed choice of right inverse of Φ_A.

Lemma 16.3 (Decomposition property)**.** *If A is injective, then for every $X \in M_D(\mathbb{C})$ there exist coefficients $c \in \mathbb{C}^d$ such that*

$$X = \sum_{i=0}^{d-1} c_i\, A^i.$$

Proof. Take $c = \Psi_A(X)$. Then $\Phi_A(c) = \Phi_A(\Psi_A(X)) = X$. □

Remark 16.4 (Non-uniqueness of the decomposition). When $d > D^2$, the map Φ_A has a nontrivial kernel, so the section Ψ_A is not unique. The statements that follow depend only on $\Phi_A \circ \Psi_A = \mathrm{id}$, so the choice does not affect the conclusions.

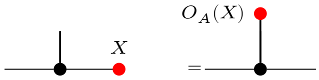
16.2 Physical realisation of virtual insertions

For an injective tensor A, the physical realisation map encodes the effect of inserting a virtual matrix X on the bond as a $d \times d$ operation on the physical indices.

Definition 16.5 (Physical realisation map)**.** Let A be an injective tensor. For $X \in M_D(\mathbb{C})$, define the $d \times d$ matrix $O_A(X) \in M_d(\mathbb{C})$ by

$$O_A(X)_{ij} := (\Psi_A(A^i X))_j,$$

so that row i of $O_A(X)$ contains the decomposition coefficients of $A^i X$ in the spanning set $\{A^j\}$. Diagrammatically, the inserted virtual matrix is re-expressed as a physical operation on the same local tensor:



The map $X \mapsto O_A(X)$ is $\mathbb{C}$-linear.

Lemma 16.6 (Defining property of the physical realisation map)**.** *For every $X \in M_D(\mathbb{C})$ and every physical index i,*

$$A^i\, X = \sum_{j=0}^{d-1} O_A(X)_{ij}\, A^j.$$

Proof. By definition, $O_A(X)_{ij} = (\Psi_A(A^i X))_j$. Applying Φ_A gives $\sum_j O_A(X)_{ij} A^j = \Phi_A(\Psi_A(A^i X)) = A^i X$, where the last step uses $\Phi_A \circ \Psi_A = \mathrm{id}$. □

Theorem 16.7 (Physical realisation is multiplicative)**.** *Let A be an injective MPS tensor. For all $X, Y \in M_D(\mathbb{C})$,*

$$O_A(XY) = O_A(X)\, O_A(Y).$$

Proof. Applying Lemma 16.6 to X yields

$$A^i X = \sum_{j=0}^{d-1} O_A(X)_{ij}\, A^j.$$

Multiply by Y on the right and apply the linear map Ψ_A:

$$\Psi_A(A^i XY) = \sum_{j=0}^{d-1} O_A(X)_{ij}\, \Psi_A(A^j Y).$$

Taking the kth component of both sides gives

$$O_A(XY)_{ik} = \sum_{j=0}^{d-1} O_A(X)_{ij}\, O_A(Y)_{jk} = (O_A(X)\, O_A(Y))_{ik}.$$

Since this holds for all i and k, the matrices are equal. □

Remark 16.8 (General injective case). When $d > D^2$, the decomposition map Ψ_A need not be unique. Nevertheless, Theorem 16.7 is already the general injective statement: for the fixed choice of Ψ_A in Definition 16.2, the associated physical realisation map O_A is multiplicative. This supersedes the earlier basis-only formulation.

16.3 Intermediate lemmas

Lemma 16.9 (Blocking preserves injectivity)**.** *Let $A_0, \dots, A_{m-1}$ be injective MPS tensors with compatible bond dimensions. Then the blocked tensor $(A_0 \otimes \cdots \otimes A_{m-1})^{(i_0,\dots,i_{m-1})} := A_0^{i_0} A_1^{i_1} \cdots A_{m-1}^{i_{m-1}}$ is again injective. (In Lean, injectivity is proved for the repackaged combined tensor that collects all site tensors into a single family, not for the physically blocked product tensor.)*

Proof. The span of the blocked matrices contains all products $A_0^{i_0} \cdots A_{m-1}^{i_{m-1}}$. Since each $\{A_k^i\}_i$ spans $M_D(\mathbb{C})$, these products span all of $M_D(\mathbb{C})$. ☐

Lemma 16.10 (Centre of $M_D(\mathbb{C})$)**.** *If $Z \in M_D(\mathbb{C})$ commutes with every element of $M_D(\mathbb{C})$, then $Z = \lambda \mathbb{1}_D$ for some $\lambda \in \mathbb{C}$.*

Proof. Standard: Z commutes with every matrix unit E_{ij}, which forces all off-diagonal entries to vanish and all diagonal entries to agree. ☐

16.4 Same state forces a virtual bond gauge

This section proves Lemma 1 of [MGRPG+18]: if two injective chains of length $n \geq 3$ generate the same state, then on each bond the virtual insertion algebras are related by an inner automorphism.

Lemma 16.11 (Virtual bond gauge – 3-site case)**.** *Let $A = (A_0, A_1, A_2)$ and $B = (B_0, B_1, B_2)$ be two 3-site injective periodic chains with common bond dimension $D \geq 1$ and $\mathcal{V}(A) = \mathcal{V}(B)$ (same MPV family). Then at the middle bond there exists an invertible matrix $Z \in \mathrm{GL}_D(\mathbb{C})$ such that the virtual-insertion trace coefficients agree after conjugation:*

$$\mathrm{tr}(A_0^{i_0} X\, A_1^{i_1} A_2^{i_2}) = \mathrm{tr}(B_0^{i_0} Z^{-1} X Z\, B_1^{i_1} B_2^{i_2})$$

for all $X \in M_D(\mathbb{C})$ and all physical indices i_0, i_1, i_2.

In the paper [MGRPG+18], Lemma 1 is stated for arbitrary chain length $n \geq 3$ and any bond position, with uniqueness of Z up to scalar. The chain-level generalization to arbitrary $n \geq 3$ and any bond position is future work.

Proof. The $\mathcal{V}(A) = \mathcal{V}(B)$ hypothesis gives, for every $X \in M_D(\mathbb{C})$, equality of the virtual-insertion trace coefficients: $\mathrm{tr}(A_0^{i_0} X\, A_1^{i_1} A_2^{i_2}) = \mathrm{tr}(B_0^{i_0} Y\, B_1^{i_1} B_2^{i_2})$ for some Y depending on X. Since all three site tensors on each side are injective, the physical realisation maps O_A, O_B are algebra homomorphisms by Theorem 16.7. Setting $T(X) = Y$ defines a $\mathbb{C}$-linear multiplicative map, hence a nonzero $\mathbb{C}$-algebra endomorphism of $M_D(\mathbb{C})$, which is bijective by Theorem 4.8. Skolem–Noether (Theorem 4.9) gives $Z \in \mathrm{GL}_D(\mathbb{C})$ with $T(X) = Z^{-1} X Z$. ☐

16.5 Aligned gauges force proportionality

After applying Lemma 16.11 to each bond and absorbing the gauge matrices, one obtains two chains whose virtual-insertion operations agree on every bond. Lemma 2 of [MGRPG+18] shows that this agreement forces the local tensors to be proportional.

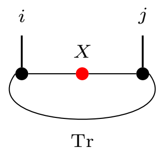
Lemma 16.12 (Aligned gauges force proportionality – Lemma 2 of [MGRPG+18])**.** *Let A_1, A_2 and B_1, B_2 be injective MPS tensors with the same bond dimension D. Suppose that for all*

$X \in M_D(\mathbb{C})$ (on the internal bond) and all $Y \in M_D(\mathbb{C})$ (on the external bond),

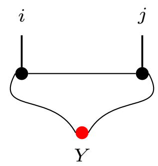
$$\mathrm{tr}(A_1^i \, X \, A_2^j) = \mathrm{tr}(B_1^i \, X \, B_2^j) \qquad \textit{for all } i, j, \tag{16.1}$$

$$\mathrm{tr}(A_2^j \, Y \, A_1^i) = \mathrm{tr}(B_2^j \, Y \, B_1^i) \qquad \textit{for all } i, j. \tag{16.2}$$

Diagrammatically, Eq. (16.1) is the equality obtained by inserting X on the bond between the two sites,



while Eq. (16.2) is the corresponding equality with the insertion Y on the outer trace bond,



so the two hypotheses distinguish the internal and external virtual loops of the two-site periodic chain.

Then there exists a nonzero scalar $\lambda \in \mathbb{C}^\times$ such that

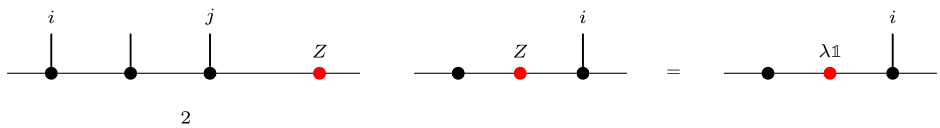
$$A_1^i = \lambda \, B_1^i \qquad \textit{and} \qquad A_2^j = \lambda^{-1} \, B_2^j$$

for all physical indices i, j.

Proof. By cyclicity and non-degeneracy of the trace pairing, the conditions in Eqs. (16.1) and (16.2) give

$$A_2^j A_1^i = B_2^j B_1^i \qquad \text{and} \qquad A_1^i A_2^j = B_1^i B_2^j \qquad \text{for all } i, j.$$

Write $\mathbb{1}_D = \sum_j c_j A_2^j$ using the decomposition map Ψ_{A_2}, and define $Z := \sum_j c_j B_2^j \in M_D(\mathbb{C})$. This is the point where the two-site comparison collapses to a single boundary matrix:



The first product identity gives $A_1^i = Z B_1^i$, and the second gives $A_1^i = B_1^i Z$. Hence Z commutes with every B_1^i; since $\{B_1^i\}$ spans $M_D(\mathbb{C})$, Z is central. By Lemma 16.10, $Z = \lambda \mathbb{1}_D$ for some $\lambda \in \mathbb{C}^\times$ (nonzero since the tensors are nonzero). Thus $A_1^i = \lambda B_1^i$, and substituting into the first product identity forces $A_2^j = \lambda^{-1} B_2^j$. □

16.6 The Fundamental Theorem for injective chains

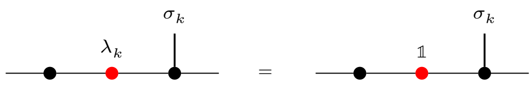
Combining Lemma 16.11 with Lemma 16.12 yields the main theorem of [MGRPG+18] for non-translation-invariant injective MPS.

Theorem 16.13 (Fundamental Theorem for injective chains – Theorem 1 of [MGRPG+18])**.** *Let* $\mathbf{A} = (A_0, \dots, A_{n-1})$ *and* $\mathbf{B} = (B_0, \dots, B_{n-1})$ *be two injective non-TI periodic chains of length* $n \geq 3$ *that generate the same chain state. Then* $\mathbf{A}$ *and* $\mathbf{B}$ *are cyclically gauge equivalent: there exist invertible matrices* $Z_0, \dots, Z_{n-1} \in \mathrm{GL}_D(\mathbb{C})$*, unique up to a common nonzero scalar factor, such that*

$$B_k^i = Z_k^{-1} A_k^i Z_{k+1}$$

for every site $k \in \{0, \dots, n-1\}$ *and every physical index* i*, with* $Z_n = Z_0$*.*

Proof. Apply Lemma 16.11 at each bond to obtain $Z_k \in \mathrm{GL}_D(\mathbb{C})$ with $O_{A,k}(X) = O_{B,k}(Z_k^{-1} X Z_k)$. Redefine $\widetilde{B}_k^i := Z_k B_k^i Z_{k+1}^{-1}$; the virtual insertion operations of $\mathbf{A}$ and $\widetilde{\mathbf{B}}$ then agree on every bond. After the bond gauges have been aligned, the argument reduces to the same local replacement step repeated around the chain:



For each adjacent pair $(k, k+1)$, block the remaining sites into a two-site chain; the agreement on both bonds gives exactly the hypotheses of Eqs. (16.1)–(16.2) in Lemma 16.12, yielding $A_k^i = \lambda_k \widetilde{B}_k^i$ for some $\lambda_k \in \mathbb{C}^\times$. Choosing a configuration σ with nonzero trace and substituting $A_k^{\sigma_k} = \lambda_k \widetilde{B}_k^{\sigma_k}$ into the chain-state identity forces $\prod_k \lambda_k = 1$. Set $\mu_k = \prod_{j<k} \lambda_j$ and replace Z_k by $\mu_k Z_k$; then $B_k^i = Z_k^{-1} A_k^i Z_{k+1}$ for all k and i, with $Z_n = Z_0$. Uniqueness up to a common scalar follows from the uniqueness clause of Lemma 16.11. ☐

Remark 16.14 (A stronger equal-MPV variant)*.* A stronger variant of the chain theorem assumes equality of all MPV coefficients for the combined tensor, rather than equality of the chain state at one fixed length. Reference [MGRPG+18] closes this gap by a blocking argument when $n \geq 3$. That reduction is stated separately from the main chain theorem.

Remark 16.15 (Relationship to the translation-invariant all-sizes theorem)*.* Theorem 16.13 handles non-TI chains at fixed length $n \geq 3$, concluding gauge equivalence up to a common scalar. The single-block Fundamental Theorem (Theorem 4.11) instead assumes agreement for every system size; the additional hypothesis eliminates the phase ambiguity, yielding exact gauge equivalence $B^i = X A^i X^{-1}$. The two results are complementary and neither implies the other in full generality.

16.6.1 Finite-length MPV agreement

Definition 16.16 (Finite-length MPV agreement)**.** Two tensors A and B of the same bond dimension satisfy *finite-length MPV agreement from* N_0 *onward* if they generate the same MPV family for all system sizes $N \geq N_0$:

$$V^{(N)}(A)_\sigma = V^{(N)}(B)_\sigma \qquad \text{for all } N \geq N_0, \sigma \in [d]^N.$$

Theorem 16.17 (Finite-length agreement implies full agreement)**.** *Let* A *be an injective tensor of bond dimension* $D \geq 1$*, and let* B *be a tensor of the same bond dimension. If* A *and* B *generate the same MPV family for every system size* $N \geq N_0$*, then they generate the same MPV family for every system size.*

Proof. Write $1 = \sum_i c_i A^i$ (by injectivity) and set $S = \sum_i c_i B^i$.

1. Show that the length-N_0 word span of B is all of $M_D(\mathbb{C})$ via the composition identity $\Phi_A \circ \ell_A = \Phi_B \circ \ell_B$ (from trace agreement at length N_0+1) and a finrank transfer argument.
2. Derive $S = 1$ by trace nondegeneracy: for every $M \in M_D(\mathbb{C})$, $\mathrm{tr}(M(S-1)) = 0$.
3. Downward induction on $k = N_0 - |w|$: decompose $\mathrm{tr}(w_A)$ and $\mathrm{tr}(w_B)$ using $1 = \sum c_i A^i$ and $S = 1$ respectively, then apply the inductive hypothesis at length $|w| + 1$.

$\square$

Theorem 16.18 (Strengthened single-block FT (finite-length))**.** *Let A be an injective tensor of bond dimension $D \geq 1$, and let B be a tensor of the same bond dimension. If A and B agree on all MPV coefficients for system sizes $N \geq N_0$ (for any threshold N_0), then A and B are gauge equivalent.*

Proof. Apply Theorem 16.17 to obtain equality of the MPV families for all system sizes, then invoke the single-block Fundamental Theorem (Theorem 4.11). $\square$

16.7 Corollaries

16.7.1 Translation-invariant chains

Corollary 16.19 (TI case)**.** *Let A and B be injective MPS tensors of bond dimension D. Suppose the two translation-invariant chains $(A, A, \dots, A)$ and $(B, B, \dots, B)$ of length $n \geq 3$ generate the same chain state. Then there exists an invertible $Z \in \mathrm{GL}_D(\mathbb{C})$, unique up to a scalar, and a scalar $\lambda \in \mathbb{C}^\times$ with $\lambda^n = 1$, such that*

$$B^i = \lambda\, Z^{-1} A^i Z$$

for every physical index i.

Proof. Apply Theorem 16.13 to the two constant chains to obtain $Z_0, \dots, Z_{n-1}$ with $B^i = Z_k^{-1} A^i Z_{k+1}$ for all k and i. Setting $G_k = Z_k Z_0^{-1}$, the relation $Z_k^{-1} A^i Z_{k+1} = Z_{k+1}^{-1} A^i Z_{k+2}$ shows $A^i = G_k^{-1} A^i G_k$ for each k. Since $\{A^i\}$ spans $M_D(\mathbb{C})$, each G_k commutes with all of $M_D(\mathbb{C})$, so $G_k = \mu_k \mathbb{1}$ by Lemma 16.10. Hence $Z_k = \mu_k Z_0$, and the cyclic condition $Z_n = Z_0$ forces $(\mu_1)^n = 1$. Setting $Z = Z_0$ and $\lambda = \mu_1$ yields the result. $\square$

16.7.2 Non-TI descriptions of TI states

Corollary 16.20 (Non-TI realisation of a TI state)**.** *Let $\mathbf{A} = (A_0, \dots, A_{n-1})$ be an injective non-TI chain of length $n \geq 3$. Suppose the state generated by $\mathbf{A}$ is translation-invariant, i.e. it equals the state generated by the cyclically shifted chain $(A_1, \dots, A_{n-1}, A_0)$. Then $\mathbf{A}$ admits a translation-invariant description with the same bond dimension: there exist an injective tensor B such that the TI chain $(B, \dots, B)$ generates the same state as $\mathbf{A}$. Explicitly, $B^i = A_0^i Z_0^{-1}$ (right multiplication,* not *conjugation), where Z_0 is defined below.*

Proof. The cyclic shift $\sigma(\mathbf{A}) = (A_1, \dots, A_{n-1}, A_0)$ generates the same state by hypothesis. Apply Theorem 16.13: there exist invertible $Z_0, \dots, Z_{n-1}$ with $A_{k+1}^i = Z_k^{-1} A_k^i Z_{k+1}$ for all k and i (indices mod n). Define $L_k = Z_0 Z_1 \cdots Z_{k-1}$ and $R_k = Z_1 Z_2 \cdots Z_k$ for $k \geq 1$, with $L_0 = R_0 = \mathbb{1}$. Iterating the gauge relation gives $A_k^i = L_k^{-1} A_0^i R_k$. Since $R_k L_{k+1}^{-1} = Z_0^{-1}$ for every k, substituting into the trace yields

$$\mathrm{tr}(A_0^{\sigma_0} \cdots A_{n-1}^{\sigma_{n-1}}) = \mathrm{tr}(A_0^{\sigma_0} Z_0^{-1} A_0^{\sigma_1} Z_0^{-1} \cdots A_0^{\sigma_{n-1}} Z_0^{-1}).$$

Hence the state is generated by the TI chain with tensor $B^i := A_0^i\, Z_0^{-1}$. Since A_0 is injective and Z_0 is invertible, B is injective. □

Symmetry-theoretic consequences of the Fundamental Theorem, including virtual representations and string order, are developed in Chapter 17.

16.7.3 Normal MPS (blocked injectivity)

Corollary 16.21 (Normal MPS Fundamental Theorem). *Let $\{A_k\}_{k=0}^{n-1}$ and $\{B_k\}_{k=0}^{n-1}$ be two non-TI periodic MPS on n sites. Suppose that blocking any L consecutive sites in either chain yields an injective tensor, and that $n \geq 3L$. If both chains generate the same state, then there exist invertible matrices $Z_0, \dots, Z_{n-1} \in \mathrm{GL}_D(\mathbb{C})$, unique up to a common scalar, such that for every site k and physical index i,*

$$B_k^i = Z_k^{-1}\, A_k^i\, Z_{k+1},$$

with $Z_n = Z_0$. That is, gauge equivalence holds for the original *(unblocked) tensors. The reduction to the injective case proceeds by grouping the chain into consecutive length-L blocks and then applying the injective-chain theorem to the blocked local tensors. The remaining passage from blocked gauges back to the original site tensors is the reason this corollary remains marked .*

Proof. Block every L consecutive sites to form a chain of length $m = \lfloor n/L \rfloor \geq 3$ with injective local tensors $A^{[L]}$ and $B^{[L]}$ (injective by hypothesis). The blocked chains generate the same state, so Theorem 16.13 gives invertible gauges $\widehat{Z}_0, \dots, \widehat{Z}_{m-1}$ with $(B^{[L]})_k = \widehat{Z}_k^{-1}\,(A^{[L]})_k\,\widehat{Z}_{k+1}$. Applying Lemma 16.12 within each L-block then decomposes the blocked gauge equivalence into site-by-site gauges, yielding $Z_0, \dots, Z_{n-1}$ for the original tensors. □

Remark 16.22. The result above establishes gauge equivalence of blocked constant chains; recovering gauges for the original unblocked site tensors is future work.

16.8 Product algebra construction and per-block gauge

The chain Fundamental Theorem (Theorem 16.13) handles non-TI periodic chains. A parallel line of argument addresses block-diagonal tensors: given a family of injective block tensors $\{A_k^i\}_{k=0}^{r-1}$ with block dimensions D_k and a second family $\{B_k^i\}$ with per-block $\mathcal{V}(A_k) = \mathcal{V}(B_k)$, one seeks per-block gauge equivalence $B_k^i = X_k\, A_k^i\, X_k^{-1}$ for each k independently, and a global block-permutation plus inner-automorphism decomposition.

16.8.1 Per-block linear extension

Definition 16.23 (Per-block linear extension). Let A_k, B_k be MPS tensors of bond dimension D_k with A_k injective and $\mathcal{V}(A_k) = \mathcal{V}(B_k)$ for each block $k \in \{0, \dots, r-1\}$. The *per-block linear extension* $T_k : M_{D_k}(\mathbb{C}) \to M_{D_k}(\mathbb{C})$ is the unique $\mathbb{C}$-linear map satisfying $T_k(A_k^i) = B_k^i$ for every physical index i.

Lemma 16.24 (Per-block multiplicativity). *For each block k, the per-block linear extension T_k is multiplicative: $T_k(MN) = T_k(M)\,T_k(N)$ for all $M, N \in M_{D_k}(\mathbb{C})$.*

Proof. Since $\{A_k^i\}$ spans $M_{D_k}(\mathbb{C})$, it suffices to check on spanning elements, where the result follows from the MPV identity. □

Lemma 16.25 (Per-block bijectivity)**.** *Each T_k is bijective.*

Proof. A nonzero multiplicative linear endomorphism of $M_{D_k}(\mathbb{C})$ is automatically bijective: injectivity follows from the simplicity of $M_{D_k}(\mathbb{C})$ (the kernel is a proper ideal, hence zero), and surjectivity from finite dimension. Non-vanishing of T_k is established via the trace pairing: if $T_k = 0$, then all $B_k^i = 0$, contradicting the non-vanishing of the trace pairing for injective tensors. □

16.8.2 Product algebra automorphism

Definition 16.26 (Product algebra automorphism)**.** The *product algebra automorphism* is the $\mathbb{C}$-algebra equivalence

$$T : \prod_{k=0}^{r-1} M_{D_k}(\mathbb{C}) \xrightarrow{\sim} \prod_{k=0}^{r-1} M_{D_k}(\mathbb{C})$$

defined by $T(M)_k = T_k(M_k)$, where each T_k is the per-block linear extension. It is an algebra automorphism since each T_k is a multiplicative, unital, bijective linear map. Diagrammatically, this is the blockwise map on the product algebra obtained by applying the corresponding per-block extension to each summand.

Theorem 16.27 (Block-permutation decomposition)**.** *The product algebra automorphism T decomposes as a block permutation composed with per-block inner automorphisms: there exist a permutation $\sigma \in S_r$ (with $D_{\sigma(k)} = D_k$ for all k) and invertible matrices $X_k \in \mathrm{GL}_{D_k}(\mathbb{C})$ such that*

$$(T(M))_{\sigma(k)} = X_k \, M \, X_k^{-1}$$

for all $M \in M_{D_k}(\mathbb{C})$ (viewed as supported in block k) and all k. Equivalently, the product-algebra automorphism first permutes the simple matrix blocks and then acts inside each block by an inner conjugation.

Proof. The automorphism T permutes the block ideals of $\prod_k M_{D_k}(\mathbb{C})$ (each block ideal is simple, and an automorphism sends simple ideals to simple ideals of the same dimension). This gives the permutation σ. On each block, the restricted map is a $\mathbb{C}$-algebra automorphism of $M_{D_k}(\mathbb{C})$; by Skolem–Noether (Theorem 4.9), it is inner. Thus T splits algebraically into the permutation of block ideals followed by inner conjugation on each surviving block. □

16.8.3 Nondegeneracy of the product trace pairing

Lemma 16.28 (Nondegeneracy of the product trace pairing)**.** *Let $M = (M_0, \dots, M_{r-1}) \in \prod_k M_{D_k}(\mathbb{C})$. If $\sum_k \mathrm{tr}(M_k N_k) = 0$ for all $N = (N_0, \dots, N_{r-1})$, then $M = 0$.*

Proof. Choose N supported on a single block k (all other components zero). The hypothesis reduces to $\mathrm{tr}(M_k N_k) = 0$ for all N_k, so $M_k = 0$ by nondegeneracy of the matrix trace pairing on $M_{D_k}(\mathbb{C})$. Since k was arbitrary, $M = 0$. □

Lemma 16.29 (Injectivity of the product Gram map)**.** *Let $\{A_k\}$ be a family of injective block tensors. The product Gram map $M \mapsto (k, i \mapsto \mathrm{tr}(M_k \cdot A_k^i))$ is injective.*

Proof. Reduces to per-block injectivity of the Gram map $M_k \mapsto (i \mapsto \mathrm{tr}(M_k \cdot A_k^i))$, which holds by injectivity of A_k (the matrices $\{A_k^i\}$ span $M_{D_k}(\mathbb{C})$, so the trace pairing against them separates points). □

16.9 The multi-block Fundamental Theorem

Theorem 16.30 (Multi-block Fundamental Theorem)**.** *Let* $\{A_k\}_{k=0}^{r-1}$ *and* $\{B_k\}_{k=0}^{r-1}$ *be two families of MPS tensors with block dimensions* D_k*, each* A_k *injective, and* $\mathcal{V}(A_k) = \mathcal{V}(B_k)$ *for all* k*. Then:*

1. *Per-block gauge equivalence:* $B_k^i = X_k\, A_k^i\, X_k^{-1}$ *for some* $X_k \in \mathrm{GL}_{D_k}(\mathbb{C})$*, for every* k *and* i*.*
2. *Global gauge equivalence: the block-diagonal tensors* $\bigoplus_k \mu_k A_k$ *and* $\bigoplus_k \mu_k B_k$ *are gauge equivalent.*

The second clause is obtained from the first by assembling the per-block gauges into one block-diagonal gauge.

Proof. Per-block gauge equivalence follows from applying the single-block Fundamental Theorem (existing) to each block, using injectivity of A_k and $\mathcal{V}(A_k) = \mathcal{V}(B_k)$. Global gauge equivalence is assembled from the per-block gauges via the block-diagonal structure. □

Theorem 16.31 (Per-block equality of MPV families and per-block gauge equivalence)**.** *Let* $\{A_k\}$ *be a family of injective block tensors. Then* $\mathcal{V}(A_k) = \mathcal{V}(B_k)$ *for all* k *if and only if* A_k *and* B_k *are gauge equivalent for all* k*.*

Proof. The forward direction is the single-block Fundamental Theorem applied blockwise. The reverse direction holds because gauge equivalence preserves all traces of word evaluations. □

16.10 Block separation from canonical form

The multi-block Fundamental Theorem (Theorem 16.30) assumes per-block $\mathcal{V}(A_k) = \mathcal{V}(B_k)$. In practice, one starts from the weaker global hypothesis that the block-diagonal tensors satisfy the same MPV family: $\mathcal{V}(\bigoplus_k \mu_k A_k) = \mathcal{V}(\bigoplus_k \mu_k B_k)$. Block separation extracts the per-block identities from the global one under canonical-form normalization. The first step is the weighted block-sum identity

$$\sum_k \mu_k^N V^{(N)}(A_k)_\sigma = \sum_k \mu_k^N V^{(N)}(B_k)_\sigma,$$

which is then combined with overlap estimates and the leading-block peel.

Definition 16.32 (Canonical form)**.** A family of block tensors $\{A_k\}_{k=0}^{r-1}$ with scaling weights $\mu_0, \dots, \mu_{r-1} \in \mathbb{C}^\times$ is in *canonical form* if:

1. each A_k is injective,
2. each A_k is left-canonical: $\sum_i (A_k^i)^\dagger A_k^i = \mathbb{1}_{D_k}$,
3. the weights are strictly ordered by norm: $|\mu_0| > |\mu_1| > \cdots > |\mu_{r-1}| > 0$,
4. the self-overlaps are normalised: $O_{A_k A_k}(N) \to 1$ as $N \to \infty$.

Definition 16.33 (Normal canonical form)**.** A family is in *normal canonical form* if it satisfies the canonical-form conditions with each block additionally *irreducible* (no nontrivial invariant projection) and *primitive* (the transfer map has spectral gap).

Theorem 16.34 (Block separation from canonical form)**.** *Let* $\{A_k\}$ *be in canonical form with weights* $\{\mu_k\}$*, and let* $\{B_k\}$ *be another family with each* B_k *injective and left-canonical. If* $\mathcal{V}(\bigoplus_k \mu_k A_k) = \mathcal{V}(\bigoplus_k \mu_k B_k)$*, then* $\mathcal{V}(A_k) = \mathcal{V}(B_k)$ *for each block* k *individually. The proof isolates the leading block and treats the remaining blocks as an exponentially decaying tail with* $\rho = |\mu_1|/|\mu_0| < 1$*.*

Proof. The proof proceeds by strong induction on the number of blocks r. For $r = 1$, the weighted sum degenerates and the result is immediate after cancelling $\mu_0^N \neq 0$.

For $r \geq 2$, the global identity $\sum_k \mu_k^N \left(V^{(N)}(A_k)_\sigma - V^{(N)}(B_k)_\sigma\right) = 0$ holds for all N and σ.

$$\mu_0^N (V^{(N)}(A_0)_\sigma - V^{(N)}(B_0)_\sigma) = -\sum_{k\geq 1} \mu_k^N (V^{(N)}(A_k)_\sigma - V^{(N)}(B_k)_\sigma).$$

Dividing by μ_0^N shows that the right-hand side is $O(\rho^N)$ with $\rho = |\mu_1|/|\mu_0| < 1$. Multiplying by the MPV components of A_0 and summing over σ converts this to an overlap identity. Dividing by $|\mu_0|^{2N}$ and using the strict norm ordering $|\mu_1|/|\mu_0| < 1$, the tail terms ($k \geq 1$) decay geometrically with ratio $\rho = |\mu_1|/|\mu_0|$. The leading term ($k = 0$) therefore converges: $O_{A_0 B_0}(N) \to 1$ as $N \to \infty$. Combined with the normalised self-overlap $O_{A_0 A_0}(N) \to 1$, this gives $\mathcal{V}(A_0) = \mathcal{V}(B_0)$ via the gauge-phase equivalence criterion. Subtracting the leading block and applying the inductive hypothesis yields $\mathcal{V}(A_k) = \mathcal{V}(B_k)$ for all remaining k. □

Theorem 16.35 (Fundamental Theorem from canonical form)**.** *Under the hypotheses of Theorem 16.34, every block pair A_k, B_k is gauge equivalent, and the block-diagonal tensors are globally gauge equivalent.*

Proof. Theorem 16.34 gives per-block $\mathcal{V}(A_k) = \mathcal{V}(B_k)$. Theorem 16.30 converts this to per-block and global gauge equivalence. □

Theorem 16.36 (Explicit gauge from canonical form)**.** *Under the same hypotheses, there exist invertible matrices $X_k \in \mathrm{GL}_{D_k}(\mathbb{C})$ such that $B_k^i = X_k A_k^i X_k^{-1}$ for every block k and physical index i.*

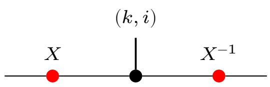
Proof. Extract the gauge matrices from Theorem 16.35 by choice. □

16.11 TI reduction corollary

Theorem 16.37 (TI reduction corollary)**.** *Let A and B be MPS tensors of bond dimension D with A injective. Consider the two constant (translation-invariant) chains $(A, \dots, A)$ and $(B, \dots, B)$ of length $n \geq 1$. If the combined tensors of the two chains generate the same MPV family, then:*

1. *B is gauge equivalent to A: there exists $X \in \mathrm{GL}_D(\mathbb{C})$ with $B^i = X A^i X^{-1}$ for all i.*
2. *B is injective.*

Proof. Block the constant chain into a combined tensor and apply Corollary 16.19 to obtain a single uniform gauge. On the combined-site index (k, i) this is the local relation



and reading this back on each site gives $B^i = X A^i X^{-1}$. Injectivity of B then follows from gauge equivalence to the injective tensor A. □

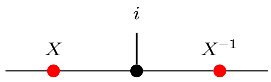
Theorem 16.38 (TI reduction from same state)**.** *Under the same setup, if $n \geq 3$ and the two constant chains generate the same chain state (rather than the same MPV family on the combined tensor), the same conclusion holds, provided one can pass from equality of chain states to equality of MPV families on the combined tensor.*

Proof. The additional hypothesis converts equality of chain states at length $n \geq 3$ to equality of MPV families on the combined tensor. Then Theorem 16.37 applies. □

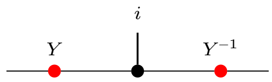
16.11.1 Global symmetry via gauge composition

The virtual representation theorem (Theorem 17.21) relies on the fact that composing two gauge transformations yields another gauge up to a scalar factor.

Lemma 16.39 (Gauge ratio commutes with tensor matrices)**.** *Diagrammatically, the hypothesis is that the two local gauges*



and



produce the same target tensor B^i from the same source tensor A^i. If $X, Y \in \mathrm{GL}_D(\mathbb{C})$ both conjugate an MPS tensor A to the same tensor B (i.e. $B^i = XA^iX^{-1} = YA^iY^{-1}$ for all i), then $Y^{-1}X$ commutes with every A^i.

Proof. Direct matrix calculation: conjugating both gauge relations and composing gives $(Y^{-1}X)A^i = A^i(Y^{-1}X)$. $\square$

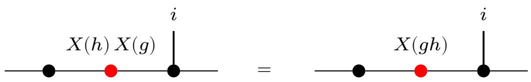
Lemma 16.40 (Gauge ratio is scalar for injective tensors)**.** *Under the same hypotheses, if A is additionally injective, then $Y^{-1}X = \lambda \mathbb{1}_D$ for some scalar $\lambda \in \mathbb{C}$.*

Proof. The same two-gauge picture shows that the ratio $Y^{-1}X$ acts invisibly on the local tensor. By Lemma 16.39, $Y^{-1}X$ commutes with all A^i. Since $\{A^i\}$ spans $M_D(\mathbb{C})$, the matrix $Y^{-1}X$ commutes with all of $M_D(\mathbb{C})$. By Lemma 16.10, it is a scalar matrix. $\square$

Theorem 16.41 (Projective multiplication law for gauge matrices)**.** *Let A be an injective MPS tensor with on-site symmetry under a group representation $U : G \to \mathrm{GL}_d(\mathbb{C})$. For each $g \in G$, let $X(g) \in \mathrm{GL}_D(\mathbb{C})$ satisfy $\widetilde{A}^i_g = X(g)\, A^i\, X(g)^{-1}$ for all i. Then for all $g, h \in G$, there exists $c(g,h) \in \mathbb{C}$ such that*

$$X(h)\, X(g) = c(g,h)\, X(g \cdot h).$$

In fact $c(g,h) \neq 0$ since both sides are invertible, so $c(g,h) \in \mathbb{C}^\times$. Diagrammatically, the two candidate gauges are competing local replacements for the same twisted tensor:



and both sides conjugate A^i to the same twisted tensor $\widetilde{A}^i_{gh}$.

Proof. By the composition law $\widetilde{A}_{gh} = \widetilde{(\widetilde{A}_h)}_g$ (Lemma 17.13), both $X(h)\,X(g)$ and $X(gh)$ conjugate A to $\widetilde{A}_{gh}$. Lemma 16.40 then gives the scalar factor. $\square$

16.12 Fundamental Theorem for injective PEPS

The following definitions and results extend the Fundamental Theorem to injective PEPS on simple graphs, following Theorem 2 of [MGRPG+18].

Definition 16.42 (Edge gauge at a vertex)**.** For an edge $e = (u, w)$ with $u < w$ and gauge $X_e \in \mathrm{GL}(D_e)$, the gauge matrix applied at vertex v is X_e if $v = u$ and $(X_e^{-1})^\top$ if $v = w$.

Definition 16.43 (Gauge vertex action)**.** The gauged tensor at vertex v is A_v^X, where $A_v^X(\eta,\sigma) = \sum_{\eta'}(\prod_{ie} M_{ie}(\eta(ie),\eta'(ie)))A_v(\eta',\sigma)$.

Definition 16.44 (Apply gauge)**.** The PEPS tensor obtained by applying edge-gauge matrices to A, with vertex tensors A_v^X.

Definition 16.45 (Gauge equivalence for PEPS)**.** Two PEPS tensors A, B are gauge-equivalent if they have the same bond dimensions and B is obtained from A by applying invertible gauge matrices on each edge.

Theorem 16.46 (Gauge invariance of PEPS state)**.** *Applying a gauge to a PEPS tensor preserves the state coefficients.*

Theorem 16.47 (Gauge equivalence implies same state)**.** *Gauge equivalence implies that the two PEPS tensors represent the same state.*

Definition 16.48 (Local tensor evaluation)**.** The local tensor evaluated at vertex v with virtual-index weighting f, computing $\sum_\eta(\prod_{ie} f(ie)(\eta(ie))) \cdot A_v(\eta,\sigma)$. This is multilinear in the components of f, not linear in the full tuple.

Theorem 16.49 (Local gauge extraction)**.** *At a single vertex, vertex-injectivity of both tensors and the same-state condition force a local gauge relation.*

Theorem 16.50 (Gauge consistency)**.** *Local gauges extracted at adjacent vertices are consistent across shared edges, yielding a global gauge family.*

Theorem 16.51 (Fundamental Theorem for injective PEPS)**.** *If two PEPS tensors on a simple graph are both vertex-injective and generate the same state, then they are gauge-equivalent (Theorem 2 of [MGRPG+18]).*

Theorem 16.52 (Gauge uniqueness up to scalar)**.** *On a connected graph, the gauge family in the Fundamental Theorem is unique up to a global multiplicative scalar.*

16.13 Closing remarks

Remark 16.53. Appendix A of [MGRPG+18] strengthens the normal-MPS bound from $n \geq 3L$ to $n \geq 2L+1$ by a more refined blocking argument. The injective PEPS theorem and the normal PEPS theorem on arbitrary graphs both rest on the same two core lemmas (Lemma 16.11 and Lemma 16.12); the graph geometry enters only through the blocking step that reduces to a three-site chain.

Chapter 17

Symmetries and String Order

This chapter collects the symmetry-theoretic consequences of the MPS Fundamental Theorem. For injective tensors, physical on-site symmetries give virtual gauges, those gauges define projective representations on the bond space, and the same virtual data controls string-order phenomena. The algebraic theorem supplies the gauge mechanism, and this chapter develops its symmetry consequences.

17.1 Physical Symmetries Give Virtual Gauges

Corollary 17.1 (Physical symmetry implies virtual gauge)**.** *Let A be an injective MPS tensor and let $U \in \mathrm{GL}_d(\mathbb{C})$ be a unitary acting on the physical index. Fix $n \geq 3$ and suppose the n-site TI state satisfies*

$$U^{\otimes n} |\psi_n(A)\rangle = |\psi_n(A)\rangle.$$

Then there exist $V \in \mathrm{GL}_D(\mathbb{C})$ and a scalar $\lambda \in \mathbb{C}^\times$ with $\lambda^n = 1$ such that

$$\sum_{j=0}^{d-1} U_{ij}\, A^j = \lambda\, V^{-1}\, A^i\, V$$

for every physical index i.

Proof. Define $B^i := \sum_j U_{ij} A^j$. Since U is invertible, $\{B^i\}$ spans $M_D(\mathbb{C})$, so B is injective. The symmetry condition means the TI chains $(A, \dots, A)$ and $(B, \dots, B)$ of length n generate the same state. Corollary 16.19 gives $V \in \mathrm{GL}_D(\mathbb{C})$ and λ with $\lambda^n = 1$ and $B^i = \lambda\, V^{-1}\, A^i\, V$ for all i. ☐

Definition 17.2 (Physical-index rotation)**.** Given a matrix $u \in M_d(\mathbb{C})$ and an MPS tensor A, the *physical-index rotation* is the tensor $(A_u)^i := \sum_j u_{ij}\, A^j$.

Theorem 17.3 (Transfer map preserved by physical rotation)**.** *If $uu^\dagger = I$, then the transfer map is invariant under physical-index rotation: $\mathcal{E}_{A_u} = \mathcal{E}_A$.*

Proof. Unitary freedom of the Kraus representation (Theorem 2.18, Wolf *Quantum Channels & Operations*). ☐

Theorem 17.4 (Left-canonical preserved by physical rotation)**.** *If $uu^\dagger = I$ and A is left-canonical, then A_u is left-canonical.*

Proof. Expand $\sum_i B_i^\dagger B_i$ where $B_i = \sum_j u_{ij} A_j$, swap sums, apply orthogonality $u^\dagger u = I$, and collapse to $\sum_j A_j^\dagger A_j = I$. □

Theorem 17.5 (Irreducibility preserved by physical rotation). *If $uu^\dagger = I$ and A is irreducible, then A_u is irreducible.*

Proof. Contrapositive: any nontrivial invariant projection P for the rotated tensor yields $(1 - P)\, A_k\, P = 0$ for all k via the orthogonality of u, contradicting irreducibility of A. □

Theorem 17.6 (Periodicity preserved by physical rotation). *If $uu^\dagger = I$ and A is periodic with period m, then A_u is periodic with the same period.*

Proof. Assembles transfer-map invariance, left-canonical preservation, and irreducibility preservation. □

Theorem 17.7 (SameMPV preserved by physical rotation). *If $\mathcal{V}(A) = \mathcal{V}(B)$, then $\mathcal{V}(A_u) = \mathcal{V}(B_u)$.*

Proof. Induction on the word length with a strengthened hypothesis carrying an arbitrary prefix matrix, reducing each step to the hypothesis via word concatenation. □

Theorem 17.8 (Physical rotation distributes over block assembly). *The rotated block-diagonal tensor satisfies $(\bigoplus_k \mu_k A_k)_u = \bigoplus_k \mu_k (A_k)_u$.*

Proof. Pointwise matrix-entry computation using linearity of the block-diagonal embedding and reindexing. □

Theorem 17.9 (Irreducible form preserved by physical rotation). *If $uu^\dagger = I$ and A is in irreducible form II, then A_u is in irreducible form II with the same block structure and rotated blocks.*

Proof. Each block remains periodic by Theorem 17.6, the SameMPV condition transfers via Theorem 17.7, and block-assembly compatibility follows from Theorem 17.8. □

Corollary 17.10 (Periodic symmetry yields $\mathbb{Z}$-gauge equivalence). *Let A be in irreducible form II and let u be a matrix such that $B := A_u$ is also in irreducible form II and $\mathcal{V}(A) = \mathcal{V}(B)$. Then there exists $m > 0$ such that A and B are $\mathbb{Z}_m$-gauge equivalent.*

Proof. Direct application of the periodic equal-case fundamental theorem (Theorem 14.28) to A and B. □

17.2 Virtual Representation Theorem

With a full group of on-site symmetries (rather than just a single unitary), the resulting gauges determine a projective representation on the bond space.

Definition 17.11 (Twisted tensor). Given a group representation $U : G \to \mathrm{GL}_d(\mathbb{C})$ on the physical index, the *g-twisted tensor* is

$$\widetilde{A}_g^i := \sum_{j=0}^{d-1} U(g)_{ij}\, A^j \qquad \text{for each } i \in \{0, \dots, d-1\}.$$

Definition 17.12 (On-site symmetry)**.** A tensor A is *on-site symmetric* under a group representation $U : G \to \mathrm{GL}_d(\mathbb{C})$ if $\mathcal{V}(A) = \mathcal{V}(\widetilde{A}_g)$ for every $g \in G$.

Lemma 17.13 (Functoriality of twisting)**.** *The twisted tensor satisfies the composition law*

$$\widetilde{A}_{gh} = \widetilde{(\widetilde{A}_h)}_g,$$

i.e. twisting by gh is the same as first twisting by h and then by g.

Proof. Expanding both sides using $U(gh) = U(g)\,U(h)$, then swapping the order of summation. □

Lemma 17.14 (Identity twist)**.** *Twisting by the identity group element is trivial: $\widetilde{A}_1 = A$.*

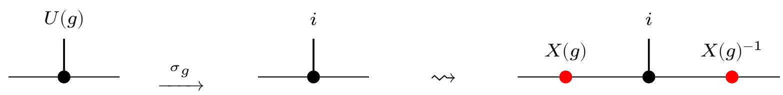
Proof. Since $U(1) = I_d$, each component reduces to $\sum_j \delta_{ij}\, A^j = A^i$. □

Lemma 17.15 (Symmetric twists are gauge equivalent)**.** *If A is injective and on-site symmetric under U, then for every $g \in G$ the twisted tensor $\widetilde{A}_g$ is gauge equivalent to A.*

Proof. On-site symmetry gives $\mathcal{V}(A) = \mathcal{V}(\widetilde{A}_g)$, and equal MPV for injective tensors implies gauge equivalence. □

Theorem 17.16 (Virtual symmetry equation)**.** *If A is injective and on-site symmetric under U, then for every $g \in G$ there exist an invertible matrix $X(g) \in \mathrm{GL}_D(\mathbb{C})$ and a nonzero scalar $\varphi(g) \in \mathbb{C}^\times$ (in fact $\varphi = 1$ in the single-block case) such that $\widetilde{A}^i_g = \varphi(g)\, X(g)\, A^i\, X(g)^{-1}$ for all i.*

Proof. The physical action is transferred to the virtual level by the same local move that replaces the twisted tensor by a gauge-conjugated one:



Apply Lemma 17.15 to obtain $X(g)$, then set $\varphi(g) = 1$. □

Lemma 17.17 (Gauge uniqueness up to scalar)**.** *Let A be an injective tensor. If $X, Y \in \mathrm{GL}_D(\mathbb{C})$ both satisfy $B^i = X\, A^i\, X^{-1} = Y\, A^i\, Y^{-1}$ for all i, then there exists a nonzero scalar $u \in \mathbb{C}^\times$ such that $Y = u \cdot X$.*

Proof. The matrix $Z = Y^{-1}X$ commutes with every A^i. Since A is injective, $\{A^i\}$ spans $M_D(\mathbb{C})$, so Z lies in the centre of $M_D(\mathbb{C})$. By Lemma 16.10, Z is a scalar matrix. Since Z is invertible, that scalar is nonzero, and $Y = u \cdot X$. □

Definition 17.18 (Scalar 2-cochain)**.** A *scalar 2-cochain* on a group G is a function $\omega : G \times G \to \mathbb{C}^\times$.

Definition 17.19 (Projective representation)**.** Given a scalar 2-cochain ω, a *projective representation* of G on $\mathbb{C}^D$ is a map $\rho : G \to \mathrm{GL}_D(\mathbb{C})$ satisfying

$$\rho(g)\, \rho(h) = \omega(g, h)\, \rho(gh) \qquad \text{for all } g, h \in G.$$

Lemma 17.20 (Associativity forces the cocycle condition)**.** *If ρ is a projective representation with factor system ω and $D > 0$, then ω satisfies the 2-cocycle identity:*

$$\omega(g, h)\, \omega(gh, k) = \omega(g, hk)\, \omega(h, k).$$

Proof. Expanding $\rho(g)\,(\rho(h)\rho(k))$ and $(\rho(g)\rho(h))\rho(k)$ using the projective multiplication law and equating via matrix associativity. The hypothesis $D > 0$ allows cancellation of the invertible matrix factor. $\square$

Theorem 17.21 (Virtual representation theorem for injective MPS)**.** *Let A be an injective MPS tensor that is on-site symmetric under a group representation $U : G \to \mathrm{GL}_d(\mathbb{C})$. Then there exist:*

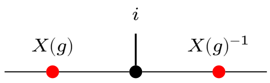
1. *a scalar 2-cochain $\omega : G \times G \to \mathbb{C}^\times$, and*
2. *a map $\rho : G \to \mathrm{GL}_D(\mathbb{C})$ satisfying*

$$\rho(g)\,\rho(h) = \omega(g,h)\,\rho(gh) \qquad \text{for all } g,h \in G,$$

such that, defining the gauge matrices $X(g) := \rho(g^{-1})$, one has

$$\widetilde{A}^i_g = X(g)\,A^i\,X(g)^{-1} \qquad \text{for every } g \in G, i \in \{0,\dots,d-1\}.$$

In tensor-network notation, for each fixed g this is the local gauge relation



with the black node denoting the tensor named in the surrounding formula and the two red side nodes acting on the virtual legs. In particular, ρ is a projective representation of G on the bond space $\mathbb{C}^D$.

Proof. Since A is injective and on-site symmetric, each twisted tensor $\widetilde{A}_g$ is gauge equivalent to A (Lemma 17.15). Choose $X(g) \in \mathrm{GL}_D(\mathbb{C})$ with $\widetilde{A}^i_g = X(g)\,A^i\,X(g)^{-1}$ for all i.

By functoriality (Lemma 17.13), the product $X(h)\,X(g)$ also intertwines A with $\widetilde{A}_{gh}$. Since $X(gh)$ does the same, gauge uniqueness (Lemma 17.17) gives a nonzero scalar $\omega'(h,g)$ with $X(h)\,X(g) = \omega'(h,g)\,X(gh)$.

Defining $\rho(g) := X(g^{-1})$ converts this to the standard law: $\rho(g)\,\rho(h) = \omega(g,h)\,\rho(gh)$ where $\omega(g,h) := \omega'(h^{-1},g^{-1})$. $\square$

Corollary 17.22 (2-cocycle identity for the factor system)**.** *If $D > 0$, the factor system ω from Theorem 17.21 satisfies the 2-cocycle identity:*

$$\omega(g,h)\,\omega(gh,k) = \omega(g,hk)\,\omega(h,k) \qquad \text{for all } g,h,k \in G.$$

Proof. Theorem 17.21 gives ρ and ω. By Lemma 17.20, the cocycle identity follows from associativity of matrix multiplication and invertibility of the representation matrices. $\square$

17.3 Cohomology Classes of Virtual Symmetry Data

Two cocycles may differ by a coboundary yet represent the same projective-representation class. The following definitions and results make this precise and establish the quotient $H^2(G,\mathbb{C}^\times)$.

Definition 17.23 (Coboundary)**.** A scalar 2-cocycle ω is a *coboundary* if there exists $\varphi \colon G \to \mathbb{C}^\times$ such that

$$\omega(g,h) = \varphi(g)\varphi(h)\varphi(gh)^{-1} \qquad \text{for all } g,h \in G.$$

Definition 17.24 (Cohomologous cocycles)**.** Two cocycles ω_1, ω_2 are *cohomologous* (written $\omega_1 \sim \omega_2$) if there exists $\varphi\colon G \to \mathbb{C}^\times$ with

$$\omega_1(g,h) = \varphi(g)\varphi(h)\varphi(gh)^{-1}\omega_2(g,h) \qquad \text{for all } g,h \in G.$$

Theorem 17.25 (Cohomologous-to is an equivalence relation)**.** *The relation "cohomologous to" is reflexive, symmetric, and transitive on the set of scalar 2-cocycles.*

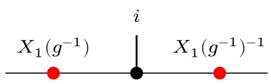
Proof. Reflexivity: take $\varphi = 1$. Symmetry: replace φ by φ^{-1}. Transitivity: compose the witnesses φ and ψ pointwise. □

Lemma 17.26 (Coboundary iff cohomologous to the trivial cocycle)**.** *A cocycle ω is a coboundary if and only if $\omega \sim \mathbf{1}$, where $\mathbf{1}(g,h) = 1$ for all g,h.*

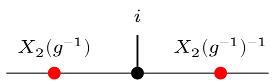
Proof. Both directions follow by absorbing or introducing the factor $\mathbf{1}(g,h) = 1$ via $x \cdot 1 = x$. □

Theorem 17.27 (Gauge independence of the cocycle class)**.** *If the bond dimension satisfies $D \geq 1$ and two projective representations ρ_1, ρ_2 (with cocycles ω_1, ω_2) both arise as virtual representations of the same injective MPS tensor A under an on-site symmetry U, then $\omega_1 \sim \omega_2$.*

Proof. For each g, gauge uniqueness gives a nonzero scalar $f(g)$ with $X_2(g^{-1}) = f(g)\, X_1(g^{-1})$. Equivalently, for each fixed g the two local gauges



and



describe the same twisted tensor, so they differ by a scalar. Substituting into the projective law and cancelling $X_1(g \cdot h)$ yields $\omega_2(g,h) = f(g)\, f(h)\, f(gh)^{-1}\omega_1(g,h)$. □

Definition 17.28 (Multiplicative 2-cocycle condition)**.** A scalar 2-cochain $\omega\colon G \times G \to \mathbb{C}^\times$ is a *2-cocycle* if it satisfies

$$\omega(g,h)\omega(gh,k) = \omega(g,hk)\omega(h,k) \qquad \text{for all } g,h,k \in G.$$

Definition 17.29 (Second cohomology quotient)**.** The second cohomology set is the quotient

$$H^2(G, \mathbb{C}^\times) := Z^2(G, \mathbb{C}^\times)/\!\sim,$$

where $\sim$ is the coboundary-equivalence relation.

Definition 17.30 (Projective equivalence)**.** Two projective representations are *projectively equivalent* when their factor systems are cohomologous.

Theorem 17.31 (Projective equivalence and cohomologous cocycles)**.** *Let ρ_1, ρ_2 be projective representations with factor systems ω_1, ω_2. Then*

$$\rho_1 \sim_{\text{proj}} \rho_2 \iff \omega_1 \sim \omega_2.$$

Proof. This is immediate from the definition of projective equivalence as cohomology-class equality for factor systems. □

17.4 Permutation-Based Physical Symmetry

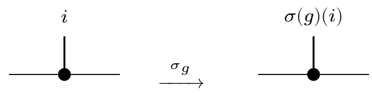
The linear-combination twist $\widetilde{A}^i_g = \sum_j U(g)_{ij}\, A^j$ is the standard form used in the virtual representation theorem. An alternative formulation replaces the matrix action by a permutation of the physical index; this is useful when the symmetry acts by reordering basis states rather than by a general linear map.

Definition 17.32 (Permutation symmetry data)**.** An *on-site symmetry datum* for a group G on a physical space of dimension d consists of a matrix-valued map $u : G \to M_d(\mathbb{C})$ and a physical-index action $\sigma : G \to \mathrm{End}(\{0,\dots,d{-}1\})$.

Definition 17.33 (Permutation-twisted tensor)**.** Given on-site symmetry data (u,σ), the *permutation-twisted tensor* at group element g is

$$(A^{\sigma_g})^i := A^{\sigma(g)(i)}.$$

In tensor-network notation the local tensor is unchanged and only the physical leg is relabelled:



Lemma 17.34 (Identity permutation twist)**.** *If $\sigma(1) = \mathrm{id}$, then $(A^{\sigma_1})^i = A^i$ for all i.*

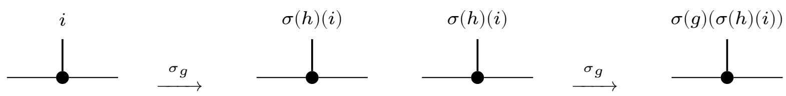
Proof. Immediate from $\sigma(1)(i) = i$ for all i. □

Composition order. The linear-combination twist (Lemma 17.13) composes as $\widetilde{(\widetilde{A}_h)}_g = \widetilde{A}_{gh}$, applying h first then g. The permutation twist below uses the same left-action convention $\sigma(gh) = \sigma(g)\circ\sigma(h)$, so $(A^{\sigma_g})^{\sigma_h} = A^{\sigma_{gh}}$; the inner permutation $\sigma(h)$ acts first on the index.

Lemma 17.35 (Composition of permutation twists)**.** *If $\sigma(gh) = \sigma(g)\circ\sigma(h)$ for all $g,h \in G$, then*

$$A^{\sigma_{gh}} = (A^{\sigma_g})^{\sigma_h}.$$

Proof. Expanding both sides: $(A^{\sigma_{gh}})^i = A^{\sigma(g)(\sigma(h)(i))} = ((A^{\sigma_g})^{\sigma_h})^i$. Diagrammatically this is just two consecutive relabellings of the same local tensor:



□

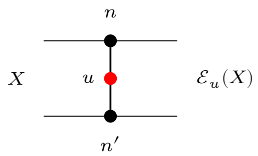
17.5 String Order and Local Symmetry Equivalence

The string order parameter [PGWS+08] detects whether an on-site unitary u acts as a local symmetry on the state generated by an injective MPS tensor.

Definition 17.36 (Twisted transfer map)**.** Given an MPS tensor A and a physical-index matrix u, the *twisted transfer map* is

$$\mathcal{E}_u(X) := \sum_{n,n'} \langle n'|u|n\rangle A_n\, X\, A^\dagger_{n'}.$$

Diagrammatically one inserts the physical operator u on the doubled local transfer cell:

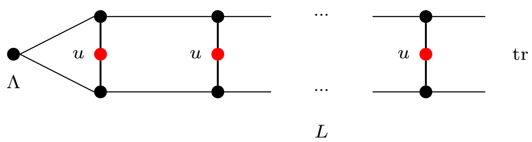


The upper and lower black nodes denote A_n and $A^\dagger_{n'}$, and the red node labelled u couples the physical legs.

Definition 17.37 (String order parameter)**.** The *string order parameter* for twist u and stationary state Λ at length L is

$$R_L(u) := \mathrm{tr}(\Lambda \cdot \mathcal{E}_u^L(\mathbb{1})).$$

In tensor-network notation this is the length-L doubled chain with a copy of u inserted on each physical rung:



The boundary matrix Λ is contracted into the virtual boundary, the red nodes labelled u sit on the physical rungs, and the right boundary is traced.

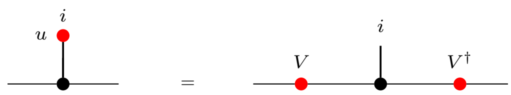
Definition 17.38 (Local symmetry)**.** An MPS tensor A has *local symmetry* under a unitary u with respect to a boundary state Λ if there exist a unitary matrix V and a phase μ with $|\mu| = 1$ such that $V^\dagger \Lambda V = \Lambda$ and

$$\sum_j u_{ij}\, A^j = \mu\, V\, A^i\, V^\dagger \quad \text{for all } i.$$

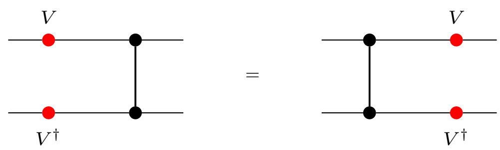
Informally, this means $\|R_L(u)\| = \|R_L(\mathbb{1})\|$ for all L.

Definition 17.39 (String order)**.** String order *exists* for u with respect to a boundary state Λ if there exist boundary matrices X, Y and a positive constant $c > 0$ such that $c \leq \|R_L(u; X, Y)\|$ for all L, where $R_L(u; X, Y)$ is the boundary-refined string-order parameter. Informally, this means the scalar string-order parameter $R_L(u)$ does not decay to zero.

Definition 17.40 (Local physical intertwining)**.** The local physical intertwining relation is $\sum_j u_{ij}\, A^j = V\, A^i\, V^\dagger$. Locally, the physical action of u can therefore be pushed to the virtual legs:



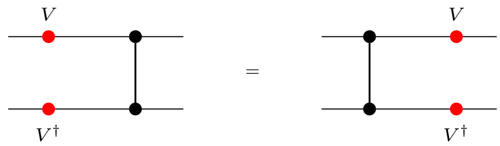
Definition 17.41 (Transfer-map covariance)**.** Transfer-map covariance is the identity that the transfer map commutes with virtual conjugation: $\mathcal{E}(V X V^\dagger) = V\, \mathcal{E}(X)\, V^\dagger$. In doubled-layer notation, the virtual operators slide through the local transfer cell:



Definition 17.42 (Commutation in the doubled transfer picture)**.** In the doubled transfer picture, one asks that the doubled transfer matrix commutes with $V \otimes \bar{V}$, expressed as $V\,\mathcal{E}_1(X)\,V^\dagger = \mathcal{E}_1(VXV^\dagger)$.

Theorem 17.43 (Transfer-map covariance and doubled commutation are equivalent)**.** *Transfer-map covariance and commutation in the doubled transfer picture are equivalent for any matrix V.*

Proof. Both sides express the same local picture with opposite orientation:



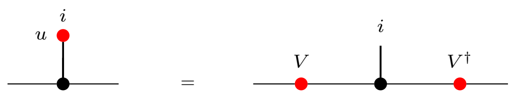
The two formulas are literally the same identity read from opposite sides. $\square$

Lemma 17.44 (Unitary Kraus mixing)**.** *If u is unitary, then replacing Kraus operators $\{A_i\}$ by their unitary mixtures $\{\sum_j u_{ij} A_j\}$ preserves the channel: $\sum_i (\sum_j u_{ij} A_j)\, Y\, (\sum_j u_{ij} A_j)^\dagger = \sum_i A_i\, Y\, A_i^\dagger$.*

Proof. This is exactly the usual invariance of a Kraus decomposition under a unitary change of Kraus operators; see Section 5.8. $\square$

Theorem 17.45 (Local physical intertwining implies transfer-map covariance)**.** *If V and u are unitary and the local physical intertwining relation holds, then the transfer map is covariant.*

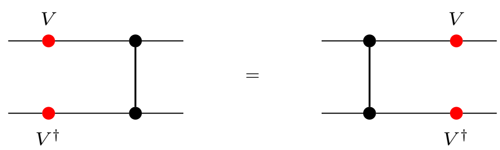
Proof. Starting from



one inserts $V^\dagger V = \mathbb{1}$ so that each local transfer cell is written in terms of conjugated Kraus operators, then replaces those operators using the intertwining relation. After that, the physical u-boxes disappear by the unitary Kraus-mixing identity. $\square$

Theorem 17.46 (Transfer-map covariance implies local physical intertwining)**.** *For an injective MPS, if V is unitary and the transfer map is covariant, then there exists a unitary u satisfying the local physical intertwining relation.*

Proof. Set $B^i := V\,A^i\,V^\dagger$. The transfer-map covariance relation



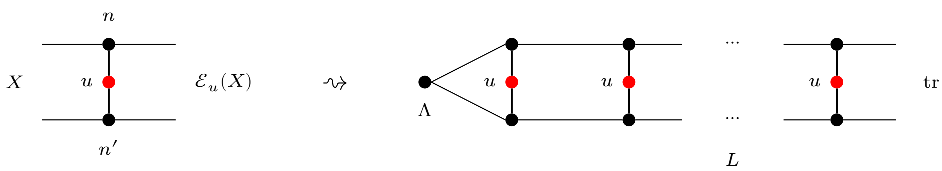
implies that the two Kraus families B and A define the same transfer map. The missing step — supplied by Kraus freedom for equal channels — converts equality of transfer maps $\mathcal{E}_B = \mathcal{E}_A$ into a unitary Kraus mixing matrix u with $B^i = \sum_j u_{ij} A^j$. $\square$

Theorem 17.47 (Spectral radius bound)**.** *For an injective MPS, assume* $uu^\dagger = \mathbb{1}$*,* $\mathcal{E}_A(\mathbb{1}) = \mathbb{1}$*,* Λ *is positive definite,* $\operatorname{tr}(\Lambda) = 1$*, and* $\mathcal{E}_A^\dagger(\Lambda) = \Lambda$*. Then every eigenvalue* λ *of* $\mathcal{E}_u$ *satisfies* $|\lambda| \le 1$*.*

Proof. Requires Cauchy–Schwarz for CP maps and the unique fixed-point property of the transfer map (Perron–Frobenius for CP maps). □

Theorem 17.48 (Local symmetry ⇔ spectral radius one)**.** *For a pure FCS, assume* $uu^\dagger = \mathbb{1}$*,* $\mathcal{E}_A(\mathbb{1}) = \mathbb{1}$*,* Λ *is positive definite,* $\operatorname{tr}(\Lambda) = 1$*, and* $\mathcal{E}_A^\dagger(\Lambda) = \Lambda$*. Then* u *is a local symmetry iff* $\rho(\mathcal{E}_u) = 1$*.*

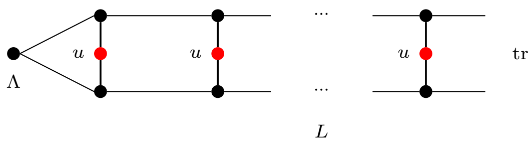
Proof. Requires spectral theory of completely positive maps (Perron–Frobenius for CP maps). Diagrammatically one is studying the asymptotics of the doubled string-order chain



by iterating the local twisted transfer cell. □

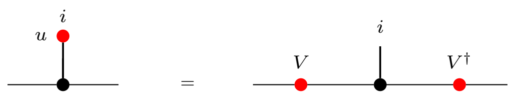
Theorem 17.49 (String order ⇔ local symmetry)**.** *For an injective MPS, assume* $uu^\dagger = \mathbb{1}$*,* $\mathcal{E}_A(\mathbb{1}) = \mathbb{1}$*,* Λ *is positive definite,* $\operatorname{tr}(\Lambda) = 1$*, and* $\mathcal{E}_A^\dagger(\Lambda) = \Lambda$*. Then string order for* u *exists iff* u *is a local symmetry.*

Proof. Requires spectral theory of completely positive maps (Perron–Frobenius for CP maps). The quantity

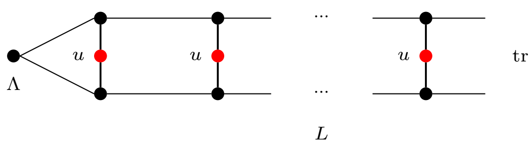


is the tensor-network expression whose non-decay is equivalent to spectral radius one. □

Theorem 17.50 (Virtual unitary from string order)**.** *For an injective MPS, assume* $uu^\dagger = \mathbb{1}$*,* $\mathcal{E}_A(\mathbb{1}) = \mathbb{1}$*,* Λ *is positive definite,* $\operatorname{tr}(\Lambda) = 1$*, and* $\mathcal{E}_A^\dagger(\Lambda) = \Lambda$*. If string order exists for* u*, then there exists a virtual unitary* V *and a unit-modulus scalar* μ *such that* $\sum_j u_{ij} A^j = \mu\, V A^i V^\dagger$ *for all* i*. Diagrammatically this is the same local intertwining relation, but only up to an overall unit scalar on the virtual side:*



Proof. Requires spectral theory of completely positive maps (Perron–Frobenius for CP maps). The tensor-network input is the non-decaying string-order chain



and the output is the phased local intertwiner shown above. □

17.6 SPT Phase Classification

Two injective symmetric tensors belong to the same symmetry-protected topological phase when their virtual representation cocycles are cohomologous [CGW11].

Definition 17.51 (Same SPT phase)**.** Injective tensors A and B, both symmetric under an on-site representation $U: G \to \mathrm{GL}_d(\mathbb{C})$, are in the *same SPT phase* if there exist virtual cocycles ω_A, ω_B with corresponding projective representations that intertwine the respective twisted tensors and satisfy $\omega_A \sim \omega_B$ (cohomologous).

Theorem 17.52 (String order holds for injective symmetric MPS)**.** *Let A be an injective tensor, symmetric under a unitary representation U. Assume $\mathcal{E}_A(\mathbb{1}) = \mathbb{1}$, Λ is positive definite, $\mathrm{tr}(\Lambda) = 1$, and $\mathcal{E}_A^\dagger(\Lambda) = \Lambda$. Then string order exists for every group element g: the twisted transfer map $\mathcal{E}_{U(g)}$ has the virtual gauge matrix as a fixed point with eigenvalue 1.*

Proof. The virtual representation theorem produces $\rho(g^{-1}) \in \mathrm{GL}_D(\mathbb{C})$ satisfying $\sum_j U(g)_{ij}\, A^j = \rho(g^{-1})\, A^i\, \rho(g^{-1})^{-1}$. A direct computation using $\mathcal{E}_A(\mathbb{1}) = \mathbb{1}$ shows that $\mathcal{E}_{U(g)}(\rho(g^{-1})) = \rho(g^{-1})$, so $\rho(g^{-1})$ is a nonzero eigenvector with eigenvalue 1. Since $|1| = 1$, the modulus-one rigidity theorem yields gauge-phase equivalence, from which one extracts a virtual unitary W and a unit-modulus phase μ satisfying the local symmetry conditions. The boundary-state invariance $W^\dagger \Lambda W = \Lambda$ then gives local symmetry, which implies string order. □

Theorem 17.53 (String order is an SPT invariant)**.** *If A and B are injective, both on-site symmetric under a unitary representation U, and if $\mathcal{E}_A(\mathbb{1}) = \mathbb{1}$, $\mathcal{E}_B(\mathbb{1}) = \mathbb{1}$, Λ_A and Λ_B are positive definite, $\mathrm{tr}(\Lambda_A) = 1$, $\mathrm{tr}(\Lambda_B) = 1$, $\mathcal{E}_A^\dagger(\Lambda_A) = \Lambda_A$, and $\mathcal{E}_B^\dagger(\Lambda_B) = \Lambda_B$, then for every $g \in G$, string order exists for A with twist $U(g)$ if and only if it exists for B. In particular, since Definition 17.51 implies on-site symmetry for both tensors, this theorem applies to tensors in the same SPT phase.*

Proof. Both directions follow from Theorem 17.52, which shows that string order holds universally for injective symmetric tensors with canonical normalisations. □

Chapter 18

Parent Hamiltonians

This chapter introduces the local ground-space construction and the parent interaction for translation-invariant periodic MPS. We follow the standard setup from [CPGSV21, PGVWC07], with the N-site Hilbert space modelled as coefficient functions

$$\{0, \dots, d-1\}^N \to \mathbb{C},$$

rather than as an explicit tensor product.

18.1 Local ground space $G_L(A)$

Fix an MPS tensor $A = \{A^i\}_{i=0}^{d-1}$ with $A^i \in M_D(\mathbb{C})$ and a block length $L \geq 1$. For a word $w = (i_1, \dots, i_L)$, write $A^w := A^{i_1} \cdots A^{i_L}$. The local ground-space map is

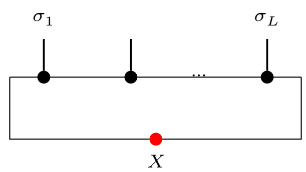
$$\Gamma_L : M_D(\mathbb{C}) \to (\{0, \dots, d-1\}^L \to \mathbb{C}), \qquad \Gamma_L(X)(\sigma) := \mathrm{tr}(A^\sigma X),$$

where A^σ abbreviates $A^{\sigma(0)} \cdots A^{\sigma(L-1)}$.

Definition 18.1 (Ground-space map)**.** The map Γ_L above is a $\mathbb{C}$-linear map

$$\Gamma_L : M_D(\mathbb{C}) \to (\{0, \dots, d-1\}^L \to \mathbb{C}).$$

In tensor-network notation,



the upper chain denotes the word tensor A^σ with open physical indices $\sigma_1, \dots, \sigma_L$, and the lower red node denotes the boundary matrix X inserted on the closing virtual bond before taking the trace. The chain length L is fixed by the surrounding formula, not by an extra label inside the diagram.

Definition 18.2 (Local ground space)**.** The *local ground space* is the image

$$G_L(A) := \mathrm{im}(\Gamma_L) \subseteq (\{0, \dots, d-1\}^L \to \mathbb{C}).$$

Lemma 18.3 (Dimension bound)**.** *For every L,*

$$\dim G_L(A) \le D^2.$$

Proof. Since $G_L(A)$ is the range of a linear map from $M_D(\mathbb{C})$,

$$\dim G_L(A) \le \dim M_D(\mathbb{C}) = D^2.$$

□

Lemma 18.4 (Ambient-space dimension)**.** *The coefficient-function model satisfies*

$$\dim(\{0, \dots, d-1\}^L \to \mathbb{C}) = d^L.$$

Lemma 18.5 (Nontriviality criterion)**.** *If $d^L > D^2$, then $G_L(A)$ is a proper subspace of the local Hilbert space:*

$$G_L(A) \ne (\{0, \dots, d-1\}^L \to \mathbb{C}).$$

Proof. If $G_L(A)$ were the whole space, its dimension would be d^L by Lemma 18.4; Lemma 18.3 gives the contradiction $d^L \le D^2$. □

18.2 Parent interaction and chain Hamiltonian

The parent interaction at range L is the orthogonal projector onto $G_L(A)^\perp$ in the L-site Euclidean space.

Definition 18.6 (Parent interaction)**.** The parent interaction at range L is the orthogonal projector

$$h_L(A) := \Pi_{G_L(A)^\perp},$$

on the local L-site space.

Lemma 18.7 (Parent interaction kills ground-space elements)**.** *For any $v \in G_L(A)$, we have $h_L(A)\, v = 0$.*

Proof. Since $h_L(A)$ is the star-projection onto $G_L(A)^\perp$, it annihilates every element of $G_L(A)$. □

For a periodic chain of length N, the parent Hamiltonian is the sum of translated copies of the local interaction.

Definition 18.8 (Translated local term)**.** For each site index $i \in \{0, \dots, N-1\}$, the translated local term h_i embeds the parent interaction into the full N-site space at position i (with periodic boundary conditions). Thus h_i acts on the cyclic interval $[i, i+L-1]$ and as the identity outside that interval.

Definition 18.9 (Parent Hamiltonian)**.** The chain Hamiltonian is

$$H_N(A, L) := \sum_{i=0}^{N-1} h_i.$$

Definition 18.10 (Frustration-free condition)**.** A state ψ is frustration-free if each local term annihilates it:

$$\forall\, i \in \{0, \dots, N-1\}, \quad h_i \psi = 0.$$

Lemma 18.11 (MPV window membership)**.** *For any window of L consecutive sites starting at position i in an N-site periodic chain ($L \le N$), the restriction of the MPV state to that window lies in $G_L(A)$.*

Proof. The witness is the product of A-matrices on the complement sites. Trace cyclicity rotates the full N-site product so that window indices come first, matching the ground-space map definition. ☐

Lemma 18.12 (Local term annihilates the MPV)**.** *For each site i, $h_i\, V^{(N)}(A) = 0$.*

Proof. Combines MPV window membership with the fact that the parent interaction kills ground-space elements. ☐

Lemma 18.13 (Parent Hamiltonian annihilates the MPV)**.** *For every N, the MPV state satisfies*

$$H_N(A, L)\, V^{(N)}(A) = 0.$$

Proof. Sum of zeros, since each local term annihilates the MPV. ☐

Lemma 18.14 (Frustration-freeness of the MPV)**.** *The MPV state is frustration-free for the parent model.*

Proof. Immediate from Lemma 18.12. ☐

18.3 Intersection property and unique ground state

This section develops the intersection property of local ground spaces for injective MPS and the resulting uniqueness of the ground state on the periodic chain.

18.3.1 Restriction maps

Given a state ψ on $L+1$ sites, we define two restriction maps.

Definition 18.15 (Left restriction)**.** Fix the last physical index j:

$$(\mathrm{restrictLast} \psi j)(\sigma) := \psi(\sigma_1, \dots, \sigma_L, j).$$

Definition 18.16 (Right restriction)**.** Fix the first physical index i:

$$(\mathrm{restrictFirst} \psi i)(\sigma) := \psi(i, \sigma_1, \dots, \sigma_L).$$

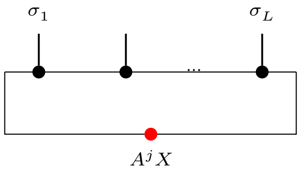
Definition 18.17 (Left ground membership)**.** A state ψ on $L+1$ sites satisfies the left ground condition if $\mathrm{restrictLast}\psi j \in G_L(A)$ for every j.

Definition 18.18 (Right ground membership)**.** A state ψ on $L+1$ sites satisfies the right ground condition if $\mathrm{restrictFirst}\psi i \in G_L(A)$ for every i.

18.3.2 Forward direction: ground space restricts

Lemma 18.19 (Left restriction)**.** *If $\psi \in G_{L+1}(A)$, i.e. $\psi(\sigma) = \mathrm{tr}(A^\sigma X)$ for some X, then* restrictLast$\psi j \in G_L(A)$ *with witness $A^j X$.*

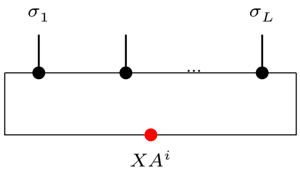
Proof. The restriction simply absorbs the last tensor into the boundary matrix:



Direct computation: $\psi(\sigma_1, \ldots, \sigma_L, j) = \mathrm{tr}(A^{\sigma_1} \cdots A^{\sigma_L} \cdot A^j \cdot X) = \Gamma_L(A^j X)(\sigma)$. □

Lemma 18.20 (Right restriction)**.** *If $\psi \in G_{L+1}(A)$ with $\psi(\sigma) = \mathrm{tr}(A^\sigma X)$, then* restrictFirst$\psi i \in$ *$G_L(A)$ with witness XA^i.*

Proof. On the right restriction one first rotates the trace and then reads the resulting boundary matrix on the shortened window:



By trace cyclicity: $\psi(i, \sigma_1, \ldots, \sigma_L) = \mathrm{tr}(A^i A^\sigma X) = \mathrm{tr}(A^\sigma \cdot XA^i) = \Gamma_L(XA^i)(\sigma)$. □

18.3.3 Injectivity of the ground-space map

Theorem 18.21 (Ground-space map injectivity)**.** *If A is injective and $L \geq 1$, then Γ_L is injective.*

Proof. If $\Gamma_L(X) = 0$ then $\mathrm{tr}(A^\sigma X) = 0$ for every length-L word σ. Since A is injective, the products $\{A^\sigma\}_\sigma$ span $M_D(\mathbb{C})$, so the trace pairing gives $X = 0$. □

Theorem 18.22 (Ground-space dimension)**.** *If A is injective and $L \geq 1$, then* $\dim G_L(A) = D^2$*.*

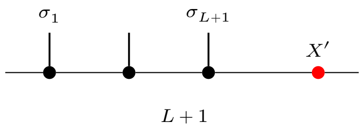
Proof. The upper bound $\dim G_L(A) \leq D^2$ is Lemma 18.3. Injectivity of Γ_L gives $\dim G_L(A) = \dim M_D(\mathbb{C}) = D^2$. □

18.3.4 The intersection property

Theorem 18.23 (Intersection property)**.** *If A is injective and $L \geq 2$, then a state ψ on $L+1$ sites satisfying both the left and right ground conditions lies in $G_{L+1}(A)$:*

$$\psi \in G_L^{\mathrm{left}} \cap G_L^{\mathrm{right}} \Longrightarrow \psi \in G_{L+1}(A).$$

Proof. The "invert-and-regrow" argument compares the left and right restrictions on their common $(L-1)$-site overlap and then reconstructs the full $(L+1)$-site word from a single boundary matrix:



1. From the right ground condition, for each physical index i there exists a unique $Y_i \in M_D(\mathbb{C})$ (by injectivity of Γ_L) such that $\psi(i,\sigma) = \mathrm{tr}(A^\sigma Y_i)$.

2. Similarly, from the left ground condition, for each j there exists a unique Z_j with $\psi(\sigma, j) = \mathrm{tr}(A^\sigma Z_j)$.

3. Matching on the $(L-1)$-site overlap and using nondegeneracy of the trace pairing: $A^j Y_i = Z_j A^i$ for all i, j.

4. Since A is injective, the matrices $\{A^j\}$ span $M_D(\mathbb{C})$ (Definition 2.20), so $\mathbb{1} = \sum_j c_j A^j$ for some coefficients $c_j \in \mathbb{C}$. Then $Y_i = \mathbb{1} \cdot Y_i = \sum_j c_j A^j Y_i = \sum_j c_j Z_j A^i = X' A^i$, where $X' := \sum_j c_j Z_j$.

5. By trace cyclicity, $\psi(\sigma) = \mathrm{tr}(A^\sigma X')$, so $\psi \in G_{L+1}(A)$.

□

Theorem 18.24 (Intersection characterization)**.** *For injective A and $L \geq 2$: $\psi \in G_{L+1}(A) \iff \psi \in G_L^{\text{left}} \cap G_L^{\text{right}}$.*

Proof. Immediate from the two forward-direction lemmas and Theorem 18.23. □

18.3.5 Unique ground state

Lemma 18.25 (Iterated contiguous-window intersection)**.** *If an N-site state satisfies the L-site ground condition on every non-wrapping contiguous window (with $L \geq 2$), then it lies in $G_N(A)$.*

Proof. The induction repeatedly merges two neighbouring admissible windows $[i, i+L-1]$ and $[i+1, i+L]$ into the larger window $[i, i+L]$ by applying Theorem 18.23 to their common overlap $[i+1, i+L-1]$. Strong induction on window size, combining adjacent windows via Theorem 18.23 at each step. □

Definition 18.26 (MPV submodule)**.** The subspace spanned by the MPV state $V^{(N)}(A)(\sigma) = \mathrm{tr}(A^{\sigma_0} \cdots A^{\sigma_{N-1}})$.

Lemma 18.27 (MPV lies in the ground space)**.** *$V^{(N)}(A) \in G_N(A)$ for every N, with boundary matrix $X = I$.*

Proof. $V^{(N)}(A)(\sigma) = \mathrm{tr}(A^\sigma) = \mathrm{tr}(A^\sigma \cdot I) = \Gamma_N(I)(\sigma)$. □

Definition 18.28 (Unique ground state)**.** A submodule S has a unique ground state if $\dim S = 1$.

Theorem 18.29 (One-dimensionality criterion)**.** *For finite-dimensional S,* $\dim S = 1$ *iff there exists a nonzero $\psi_0 \in S$ such that every $\psi \in S$ is of the form $c\,\psi_0$.*

Proof. Apply the standard finite-dimensional criterion that a nonzero space has dimension one exactly when all of its vectors are proportional to a single nonzero vector. ☐

Definition 18.30 (Periodic chain ground space)**.** For $N > 0$ and $L \leq N$, define the *periodic chain ground space*

$$\mathcal{G}_{N,L}(A) := \{\psi \in (\{0, \dots, d-1\}^N \to \mathbb{C}) : \text{for every cyclic window } [i, i+L-1], \psi|_{[i,i+L-1]} \in G_L(A)\},$$

where the condition $\psi|_{[i,i+L-1]} \in G_L(A)$ means that for every choice of physical indices outside the window, the restriction of ψ to that window lies in $G_L(A)$. When $N = 0$ or $L > N$, set $\mathcal{G}_{N,L}(A) := \top$ by convention.

Theorem 18.31 (MPV nonvanishing for block-injective tensors)**.** *If A is L_0-block-injective with $L_0 > 0$, $D \geq 1$, and $N \geq L_0 + 1$, then $V^{(N)}(A) \neq 0$ in the N-site space.*

Proof. Suppose for contradiction that $V^{(N)}(A) = 0$, i.e. $\operatorname{tr}(A^\sigma) = 0$ for every word σ of length N. For any words w and u with $|w| = N - L_0$ and $|u| = L_0$, we have $\operatorname{tr}(A^w A^u) = 0$. Since A is L_0-block-injective, $\{A^u : |u| = L_0\}$ spans $M_D(\mathbb{C})$. By nondegeneracy of the trace pairing, $A^w = 0$ for every $|w| = N - L_0$. Repeating the argument with words of length $N - 2L_0$ (since $|w'| + L_0 = N - L_0$, the product $A^{w'} A^{u_1}$ has length $N - L_0$, so $A^{w'} A^{u_1} = 0$ by the previous step) and iterating, we eventually reach $\operatorname{tr}(A^u) = 0$ for $|u| = L_0$, which gives $I = 0$, a contradiction. ☐

Theorem 18.32 (Chain ground space for block-injective tensors)**.** *If A is L_0-block-injective, $D \geq 1$, and the window size satisfies $L \geq 2L_0$, then on the periodic chain with $N \geq 2$ sites, the chain ground space equals the MPV submodule: $\mathcal{G}_{N,L}(A) = \operatorname{span}\{V^{(N)}(A)\}$.*

Remark. The injective special case ($L_0 = 1$) is proved; the block-injective generalization is future work.

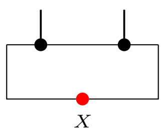
Theorem 18.33 (Chain ground space for normal tensors)**.** *If A is normal, L_0-block-injective, and $D \geq 1$, the window requirement is reduced from $2L_0$ to $L_0 < L$. The normality hypothesis enables this reduction via the structure theory of normal MPS (peripheral spectrum, canonical form); see Ref. [CPGSV21, §IV.C]. Concretely, the window range can be reduced from $2L_0$ to $L_0 + 1$ via the peripheral-spectrum structure of the transfer map (Theorem 11.31) and the block-separation argument (Theorem 12.16).*

Theorem 18.34 (Unique ground state on the periodic chain)**.** *For an injective tensor A with $D \geq 1$ on a periodic chain of $N \geq L$ sites with interaction range $L \geq 2$, the chain ground space is one-dimensional, spanned by the MPV.*

Proof. **Step 1 (iterate intersection).** By Lemma 18.25, any state ψ in the chain ground space lies in $G_N(A)$, so there exists $X \in M_D(\mathbb{C})$ with

$$\psi(\sigma_1, \dots, \sigma_N) = \operatorname{tr}(A^{\sigma_1} \cdots A^{\sigma_N} X).$$

In tensor-network form, the boundary operator sits on the closing virtual bond:



Step 2 (wrapping window forces X scalar). The chain ground space condition from Definition 18.30 requires $\psi|_W \in G_L(A)$ for every cyclic window W, including those that wrap around the boundary. In particular, the window starting at site $N-1$ gives

$$(\psi|_W)(i_N, i_1, \dots, i_{L-1}) = \mathrm{tr}(A^{i_1} \cdots A^{i_{L-1}} \cdot X A^{i_N}),$$

for some boundary matrix depending on X. Comparing this with the representation $\psi = \Gamma_N(X)$ from Step 1 and using trace cyclicity gives $A^j X = X A^j$ for every j. Since A is injective, the matrices $\{A^j : 0 \le j \le d-1\}$ span $M_D(\mathbb{C})$ (Definition 2.20), so X commutes with every element of $M_D(\mathbb{C})$. By Schur's lemma for the full matrix algebra, $X = \lambda I$ for some $\lambda \in \mathbb{C}$.

Step 3 (conclude). $\psi(\sigma) = \lambda\ \mathrm{tr}(A^\sigma) = \lambda\, V^{(N)}(A)(\sigma)$, so every ground state is proportional to the MPV. Since the MPV is nonzero by Theorem 18.31 applied with $L_0 = 1$, the ground space is one-dimensional. □

Theorem 18.35 (Unique ground state for block-injective tensors)**.** *If A is L_0-block-injective with $L_0 > 0$ and $D \ge 1$, the parent Hamiltonian with interaction range $2L_0$ has a unique ground state on the periodic chain.*

Theorem 18.36 (Optimal unique ground state for normal tensors)**.** *If A is normal (hence L_0-block-injective for some $L_0 > 0$), $D \ge 1$, and in normal form, the interaction range can be reduced to $L_0 + 1$.*

18.4 Commuting parent Hamiltonians

A parent Hamiltonian has *commuting* local terms when all translated interaction projectors commute with each other.

Definition 18.37 (Commuting parent Hamiltonian)**.** The parent Hamiltonian with block length L on N sites has *commuting* local terms if $h_i h_j = h_j h_i$ for all $i, j \in \{0, \dots, N-1\}$. See Ref. [CPGSV21, Definition 3.9].

Definition 18.38 (Nearest-neighbor commuting parent Hamiltonian)**.** A *nearest-neighbor commuting parent Hamiltonian* (NNCPH) is a commuting parent Hamiltonian with block length $L = 2$. See Ref. [CPGSV21, Definition 3.9].

Theorem 18.39 (RFP implies NNCPH)**.** *If A is a renormalization fixed point (RFP), then its parent Hamiltonian with $L = 2$ has commuting local terms. **Status:** A previous proof attempt was found incorrect and removed; this result requires the structural theory of renormalization fixed points needed for [CPGSV21, Theorem 3.10].*

18.5 Decorrelation and commuting parent Hamiltonians

This section develops the abstract operator-algebraic formulation of decorrelation and its connection to commuting parent Hamiltonians, following Appendix D, §D.2 of [CPGSV21].

Definition 18.40 (Decorrelation)**.** Given an idempotent endomorphism P_K and sets of observables $\mathcal{O}_A$, $\mathcal{O}_B$, regions A and B are *decorrelated* w.r.t. K when $P_K \circ O_A \circ (1 - P_K) \circ O_B \circ P_K = 0$ for all $O_A \in \mathcal{O}_A$, $O_B \in \mathcal{O}_B$.

Definition 18.41 (Commuting parent Hamiltonian structure)**.** A subspace K (given by its idempotent endomorphism P_K) has a *commuting parent Hamiltonian structure* if there exist idempotent endomorphisms P_{AX} and P_{XB} that commute and satisfy $P_{AX} \circ P_{XB} = P_K$.

Theorem 18.42 (Commuting parent Hamiltonian implies decorrelation)**.** *If a subspace K has a commuting parent Hamiltonian decomposition $P_K = P_{AX} \circ P_{XB}$ with $[P_{AX}, P_{XB}] = 0$, and A-observables commute with P_{XB} while B-observables commute with P_{AX}, then regions A and B are decorrelated w.r.t. K. This is the backward direction of Proposition D.3 from [CPGSV21], Appendix D, §D.2.*

Proof. The key identity is $P_{XB} \circ (1 - P_{AX} \circ P_{XB}) \circ P_{AX} = 0$. Using the commutation hypotheses to slide O_A past P_{XB} and O_B past P_{AX}, the five-fold composition factors through this zero term. □

Theorem 18.43 (Commuting parent Hamiltonian implies decorrelation (bundled))**.** *Corollary of Theorem 18.42: if P_K has a commuting parent Hamiltonian decomposition and observables respect locality, then regions A and B are decorrelated w.r.t. K.*

Lemma 18.44 (Frustration-free identity)**.** *For any endomorphisms P and Q,*

$$(1 - P) + (1 - Q) - (1 - P) \circ (1 - Q) = 1 - P \circ Q.$$

Proof. Expanding the left-hand side and cancelling gives the result. □

Theorem 18.45 (Commuting parent Hamiltonian identity)**.** *If $P_K = P_{AX} \circ P_{XB}$ is a commuting parent Hamiltonian decomposition, then*

$$Q_{AX} + Q_{XB} - Q_{AX} \circ Q_{XB} = 1 - P_K,$$

where $Q_{AX} = 1 - P_{AX}$ and $Q_{XB} = 1 - P_{XB}$ are the complementary projectors. This is the frustration-free form of the parent Hamiltonian from [CPGSV21], Appendix D, §D.2.

Proof. Applying Lemma 18.44 to P_{AX} and P_{XB}, and substituting $P_{AX} \circ P_{XB} = P_K$. □

Theorem 18.46 (Ground-space membership)**.** *If $P_K = P_{AX} \circ P_{XB}$ is a commuting parent Hamiltonian decomposition, then*

$$P_K v = v \quad \iff \quad P_{AX} v = v \text{ and } P_{XB} v = v.$$

This is the algebraic form of equation (D.2) from [CPGSV21]: $K_{AXB} = (K_{AX} \otimes H_B) \cap (H_A \otimes K_{XB})$.

Proof. ($\Rightarrow$) If $P_K v = v$, then $P_{AX}(P_K v) = (P_{AX} \circ P_K)v = P_K v = v$ by left absorption, and similarly for P_{XB}.

($\Leftarrow$) If $P_{AX} v = v$ and $P_{XB} v = v$, then $P_K v = (P_{AX} \circ P_{XB})v = P_{AX}(P_{XB} v) = P_{AX} v = v$. □

18.6 Spectral gap

A frustration-free Hamiltonian is *gapped* if the first excited eigenvalue is bounded away from zero uniformly in the chain length N. For MPS parent Hamiltonians, the spectral gap follows from a martingale criterion due to Kastoryano and Lucia [KL18].

Theorem 18.47 (Martingale criterion for spectral gap)**.** *Consider a frustration-free Hamiltonian*

$$H = \sum_{i=1}^{n} h_i$$

with projector interaction terms ($h_i^2 = h_i$). Suppose that for every pair of overlapping terms h_i, h_j there exist constants $c_{ij} \geq 0$ and $\gamma > 0$ satisfying

$$h_i h_j + h_j h_i \geq -c_{ij}(1-\gamma)(h_i + h_j),$$

and suppose further that $c_{ij} = c_{ji}$ for every overlapping pair (i,j) and that $\sum_{j\colon j\neq i} c_{ij} \leq 1$ for every i. Then the smallest positive eigenvalue of H satisfies $\lambda^+_{\min}(H) \geq \gamma$.

Proof. The relevant geometry is a family of translated length-L projector windows on the ring; two terms h_i and h_j interact only when their supports overlap, equivalently when the cyclic distance $d_N(i,j)$ is smaller than L. From $h_i^2 = h_i$,

$$H^2 = \sum_i h_i + \sum_{\substack{i<j \\ \text{overlapping}}} (h_i h_j + h_j h_i) + \sum_{\substack{i<j \\ \text{non-overlapping}}} (h_i h_j + h_j h_i).$$

Non-overlapping projectors commute ($h_i h_j = h_j h_i$), so their product $h_i h_j$ is itself a projector and hence PSD; thus the last sum is ≥ 0. For the overlapping sum, the martingale condition gives $h_i h_j + h_j h_i \geq -c_{ij}(1-\gamma)(h_i + h_j)$. Summing over unordered pairs $\{i,j\}$:

$$\sum_{\{i,j\},\text{overlap}} (h_i h_j + h_j h_i) \geq -(1-\gamma) \sum_{\{i,j\},\text{overlap}} c_{ij}(h_i + h_j).$$

Collecting terms for each h_i and using the symmetry assumption $c_{ij} = c_{ji}$:

$$\sum_{\{i,j\},\text{overlap}} c_{ij}(h_i + h_j) = \sum_i \Big(\sum_{\substack{j\neq i \\ \text{overlap}}} c_{ij} \Big) h_i \leq H,$$

where the last inequality uses the row-sum hypothesis $\sum_{j\neq i} c_{ij} \leq 1$. Combining, $H^2 \geq H - (1-\gamma)H = \gamma H$. For any eigenvector with $H\psi = \lambda\psi$ and $\lambda > 0$, this gives $\lambda^2 \geq \gamma\lambda$, hence $\lambda \geq \gamma$ [KL18]. ☐

Lemma 18.48 (Martingale condition for MPS)**.** *Let A be injective and fix an interaction range $L \geq 2$ as in Definition 18.6. Let $h_L(A) := \Pi_{G_L(A)^\perp}$ be the L-site parent interaction, and for each chain length and site i let h_i denote its translate as in Definition 18.8. Then for every pair of overlapping terms h_i and h_j with $0 < d_N(i,j) < L$, where $d_N(i,j) := \min(|i-j|, N - |i-j|)$ is the cyclic distance on the ring (i.e. whose L-site supports share at least one site under periodic boundary conditions), the martingale condition of Theorem 18.47 holds with constants c_{ij} and $\gamma > 0$ depending only on A (not on N).* Note: *The proof below fully establishes the martingale constants only for $L = 2$; for $L > 2$ the existence of suitable constants is asserted without complete proof (see the remark at the end of the proof). Only the $L = 2$ case is used in this chapter (Theorem 18.49). (When $L = 1$ the interaction terms have disjoint supports, so there are no overlapping pairs and the spectral gap follows directly without the martingale machinery.)*

Proof. Two terms h_i, h_j with $0 < d_N(i,j) < L$ have supports overlapping in $L - d_N(i,j)$ sites (where d_N is the cyclic distance on the N-site ring). Since A is injective, the intersection property (Theorem 18.23) gives the qualitative identity $\ker(h_i) \cap \ker(h_j) = G_{[i,j+L-1]}(A)$. Because the local Hilbert space is finite-dimensional, the subspaces $\ker(h_i)$ and $\ker(h_j)$ live in a fixed finite-dimensional space. The intersection identity shows that $\ker(h_i) \cap \ker(h_j)$ is exactly $G_{[i,j+L-1]}(A)$; in particular there is no non-trivial vector in one kernel that is orthogonal to the other yet lies in both. Hence the Friedrichs angle $\theta_{d_N(i,j)}$ between $\ker(h_i)$ and $\ker(h_j)$ is strictly positive (for the fixed injective tensor A). Because the angle depends only on the overlap pattern $d_N(i,j) \in \{1,\dots,L-1\}$ and not on the absolute position, the number of distinct angles is at most $L-1$. Let $\alpha := \max_{1\le s<L} \cos\theta_s < 1$ be the worst-case cosine of the Friedrichs angles.

Row-sum bound. Since $L \ge 2$, each site i overlaps with at most $2(L-1)$ neighbours (those j with $0 < d_N(i,j) < L$). Set $c_{ij} := \frac{1}{2(L-1)}$ for every overlapping pair and $c_{ij} := 0$ otherwise. These constants depend only on the cyclic distance $d_N(i,j)$ and therefore satisfy $c_{ij} = c_{ji}$. Then $\sum_{j\neq i} c_{ij} \le 2(L-1)\cdot\frac{1}{2(L-1)} = 1$, satisfying the row-sum hypothesis of Theorem 18.47.

Positivity of γ. The Friedrichs angle bound gives, for each overlapping pair, $h_ih_j + h_jh_i \ge -\alpha\,(h_i+h_j)$. The martingale inequality holds with $\gamma = 1-2(L-1)\alpha$, which requires $c_{ij}(1-\gamma) = \frac{1}{2(L-1)}\cdot 2(L-1)\alpha = \alpha$ (matching the Friedrichs bound exactly). Since $\alpha < 1$ (the Friedrichs angle is strictly positive), it remains to show $\gamma > 0$, i.e. $\alpha < \frac{1}{2(L-1)}$. In the blocked picture of Theorem 18.49, $L = 2$ and each site overlaps with at most 2 neighbours, so the condition reduces to $\alpha < \frac{1}{2}$. For $L = 2$ with an injective tensor, the two projectors h_i and h_{i+1} act on a finite-dimensional space, and the intersection property (Theorem 18.23) guarantees that $\ker(h_i)\cap\ker(h_{i+1})$ is exactly $G_{[i,i+2]}(A)$; hence $\|P_iP_{i+1}\| < 1$ (the Friedrichs cosine is strictly less than 1). On a finite-dimensional space with two projectors, the Friedrichs cosine satisfies the quantitative bound $\alpha \le 1-c_{\min}$ where $c_{\min} > 0$ depends on the minimal principal angle between the two kernels. The bound $\alpha < \frac{1}{2}$ then follows from the Nachtergaele finite-size criterion [Nac96], which establishes that the overlap $\|P_iP_{i+1}\|$ for injective MPS is bounded away from 1 by a constant depending only on the tensor A. For general $L > 2$, the uniform constants $c_{ij} = \frac{1}{2(L-1)}$ make the condition $\alpha < \frac{1}{2(L-1)}$ increasingly restrictive as L grows. A more refined choice of separation-dependent constants $c_{ij} = c_{d_N(i,j)}$ that weight small-separation (large-overlap) pairs more heavily can exploit the exponential decay of the Friedrichs cosine α_s in s to satisfy the row-sum constraint. However, this refinement is not needed for the main result: in Theorem 18.49 below, the argument is applied only with $L = 2$ in the blocked picture. In all cases, the constants depend only on A and are independent of N [FNW92, KL18]. ☐

Theorem 18.49 (Spectral gap for MPS parent Hamiltonians)**.** *For an L_0-block injective MPS tensor A, let $L := 2L_0$ be the interaction range (two blocked sites in the L_0-blocked picture). There exists $\gamma > 0$ such that for all $N \ge 2L$ with $L_0 \mid N$,*

$$\lambda^+_{\min}(H_N(A,L)) \ge \gamma,$$

i.e. the parent Hamiltonian is uniformly gapped.

Proof. Since A is L_0-block injective, the blocked tensor $A^{[L_0]}$ is injective with physical dimension d^{L_0} and bond dimension D. Each blocked site represents L_0 original sites, so the range-$L = 2L_0$ interaction on the original chain corresponds exactly to a range-2 interaction on the blocked chain (two blocked sites). Thus one passes to the blocked chain and applies the martingale estimate there with interaction range 2. *Remark on interaction range:* Theorem 18.35 establishes ground-state uniqueness at range $L_0{+}1$, whereas here we use range $2L_0$. The ground spaces are nested: increasing the interaction range can only shrink (or preserve) the ground space, so uniqueness at range $L_0{+}1$ implies uniqueness at range $2L_0$. The larger range is needed here so that, after L_0-blocking, each blocked interaction term spans exactly 2 blocked sites, enabling the martingale machinery with $L = 2$. The divisibility condition $L_0 \mid N$ ensures the blocking is compatible with the periodic chain. Apply Lemma 18.48 to the injective tensor $A^{[L_0]}$ with range 2 to obtain the martingale constants c_{ij} and $\gamma > 0$. These constants depend only on $A^{[L_0]}$ (hence only on A and L_0), not on N. Theorem 18.47 then gives the uniform gap $\lambda^+_{\min}(H_N) \ge \gamma$ [FNW92, Nac96]. ☐

18.7 Ground space of non-injective MPS

For a general (non-injective) MPS tensor in block-diagonal canonical form, the ground space is no longer one-dimensional. Instead, it is spanned by the MPVs of the individual normal blocks — the basis of normal tensors (BNT) from Chapter 13 [FNW92, PGVWC07].

Lemma 18.50 (Each BNT MPV lies in the Hamiltonian kernel)**.** *Let $A = \bigoplus_{k=1}^{g} \mu_k A_k$ be in canonical form with BNT separation (Definition 13.3), where $\{A_1, \dots, A_g\}$ is a basis of normal tensors (Definition 13.1). For $N \geq 2(L_0 + 1)$ (where L_0 is the maximum injectivity length of the blocks) and each $k = 1, \dots, g$,*

$$V^{(N)}(A_k) \in \ker(H_N(A, L_0{+}1)).$$

Proof. Each A_k is normal, so Lemma 18.13 (applied to the block-diagonal structure) gives

$$H_N(A, L_0{+}1)\, V^{(N)}(A_k) = 0$$

for every $k = 1, \dots, g$. Hence $V^{(N)}(A_k) \in \ker(H_N(A, L_0{+}1))$. □

Theorem 18.51 (Ground space equals span of BNT)**.** *Let $A = \bigoplus_{k=1}^{g} \mu_k A_k$ be in canonical form with BNT separation (Definition 13.3), where $\{A_1, \dots, A_g\}$ is a basis of normal tensors (Definition 13.1). For $N \geq 2(L_0 + 1)$ (where L_0 is the maximum injectivity length of the blocks), the ground space of the parent Hamiltonian satisfies*

$$\ker(H_N(A, L_0{+}1)) = \operatorname{span}\{V^{(N)}(A_k) : k = 1, \dots, g\}.$$

Proof. By Lemma 18.50 and linearity of the kernel, we have

$$\operatorname{span}\{V^{(N)}(A_k) : k = 1, \dots, g\} \subseteq \ker(H_N(A, L_0{+}1)).$$

For the reverse inclusion, take $\psi \in \ker(H_N(A, L_0{+}1))$. The block-diagonal structure of A gives, at the interaction range $L_0{+}1$ used in the Hamiltonian, the containment $G_{L_0+1}(A) \subseteq \sum_k G_{L_0+1}(A_k)$: any local ground state of A on $L_0{+}1$ sites is a sum of local ground states of the individual blocks, because the off-diagonal blocks vanish in the transfer map (the canonical form and block separation (Theorem 12.16, Chapter 12) ensure that distinct blocks have orthogonal images under Γ_{L_0+1} for $L_0{+}1 \geq L_0$, so the sum is in fact direct). Each normal block A_k has injectivity length $L_k \leq L_0$, so the L_k-blocked tensor $A_k^{[L_k]}$ is injective. Applying Theorem 18.23 to this injective blocked tensor (in unblocked coordinates, as in Theorem 18.35) and iterating across the chain for each block independently, and using Theorem 18.36 for each normal block, the component of ψ associated with block A_k must be proportional to $V^{(N)}(A_k)$. This gives the reverse inclusion [FNW92, PGVWC07]. □

Remark. The previous lemma and theorem are stated in terms of $\ker(H_N(A, L_0{+}1))$. An equivalent formulation uses the chain ground space (Definition 18.30); a bridge theorem equating these two objects is future work.

Chapter 19

Exponential Decay of Correlations

This chapter records the transfer-matrix expression for thermodynamic-limit correlators of a normal MPS and the resulting exponential decay estimate. The key mechanism is spectral: after removing the leading eigenvalue contribution, the connected part is governed by subleading eigenvalues of the transfer map.

When the MPS tensor generates a parent Hamiltonian (Chapter 18), the correlator quantifies spatial correlations in the ground state. The exponential decay rate is set by the spectral properties of the transfer map, i.e. by the modulus of its subleading eigenvalues.

The spectral analysis of the transfer map and the mixed transfer operator, including the eigenvalue bound $\rho(F_{AB}) \leq 1$, the strict spectral gap $\rho(F_{AB}) < 1$ for non-equivalent blocks, and the resulting overlap decay, is developed in Chapter 9. This chapter focuses on the *correlator* side: one-point and two-point expectations, their spectral decomposition, and quantitative bounds on decay rates.

19.1 Connected correlators from the transfer map

Definition 19.1 (One-point expectation)**.** Fix an MPS tensor and a right fixed point of its transfer map. Write the tensor as A and the fixed point as ρ^R. Under the hypotheses of Theorem 8.9, that fixed point exists and is unique up to scaling; we normalize it so that $\mathrm{tr}(\rho^R) = 1$. For a local observable $X \in M_D(\mathbb{C})$ define

$$\langle X_0 \rangle := \mathrm{tr}(X\rho^R).$$

Definition 19.2 (Two-point expectation at distance n)**.** For observables $X, Y \in M_D(\mathbb{C})$ and $n \geq 0$,

$$\langle X_0 Y_n \rangle := \mathrm{tr}\big(Y\, \mathcal{E}_A^n(X\rho^R)\big)\,.$$

Equivalently: insert X at site 0, propagate by $\mathcal{E}_A^n$, then insert Y and take the trace.

Definition 19.3 (Connected correlator)**.** The connected two-point function is

$$C(X, Y; n) := \langle X_0 Y_n \rangle - \langle X_0 \rangle \langle Y_0 \rangle.$$

19.2 Spectral expansion and exponential decay

Theorem 19.4 (Sum of exponentials)**.** *For a normal MPS, the connected correlator admits a spectral expansion of the form*

$$C(X,Y;n) = \sum_{j=1}^{D^2-1} c_j\,\lambda_j^n,$$

where λ_j are subleading transfer-matrix eigenvalues.

Proof. The leading eigenvalue contribution ($\lambda_1 = 1$) equals $\langle X_0\rangle\langle Y_0\rangle$ and cancels in the connected part. Apply spectral decomposition to $\mathcal{E}_A^n$. □

Theorem 19.5 (Exponential decay bound)**.** *If $|\lambda_j| \le |\lambda_2| < 1$ for all subleading eigenvalues, then there exists $C_{XY} \ge 0$ such that*

$$|C(X,Y;n)| \le C_{XY}\,|\lambda_2|^n.$$

Proof. Combine the sum-of-exponentials expansion with the triangle inequality. □

Remark 19.6 (Spectral hypothesis and the spectral gap). The hypothesis "$|\lambda_2| < 1$" in Theorem 19.5 is precisely the *spectral gap* condition for the transfer map $\mathcal{E}_A$. For primitive channels, the complementary spectral gap $\rho(\mathcal{E}_A - P) < 1$ is established in Theorem 5.39 (Chapter 5); for irreducible trace-preserving tensors, the analogous result is Theorem 9.26. Thus the exponential decay of connected correlations follows directly from the block separation theory once the spectral decomposition is in hand.

Definition 19.7 (Correlation length)**.** The correlation length associated with λ_2 is

$$\xi := -\frac{1}{\log|\lambda_2|}.$$

Lemma 19.8 (Correlation length is positive)**.** *If $|\lambda_2| < 1$ and $\lambda_2 \neq 0$, then $\xi > 0$.*

Proof. Since $0 < |\lambda_2| < 1$, we have $\log|\lambda_2| < 0$, so $\xi = -1/\log|\lambda_2| > 0$. □

19.3 Quantitative spectral gap bounds

The spectral gap theorems in Chapter 5 establish $\rho(\mathcal{E}_A - P) < 1$ for primitive channels qualitatively. This section records the *quantitative* refinements that give explicit constants and rates for the single-tensor transfer map $\mathcal{E}_A$.

Theorem 19.9 (Exponential convergence of injective primitive channels)**.** *Fix an injective, trace-preserving MPS tensor A with positive definite fixed point ρ. Let*

$$P(X) = \frac{\mathrm{tr}(X)}{\mathrm{tr}(\rho)}\rho$$

be the corresponding fixed-point projection. Then there exist $C > 0$ and $0 < \delta \le 1$ such that for all $n \ge 0$ and $X \in M_D(\mathbb{C})$,

$$\|\mathcal{E}_A^n(X) - P(X)\| \;\le\; C\,(1-\delta)^n\,\|X\|.$$

The spectral gap δ exists by primitivity (Theorem 5.39, the complementary spectral gap).

Proof. Injectivity implies primitivity of the transfer map. The complementary spectral radius $\rho(\mathcal{E}_A - P) < 1$ then follows from the primitive spectral gap theorem. The Gelfand formula yields a geometric bound $\|(\mathcal{E}_A - P)^n\| \leq C\,(1-\delta)^n$ for suitable constants, giving the claimed estimate. □

Theorem 19.10 (Correlation length bound)**.** *For an injective trace-preserving MPS tensor, there exist $C > 0$ and $\xi > 0$ such that for all $n \geq 0$ and all traceless $X \in M_D(\mathbb{C})$ (i.e. $\mathrm{tr}(X) = 0$),*

$$\|\mathcal{E}_A^n(X)\| \;\leq\; C\,e^{-n/\xi}\,\|X\|.$$

Traceless matrices lie in $\ker P$*, where P is the fixed-point projection. Since $\mathcal{E}_A - P$ has spectral radius strictly less than* 1*, the Gelfand formula gives exponential decay at rate $\xi = -1/\log\rho(\mathcal{E}_A - P)$.*

Proof. For traceless X we have $P(X) = 0$, so $\mathcal{E}_A^n(X) = (\mathcal{E}_A - P)^n(X)$. The exponential convergence theorem gives a geometric bound $C\,(1-\delta)^n\,\|X\|$. Rewriting $(1-\delta)^n = e^{-n/\xi}$ with $\xi = -1/\log(1-\delta)$ yields the claimed estimate. □

Theorem 19.11 (Spectral gap from injectivity)**.** *For an injective trace-preserving MPS tensor, there exists $\delta > 0$ such that every eigenvalue $\mu \neq 1$ of the transfer map satisfies $|\mu| \leq 1 - \delta$.*

Proof. Injectivity implies eventual full Kraus rank. This yields primitivity, hence a spectral gap for the complementary map. Since the transfer map has only finitely many eigenvalues, one may choose a uniform $\delta > 0$ separating 1 from the remaining spectrum. □

19.4 Relation to parent Hamiltonians

Remark 19.12. When the MPS tensor A generates a parent Hamiltonian $H_N(A, L)$ (Definition 18.9) whose ground state is the matrix product vector (Lemma 18.13), the exponential decay bound (Theorem 19.5) shows that ground-state correlations decay with a finite correlation length ξ.

19.5 Renormalization fixed points and zero correlation length

Definition 19.13 (Renormalization fixed point)**.** An MPS tensor A is a *renormalization fixed point* (RFP) when its transfer map is idempotent, $\mathcal{E}_A \circ \mathcal{E}_A = \mathcal{E}_A$. See [CPGSV21, Definition 3.2].

Definition 19.14 (Local orthogonality)**.** A single BNT block is *locally orthogonal* when its self-transfer map is idempotent. For a single tensor this coincides with the RFP condition (Definition 19.13); in the full multi-block BNT setting, local orthogonality additionally requires that mixed transfer operators vanish. See [CPGSV21, Definition 3.5].

Theorem 19.15 (RG flow convergence for canonical-form tensors)**.** *For a tensor in canonical form with blocks (A_k) and weights (μ_k), the iterated blocking $\mathcal{E}_{A_k}^{2^n}$ converges pointwise to an idempotent linear map $\mathcal{E}_\infty$ as $n \to \infty$. That is, there exists $\mathcal{E}_\infty$ with $\mathcal{E}_\infty^2 = \mathcal{E}_\infty$ such that $\mathcal{E}_{A_k}^{2^n}(\rho) \to \mathcal{E}_\infty(\rho)$ for all density matrices ρ. This is [CPGSV21, Appendix B].*

Proof. Each block is injective and left-canonical, so its transfer map is a primitive channel with a unique positive-definite fixed point ρ_0. The exponential convergence bound $\|\mathcal{E}^n(X) - P(X)\| \leq C(1-\delta)^n\|X\|$ (Theorem 19.9) then gives pointwise convergence $\mathcal{E}^n(X) \to P(X)$, and composing with the subsequence $2^n \to \infty$ yields the result. The witness $\mathcal{E}_\infty = P$ is the rank-one fixed-point projection, which is idempotent. □

Remark 19.16. When the transfer map is idempotent ($\mathcal{E}^2 = \mathcal{E}$), connected correlations become separation-independent for all separations $n \geq 1$. Informally, this corresponds to zero correlation length ($\xi = 0$). The full spectral equivalence (idempotence iff vanishing subleading spectrum) is not yet formalized.

Definition 19.17 (Correlations independent of distance)**.** For a positive definite right fixed point $\rho^R > 0$, an MPS tensor has *correlations independent of distance* (CID) when for all local observables X and Y and all separations $n, m \geq 1$,

$$C(X, Y; n) = C(X, Y; m).$$

Theorem 19.18 (CID implies RFP)**.** *If an MPS tensor admits a positive definite right fixed point of its transfer map and has correlations independent of distance, then its transfer map is idempotent. This is the reverse direction of [CPGSV21, Theorem 3.8].*

Proof. Trace nondegeneracy and invertibility of the fixed point reduce idempotence to equality of the connected correlator at separations $n = 2$ and $n = 1$. ☐

Theorem 19.19 (Zero correlation length iff idempotent transfer)**.** *An MPS tensor has zero correlation length (i.e. is both locally orthogonal and CID) if and only if its transfer map is idempotent. This is [CPGSV21, Theorem 3.8].*

Proof. The forward direction extracts local orthogonality (which equals idempotence by definition). The reverse uses $\mathcal{E}^n = \mathcal{E}$ for $n \geq 1$ to show the connected correlator is separation-independent. ☐

Chapter 20

Concrete Examples

This chapter collects concrete MPS tensors that illustrate the general theory developed in earlier chapters. Each example is specified by its physical dimension d, bond dimension D, and explicit matrix family $\{A^i\}$. Key structural properties (injectivity, RFP status, symmetry) are stated as propositions linking back to the general definitions.

References follow [CPGSV21], Section III.

20.1 The GHZ state

The Greenberger–Horne–Zeilinger (GHZ) state is the prototypical non-injective, renormalization-fixed-point MPS.

Definition 20.1 (GHZ tensor)**.** Set $d = D = 2$. The GHZ tensor is the family $\{A^0, A^1\} \subset M_2(\mathbb{C})$ defined by

$$A^0 = |0\rangle\langle 0| = \mathrm{diag}(1, 0), \qquad A^1 = |1\rangle\langle 1| = \mathrm{diag}(0, 1).$$

For a system of N sites the resulting MPV is the GHZ state

$$|\mathrm{GHZ}_N\rangle = |0\rangle^{\otimes N} + |1\rangle^{\otimes N}.$$

Theorem 20.2 (GHZ transfer map is idempotent)**.** *The transfer map of the GHZ tensor satisfies* $\mathcal{E}_A^2 = \mathcal{E}_A$.

Proof. The transfer map acts by $\mathcal{E}_A(X)_{ij} = \delta_{ij} X_{ij}$ (extraction of the diagonal). This operation is manifestly idempotent. □

Theorem 20.3 (GHZ tensor is an RFP)**.** *The GHZ tensor is a renormalization fixed point.*

Proof. Immediate from Theorem 20.2. □

Theorem 20.4 (GHZ tensor is not injective)**.** *The GHZ tensor is not injective:* $\mathrm{span}_{\mathbb{C}}\{A^0, A^1\}$ *is the two-dimensional space of diagonal matrices in* $M_2(\mathbb{C})$*, which does not equal* $M_2(\mathbb{C})$*.*

Proof. Every element of $\mathrm{span}_{\mathbb{C}}\{A^0, A^1\}$ has zero off-diagonal entries, so the matrix with all entries equal to 1 is not in the span. □

Theorem 20.5 ($\mathbb{Z}_2$ on-site symmetry of the GHZ tensor)**.** *Let* $\mathbb{Z}_2 = \{1, \sigma_x\}$ *act on* $\mathbb{C}^2$ *by the Pauli-X matrix* σ_x*. The GHZ tensor is on-site symmetric under this action: for each* $g \in \mathbb{Z}_2$*, the twisted tensor* $\widetilde{A}_g$ *defines the same MPV family as* A*, i.e.* $\mathcal{V}(A) = \mathcal{V}(\widetilde{A}_g)$*.*

Proof. The identity element is trivial. For the generator $g = \sigma_x$, the twisted tensor satisfies $\widetilde{A}^i_g = A^{1-i}$, i.e. the two matrices are swapped. Conjugation by σ_x implements the same swap, giving gauge equivalence and hence the same MPV family. $\square$

20.2 The AKLT state

The Affleck–Kennedy–Lieb–Tasaki (AKLT) state is the canonical example of a normal (block-injective) MPS with non-trivial symmetry-protected topological (SPT) order.

Definition 20.6 (AKLT tensor)**.** Set $d = 3$ (spin-1) and $D = 2$. The AKLT tensor is the family $\{A^0, A^1, A^2\} \subset M_2(\mathbb{C})$ defined via the Pauli matrices by

$$A^0 = \frac{1}{\sqrt{3}}\,\sigma_z, \qquad A^1 = \frac{\sqrt{2}}{\sqrt{3}}\,|0\rangle\langle 1|, \qquad A^2 = -\frac{\sqrt{2}}{\sqrt{3}}\,|1\rangle\langle 0|,$$

where $\sigma_z = |0\rangle\langle 0| - |1\rangle\langle 1|$, so equivalently $A^1 = \frac{1}{\sqrt{3}}\,\sigma_+$ and $A^2 = -\frac{1}{\sqrt{3}}\,\sigma_-$ for $\sigma_+ = \sqrt{2}\,|0\rangle\langle 1|$ and $\sigma_- = \sqrt{2}\,|1\rangle\langle 0|$. Equivalently, the matrices are the Clebsch–Gordan coefficients coupling spin-1 and spin-$\frac{1}{2}$ to spin-$\frac{1}{2}$.

Theorem 20.7 (AKLT tensor is not injective)**.** *The AKLT tensor is not injective at a single site:* $\mathrm{span}_{\mathbb{C}}\{A^0, A^1, A^2\}$ *is the three-dimensional space of traceless matrices* $\mathfrak{sl}(2, \mathbb{C})$*, which is a proper subspace of* $M_2(\mathbb{C})$*.*

Theorem 20.8 (AKLT tensor is normal)**.** *The transfer map of the AKLT tensor has a unique eigenvalue of maximal modulus (a simple spectral-radius eigenvalue); in particular the AKLT tensor is normal. Concretely, the length-*2 *blocked tensor is injective.*

Theorem 20.9 ($\mathrm{SO}(3)$ on-site symmetry of the AKLT tensor)**.** *The AKLT tensor is on-site symmetric under the spin-*1 *representation of* $\mathrm{SO}(3)$ *(the spin-*1 *representation of* $\mathrm{SU}(2)$ *factors through* $\mathrm{SO}(3)$*), with the virtual representation being the projective spin-*$\frac{1}{2}$ *representation. The projective phase is the non-trivial element of* $H^2(\mathrm{SO}(3), \mathrm{U}(1)) \cong \mathbb{Z}_2$*, witnessing a non-trivial SPT phase. The* $\mathbb{Z}_2$ *subgroup symmetry is formalized; the full* $\mathrm{SO}(3)$ *projective representation is deferred.*

20.3 The cluster state

The cluster state is a universal resource for measurement-based quantum computation and provides another example of a non-trivial SPT phase.

Definition 20.10 (Cluster-state tensor)**.** Set $d = D = 2$. The cluster-state tensor is the family $\{A^0, A^1\} \subset M_2(\mathbb{C})$ defined by

$$A^0 = |+\rangle\langle 0| = \frac{1}{\sqrt{2}}\begin{pmatrix} 1 & 0 \\ 1 & 0 \end{pmatrix}, \qquad A^1 = |-\rangle\langle 1| = \frac{1}{\sqrt{2}}\begin{pmatrix} 0 & 1 \\ 0 & -1 \end{pmatrix},$$

where $|\pm\rangle = \frac{1}{\sqrt{2}}(|0\rangle \pm |1\rangle)$.

Theorem 20.11 (Cluster-state tensor is not injective)**.** *The cluster-state tensor is not injective:* $\mathrm{span}_{\mathbb{C}}\{A^0, A^1\} = \mathrm{span}_{\mathbb{C}}\{|+\rangle\langle 0|, |-\rangle\langle 1|\}$ *has dimension* 2*, not* $\dim M_2(\mathbb{C}) = 4$*.*

Proof. Both A^0 and A^1 are rank-one matrices with linearly independent column spaces, so they are linearly independent. Their span is two-dimensional, which is strictly less than $\dim M_2(\mathbb{C}) = 4$. $\square$

Remark 20.12. Although the cluster-state tensor is not injective at length 1, the length-2 blocked tensor is injective. Using $\langle 0|\pm\rangle = \frac{1}{\sqrt{2}}$, $\langle 1|+\rangle = \frac{1}{\sqrt{2}}$, and $\langle 1|-\rangle = -\frac{1}{\sqrt{2}}$, the four products $A^{ij} = A^i A^j$ are

$$A^{00} = \tfrac{1}{\sqrt{2}}|+\rangle\langle 0|, \quad A^{01} = \tfrac{1}{\sqrt{2}}|+\rangle\langle 1|, \quad A^{10} = \tfrac{1}{\sqrt{2}}|-\rangle\langle 0|, \quad A^{11} = -\tfrac{1}{\sqrt{2}}|-\rangle\langle 1|.$$

Since $\{|+\rangle, |-\rangle\}$ and $\{\langle 0|, \langle 1|\}$ are bases, these four rank-one matrices are linearly independent and span $M_2(\mathbb{C})$.

Theorem 20.13 ($\mathbb{Z}_2 \times \mathbb{Z}_2$ on-site symmetry of the blocked cluster-state tensor)**.** *The length-*2 *blocked cluster-state tensor is on-site symmetric under $\mathbb{Z}_2 \times \mathbb{Z}_2$. The two generators act on the* 4*-dimensional blocked physical space $(\mathbb{C}^2)^{\otimes 2}$ by $\sigma_x \otimes I$ and $I \otimes \sigma_x$, which commute and each square to the identity, defining a linear representation of $\mathbb{Z}_2 \times \mathbb{Z}_2$. The corresponding virtual matrices are $V_1 = \sigma_z$ and $V_2 = \sigma_x$, which anticommute ($V_1 V_2 = -V_2 V_1$). This projective phase is the non-trivial element of $H^2(\mathbb{Z}_2 \times \mathbb{Z}_2, \mathrm{U}(1)) \cong \mathbb{Z}_2$, witnessing a non-trivial SPT phase.*

Bibliography

[CGW11] Xie Chen, Zheng-Cheng Gu, and Xiao-Gang Wen. Classification of gapped symmetric phases in one-dimensional spin systems. *Physical Review B*, 83(3):035107, 2011.

[CPGSV17a] J. I. Cirac, D. Pérez-García, N. Schuch, and F. Verstraete. Matrix product density operators: Renormalization fixed points and boundary theories. *Ann. Phys.*, 378:100–149, March 2017.

[CPGSV17b] J. Ignacio Cirac, David Pérez-García, Norbert Schuch, and Frank Verstraete. Matrix product unitaries: Structure, symmetries, and topological invariants. *Journal of Statistical Mechanics: Theory and Experiment*, 2017(8):083105, 2017.

[CPGSV21] J. Ignacio Cirac, David Pérez-García, Norbert Schuch, and Frank Verstraete. Matrix product states and projected entangled pair states: Concepts, symmetries, theorems. *Reviews of Modern Physics*, 93(4):045003, 2021.

[DlCCSPG17] Gemma De las Cuevas, J. Ignacio Cirac, Norbert Schuch, and David Pérez-García. Irreducible forms of matrix product states: Theory and applications. *Journal of Mathematical Physics*, 58(12):121901, 2017.

[EHK78] David E. Evans and Raphael Høegh-Krohn. Spectral properties of positive maps on C^*-algebras. *Journal of the London Mathematical Society*, 17(2):345–355, 1978.

[FNW92] M. Fannes, B. Nachtergaele, and R. F. Werner. Finitely correlated states on quantum spin chains. *Communications in Mathematical Physics*, 144(3):443–490, 1992.

[Jac09] Nathan Jacobson. *Basic Algebra II.* Dover Publications, 2009. Unaltered reprint of the 2nd edition (W. H. Freeman, 1989).

[KL18] Michael J. Kastoryano and Angelo Lucia. Divide and conquer method for proving gaps of frustration free spin systems. *Journal of Statistical Mechanics: Theory and Experiment*, 2018(3):033105, 2018.

[MGRPG+18] András Molnár, José Garre-Rubio, David Pérez-García, Norbert Schuch, and J. Ignacio Cirac. Normal projected entangled pair states generating the same state. *New Journal of Physics*, 20(11):113017, 2018.

[Nac96] Bruno Nachtergaele. The spectral gap for some spin chains with discrete symmetry breaking. *Communications in Mathematical Physics*, 175(3):565–606, 1996.

[PGVWC07] D. Pérez-García, F. Verstraete, M. M. Wolf, and J. I. Cirac. Matrix product state representations. *Quantum Information & Computation*, 7(5):401–430, 2007.

[PGWS+08] D. Pérez-García, M. M. Wolf, M. Sanz, F. Verstraete, and J. I. Cirac. String order and symmetries in quantum spin lattices. *Physical Review Letters*, 100(16):167202, 2008.

[SPGWC10] M. Sanz, D. Pérez-García, M. M. Wolf, and J. I. Cirac. A quantum version of Wielandt's inequality. *IEEE Transactions on Information Theory*, 56(9):4668–4673, 2010.

[Wol12] Michael M. Wolf. Quantum channels & operations: Guided tour. Lecture notes, 2012. Originally distributed via TU Munich foswiki; archival PDF at https://mediatum.ub.tum.de/doc/1099654/1099654.pdf.